STATE OF NEW YORK
PUBLIC SERVICE COMMISSION

Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of The Brooklyn Union Gas Company d/b/a National Grid NY for Gas Service	**)**	**Case 19-G-0309**
Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of KeySpan Gas East Corp. d/b/a National Grid for Gas Service	**)**	**Case 19-G-0310**

DIRECT TESTIMONY

OF

GREGORY LANDER

ON BEHALF OF

ENVIRONMENTAL DEFENSE FUND

Dated: August 30, 2019

I. Introduction and Qualifications

Q. Please state your name and business address.

A. My name is Gregory M. Lander. My business address is 83 Pine Street, Suite 101, West Peabody, MA 01960.

Q. What is your occupation and by whom are you employed?

A. I am President of Skipping Stone, LLC ("Skipping Stone").

Q. Please state your educational background and experience.

A. I graduated from Hampshire College in Amherst, Massachusetts, in 1977, with a Bachelor of Arts degree. In 1981, I began my career in the energy business at Citizens Energy Corporation in Boston, Massachusetts ("Citizens Energy"). I became involved in the natural gas business of Citizens Energy in 1983. Between 1983 and 1989, I served as Manager, Vice President, President and Chairman of Citizens Gas Supply Corporation (a subsidiary of Citizens Energy).

I started and ran an energy consulting firm, Landmark Associates, from 1989 to 1993, during which time I consulted on numerous pipeline open access matters, a number of Federal Energy Regulatory Commission ("FERC") Order No. 636 rate cases, pipeline certificate cases, fuel supply and gas transportation issues for independent power generation projects, international arbitration cases involving renegotiation of pipeline gas supply contracts, and natural gas market information requirements cases (FERC Order Nos. 587 *et seq.*). In 1993, I founded TransCapacity LP, a software and natural gas information services company.

Since 1994, I have also been a Services Segment board member of the Gas Industry Standards Board ("GISB") and its successor organization, the North American Energy Standards Board ("NAESB"). During the period 1994 to 2002, I served as a Chairman of the Business Practices Subcommittee, the Interpretations Committee, the Triage Committee, and several GISB/NAESB Task Forces. I am currently a Board Member of NAESB and have served continuously in that capacity since 1997.

Skipping Stone, Inc. acquired TransCapacity in 1999, and since that time I have headed up Skipping Stone's Energy Logistics practice, where my specialization has been interstate pipeline capacity issues, information, research, pricing, acquisition due diligence and planning. In 2001, Skipping Stone launched CapacityCenter.com, a pipeline capacity information service. In 2004, Skipping Stone was acquired by Commerce Energy Group, a national retail energy services provider. In 2005, I was appointed President of Skipping Stone, which operated as a wholly owned subsidiary of Commerce Energy Group. In 2008, I purchased substantially all of the assets of Skipping Stone and now operate essentially the same business as before the Commerce Energy transaction as Skipping Stone, LLC.

From 1984 to present, I have maintained a deep familiarity with a wide range of pipeline transportation issues, beginning with access to pipeline capacity to make competitive sales, resolution of the pipeline take-or-pay contracting regime, pipeline affiliate marketer concerns, restructuring of the pipelines from merchants

to transporters and thereafter, and definitions of what constituted a pipeline capacity "right" for the purposes of formulating the then newly commenced capacity release and capacity rights trading business process. I continue to be involved in nearly all facets of the capacity information and trading business as part of my duties at Skipping Stone. In addition, I have been the lead principal on all 50+ pipeline and storage mergers and acquisitions transactions as well as all pipeline and storage facility expansion projects for which Skipping Stone has been retained by potential purchasers and project sponsors to provide economic due diligence consulting and market analysis. In addition, I have testified before, participated in or assisted with proceedings before, state public utilities commissions and/or their staffs in the states of Maine, Massachusetts, Missouri, Virginia, South Carolina, California, Rhode Island, New York and New Jersey with respect to infrastructure matters, integrated resource plans, and fuel cost recovery proceedings. Please refer to Exhibit __ (GL-1), which contains my current CV.

Q. Have you previously filed testimony before regulatory commissions?

A. I have filed testimony before the New York Public Service Commission ("Commission"), Massachusetts Department of Public Utilities, the Maine Public Utilities Commission, the Virginia State Corporation Commission, the Missouri Public Service Commission, the California Public Utilities Commission, and the South Carolina Public Service Commission. I have also filed testimony in several

FERC proceedings. Please refer to Exhibit __ (GL-1), which contains a full list of case names and docket numbers in which I have participated as a witness.

Q. On whose behalf are you testifying in this proceeding?

A. I am submitting testimony on behalf of the Environmental Defense Fund ("EDF").

II. Purpose of Testimony and Recommendations

Q. What is the purpose of your testimony?

A. The purpose of my testimony is to provide recommendations regarding how the gas supply planning process of The Brooklyn Union Gas Company d/b/a National Grid NY ("KEDNY") and KeySpan Gas East Corporation d/b/a National Grid ("KEDLI") (collectively, "Companies") could be improved.

Q. Please provide a summary of your testimony and recommendations.

A. My testimony first details the changed market conditions and new developments that compel the need for revisions to the Companies' gas supply planning process. I then provide an overview of the current gas supply planning process in New York and compare this process to other states. To improve the Companies' gas supply planning process, I propose three recommendations. I first suggest that the Companies should be subject to an annual gas supply process, with discovery rights for intervenors and technical conferences as needed. I next suggest that the Companies should be required to submit a long-range plan in a format, which would become the basis for evaluating future cost recovery. The long-range plan would set forth projections of demand and resources to meet that demand,

identifying the "all-in" cost for each resource. Finally, I suggest additional categories of information the Companies should submit in order to assist the Commission in its review of gas supply issues. Taken together, I explain how these refinements will help identify potential issues well in advance of experiencing demand/supply mismatches; manage the fixed costs commitments made by the Companies; and provide a more thorough framework for the Companies to consider alternatives that would have lower all-in costs to customers.

Q. Was your testimony prepared by you or under your supervision?

A Yes.

Q. Are you sponsoring any exhibits?

A. Yes, I am sponsoring the following exhibits:

Exhibit __ (GL-1): CV and List of Expert Testimony of Gregory M. Lander

Exhibit __ (GL-2): Winter Supply Review Data Request; Case No. 18-M-0272

Exhibit __ (GL-3): Rhode Island Joint Memorandum; Docket 4816

Exhibit __ (GL-4): Con Edison Response to EDF 6-48; Case No. 19-G-0066 *et al.*

Exhibit __ (GL-5): Con Edison Response to EDF 6-50; Case No. 19-G-0066 *et al.*

III. Need for Revisions

Q. Have there been changes in market conditions over the last decade that have affected natural gas utilities?

A. Yes. Over the past decade, two major trends have marked the evolution of the natural gas business. These two major trends, viewed generally, have had and will

continue to have profound impacts on gas distribution utilities, especially in the northeastern United States. The first of these trends is the explosive growth in natural gas supply from the Marcellus and Utica shales in the regions of northeastern and southwestern Pennsylvania, southeastern Ohio and northern West Virginia. The second is the dramatic increase in gas-fired electric generation, especially in Ohio, Pennsylvania, New Jersey, New York and New England. The second of these trends, driven by plentiful, low-priced natural gas has placed strains on the legacy pipeline infrastructure serving the same region. The dramatic growth in gas-fired generation has had a particular impact on the legacy pipeline infrastructure especially during winter periods when both domestic gas fueled heating needs and electric demands have placed simultaneous demands on pipelines. In the past, Local Distribution Companies ("LDCs") like the Companies were the primary users of pipelines during the winter. Now, LDCs and generators seek to use the same pipelines at the same hours of the day. As a result, the flexibility that the LDCs used to have with respect to hourly takes is being much more frequently restricted by the pipelines to the LDCs' specified contract rights. At the same time, because the loads of electric generators can be as large as (or larger) than many of the LDCs' take stations for whole swaths of their distribution territories, these combined demands are a new and growing factor that the industry has to cope with. Thus, LDC capacity planning that previously may have viewed its planning criteria as a daily load, now has to increasingly plan for peak hours of many more days in the winter than its "design

day." While in the past LDCs' on-system peaking resources may have been used, or planned to be used, during three to twenty days per winter, now may need to be called upon throughout the winter to meet hourly demands that are in excess of the LDCs' hourly contract rights. The significance of this is that even though the hours of use of on-system resources may increase, when looked at from an annual utilization basis, may not exceed 3% to 6% of 8760 hours per year (i.e., 250 to 500 hours per year). Given such low load factors and given the cost of incremental pipeline capacity, it simply may not make sense to commit to long-term capacity contracts in light of significant costs, the State's climate policies, and the Commission's findings on non-pipeline alternatives.[1]

Q. Please explain the consequences of this market evolution.

A. In addition to the tightening of pipeline operations which have reduced LDCs' previous flexibility to take gas at a greater hourly rate than their strict contract entitlements, producers and marketers have subscribed to large amounts of capacity from supply areas both to and past large metropolitan demand centers, and, in many cases, all the way from the supply areas to the U.S. Gulf coast to supply Liquefied Natural Gas ("LNG") export facilities. This increase in pipeline capacity is a resource, which, if it can flow into (as opposed to past), LDC distribution locations can be contracted for by LDCs based upon a price which

[1] *Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Central Hudson Gas & Electric Corporation for Gas Service*, Order Adopting Terms of Joint Proposal and Establishing Electric and Gas Rate Plan, Case No. 17-G-0460 *et al.* at page 75 (June 14, 2018) (finding that that non-pipeline alternatives should be "explored as a universal practice as an alternative to traditional investments.").

makes it in the supplier's interest to supply the LDC instead of taking the gas to export. Such arrangements would have the effect of reducing the need to build additional capacity from supply areas to market, and instead provide and plan for the capacity to exist along the "last mile" (i.e., into and from the LDC take stations). These combined facts—LDC short duration needs coupled with existing capacity moving gas past the LDC demand centers—present an opportunity to view capacity planning in a different way than ever before. Thus, a portfolio approach, a combination of on-system resources to meet needle hourly peaks plus contracted supply otherwise bound for export, have the effect of making LDC capacity planning a multi-dimensional exercise.

Q. Has National Grid acknowledged the need for refinements to its gas supply planning in other jurisdictions?

A. Yes. In a February 20, 2019 Joint Memorandum submitted to the Rhode Island Public Utilities Commission, National Grid stated:

> In the past, the [Long Range Plan] filings were not controversial and tended to raise few complicated issues. But now, the Company needs to plan in a way that assures adequate capacity and delivery security under supply contracts, the magnitude and implications of which have grown substantially. As a result, the current framework and template for the Company's long-range planning is no longer sufficient for an appropriate regulatory review.[2]

Q. Would you agree that these same concerns are relevant to National Grid's New York utilities?

A. Yes, the two trends I detailed above have profound impacts on gas distribution utilities, both in the northeastern United States and in New York State.

[2] Exhibit__(GL-3) at page 2.

Q. Has the Commission recognized the need to review and assess policies, programs, and regulations in light of these changes?

A. Yes. In a January 28, 2019 Statement, Chair John Rhodes directed DPS Staff to "develop a report to review the changing market conditions that gave rise to Con Edison's decision [to institute a moratorium] and to develop recommendations to make certain that utilities across the state are able to meet customer needs in a manner consistent with the State's energy goals."[3] The Statement also acknowledges the need for utilities to help consumers conserve energy and to solicit for natural gas supply alternatives. My recommendations set forth below will help the Commission and the Companies satisfy these objectives.

Q. Have there been any other recent developments in New York that compel utilities to revisit their gas supply planning?

A. Yes. On July 18, 2019, Governor Cuomo signed into law the Climate Leadership and Community Protection Act.[4] The Act requires a reduction in statewide greenhouse gas emissions by 85 percent by 2050 from 1990 levels, with an additional goal of achieving net zero emissions across all sectors of the economy by that time. The Act provides that in issuing approvals and decisions, *all* State agencies, offices, authorities, and divisions "shall consider whether such decisions

[3] *In the Matter of Staff Investigation into a Moratorium on New Natural Gas Services in the Consolidated Edison Company of New York, Inc. Service Territory*, Case No. 19-G-0080, Statement from Public Service Commission Chair John B. Rhodes on Consolidated Edison's Decision to Stop Accepting New Gas Customers on a Temporary Basis in Westchester County (January 28, 2019).

[4] New York State Climate Leadership and Community Protection Act (CLCPA), https://legislation.nysenate.gov/pdf/bills/2019/S6599.

are inconsistent with or will interfere with the attainment of the statewide greenhouse gas emissions limits established in Article 75 of the Environmental Conservation Law…."[5]

Q. Please explain the impact of this new legislation on gas supply planning.

A. This Act underscores the importance of thoughtful, transparent, and robust planning. As New York proceeds forward on its path outlined in the legislation there is a need for a transparent demonstration of the true demands of the system and the all-in costs of meeting that demand with various resources, being mindful not to lock-in greenhouse gas emissions from unnecessary long-lived and possibly stranded infrastructure.

IV. Suggested Improvements to Gas Supply Planning

Q. Please provide an overview of the current gas supply planning process in New York.

A. It is my understanding that the Commission opens a docket every year in order to examine each gas utility's supply plans. Each utility provides information regarding their portfolio and purchasing strategies, anticipated portfolio changes, gas supply portfolio details, among other issues.

Q. Have you reviewed the Companies' gas supply submittal in Case No. 18-M-0272?

A. Yes, I have reviewed the Companies' July 16, 2018 response to Staff's Winter Supply Review Data Request, attached to my testimony as Exhibit __ (GL-2), and

[5] CLCPA, Section 7(2).

the various updates and reports submitted by the Companies in that same proceeding. In addition, I have reviewed the testimony of Elizabeth Arangio submitted in these proceedings.

Q. Do you have any observations regarding these annual gas supply review proceedings?

A. Yes. Although the Commission has broad discretion to review the long-range programs of gas utilities,[6] the Commission historically has not made any formal public findings regarding the sufficiency of each gas utility's supply plan in these proceedings. Rather, it seems to be just a process between the utility and DPS Staff. Moreover, they are not formatted in such a manner as would make it possible to compare what is set forth in these planning documents and what customers ultimately will be asked to pay. Intervenors historically have not had discovery rights in these proceedings and there are no transcribed, on the record, technical conferences, let alone a formal evidentiary hearing.

Q. How does this process in New York compare to other state commissions' review of gas utility supply plans, of which you are aware?

A. At present, based upon my knowledge, there is no "perfect" state process. However, there are several states which require submissions of data, which if combined, come close to the type of transparency that could and should be the hallmark of an ideal process. For instance, in North Carolina, Piedmont Natural

[6] Public Service Law § 5(2) (the Commission has broad discretion to "encourage all persons and corporations subject to its jurisdiction to formulate and carry out long-range programs, individually or cooperatively, for the performance of their public service responsibilities with economy, efficiency, and care for the public safety….").

Gas files historic and projected load duration curves and against such curves presents its "resource stack" of pipeline capacity and on-system supplementals (e.g., LNG and CNG) to demonstrate its resource sufficiency.[7] In Massachusetts, the utilities present design day demand (net of conservation and energy efficiency) and against that they present their contractual[8] and on-system resource stack and identify surplus or deficit conditions with respect to that matching of forecasted demand to contracted resources.[9] In addition, many utilities that I am aware of also conduct post-winter analyses of peak hourly takes at the take station level and compare these experienced takes to their system model's predictions of takes under the experienced degree day conditions. In many cases this "post-action" analysis coupled with future demand growth projections identifies potential hourly demand versus hourly contracted resource deficits.

In Rhode Island, the Division of Public Utilities and Carriers (i.e., the Staff) and Narragansett Electric Company (a National Grid company) have proposed to

---

7 Annual Review of Gas Costs Pursuant to G.S. 62-133.4(c) and Commission Rule R1-17(k)(6), North Carolina Utilities Commission Docket No. G-9, Sub 727, Testimony and Exhibits of Gennifer Raney on behalf of Piedmont Natural Gas Company, Inc. (August 1, 2018), https://starw1.ncuc.net/NCUC/ViewFile.aspx?Id=feb95b8f-afe1-4fab-8040-edd252c431a3.

8 Here, contractual includes both utility held pipeline capacity and Delivered Service contracts, the description of the service provided, and the daily quantity available under the contract.

9 *See, e.g.*, Boston Gas Company d/b/a National Grid Long-Range Resource and Requirements Plan (November 1, 2018), https://fileservice.eea.comacloud.net/FileService.Api/file/FileRoom/10008562.

improve the information available in the gas supply planning process.[10] One such improvement is the presentation of the effective per unit fixed costs based upon the planned load factor utilization of the individual transportation resource paths. Adding to this effective (i.e., load factor sensitive) fixed cost, the modeled cost of gas, generates an "all-in" cost presentation. The revised proposed protocol, set forth in a February 20, 2019 Joint Memorandum, is attached to my testimony as Exhibit__(GL-3).

Q. Do you have any recommendations to improve the Companies' gas supply planning process?

A. Yes, my recommendations fall into three general categories. First, I propose changes to the process by which gas supply issues are addressed. These changes would augment and supplement the current annual Staff review of gas supply plans. Second, I propose changes to how the Commission should review and consider gas supply information. In essence, each Company would be required to submit a long range plan, which would form the basis for future cost recovery in a gas cost reconciliation proceeding. Third, I propose changes to the types of information that each Company should submit to better inform gas supply decision making.

Q. Please explain your first category of changes to the process by which gas supply issues are addressed.

[10] At present, in part because the protocol is pending before the Rhode Island Public Utilities Commission, the information is not available outside of the Division and the Company.

A. First of all, the process should be an annual process. It should be an open process with intervenors having discovery rights, and to the extent it is more administratively efficient to have a series of technical conferences (following discovery), then those should be transcribed. If needed, an evidentiary hearing should be established. As the Commission has recently acknowledged, "[g]as supply constraint solutions will need to involve greater visibility of the distribution planning process to stakeholders and local communities, to enable joint problem solving."[11]

Q. Please explain your second category of changes to how the Commission should review and consider gas supply information.

A. As a starting place, each Company should submit a long-range plan, which would set forth projections of demand, by peak hour by Company operational "division" and by day by "division." Against that demand, the resources to meet that demand should be set based upon the contracts and the on-system supply capabilities of each Company. Each Company should then identify the cost of each resource (fixed cost and projected[12] or known[13] variable costs) and the projected load

---

[11] *Petition of Consolidated Edison Company of New York, Inc. for Approval of the Smart Solutions for Natural Gas Customers Program*, Case No. 17-G-0606 at 35, Order Approving with Modification the Non-Pipeline Solutions Portfolio (February 7, 2019).

[12] Here, "projected" would be in those cases where future costs are based upon market indices and projections of those future values.

[13] Here, "known" would exclude instances where negotiations have not been completed or where revealing "known" would impact negotiations for similarly situated transactions.

factor utilization of the resources so that all-in costs can be reviewed and alternatives that might result in lower all-in cost be evaluated.

Q. Do the Companies provide any of this information already?

A. In Ms. Arangio's testimony, this type of presentation was provided for certain peak period resources. My recommendation is that it be done for all transportation paths based upon modeled utilizations of the capacity paths. In addition, it would be helpful to provide this presentation for design day, normal winter weather and design winter weather.

Q. Once each Company submits a long range plan, what would follow?

A. An agreed-upon long range plan would become the basis for both the annual (and/or semi-annual) gas cost reconciliation proceedings; and, for rate case revenue requirement development. In the gas cost reconciliation proceedings, the long range plan would provide the baseline. Differences between the baseline and the actuals in the gas cost reconciliation proceeding would be evaluated as "variances from plan." This is in contrast to the Commission having to review the gas cost reconciliation proceedings with information, resources, structures, and costs that may not have been revealed previously.

Q. Please explain what you mean by "all-in cost."

A. The "all-in cost" is determined by looking at the annual facilities' fixed costs plus commodity/O&M cost per unit of demand met taking into account the load factor of the annual demand to be met. Below I provide an example of "all-in" cost calculations.

	Annual Facilities' / Fixed Costs	Annual O&M / Commodity Costs	Peak Hour Demand (Dth/Hr)	Annual Incremental Demand Met	All-in Cost ($/Dth)
Ex. 1	$5,000,000	$1,800,000	1,000	150,000	$45.33
Ex. 2	$15,768,000	$420,000	1,000	150,000	$107.92

Ex. 1 Assumptions: Annual Cost of CNG Facility is $5 MM; CNG $/Dth $12;
Ex. 2 Assumptions: Annual Cost of New build PL Capacity at $1.80/Dthd; $/Dth $2.80;

Common Asssumptions: 1,000 Dth/Hr (24,000 Dthd); and 150 Hours/Yr Equivalent Full use.

Q. Are you saying that the figures above are representative of actual costs that the Companies might incur?

A. Not necessarily. This is an illustrative example demonstrating how each Company would calculate the "all in" cost of various alternatives.

Q. Please explain your third category of changes as to the types of information that the Companies should submit.

A. I recommend the Companies file the material they file today in the gas supply planning dockets reviewed by Staff, a long range plan as outlined above, and the following, available to intervenors:[14]

1) Historic (i.e., actual) daily winter period demand curves for the prior 5 years by class along with the prior demand forecasts for the same periods;

2) Historic (i.e., actual) daily non-winter period demand curves for the prior 5 years by class along with the prior demand forecasts for the same periods;

[14] At present, the Companies provide this material to Staff but it is redacted when posted to the Commission's website.

3) Historic (i.e,, actual) system winter period demand curves, (hourly and daily) for each of the Company's take stations for the prior five years along with the demand forecasts for the same periods;

4) Historic (i.e., actual) system non-winter period demand curves, (hourly and daily) for each of the Company's take stations for the prior five years along with the demand forecasts for the same periods;

5) The historic (i.e., actual) resource stacks of the Company employed to meet those historic demand curves;[15]

6) The Company's forecasted winter period system demand duration curves for the next five years;

7) The Company's forecasted non-winter period system demand duration curves for the next five years;

8) The Company's forecasted winter period demand curves, (hourly and daily) for each of the Company's take stations;

9) The Company's forecasted non-winter period demand curves, (hourly and daily) for each of the Company's take stations;

10) The resource stacks (including separate presentation of their respective fixed and projected variable costs and projected load factor utilization) the

[15] Note that for the hourly resources by take station, the Company should indicate a) its hourly contract rights, b) any overrun services it received to meet hourly demands; and c) Company-operated facilities employed to meet demands not otherwise met by contract rights or overruns service(s). In addition, the Company should note the extent to which delivered services (either to the Company or for the benefit of the Company's transportation customers) contributed to meeting demands by take station.

Company has under contract to meet the Company's forecasted forward period demand curves;[16] and

11) For those forecasted demands not met by existing contract rights plus Company-operated facilities, the Company should identify all potential resources (including non-pipeline solutions) under consideration and each such resource's forecasted all-in cost (as defined above) and provide the detailed analysis and assumptions used for the build-up of such resources' all-in costs presented by the Company. In addition, the Company should identify potential non-pipeline solutions not under consideration for each forecasted period, and the detailed analysis performed as to why the particular potential non-pipeline solutions are not under consideration for the subject period(s).

Q. How would this information assist the Commission?

A. The historic information would be used to evaluate previously forecasted demands to actual realized demands and the resources used to meet those actual demands. The forecasted demand information would identify gaps, if any, between forecasted demands and resources. The presentation of potential resources, their timing, all-in costs, and capabilities would assist the Commission in both understanding the available alternatives and the trade-offs involved with each. I recommend that the culmination of this planning, presentation and

[16] Note that for the hourly resources, by take station, the Company should indicate then-existing contract rights and Company-operated facilities' hourly (and total) capabilities to be employed to meet forecasted demands not met by take station contract rights. In addition, the Company should note the extent to which delivered services (either to the Company or for the benefit of the Company's transportation customers) contributed to meeting demands by take station.

consensus building process be Commission approval, based on a traditional review of reasonableness.

Lastly, the consensus resource stack on contracts, costs and capabilities would form the baseline against which subsequent gas reconciliation proceedings are conducted and against which variances from plan are identified and worked through.

Q. Would these refinements lead to benefits?

A. A more robust planning process would lead to multiple benefits, including: (1) identifying potential issues well in advance of experiencing demand/supply mismatches; (2) a way to manage and contain the fixed cost commitments made by the Companies, which will protect ratepayers against unreasonable financial risk and protect the Companies against prudency risk from after-the-fact regulatory challenges; and (3) a more thorough framework for the Companies to consider alternatives that would have lower all-in costs to customers.

V. New York Facilities System

Q. Please provide an overview of the New York Facilities System.

A. As Ms. Arangio explains:

> KEDNY and KEDLI are parties to an agreement with Consolidated Edison Company of New York, Inc. ("Con Edison") concerning the ownership and operation of the New York Facilities System ("NYFS"), the high pressure gas transmission system serving the three downstate New York distribution companies. The agreement permits the parties to contract for the transportation and receipt of gas from various interstate pipelines that interconnect with the NYFS, including Transcontinental Gas Pipeline Company LLC ("Transco"), Texas Eastern Transmission LP ("Texas Eastern"), Iroquois Gas Transmission System LP

("Iroquois"), and Tennessee Gas Pipeline Company ("Tennessee").[17]

Q. How do the Companies and Con Edison plan for the needs of the NYFS?

A. My understanding is that the Companies and Con Edison prepare joint hydraulic models of the NYFS,[18] file an updated appendix annually with the Commission, and discuss the modeling results from time-to-time with Staff at their request.[19]

Q. Do you have any recommendations to improve the planning process for the NYFS?

A. Yes. I recommend that the Commission undertake a proceeding where the planning assumptions of the NYFS are formalized and integrated into the more general, downstate energy planning process for the NYFS members. The Commission may even find that it appropriate to view and regulate the NYFS as a functioning independent not-for-profit gas grid operator. This would entail formally identifying which facilities of the downstate gas utilities be contained within such organization, defining cost allocation and cost recovery mechanisms for the organization and finally, defining the goals and charges of such organization in the planning, operating, and optimal execution of the State's energy and climate policies.

Q. Does this conclude your testimony?

A. Yes.

---

[17] Arangio Testimony at page 5.

[18] Exhibit___(GL-4).

[19] Exhibit___(GL-5).

Exhibits of Gregory Lander

Exhibits __ (GL-1 through GL-5)

Greg Lander, President
Skipping Stone LLC

Professional Summary:

As President of Skipping Stone Inc., Greg Lander is responsible for Strategic Consulting in the mergers and acquisition arena with numerous clients within the energy industry. Generally recognized in the energy industry as an expert, he has advised and/or given testimony at numerous Federal Energy Regulatory Commission (FERC), State, arbitration, and legal proceedings on behalf of clients and has advised as well as initiated standards formation before the Gas Industry Standards Board (GISB) (predecessor to the North American Energy Standards Board (NAESB)). As Founder, President, and Chief Technology Officer of TransCapacity Limited Partnership, he was responsible for conceiving, planning, managing, and designing Transaction Coordination Systems utilizing Electronic Data Interchange (EDI) between trading partners. As a founding member of GISB, he assisted in establishing protocols and standards within the Business Practices, Interpretations and Triage Subcommittees.

Professional Accomplishments:

- Handled all Due Diligence for purchaser (Loews Corp) in acquisitions of two interstate pipelines, one natural gas storage complex, and ethylene distribution and transmission systems (Texas Gas Transmission, Gulf South Pipeline, Petal Storage, Petrologistics, and Chevron Ethylene Pipeline) most in excess of $1 Billion. Developed purchaser's business case model, including rate/revenue models, forward contract renewal models, export basis modeling and revenue models, and operating cost and capex models. Coordinated Engineering and Environmental Due Diligence Teams integrating findings and assessments into final Diligence Reports.
- Assisted major electric retailer in 9 states with business case development for entry into North Eastern U.S. Commercial & Industrial natural gas marketing business. Identified market share of incumbents; retail registration process, billing processes; utility data exchange rules and procedures and developed estimates of addressable market by utility.
- Handled all economic Due Diligence for purchaser of large minority stake in Southern Star Gas Pipeline. Developed purchaser's business case model, including rate/revenue models and forward contract renewal models, assessed potential competitive by-pass of asset located in "pipeline alley", developed revenue models and operating cost and capex models. Coordinated Engineering, Pipeline Integrity, and Environmental Due Diligence Teams integrating findings and assessments into final Diligence Reports.
- Developed post-acquisition integration plans for inter-operability and alterations to system operations to take advantage of opportunities presented by

synergistic facilities' locations and functions and complimentary contractual requirements. Implementation of plan resulted in fundamental changes to systems operations and improvement in systems, net revenues, capacity capabilities, and facilities utilization.

- Handled all economic analysis, modeling, and systems capability due diligence for potential purchaser in several preliminary or completed yet un-consummated pre-transaction investigations involving Panhandle Eastern, Northern Border, Bear Paw, Florida Gas, Transwestern, Great Lakes, Guardian, Midwestern, Viking, Southern Star, Columbia Gas, Midla, Targa (No. Texas), Ozark, ANR, Falcon Gas Storage, Tres Palacios, Rockies Express, Norse Pipelines, Southern Pines, Leaf River, LDH (Mont Belvieu), Kinder Morgan Interstate, Trailblazer, Rockies Express and South Carolina Gas Transmission.
- Post Texas Gas Transmission and Gulf South Pipe Line acquisitions, assisted with all investigations involving assessments and proposals for realizing potential synergies with/from asset portfolio; rate case strategy development and alternate case development; and strategies around contract renewal challenges.
- Headed up due diligence team in acquisition of multi-state retail (residential) natural gas and electric book by Commerce Energy.
- Headed up due diligence team in acquisition of multi-state retail (C&I) natural gas book by Commerce Energy.
- Served as lead consultant for consortium of end-users, Local Distribution Companies, Power Generators, and municipalities in several major FERC Rate Cases, service restructuring, and capacity allocation proceedings involving a major Southwestern U.S. Pipeline.
- Served as lead consultant and expert witness for consortium of end-users, Local Distribution Companies, Power Generators, and municipalities in major FERC rate case under litigation involving decades-long disputes over service levels, cost allocation, and rate levels.
- Served as lead consultant for consortium of end-users and municipalities in major FERC rate case involving implementation of proposed rate design, cost allocation, and rate level changes.
- Expert witness in numerous gas and electric utility rate cases; integrated resource plans; litigated service offerings and cost approval and allocation proceedings for public interest clients. Controversies, often involving hundreds of millions to billions of dollars over cases' time horizons, are common.
- Developed and critiqued Rate Case Models for several pipeline proceedings and proposed proceedings (as consultant variously to both pipeline and shippers). Activities included modeling (and critiquing) new services' rates, costs, and revenues; responsibilities included development of various alternative cost allocation/rate designs and related service delivery scenarios.

- Handled all market assessment, forward basis research, and transportation competition modeling for several proposed major pipelines and laterals, including two $1 Billion+ Greenfields projects that went into construction and operation providing new outlets for growing southwestern shale production. (Gulf Crossing and Fayetteville Lateral).
- Assessed supply and demand balance for Southwestern US (OK, TX, Gulf Coast and LA) including assessment of future demand and supply displacement associated with West Texas wind power development and its likely impact on pipeline export capacity from region.
- Assessed supply and demand balance for Northeast to Gulf Coast capacity additions including assessment of Gulf Coast demand and export growth and its likely impact on forward basis.
- Assessed start-up gas supply needs for Appalachian coal fired power plant, resulting in installation of on-site LNG storage and gasification to address lack of enough firm pipeline capacity to meet need.
- Assessed installed and projected wind-turbine capacity in ERCOT and its eventual impact on Texas electric market as wind power output approaches minimum ERCOT load levels.
- Designed and developed EDI based data collection system, data warehouse and web-based delivery system (www.capacitycenter.com) for delivering capacity data collected from pipelines to shippers, marketers, traders, and others interested in capacity information to support business operations and risk-management requirements.
- Assisted client in developing proposals to increase pipeline capacity responsiveness and proposed market fixes that would create price signals around sub-day non-ratable flows, including rate proposals, sub-day capacity release markets, and measures to address advance reservation of capacity for electric generation fuel to meet sub-day generation demands.
- Developed “universal capacity contract” data model for storage of all interstate capacity contract transactions from all interstates in single database.
- Led design effort culminating in FERC-mandated datasets defining pipeline capacity rights, (including receipt capacity, mainline capacity, delivery capacity, segmentation rights, in and out of path capacity rights), Operationally Available Capacity, Index of Customers, and Transactional Capacity Reports (through GISB).
- Assembled consortium of utilities to investigate and develop large high-deliverability salt storage cavern in desert southwest (Desert Crossing). As LLC’s Acting Manager, was responsible for developing business case and economic models; handling all partner issues and reporting; coordinating all field engineering, facilities design, planning and siting; and managing all environmental, legal, engineering and regulatory activities. Wrote FERC Tariff. Brought project to NEPA Pre-Filing Stage and conducted non-binding Open

Season, as well as assisted with prospective shipper negotiations. Project cancelled due to 2001 "California Energy Crisis" and contemporaneous Enron and energy trading sector implosions.

- Designed comprehensive retail energy transaction and customer acquisition data model, process flow, and transaction repository for web-based customer acquisition and customer enrollment intermediary.
- Experienced in negotiation and drafting (from both seller side and buyer side) of firm supply, firm transportation, firm storage, and power supply and capacity agreements for numerous entities including project financed IPPs and for new greenfield pipeline and expansion of storage system.
- Provided market entry assessment for large international manufacturing and service company seeking to enter U.S. micro-grid, combined heat and power, and integrated solar, gas & battery markets.
- Conducted interstate pipeline capacity utilization analysis for New England following winter of 2013/2014 price fly-up.
- Conducted PJM East interstate gas pipeline capacity utilization and comparative analysis between pipelines with standard NAESB nominating cycles versus those with near hourly scheduling practices.
- Conducted requirements analysis for several firms pursuing software selection of energy transaction systems.
- Instrumental in the formation of the GISB. Member of industry team that lead the development of the proposal for and bylaw changes related to the formation of NAESB.
- Provided support to numerous clients and clients' attorneys in disputes involving capacity contracts, capacity rights allocations, tariffs, rate cases, intellectual property rights cases, and supply contract proceedings as both up-front and behind the scenes expert.

Associations and Affiliations:

Longest serving Member of Board of Directors for NAESB and prior to that GISB – 23 years.

GISB Committees: Former Chairman, Business Practices Subcommittee – drafted approximately 450+ initial industry standards that are now codified FERC regulations (Order 567); Former Chairman, Interpretations Subcommittee – drafted and led adoption process for first 50+ standards interpretations; Former Chairman, Triage Subcommittee; Title Transfer Tracking Task Force; Order 637 GISB Action Subcommittee; and industry Common Codes Subcommittee. Currently member of NAESB Wholesale Gas Quadrant Executive Committee and of NAESB Parliamentary Committee

Past and Affiliations and Associated Accomplishments:

1981-1989: One of five initial employees of Citizens Energy Corporation, Boston Mass. Responsible for starting and growing Citizens Gas Supply, one of the first independent gas marketers of the early 1980's, into $200MM+ annual operation. Successfully lobbied for pipeline Open Access (Orders 436 and 636), introduction of pipeline Affiliated Marketer rules of conduct (Order 497), and Open Access to pipeline operational information (Order 563).

1989-1993: Independent Consultant - Natural Gas Projects, Pipeline Rate Cases, Project Financed Contract negotiations, and Independent Power markets

1993 – 1999: Founder and President, TransCapacity Service Corp – Software products and services related to pipeline capacity trading, nomination, and contracting. Raised $17 MM from industry player to establish TransCapacity. Successfully lobbied for Pipeline restructuring and formation of capacity release market (Order 636). Sold to Skipping Stone.

1999 – 2004: Principal and Partner, Skipping Stone – Energy market consultants

2004 – 2008: President of Skipping Stone following purchase of Skipping Stone by Commerce Energy, Inc.

2008: Repurchased Skipping Stone from Commerce Energy, Reformulated Skipping Stone as LLC with Peter Weigand

2008 to Present: President and Partner, Skipping Stone. In addition to handling book of clients, responsible for all Banking, Accounting, Operations, Risk Management and contract matters for Skipping Stone.

Education:

1977: Hampshire College, Amherst, MA; Bachelor of Arts

Publication:

2013: Synchronizing Gas & Power Markets - Solutions White Paper

Expert Testimony of Gregory M. Lander

Name of Case	Jurisdiction	Docket Number	Date
El Paso Natural Gas Company	Federal Energy Regulatory Commission	RP04-251-000	May 3, 2004 (Testimony)
El Paso Natural Gas Company	Federal Energy Regulatory Commission	RP08-426-000	May 19, 2009 (Answering Testimony) June 2, 2010 (Supplemental Answering Testimony)
El Paso Natural Gas Company	Federal Energy Regulatory Commission	RP10-1398-000	June 28, 2011 (Answering Testimony) March 4, 2014 (Answering Testimony)
Petition of Boston Gas Company and Colonial Gas Company, each d/b/a National Grid for Approval by the Department of Public Utilities for a Firm Transportation Contract with Algonquin Gas Transmission Company	Massachusetts Department of Public Utilities	13-157	December 12, 2013 (Direct Testimony)
Petition of Boston Gas Company d/b/a National Grid for Approval by the Department of Public Utilities of a twenty-year Firm Transportation Agreement with Tennessee Gas Pipeline Company, involving an expansion of Tennessee's interstate pipeline running from Wright, New York to	Massachusetts Department of Public Utilities	15-34	June 5, 2015 (Direct Testimony)

Dracut, Massachusetts, known at the Northeast Energy Direct Project			
Petition of Bay State Gas Company d/b/a Columbia Gas of Massachusetts for Approval by the Department of Public Utilities of a twenty-year Firm Transportation Agreement with Tennessee Gas Pipeline Company, involving an expansion of Tennessee's interstate pipeline running from Wright, New York to Dracut, Massachusetts, known at the Northeast Energy Direct Project	Massachusetts Department of Public Utilities	15-39	June 5, 2015 (Direct Testimony)
Petition of The Berkshire Gas Company for Approval of a Precedent Agreement with Tennessee Gas Pipeline Company, LLC, pursuant to G.L. c. 164, § 94A	Massachusetts Department of Public Utilities	15-48	June 5, 2015 (Direct Testimony)
Investigation of Parameters for Exercising Authority Pursuant to Maine Energy Cost Reduction Act, 35-A M.R.S.A. Section 1901	Maine Public Utilities Commission	2014-00071	July 11, 2014 (Direct Testimony)
Virginia Electric and Power Company's Integrated Resource Plan filing pursuant to Va. Code § 56-597 *et seq*.	Virginia Corporation Commission	PUR-2017-00051	August 11, 2017 (Direct Testimony)
In the Matter of the Laclede Gas Company's Request to Increase Its Revenues for Gas Service	Missouri Public Service Commission	File No. GR-2017-0215	September 8, 2017 (Direct Testimony) November 21, 2017 (Surrebuttal

In the Matter of the Laclede Gas Company d/b/a Missouri Gas Energy's Request to Increase Its Revenues for Gas Service		File No. GR-2017-0216 (Consolidated)	Testimony)
Application of San Diego Gas & Electric Company (U902M) for Authority, Among Other Things, to Update its Electric and Gas Revenue Requirement and Base Rates Effective on January 1, 2019. Application of Southern California Gas Company (U904G) for Authority, Among Other Things, to Update its Gas Revenue Requirement and Base Rates Effective on January 1, 2019.	California Public Utilities Commission	Application 17-10-007 Application 17-10-008 (Consolidated)	May 14, 2018 (Direct Testimony) June 8, 2018 (Rebuttal Testimony)
Application of Virginia Electric and Power Company to revise its fuel factor pursuant to § 56-249.6 of the Code of Virginia	Virginia State Corporation Commission	PUR-2018-00067	June 14, 2018 (Direct Testimony)
Application of Southern California Gas Company (U 904 G) and San Diego Gas & Electric Company (U 902 G) Regarding Feasibility of Incorporating Advanced Meter Data Into the Core Balancing Process	California Public Utilities Commission	Application 17-10-002	July 2, 2018 (Direct Testimony)
Virginia Electric and Power Company's Integrated Resource Plan filing pursuant to Va. Code § 56-597 *et seq*.	Virginia Corporation Commission	PUR-2018-00065	August 13, 2018 (Direct Testimony)
In the Matter of Constellation Mystic Power, LLC	Federal Energy Regulatory Commission	ER18-1639	September 4, 2018 (Cross Answering Testimony)
South Carolina Electric and	South Carolina Public	Docket Nos.	September 24, 2018

Gas Company Application for Approval of Merger with Dominion Resources	Service Commission	2017-370-E; 2017-305-E; and 2017-207-E	(Direct Testimony)
In re: Annual Review of Base Rates for Fuel Costs of South Carolina Electric and Gas Company	South Carolina Public Service Commission	Docket 2019-2-E	March 19, 2019 (Direct Testimony)
Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Consolidated Edison Company for Gas Service	New York Public Service Commission	Case 19-G-0066	May 24, 2019 (Direct Testimony) August 19, 2019 (Surrebuttal Testimony)
Application of Virginia Electric and Power Company to revise its fuel factor pursuant to VA Code § 56-249.6.	Virginia State Corporation Commission	Case No. PUR-2019-00070	June 19, 2019 (Direct Testimony)
In the Matter of Annual Review of Base Rates for Fuel Costs for Duke Energy Carolinas, LLC, Increasing Residential and Non-Residential Rates	South Carolina Public Service Commission	Docket 2019-3-E	August 19, 2019 (Direct Testimony)

Case 18-M-0272 - Winter Supply Review Data Request

The Brooklyn Union Gas Company d/b/a National Grid NY ("KEDNY"), KeySpan Gas East Corporation d/b/a National Grid ("KEDLI"), and Niagara Mohawk Power Corporation d/b/a National Grid ("NMPC") (collectively, "National Grid" or "Company") submit the following information with regard to National Grid's gas supply readiness for the 2018/19 winter season.

Issues 1 and 2

Please provide the following information related to your company's portfolio and purchasing strategy for the upcoming 2018/19 send out year and anticipated portfolio changes over the next five years, and:

1. **Table 1: System design day capacity capability by service area (or gate when indicated) and peak design day demand by service area (or gate when indicated). Please include all capacity volumes, including all recallable capacity assets that are available or needed for peak design day and specify the volumes supporting sales, transportation customers and retail access capacity release. Include last year's 2017-18 Table 1 (final update) data for purposes of comparison. The total capacity capability must meet or exceed the design peak day demand value provided. Identify any projected capacity assets that are not yet finalized but will be prior to the upcoming winter heating season, including both a description and projected completion date.**

   Upstate NY has delivery point entitlement redundancy on two of the contracts referenced in these tables; DTI contracts 100001 and 700001 each have deliverability to both the east gate and the west gate that, when added together, exceed the contract MDQ. Therefore, the Company has allocated the MDQ of each contract to the east gate and west gate in accordance with its design day customer requirements. The total amount the Company may transport to the DTI east gate and DTI west gate using all of its DTI transportation contracts is limited by the Maximum Daily Delivery Obligation ("MDDO") for each region.

   Please see the following attachments:
   - Tables 1a, 1b, 1c, 1d: 2018/19 Total System Firm Peak Day Capacity
   - Exhibits 1, 2, 3, 4: 2017/18 Total System Firm Peak Day Capacity

2. **Table 2: Estimated annual, winter season, and daily requirements by service area (or gate when indicated) for last year and the next five years, using design weather. Include a description of the design weather criteria and explain any changes from the previous year. Specifically, since many areas of the state experienced an extended period of colder than normal weather last winter, how does this experience impact your daily or winter season design parameters and are there any changes required? Also, how is this affecting your storage injection season and how will it be addressed? The 2017-18 actual data experienced last year is to be included for purposes of comparison. Also include any and all service areas (or gate stations when indicated) where**

moratoriums have been put into place or have the possibility of being instituted in the next five years. Identify where any curtailments to firm customers may occur.

Although the Company experienced an extended period of colder-than-normal weather last winter, it did not make any changes the daily or winter season design parameters as a result of the weather.

Niagara Mohawk

NMPC typically begins injections in April and continues through October as described in the Company's response to Question 9. However, due to a colder-than-design April, the Company withdrew more than it injected. Therefore, the Company started May behind its plan. In response, daily injection targets for the remainder of the off-peak season were adjusted to be more aggressive. Additionally, the Company executed an inventory transfer to help ensure its targets are met. The Company is currently on track to meet its October 31st target inventory.

Please see the Company's response to question 28c regarding moratoriums.

Downstate NY (HDD)

	Design Year	June 2017 – May 2018 Actuals	Actual / Design
Year	5,370	4,669	86.9%
Winter (Nov-Mar)	4,471	3,940	88.1%
Peak Day	65	57	87.7%

The design weather criteria have not changed from the Company's 2017 submission for Downstate NY.

Upstate NY (HDD)

	Design Year	June 2017 – May 2018 Actuals	Actual / Design
Year	7,431	6,503	87.5%
Winter (Nov-Mar)	5,817	5,210	89.6%
Peak Day	75	66	88%

The Upstate NY design year was adjusted forward from 7,397 to 7,431 HDD. The Upstate NY design November-March was adjusted downward from 5,832 to 5,817 HDD. The Upstate NY Peak Day remains 75 HDD.

Please see the following attachments:

- Tables 2a, 2b, 2c, 2d: 2017/18 Estimated Requirements – Design Weather

3. Table 3: Same information as requested in (2), but using normal weather. Include a description of the normal weather criteria and the calculation methodology. The 2017-18 data submitted last year is to be included for purposes of comparison. Please Note: if Table 3 is based on a sales forecast using anything less than 30 years of weather data, no part of Table 3 may be used to develop any part of Table 2.

Downstate NY (HDD)

	Normal Year	June 2017 – May 2018 Actuals	Actual / Normal
Year	4,587	4,669	101.8%
Winter (Nov-Mar)	3,819	3,940	103.2%
Peak Day		57	

The Downstate NY Normal Heating Degree Days for any given calendar day within a month, are based on the average of the degree days for that calendar day over the thirty year period ending June 30, 2015.

Upstate NY (HDD)

	Normal Year	June 2017 – May 2018 Actuals	Actual / Normal
Year	6,564	6,503	99.1%
Winter (Nov-Mar)	5,135	5,210	101.5%
Peak Day		66	

In prior years, NMPC used the arithmetic average of daily weather data from the KSYR, Syracuse, NY and the KALB, Albany, NY weather stations. Beginning with this submission, NMPC developed its normal years (6,588 HDD for Syracuse and 6,539 HDD for Albany) based on 30 years of weather data from July 1986 to June 2016. This separation of weather station data was implemented as part of the Joint Proposal in Case 17-G-0239.

The Upstate NY normal year was revised downward from 6,571 to 6,550 HDD. The Upstate NY normal November-March was revised downward from 5,151 to 5,128 HDD.

Please see the following attachments:

- Tables 3a, 3b, 3c, 3d: 2018/19 Estimated Requirements – Normal Weather
- Exhibits 5, 6, 7, 8: 2017/18 Estimated Requirements – Normal Weather

4. Identify your source for heating degree day (HDD) data, including the specific weather data points used for forecast purposes. Describe your source and/or your calculation of design day and design winter data (i.e. calculated from normal usage or an actual historic period). Identify the time periods used to develop usage per HDD for both design and normal usage, and explain the frequency of updates. If 30 years of data is not being used for design, please explain why. Please explain how usage per HDD for the peak period is calculated and verified.

Downstate NY (KEDNY and KEDLI)

The Downstate NY (KEDNY and KEDLI) design standards are based on the KNYC Central Park, NYC, NY weather station. The Company developed its normal year (4,587 HDD) based on 30 years of weather data from July 1985 to June 2015. This normal year was approved by the Commission in Cases 16-G-0058 and 16-G-0059.

To develop its normal year, the Company calculated the monthly HDD values as well as the standard deviation of the daily HDDs in each month. The Company then developed its normal year ('Typical Meteorological Year') by choosing historical months whose monthly totals and standard deviations approximated the mean monthly HDD and mean standard deviations observed. The Company then tuned each month's data to achieve the exact mean monthly total and mean standard deviation.

Using the mean and standard deviation of the annual HDD data, the Company established its design year standard of 5,370 HDD as a once-in-40-year probability of occurrence. The Downstate NY Peak Day remains at 65 HDD.

Upstate NY (NMPC)

In prior years, the Company has used the arithmetic average of daily weather data from the KSYR, Syracuse, NY and the KALB, Albany, NY weather stations. Beginning with this submission, the NMPC design standards are based on the KSYR, Syracuse, NY weather station for its Central division and the KALB, Albany, NY weather station for its Eastern division. The Company developed its normal years (6,588 HDD for Syracuse and 6,539 HDD for Albany) based on 30 years of weather data from July 1986 to June 2016. This separation of weather station data was implemented as part of the Joint Proposal in Case 17-G-0202.

The Company uses the same approach as described above in the Downstate NY description to establish its Upstate NY Typical Meteorological Years.

Using the mean and standard deviation of the annual HDD data, the Company established its design year standard of 7,462 HDD for Syracuse and 7,400 HDD for Albany using the same probability of occurrence used in prior years' KSYR and KALB average data: once-in-19.07 years. The Upstate NY Peak Days for both Syracuse and Albany remain at 75 HDD.

Usage per HDD

The Company's wholesale (city gate) forecast begins with aligning its regression equations relating its actual daily send out observations to actual HDD for the most recent 12-month period. For this forecast, that period is April 2017 – March 2018. There are independent regression equations for KEDNY, KEDLI, NMPC-Eastern, and NMPC-Central. Each regression equation is a segmented equation that identifies the point at which space heating equipment begins to be used. Each regression equation also captures the observed usage changes between weekday and weekend, as well as the effect of prior days' HDD on each day's send out. The accuracy of the Company's wholesale forecasts is then tracked throughout the heating season in the event that actual customer demand deviates from forecasted demand so that appropriate actions can be taken.

5. Describe the load forecasting tools used to develop the above forecasts. Indicate how all natural gas efficiency programs, Demand Response Programs, Microgrids, and Non-Pipe Alternatives (NPA) conducted by your company, contractors or the New York State Energy Research and Development Authority (NYSERDA) have been incorporated into these forecasts and your capacity planning. Provide a summary of the projected energy savings and the actual savings realized to date. How are these savings translated into the normal usage projection in Table 3?

The Company's load growth forecasts are based on econometric/statistical forecast models developed for each residential, commercial/industrial, and multifamily rate class. Two different models were developed for each class: (1) a model to forecast the number of customers, and (2) a model to forecast use per customer. In the number of customer models, the independent variables are a combination of time trends, population, households, employment, and gas and oil prices. In the models calculating use per customer, the independent variables are a combination of time trends, heating degree days, population, housing stock, income, employment, unemployment, gross domestic product, and gas and oil prices. The Company obtained the historical data and forecasts of the independent variables from a leading economic research and forecasting firm and energy prices from the U.S. Department of Energy, Energy Information Administration, and used this data to develop the econometric models and generate load forecasts.

The Company's historical sales data includes the impact of actual energy efficiency savings from both NYSERDA and Company-sponsored energy efficiency programs.

In 2016, the Company's Energy Efficiency department began its ETIP ("Energy Efficiency Transition Implementation Plan). In prior years, the Company ran its EEPS ("Energy Efficiency Portfolio Standard") programs. The table below provides the Company's annual savings under EEPS for 2009-2015 and its ETIP program actuals for 2016 and 2017 are presented.

Energy Efficiency Gas Savings - Annual Therm Savings

	Actual Savings (therms)								
TOTAL	***2009***	***2010***	***2011***	***2012***	***2013***	***2014***	***2015***	***2016***	***2017***
KEDNY	9,200	587,954	1,530,262	1,481,435	1,132,110	1,874,514	2,544,660	***3,502,389***	***3,847,415***
KEDLI	53,726	654,634	1,158,240	1,098,762	1,509,023	1,718,288	1,501,390	***1,420,901***	***2,500,703***
NMPC	305,798	1,651,622	3,397,947	4,021,035	4,234,055	4,517,154	4,504,020	***5,331,860***	***4,725,877***

Residential

	2009	***2010***	***2011***	***2012***	***2013***	***2014***	***2015***	***2016***	***2017***
KEDNY	9,200	111,957	287,168	308,576	269,989	478,774	665,900	1,767,758	-
KEDLI	53,726	308,526	373,641	323,679	656,365	740,651	578,250	785,138	-
NMPC	305,798	1,418,699	2,432,972	2,714,493	3,016,045	3,371,238	3,499,230	3,356,506	-

C&I

	2009	***2010***	***2011***	***2012***	***2013***	***2014***	***2015***	***2016***	***2017***
KEDNY	0	475,997	1,243,094	1,172,859	862,121	1,395,740	1,878,760	1,734,631	-
KEDLI	0	346,108	784,599	775,082	852,658	977,636	923,140	635,764	-
NMPC	0	232,922	964,975	1,306,541	1,218,010	1,145,917	1,004,790	1,975,354	-

The forecast includes this trend in continuing load reduction based on the historical successes in energy efficiency reductions in load. No further adjustments were made to the forecast.

The Company used the demand forecasts from the above methodology to calculate monthly and annual load forecasts by class and in total. The Company then distributed the monthly total forecasts to daily levels to determine the normal and design customer requirements based on the regression analysis of daily-metered send out and HDD over the most recent 12-month period ending March 31st (April 2017 – March 2018) described by the Company in its response to Question 4 above and presented in Table 3.

6. What is your current forecast/planning horizon for supply and capacity purposes and explain why it is used? If you aren't using a minimum of five years for system planning, please explain.

The Company adopted a ten-year forecast/planning horizon in 2015 and will continue this practice going forward. The ten-year horizon will better position the Company to make long-term supply decisions, including the need for pipeline projects that require significant lead time to be placed in service.

(Business Confidential)

Downstate NY Supply/Demand Balance



Upstate NY Supply/Demand Balance



7. A winter season load duration curve for 2018-19 send out year that shows how supplies can meet a severe winter season and peak design day. This should be provided for each

service area (or gate when indicated). Include all data in an unlocked digital Microsoft Excel File.

(Business Confidential)

Please see the excel workbooks for the backup to the exhibits below.

Please see the following exhibits:

- Exhibit 9: Downstate New York 2018/19 Winter Load Duration Curve
- Exhibit 10: Upstate New York 2018/19 Winter Load Duration Curve
- Exhibit 11: Upstate New York 2018/19 Winter East Gate Load Duration Curve
- Exhibit 12: Upstate New York 2018/19 Winter West Gate Load Duration Curve

8. Provide this year's and last year's planned storage curves versus actual storage curves for injections and withdrawals.

The tables below provide rule curves for the 2017/18 and 2018/19 winter seasons and actual end of month storage balances for the 2017/18 winter season.

Downstate NY:

	Rule Curve 2018/19	Rule Curve 2017/18	Actual Inventory 2017/18
Nov	58,032,005	53,854,321	56,820,536
Dec	46,337,395	45,212,651	46,027,732
Jan	27,312,227	25,698,591	31,969,880
Feb	9,783,074	10,850,105	20,845,237
Mar	646,567	702,917	9,552,868

Upstate NY:

	Rule Curve 2018/19	Rule Curve 2017/18	Actual Inventory 2017/18
Nov	15,349,325	14,614,211	15,079,558
Dec	12,244,967	11,809,064	12,345,363
Jan	7,933,359	7,871,114	9,136,281
Feb	4,484,073	3,963,050	5,613,180
Mar	2,069,572	0	2,049,383

Please note that due to the new Upstate NY Customer Choice Program, the Company expects to manage an additional 1.5 Bcf of storage. These rule curve volumes do not reflect that to aid in comparison to 2017/18.

Please see the response to Question 9 for the Company's injection plan.

9. Describe the storage injection plan for this injection season and highlight any modifications from the prior year plan.

The Company's basic injection strategy is to begin injecting small volumes in the month of April followed by levelized injections from May through September, then tapering down injections in October. The Company's transportation assets available for injections may be limited in April and October, as they may be needed to meet customer requirements. Additionally, the Company requires flexibility in these shoulder months to adjust to changes in weather.

After each month, inventories and potential ratchets are reviewed and adjustments are made to continue injections. The target inventory level for most storage fields is 100% with some exceptions found in the following table. Additional injections ("downswing") may be needed to accommodate warm days in November. Below is a summary of targets for those storage fields that do not get filled to 100% along with changes from 2017 to 2018.

LDC	Field	2017 Target	2018 Target	Reason for less than 100% target
KEDNY	Transco GSS	95%	95%	Downswing/November Injections allowed
KEDNY	Transco S-2	95%	95%	Downswing/November Injections allowed
KEDLI	Transco WSS	91%	91%	As per Force Majeure Plan, updated annually
KEDNY	Transco LSS	100%	99%	Operational flexibility
KEDNY	Tetco SS-1	97%	97%	Downswing/November Injections allowed
KEDNY	Honeoye	85%	85%	Avoid non-dispatchable volumes
KEDLI	Transco GSS	95%	95%	Downswing/November Injections allowed
KEDLI	Transco LSS	100%	99%	Operational flexibility
KEDLI	Transco SS-2	100%	99%	Operational flexibility
NMPC	Dominion GSS	95%	93.5%	Downswing/November Injections allowed

The KEDNY and KEDLI LSS target as well as the KEDLI SS-2 target were reduced by 1% to allow for maximum injection capacity through the end of October. The NMPC GSS target is 93.5% to allow for additional downswing going into November.

10. Provide the method(s) used by the Company to determine or ensure the least cost dispatch when the heating season is warmer than normal.

The Company's fundamental approach to supply procurement (*i.e.,* serving customers at the lowest possible cost) is the same under design, normal, and warmer-than-normal conditions. On a monthly basis, the Company first estimates minimum daily customer requirements from a combination of the annual gas load forecast and actual usage over the past several years. The Company then dispatches a least-cost combination of storage withdrawals and base load flowing supplies to meet, but not exceed, minimum customer requirements, while also satisfying anticipated operational needs. The Company retains enough capacity to meet normal and maximum daily customer requirements, while also incorporating flexibility to meet operational requirements and to adapt to changing pricing dynamics.

The Company seeks to economically displace storage withdrawals with flowing supplies when the costs of flowing supplies are less than the storage inventory cost. The opportunity to economically displace storage withdrawals with less expensive flowing supplies can occur when the heating season is warmer than normal, normal, or colder than normal. The Company will only displace storage withdrawals to the extent that it can still satisfy must turn requirements pursuant to its storage agreements. Please see the Company's response to Question 11 regarding must turn requirements.

Warmer-than-normal weather may also present the Company with additional opportunities for optimization of excess capacity. The Company's response to Question 22 details how the Company makes use of this excess capacity to reduce customer costs.

11. Provide the Company must turn requirements for each storage facility(s).

Depending on the field, the must turn requirements must be met by March 31st or April 15th and are equal to the amount that the storage gas balance exceeds the must turn percentage of storage capacity as of November 1st. A summary of must turn requirements by field is listed below:

LDC	Storage Field	Must Turn %	Must Turn Required By
KEDNY	Transco GSS	35%	April 15
KEDNY	Transco LSS	35%	March 31
KEDNY	Dominion GSS	35%	April 15
KEDNY	Dominion GSS-TE	35%	April 15
KEDNY	Equitrans Keystone	25%	March 31
KEDLI	Transco GSS	35%	April 15
KEDLI	Transco LSS	35%	March 31
KEDLI	Dominion GSS	35%	April 15
KEDLI	Dominion GSS-APEC	35%	April 15
KEDLI	Dominion GSS-TE	35%	April 15
KEDLI	Dominion GSS (6 Bcf field)	35%	April 15
KEDLI	Dominion GSS-North Summit	35%	April 15
NMPC	Dominion GSS	35%	April 15

Issue 3

Please provide the following information related to your company's operations and optimization procedures:

12. A send out schedule (or curve) for forecasting requirements under the varying conditions that are considered in developing the estimates (e.g., temperature, wind, weekend/weekday, etc.).

The Company performs linear regression analysis with three years of historical data for Downstate NY and four years of historical data for Upstate NY to develop predictive send out equations each month. The Company has found that limiting regressions to three years of data provides the most accurate estimate of send out for Downstate NY, as using additional years of data tends to skew the send out lower due to system growth. Upstate NY has not experienced the same level of growth as compared to Downstate NY, so regressing with an additional year of data provides for a more accurate forecast for this territory.

The send out equations correlate daily firm demand (sales and transportation) for all Companies and TC/Interruptible demand (sales and transportation) for Downstate NY to the following variables:

- The gas day's predicted 24-hour average temperature
- Prior and two prior gas days' 24-hour average temperatures
- Wind speed
- Day type (to differentiate a weekday from a weekend day or holiday)
- Day of month (to differentiate the beginning-of-month versus end-of-month for some shoulder period months)
- Cloud Cover (Upstate NY only)

Not all of the factors listed above are needed to predict daily send out in each month. For example, during the summer, send out may not correlate strongly to wind speed. The use of cloud cover is unique to the Upstate NY equation, as it has proven to be a material variable in determining load during the winter season. The Downstate NY region does not experience the same material impact of cloud cover.

Gas Supply Planning monitors performance of the equations on an ongoing basis and makes adjustments as required to ensure the continued accuracy of the forecast.

Please see the following attachment:

- Exhibit 13: Equations For Predicting November 2018 Firm & TC Throughput

13. Gas supply portfolio information (highlight changes as indicated on the charts, including all capacity or supply contracts remaining to be finalized prior to the winter heating season):

a. Table 4: Transportation capacity data including contract volumes and expiration dates. Please be prepared to discuss how the capacity is actually used during our meeting.

Please see the following tables:
- Table 4a: The Brooklyn Union Gas Company Firm Transportation Capacity
- Table 4b: KeySpan Gas East Corporation Firm Transportation Capacity
- Table 4c: Niagara Mohawk Power Corporation Firm Transportation Capacity

b. Table 5: Storage capacity data, including contract volumes and expiration dates. At our meeting, please be prepared to provide the current average price of gas in storage and your forecast for November 1.

Please see the following tables:
- Table 5a: The Brooklyn Union Gas Company Firm Storage Capacity
- Table 5b: KeySpan Gas East Corporation Firm Storage Capacity
- Table 5c: Niagara Mohawk Power Corporation Firm Storage Capacity

c. Table 6: Gas supply contract data including contract volumes, terms and expiration dates.

Please see the following tables: **(Business Confidential)**
- Table 6a: The Brooklyn Union Gas Company Gas Supply Contracts
- Table 6b: KeySpan Gas East Corporation Gas Supply Contracts
- Table 6c: Niagara Mohawk Power Corporation Gas Supply Contracts

14. Please provide a flow diagram of the gas system showing how the assets included in Tables 4, 5 and 6 are utilized to provide service to your customers.

Please see the following exhibits:
- Exhibit 14: Downstate NY Flow Diagrams
- Exhibit 15: NMPC Flow Diagram **(Business Confidential)**

15. Are you aware of any pipeline or other capacity asset projects that will impact your ability to deliver or supply gas (i.e. Constitution Pipeline, NFGS Northern Access, Williams Northeast Supply Enhancement Project, etc.)?

There are several pipeline projects in various stages of development that may affect the cost of delivered supply to National Grid's customers this winter heating season.

For its Downstate NY portfolio, the Company expects the most significant impacts are likely to result from Transco's Atlantic Sunrise. Transco's Atlantic Sunrise project will add 1.7 Bcf

of capacity from Northeast PA south to Station 85. National Grid is not a shipper on this project, but it will provide incremental supplies to Station 195 and will present opportunities to fill the Company's Transco capacity, including the future Northeast Supply Enhancement Project.

Customers in Downstate NY may also see impacts as a result of the Mountain Valley Pipeline project. Currently under construction, the project is expected to provide up to 2 Bcf of capacity from the Equitrans transmission system in Wetzel County, West Virginia to Transco's Station 165. Like Atlantic Sunrise, National Grid is not a shipper on this project, but expects the new supplies to present additional opportunities to fill its Transco capacity. The Mountain Valley Pipeline continues to target an in-service date of Q4 2018.

The Company recently contracted with Tennessee Gas Pipeline Company for two twenty-year agreements that in aggregate will allow NIMO to move 50,000 Dth/day from Dracut, MA to its city-gate in South Albany, NY. To address liquidity concerns at Dracut, Portland Natural Gas Transmission System ("PNGTS") has proposed a phased in project that once fully in service will deliver supplies from Dawn, Ontario into Dracut via Union and TransCanada pipeline. Service on the PNGTS expansion is expected to commence as early as Q4 2018. Although NIMO is not a shipper on the PNGTS project, the Company anticipates the PNGTS project will increase liquidity at Dracut.

16. Describe, if applicable, current practices and any anticipated changes related to on-system peaking facilities and other peak shaving techniques. If you operate LNG peaking facilities, please describe current plans for any activities at the facilities that would interrupt their availability and steps being taken to mitigate those interruptions. Please also list any anticipated permitting issues or resistance from local or other government entities

(Business Confidential)

[redacted]

Please see the Company's response to question 28(i) regarding CNG.

17. A list of the dates, times and durations of all OFOs and interruptions or curtailments on your system during the 2017-18 heating season. Please differentiate the interruptions

and curtailments between electric generator and other interruptible customer class curtailments.

(Business Confidential)

Please see the following exhibits:

- Exhibit 16: DSNY OFOs
- Exhibit 17: USNY OFOs
- Exhibit 18 Upstate NY & Downstate NY Interruption Curtailment Summary
- Exhibit 19: Downstate NY Interruptibility Matrix

18. An explanation of how the company determines capacity and peaking supplies required for each of its interruptible service classes if utilized.

National Grid does not contract for or provide capacity or peaking supply requirements for its interruptible service classes in Upstate or Downstate NY.

19. A description of the long and short term forecasting process used for gas dispatch purposes. Include all weather services and a description of any in-house software utilized. Please explain how accurate your short term forecasts were during the 2017-18 heating season by using a back cast after the actual weather is known.

Long term forecasting is done per the methodology described in response to Question 5 above.

Short term forecasting is a joint effort between Energy Procurement and Gas Control. Both areas use Weather Services International (WSI) for weather forecasts. WSI provides a 10-day forecast, which is updated five times per day. WSI also includes actual weather for the previous day. The data include hourly temperatures, high & low temperatures with the time of day, wind speed, and cloud cover. The information is transmitted via e-mail and FTP. Additionally, the Company receives zone forecasts for the Company's service areas in a text format from Meteorlogix.

Two methodologies are used for short term forecasting:

- Matching Days Selection: This is an application built into the Company's SCADA System.
- Rate Curve forecasting - Energy Procurement provides a rate curve calculation each month for each of National Grid's service areas, as described in response to Question 12 above. SCADA Support personnel enter these calculations into the SCADA System. The weather data from WSI are loaded into the system and applied to the send out equations. Send out forecasts for each area are updated five times daily with scheduled weather updates.

Gas Supply Planning ensures sufficient supplies are dispatched in a least-cost manner to meet the send out forecast from Gas Control and incorporates flexibility into the dispatch based on similar days.

Back cast

The Mean Absolute Percentage Error ("MAPE") of the 7 AM (day-ahead) weather and customer requirements forecast compared to actual weather and requirements is provided in the tables below. *Note send out is raw and not corrected for weather.

	Send out		
	KEDLI	KEDNY	NIMO
Dec	4.39%	4.23%	3.11%
Jan	3.73%	3.68%	5.31%
Feb	4.25%	4.09%	5.16%
March	3.51%	5.61%	3.57%
	Weather		
	KEDLI	KEDNY	NIMO
Dec	4.47%	4.47%	8.18%
Jan	4.39%	4.39%	6.62%
Feb	2.58%	2.58%	2.55%
March	2.71%	2.71%	3.21%

Please see the following attachment:

- Exhibit 20: Matching Days Selection **(Business Confidential)**

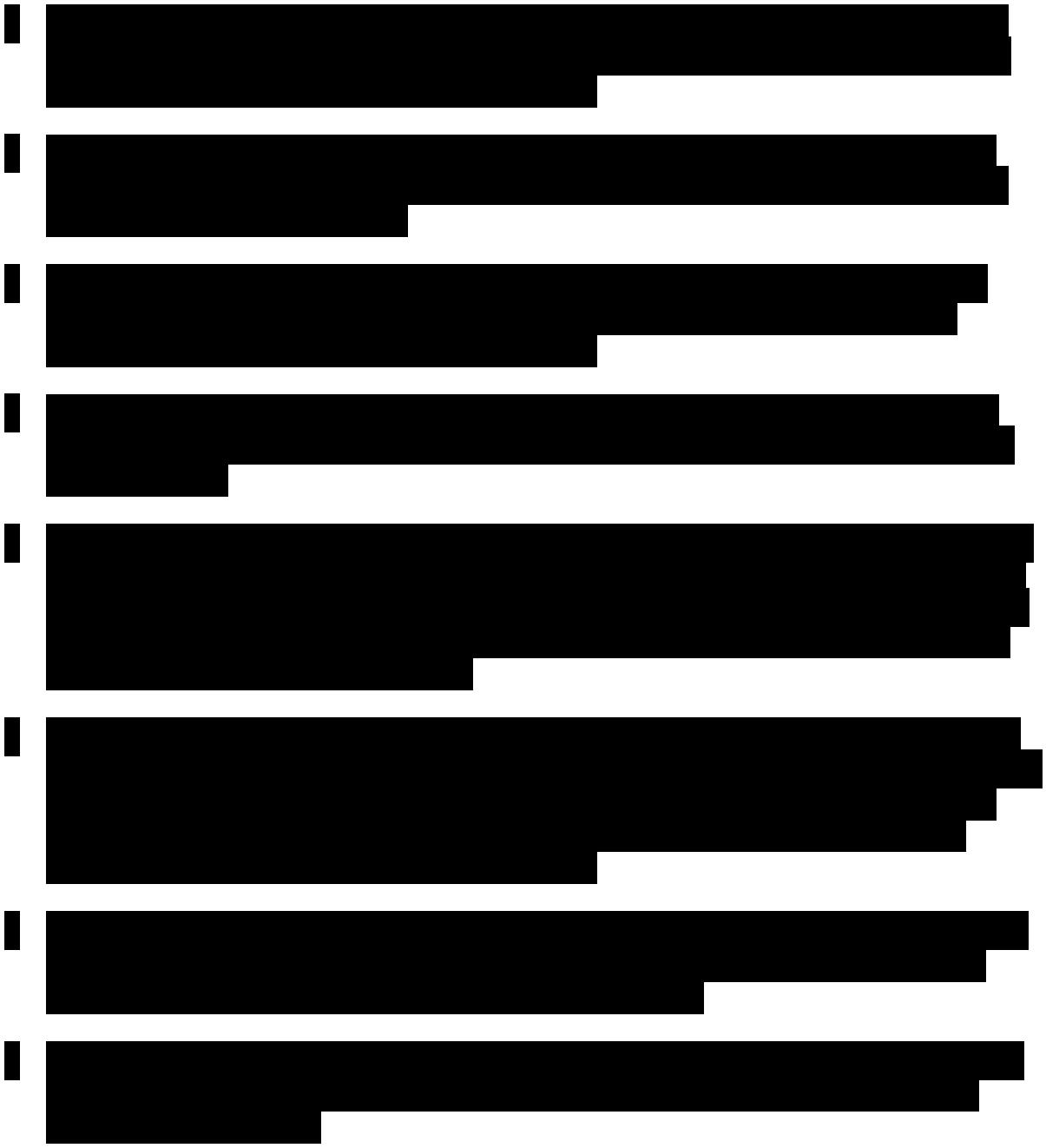
20. If your company has had a management audit within the last three years, please list any recommendations from that audit that relate to gas supply procurement, load forecasting, gas price risk management, or system planning. Explain whether or not these recommendations have been implemented, and the status of any changes.

The Company has not had a management audit in the past 3 years.

21. A detailed description of any existing asset management or asset optimization agreements, as well as any such agreements being considered or planned. All agreements that include firm capacity and/or supply that is recallable during the winter heating season should be included in table 1.

(Business Confidential)

Existing AMAs:



[redacted]

Considered/Planned AMAs:

[redacted]

22. A description of your company's plans and strategy with respect to off-system sales, capacity release and streaming arrangements for the 2018-19 winter season as well as any such transactions that extend beyond the 2018-19 winter.

(Business Confidential)

[redacted]

23. Status of mandatory capacity release and grandfathered capacity programs.

a. Status of marketer compliance with the Commission's primary point capacity requirement for grandfathered capacity. Include how much grandfathered capacity remains on your system.

All marketers serving core customers rely upon capacity released by the Company. There is no grandfathered capacity remaining on our system.

b. Please describe the methodology utilized to determine the mandatory capacity release to the marketers. Indicate how this compares with the methodology utilized to determine capacity required for firm sales customers.

The Company plans portfolios for all core customers to reliably meet their requirements under design conditions. The portfolios are comprised of year-round transportation contracts, storage contracts, and peaking contracts. The deliverability of these assets corresponds to customer requirements as closely as possible. The Company then releases capacity to marketers pursuant to the descriptions below. The remaining assets are used to meet the requirements of firm sales customers and non-core customers and to balance system load.

Downstate NY

ESCOs serving customers under the Core Monthly Balancing Program must participate in the Company's Mandatory Capacity Program subject to the terms and conditions of the KEDLI and KEDNY tariffs and the Gas Transportation Operational Procedures Manual. The Mandatory Capacity Program consists of four capacity tiers: 1) Tier 1 – Pipeline Capacity Release, 2) Tier 2A – Physical Storage Release, 3) Tier 2B – Retail Access Storage and 4) Tier 3 – Peaking Service if an ESCO's Tier 1 and 2 allocations cannot meet estimated requirements. The Company will use the aggregate Base and Slope components of each ESCO pool to provide the ESCO with sufficient portfolio assets to meet the design weather requirements of each ESCO's pool. These portfolio assets will reflect (on a proportional basis) the mix of transportation and storage assets used by the Company to serve its firm sales customers. ESCOs will also be offered access to additional storage assets every April in the form of a physical release of storage capacity from the Company's Gulf Coast Storage assets.

Niagara Mohawk Power Corporation

Each month the Marketer has the ability to enroll or drop customers from their pool. After the deadline for enrollments is done National Grid's system Transportation Services Application (TSA) receives a file from National grid's system Customer Service System (CSS). The file from CSS contains customer's base & slopes for each Marketer. TSA takes all the Base/Slopes to calculate their MPDQ and then does a calculation to sum all of the Marketers customers MPDQ and calculates the Marketer's Pool's total peak day allocation.

c. Please describe how your company keeps marketers informed of changes in procedures. Include the frequency and past/proposed dates of marketer meetings relating to the 2018-19 heating season.

The Company sends emails on a regular basis notifying marketers of any changes in procedures, operational issues, or regulatory requirements. The Company hosts an annual post-winter meeting to discuss all issues and relevant proposed changes. We also

had two meetings via WebEx to specifically discuss proposed changes to the NMPC Customer Choice Program prior to our annual meeting. The date of our last marketer meeting was May 16, 2018. Any changes to the Customer Choice program for the upcoming heating season were discussed during this meeting.

d. List the pipelines and allocation percentages being utilized for the mandatory assignment of capacity.

Downstate NY

As of November 2017, Tier 1 pipeline capacity volumes shall be released in the following manner and percentages:

Path	Allocation %	Term
Transco Long Haul	29.7%	Monthly
Transco Leidy Line	6.9%	6 Months
Transco Rockaway	17.0%	6 Months
Texas Eastern (Tetco) Long Haul	7.1%	6 Months
Texas Eastern (Tetco) Short Haul	5.6%	6 Months
Tennessee Pipeline	2.6%	6 Months
Dominion Leidy	7.0%	6 Months
Iroquois NE07	15.1%	Annually
Iroquois Dawn via Waddington	9.0%	Annually

These percentages are subject to change based upon the amount of the Company's total contracted pipeline capacity at the time of release. The Company will recalculate the ESCO's required capacity volume each month to reflect an increase or decrease in the number or demand of customers and adjust each ESCO's Tier 1 Transco Long Haul capacity release volume accordingly. Should the ESCO during the Gas Year have a decrease in the number of customers where by their transportation capacity release volumes are less than their monthly release on Transco Long Haul, adjustments will be made on the remaining pipelines in the following order: Transco Leidy, Transco Rockaway, TETCO, IROQ DAWN and IROQ NE07.

Niagara Mohawk Power Corporation

In the upstate NY service territory, Dominion Transmission, Inc. (DTI) pipeline and storage capacity comprise 100% of the capacity released to marketers. The following percentages of DTI capacity are released to the Marketers as of November 2017:

Capacity	Percent of Maximum Peak Day Quantity
DTI FTNN Pipeline Capacity	45%
DTI GSS Storage Demand	55%
DTI GSS Storage Capacity	55% MPDQ x 52.3 days
DTI FTNN-GSS Pipeline Capacity from Storage	55%

As a result of the Joint Proposal in the Company's recent rate case, beginning in November 2018, marketers will receive their pro rata share of all releasable/assignable assets in the Company's portfolio at the contractual demand rate.

e. Please provide a comparison between your company's weighted average cost of capacity and the charges paid by marketers and direct customers for released capacity. What process, if any, is utilized to true-up any differences?

Downstate NY

For the 2017/18 season (from July 2017 through June 2018), the KEDNY/KEDLI WACOC was between $0.8632 and $0.9006 per Dth. All releases to marketers were made at the WACOC. There is no true-up and/or reconciliation process in place to account for any differences in cost.

Niagara Mohawk Power Corporation

For the 2017/18 season (through June 2018), Niagara Mohawk Power Corporation's transportation WACOC is $0.2536 per Dth. Marketers were released Dominion transportation capacity at the maximum pipeline tariff rate of $0.1372 per Dth. Dominion GSS storage and FTNN-GSS transportation from storage to the city gate was released at maximum pipeline tariff rates as follows:

Capacity	Release Rate ($/Dth/month)
DTI GSS Capacity	$0.0145
DTI GSS Demand	$1.8672
DTI FTNN-GSS Transport	$4.1743

There is no true-up and/or reconciliation process in place to account for any differences in cost. Beginning in November 2018, marketers will receive their pro rata share of all releasable/assignable assets in the Company's portfolio at the contractual demand rate.

f. Please describe how your company determines the daily delivery quantities (DDQ) provided to marketers each month for their daily delivery requirements. Provide a sample calculation.

Downstate NY

A Customer's Core DDQ is calculated by multiplying forecasted HDDs by a customer's Slope and adding the product to the customer's Base. The Core DDQ is adjusted by a UFG factor as an allowance for losses incurred in the process of delivery to the customer's metered facilities. Each ESCO's aggregate Core DDQ is rounded to the nearest whole dekatherm and posted to the Electronic Bulletin Board (EBB) as the ESCO's daily requirement.

KEDNY Example:

If the Projected Temperature is 42 degrees, 65-42 = 23 HDD for that specific day. The UFG = 1.0227 and the customer's base is 50 and slope is 3, then the formula is:

DDQ = 1.0227 * (50 + (3 * (IF (Forecasted Temperature > 65, 0, 65 – Forecasted Temperature)
DDQ = 1.0227 * (50 + (3 * (23)))
DDQ = 121 Dth for that day and it's posted directly into our EBB for the specific marketer.

Niagara Mohawk Power Corporation

NMPC refers to the DDQ as the Daily Contract Quantity (DCQ).

DCQ = (HDD X (Thermal Response X Monthly Thermal Adjustment Factor)) + daily base load.

If this calculation produces DCQ's greater than a marketer's DTI FT release, National Grid will only expect a nomination that is less than or equal to their DTI FT release.

Thermal Adjustment Factor:

The percentages listed in the normal response column below are applicable when HDD are less than or equal to the HDD threshold. When HDD are greater than the threshold, the cold response percentages are utilized. The Company believes this adjustment provides a more accurate estimate of customer requirements at colder temperatures during shoulder months compared with no adjustment.

Month	*Normal Response*	*HDD Threshold*	*Cold Response*
Nov	100%	70	100%
Dec	100%	70	100%
Jan	100%	70	100%
Feb	100%	70	100%
Mar	100%	70	100%
Apr	85%	10	100%
May	50%	10	100%
Jun	25%	70	25%
Jul	0%	70	0%
Aug	0%	70	0%
Sep	40%	15	100%
Oct	85%	24	100%

Example:

If the Projected Temperature is 42 degrees, 65 - 42 = 23 HDD for that specific day. The customer's base is 50 and slope is 3, and it is September. The formula is:

DCQ = (23 X (3 X 100%)) + 50
DCQ = 119 Dth for that day

In Upstate NY, the Monthly Balancing ESCOs Unaccounted For Gas (UFG) is only included in a true up report generated by National Grid's website Transportation Service Application (TSA). The ESCO's Actual usage will be trued up to the ESCOs actual DCQs by billing cycle on a monthly basis. Cashouts will be determined using the average of the Gas Daily DTI North Point / Mid-Point, plus DTI Fuel, plus DTI's Commodity Charge for each period for which the DCQ's were delivered. When the Gas Daily DTI North Point/Mid-Point is not posted, the Gas Daily DTI South Point/Mid-Point will be used.

g. Please indicate if any marketers serving core customers on your system failed to perform as anticipated during the previous winter, and if so, what steps you took to ensure reliability of service.

No marketers failed to perform during the 2017/18 winter.

24. Description and status of efforts to verify customer alternative fuel availability and equipment testing, including:

a. Methods utilized to verify dual-fuel customers' capabilities, including power generation customers.

NMPC, KEDNY, and KEDLI hold annual tests at the beginning of each winter heating season to verify that dual-fuel customers are capable of switching from natural gas to their alternate fuel. With regards to power generators, KEDNY and KEDLI have a process to confirm that there are five days of alternate fuel on hand.

Field visits are made to all customers who fail to switch to their alternate fuel during the test to determine the cause of failure. These field visits are conducted as soon as possible following the test to ensure the necessary remedies are put in place prior to the onset of extreme winter cold.

Currently, NMPC sends an Alternative Fuel Source Verification ("AFSV") form to each non-firm customer, requiring the customer to identify their alternative fuel source. If the AFSV is not returned, NMPC presumes the customer uses No. 2 fuel oil as their alternate fuel. NMPC then sends each customer an Alternate Fuel Source Requirements ("AFSR") form that sets forth the alternative fuel source storage capacity requirements for their alternate fuel. The customer is required to return the AFSR form specifying its total on-site alternative fuel storage capacity. The customer must select one of the following three options to satisfy the alternative fuel source requirements:

1. Fill on-site storage and enter a contractual relationship with a fuel supplier (if necessary);
2. Curtail operations to meet the available on-site storage; or
3. Shut down operations

Beginning in the Winter of 2017/2018, pursuant to the terms of the Joint Proposal in Cases 16-G-0058 and 16-G-0059, KEDNY and KEDLI required customers to provide affidavits confirming that they have alternative fuel supply contracts in place for the upcoming winter heating season and that such contracts provide for alternative fuel replenishment on an as-needed basis. The affidavits also include information regarding on-site storage facilities, total storage capacity, and estimated number of peak days of storage. This requirement was the first year of a three-year phase-in period. The Company will collect affidavits at random from one-third of its non-firm customer base each year. New affidavits will be required from individual customers every three years. Customers will be responsible for providing new affidavits if there are any changes to the affidavits originally submitted prior to the three year renewal.

b. Please provide the results of compliance with the interruptible rules during last winter. Be sure to include the number of customers switched to firm service or removed from gas service due to non-compliance.

KEDNY:
November 16 Annual Test – 26 violators
December 28 – January 3 – 85 violators
January 4 – January 8 – 135 violators
Three customers were switched to firm service due to non-compliance.

KEDLI:
November 14 Annual Test – 4 violators
December 28 – January 3 – 8 violators
January 4 – January 8 – 4 violators
February 3 – 2 violators
No customers were switched to firm service due to non-compliance.

NMPC:
November 15 Annual Test – 0 violators
December 28 – December 29 – 2 violators
December 31 – January 2 – 0 violators
January 5 – January 7 – 0 violators

c. How many customers will be visited out of how many customers in total? Will all customers with non-compliance issues last winter be visited? How often will the complaint customers be visited? In the Downstate NY service territories, there are 2,596 KEDNY customers and 175 KEDLI customers on dual fuel rates. In the Upstate NY service territory, there are 27 alternate fuel customers.

KEDNY and KEDLI will conduct visits to those Downstate non-firm customers who fail to comply with the switchover test or interruptions. NMPC non-firm customers who failed to supply the alternate fuel verification forms are site visited.

d. What are the alternate fuels and how many customers are in each fuel category?

All of the Downstate NY customers have the capability of using #2 oil as their alternate fuel.

Please see below for the alternate fuels for the UNY customers.

Fuel Type	Customer Count
Distillate Fuel Oil (#2 oil or diesel)	13
Non-Human Needs - #6 Fuel Oil	4
Peak Shaving - #2 Oil	0
Peak Shaving – Kerosene	1
Human Needs - #2 Fuel Oil	1
Biodiesel	0
#4 Fuel Oil	1
Electric	2
Coal	0
Firm Service – Empire 63W	1
Not Eligible for Rule 3.4	4
Total	27

e. Are affidavits required?

Please see response to Q24.a above.

f. Outcome of review? Rechecks?

Please see response to Q24.b above

g. Provide a copy of this year's (if available) and last year's pre-season letter(s), if applicable. Have you made or are you planning to make any changes to these letters based on the events of the 2017-18 heating season? If yes, what are the changes?

Please see the following exhibits:

- Exhibit 21: KEDNY TC Letter
- Exhibit 22: KEDLI TC Letter
- Exhibit 23 : KEDLI IT Letter
- Exhibit 24: Upstate NY Letter, Upstate NY Alternate Fuel Source Verification, Upstate NY Alternate Fuel Source Requirements

h. Did your experience servicing dual fuel customers during last winter indicate the need for additional alternate fuel inventory requirements? If so, what changes do you recommend?

The Company does not think there is a need for additional alternate fuel inventory requirements.

i. Is the company aware of any issues regarding interruptible customers not receiving their oil deliveries during the winter season? If so, please provide details of when and where this occurred, as well as what the company would suggest could be done to help these customers.

One customer in UNY experienced issues getting delivery of #6 oil in late December. NYC schools experienced issues with oil deliveries during the second interruption and had to burn natural gas.

j. Will you be modifying your procedures for verifying alternate fuel inventories being held by interruptible customers (including generators and temperature-controlled customers) as a result of the winter of 2017-2018? If so, how? If affidavits are not used, explain why not?

NMPC, KEDNY, and KEDLI will not be modifying its procedures for verifying alternate fuel inventories. However, as part of the TC/IT Collaborative in Cases 16-G-0058 and 16-G-0059, KEDNY and KEDLI are discussing potential revisions to the penalty language with the collaborative parties.

25. Describe the methods used to communicate with interruptible customers, their marketers/fuel suppliers, NYSERDA and the various Oil Associations in New York prior to, and during, periods of interruption in compliances with Commission Order in Case 11-G-0543.

The Company communicates with interruptible customers, their marketers/fuel suppliers, NYSERDA and the various NY Oil Associations via numerous methods including texting, faxing, calling, and e-mailing pursuant to the Commission's Order in Case 11-G-0543.

26. Please provide the total number of firm dual fuel and interruptible customers, by service class, including how many interruptible are temperature controlled. How will the switch to their alternate fuel be accomplished and ensured? Please indicate the number of process customers that are exempt from maintaining alternate fuel supplies and have indicated intention to do so, and provide a copy of the affidavit to be submitted by those customers. Please provide a copy of the letter that will be sent to all dual fuel customers if there are five or more interruptions prior to February 15, per Commission Order in Case 11-G-0543.

KEDNY has 2,596 TC customers and one interruptible customer. KEDLI has 94 TC customers and 81 interruptible customers. NMPC has 27 non-firm customers (note that NMPC does not have a TC service). Downstate NY does not track whether a firm customer has dual fuel equipment. KEDNY and KEDLI have tariff provisions that TC and interruptible customers who switch to firm service must disconnect their alternate fuel equipment. KEDNY and KEDLI currently do not have any process customers that are exempt from maintaining alternate fuel supplies. Please see Exhibit 25 which contains the fuel replenishment letter for dual fuel customers of KEDNY and KEDLI.

KEDNY/KEDLI TC customers with automatic switchover equipment will be switched to an alternate fuel automatically by the Company's control rooms. TC customers with semiautomatic equipment will be notified to switch via an alarm from the control room. Interruptible customers will be notified via the phone calls. The Company will identify any customers who burn unauthorized gas during the interruption. Technicians will be sent to the location to identify the reason for the burn and to repair any problems with National Grid equipment. If customers' burn gas for reasons associated with customer equipment or low levels of alternate fuel, the customer will be penalty billed for unauthorized gas use. If deemed necessary for reliability purposes, such customers will be disconnected from gas service.

Please see the following exhibit:

- Exhibit 25: Universal NY Letter Replenishment

27. A current organization chart for your company's gas supply department. Please include a list of contact people for the winter season for updated storage, peaking and other supply related information. Include the chief dispatcher and telephone numbers for both weekdays and weekends.

(Business Confidential)

The primary contact for updated storage and peaking information is Dolores Petrocelli at [redacted]. Alternate contacts are Robert Moore at [redacted] and MaryBeth Carroll at [redacted]

Please see the following exhibits:

- Exhibit 26: Gas Supply Planning Organization Chart
- Exhibit 27: Gas System Control Organization Chart

28. Please provide the following information on conversions to natural gas:

a. Requests received per year, for the last five years, from customers using heating fuels other than natural gas. Provide the data broken down between residential and non-residential customers. If there has been an increase in requests, how is the company handling such an increase?

	FY14	*FY15*	*FY16*	*FY17*	*FY18*
NYC	12,433	9,261	6,575	4,160	5,258
LI	49,628	36,070	30,342	22,100	23,610
Upstate NY	1,947	1,269	1,381	978	1,082
Total	**64,008**	**46,600**	**38,298**	**27,238**	**29,950**

b. Do you see new opportunities to expand gas services, regardless of the ever changing cost differential between natural gas and its alternatives? If so, please explain any

plans and the expected number of customer conversions. Please outline coordination of these activities with Case 12-G-0297, which is investigating expansion of the natural gas system in New York State.

KEDLI sees opportunities to expand gas service in its service territory where the natural gas heating saturation level is only 47%. However, the price advantage has narrowed compared to full service oil which represents up to 60% of the market. For the 40% of the market that buy oil "cash on delivery," oil is approximately the same price of natural gas.

The Joint Proposal in Case 16-G-0058 provides for the continuation of KEDLI's Neighborhood Expansion Program. The Neighborhood Expansion program utilizes geospatial, engineering, main, supply, customer interest, customer load and other data to identify good locations for growth projects – streets or neighborhoods where prospective customer density would support entitlements-based main extensions. Once these areas are identified, KEDLI looks to secure commitments from a threshold level of customers to justify the capital investment in the infrastructure necessary to serve the area. KEDLI's Neighborhood Expansion Program results are shown in the table below:

	CY15	*CY16*	*CY17*
Number of Services	375	576	602
Feet of Main	58,000	130,000	100,183

To promote gas growth in NMPC's service territory (outside of the East Gate region), the Joint Proposal in Case 17-G-0239) provides for the continuation of NMPC's oil-to-gas conversion and Neighborhood Expansion programs.

NMPC Neighborhood Expansion Program results are shown in the table below:

	CY14	*CY15*	*CY16*	*CY17*
Number of Services	120	176	124	78
Feet of Main	12,478	14,309	14,617	15,412

c. Identify and areas within your service territory where you have placed a moratorium on the addition of new customers, new service or any restrictions on existing services. Describe the reasons why such a restriction is necessary.

Upstate NY

The Company has currently restricted the following in the NMPC East Gate region:

- Migration from Daily Balanced Transportation service to Monthly Balanced Transportation service or sales service
- All new Standby Sales Requests

The reason for the moratorium is operational in nature. During a peak hour, East Gate requirements exceed the allowable flow rate from DETI. The Company has utilized its TGP city gate in South Albany to offset DETI flows, but is limited in takeaway capacity at that location. The Albany Loop Project, scheduled to be constructed in the next 3-5 years, will increase takeaway from that location. Until then, the Company will look to CNG to help manage peak hour requirements.

The Company is currently evaluating the need to expand the moratorium to include other customer classes as well.

Downstate NY

The Company is closely monitoring the timeline for the NESE project and associated approvals. Further delays may impact new firm customers.

d. Please provide a list of Title V air permit holders in your service territory (available on the DEC website) and indicate whether there is currently natural gas service to each.

There are 405 NYS DEC Title V Air Permit holders in National Grid's service territory. Natural gas service is available to approximately 281 of these locations. Please see the attached spreadsheet for the details.

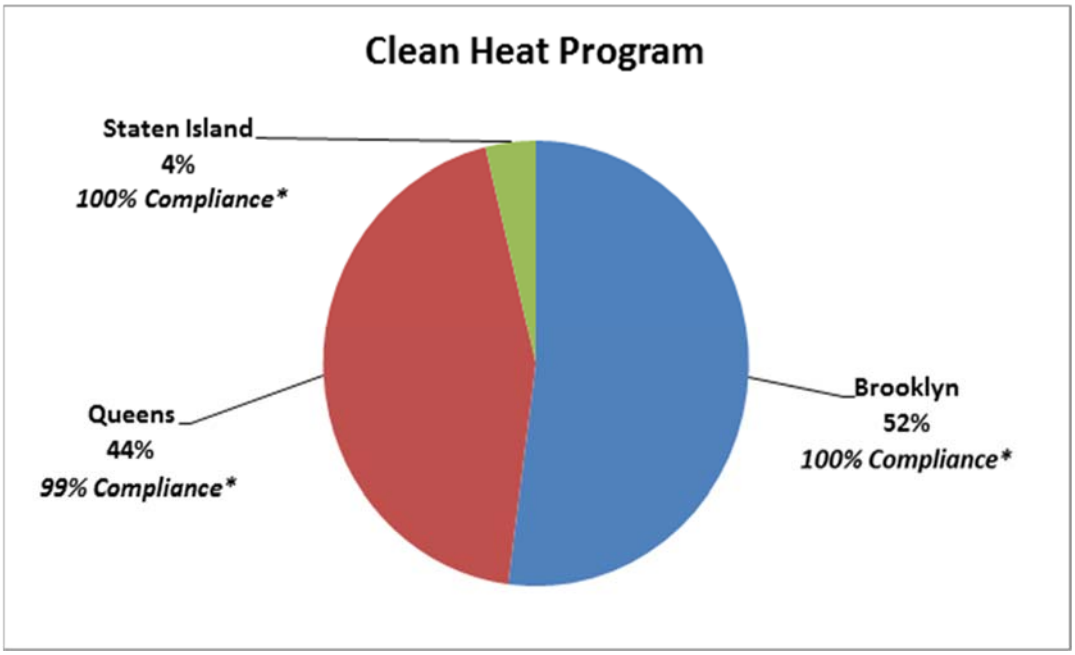
Please see the following exhibit:

- Exhibit 28: List of NYS DEC Title V Air Permit holders

e. If you serve customers in the New York City area, describe your analysis and the impacts to your system of the city's anticipated proposal to require larger buildings to convert from heavy fuel oils (#4 and #6) to lighter oil (#2 and bio-fuels) and/or natural gas for space heating.

Under the NYC Clean Heat program, National Grid has converted nearly all of the approximately 800 buildings in its service territory from No. 4/6 oil to natural gas. As a result, customer requirements have grown and infrastructure to support these conversions has been installed where needed.

The City Clean Heat program deadline to phase out of all No. 6 heating oil was June 30th, 2015. To date, the NYC Department of Environmental Protection has achieved 99.8% compliance with the regulation. The deadline for the phase out of all No. 4 heating oil is January 1st, 2030. At this point the City has achieved it goals and is focusing on pursuing other carbon reductions options.



Please note: Staten Island is the first borough to comply with LL43 and no longer burning #6 oil.

f. Please provide a list of natural gas distribution system expansion projects, including new franchise opportunities, which are being pursued in the next five years. If there are none, please explain how this is justified given the Commission's stated goal of expanding the natural gas system in New York State.

National Grid is constantly looking at ways in which to expand the availability of gas to customers in New York. The Company's capital programs in the NMPC, KEDNY, and KEDLI service areas include gas main extensions or upgrades to support new growth in accordance with tariff guidelines. The Company actively reaches out to potential customers both on and near gas mains to promote conversion to gas service and offers assistance to both building owners and contractors. In addition to its regular gas expansion, sales and marketing activities, National Grid currently has a number of projects/programs in place (or proposed) in accordance with the Gas Expansion Collaborative that are intended to develop new approaches to support the goal of expanding the natural gas system as follows.

Niagara Mohawk

CASE 12-G-0202

Over the past three years, NMPC, Department of Public Service Staff, and other interested parties have participated in the Niagara Mohawk Gas Growth Collaborative formed to consider programs to promote the use of natural gas in the Company's service

territory and address the challenges of converting new gas customers. As part of the Collaborative, the Commission approved the Company's proposed Neighborhood Expansion Program ("NEP") under which the Company utilizes customer density and propensity modeling to identify main extension opportunities in streets/neighborhoods. The NEP was continued in NMPC's recent rate case (Case 17-G-0239). A 1,450 foot NEP project with 23 customers has been sold and is in the design stage. The Company has marketed to a number of projects totaling 23,316 feet of main with 328 prospects for CY18.

KEDLI

CASE 12-G-0202 and 14-G-0214

In its December 15, 2014 order, the Commission approved KEDLI's implementation of a Neighborhood Expansion Program functionally identical to NMPC's NEP. KEDLI's NEP was approved to continue in Case 16-G-0058. The Company had identified 742 potential main projects that could serve more than 16,000 potential customers (Most of these projects were 500-1,000 feet in length). However, this approach did not yield as many results as anticipated. The Company has since identified 144 large projects that average approximately 13,000 feet of main per project. The largest completed project to date is the Village of East Hills totaling 80,000 feet of main with 500 services installed to date. The Company is actively engaged with six large projects totaling 160,000 feet of main. To complement these efforts the Company is pursuing smaller main extensions when new inquiries meet NEP density requirements.

KEDNY

The age, density, and existing KEDNY territory coverage does not lend to significant expansion projects. The Company will continue to monitor areas of the system where capacity concerns may start to have limiting impact for future reinforcements such as Northern Queens, which was indicated in last year's response.

g. Please provide a 3-year forecast of all projects advancing REV clean energy/demand response natural gas solutions.

- Please see Exhibit 29: "April 20 2018 Quarterly Report Filing- Gas Demand Response"

h. Please identify any potential use of renewable gas resources, or demand response program proposals that may address capacity or pressure constrained service areas, that could be utilized to reduce peak day demand and alleviate the need for additional interstate pipeline capacity or gas distribution network upgrades.

In addition to the Company's responses to 28g and 28i NMPC is in the middle of a collaborative to investigate non-pipe alternatives to alleviate on-system constraints and potentially provide incremental peak day/hour gas supplies.

i. Please identify any potential use of CNG or LNG as a non-pipes alternative (NPA) to alleviate the need for additional interstate pipeline capacity or gas distribution network upgrades.

(Business Confidential)

[redacted]

Issue 4

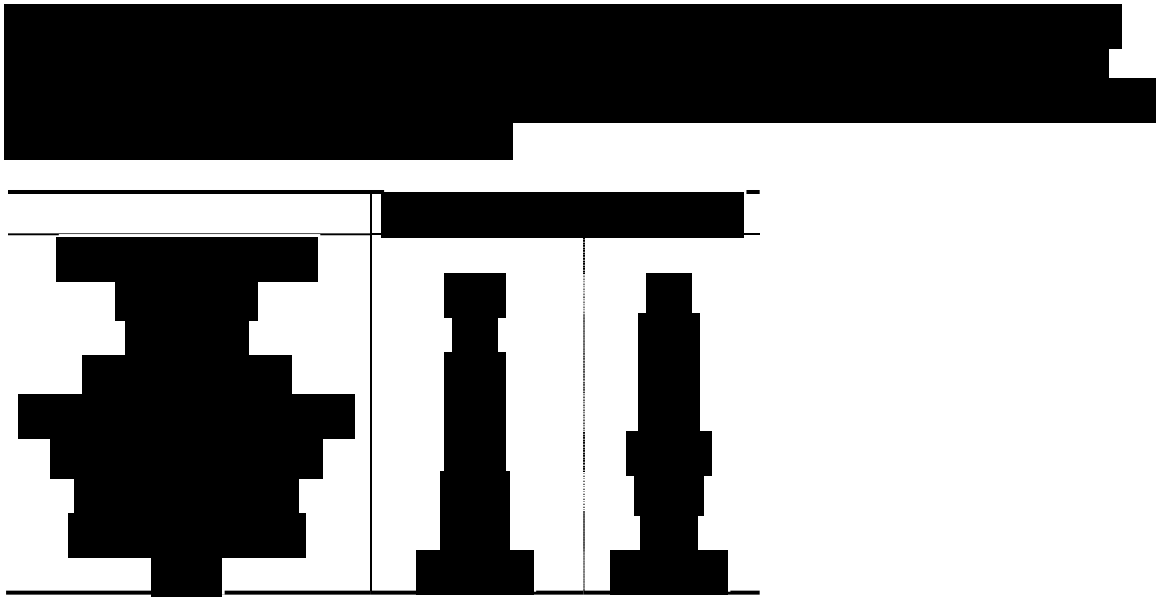
Please provide the following information related to your company's plans to diversify purchases and mange gas price risk:

29. A description of your company's gas purchasing strategy, including:

(Business Confidential)

a. Information regarding gas purchases for last year and any planned changes for this year. Did your experience during the winter of 2017-18 lead to any changes? If so, what are the changes? If not, why not? Please include an identification of the amount of Canadian, domestic gas (identify shale gas purchases if available) and CNG/LNG purchased.

[redacted]



b. The types of contracts and associated contract flexibility.



c. Spot purchases of gas supplies for a term of one month or less are made throughout the year to supplement term supplies to meet system demand, manage demand variations, and maintain planned storage inventory targets. North American Energy Standards Board (NAESB) form standard contracts have been established with a number of qualified and reliable gas suppliers, which spreads acquisition and performance risk among many counterparties, thus improving reliability. Soliciting bids from multiple counterparties also affords more competitive pricing. The extent of planned reliance on firm gas, spot gas, swing gas, etc.

[redacted]

d. The description of any triggers to purchase spot (daily) gas.

[redacted]

e. Pricing terms, indices, etc. of the contracts.

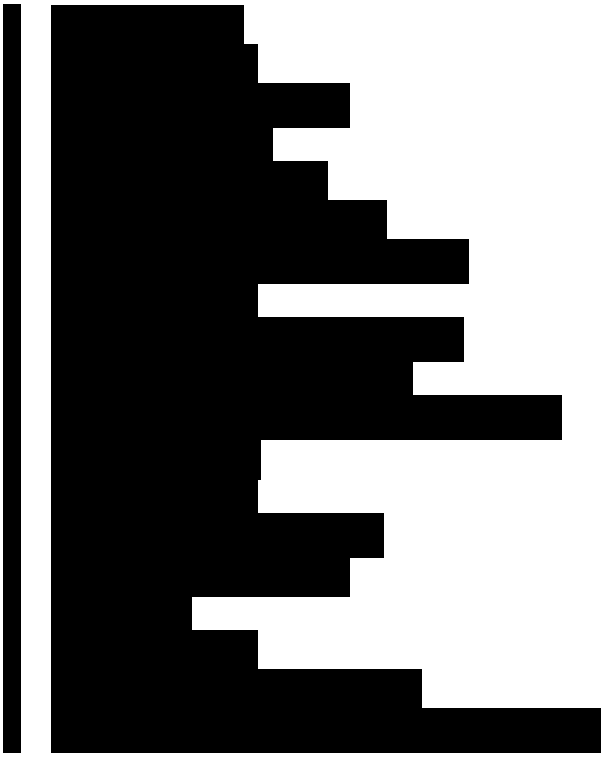
[redacted]

[redacted]

[redacted]

f. The liquid point(s) that you typically purchase at.

[redacted]



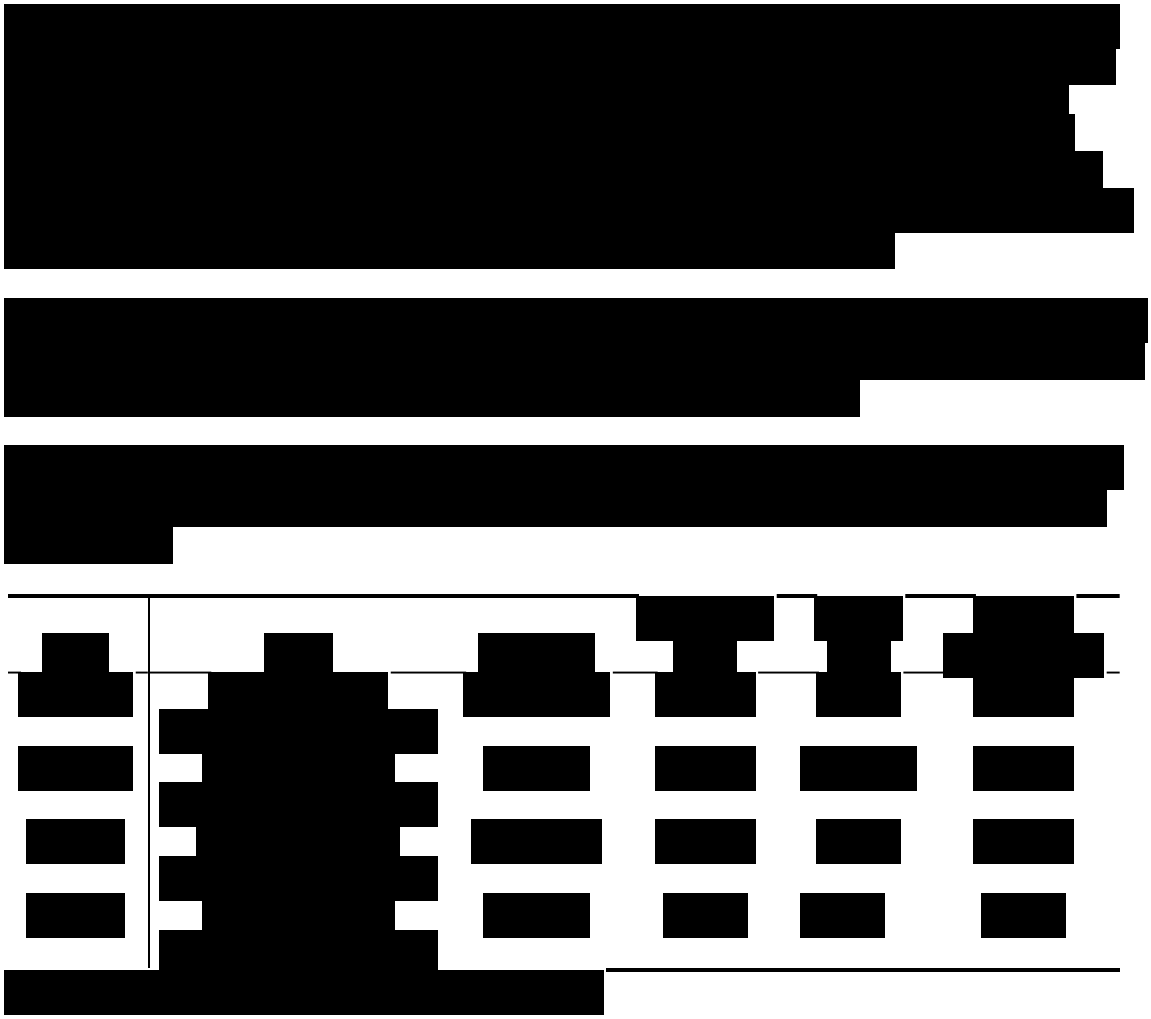
g. The effects that recent and proposed pipeline projects and new supply sources of gas have had on your current (and long-term) purchasing strategy. Include the breakdown of the volumes of gas purchased for the 2017-18 winter and projected for the 2018-19 winter purchases from the Northeast (Marcellus/Utica), mid-continent, Gulf and Canadian supply regions.

[redacted]

[redacted]

h. Strategy for using storage assets going forward in light of Marcellus area production, since what was once market area may now be considered production area.

[redacted]



(Business Confidential)



i. Any analysis of your use of long-haul capacity versus short haul and/or local production going forward, including any contracts recently or planned to be converted from long-haul to short-haul.

(Business Confidential)





[redacted]

30. A description of your company's gas price risk management strategy, including answers to the following questions:

(Business Confidential - Responses a through m, o)

a. What percentage of your gas supply do you hedge (1) physically and (2) financially?

[redacted]

b. Please break this down between storage and fixed price contracts.

[redacted]

c. If you use fixed price contracts, how, when and in what increments are they purchased?

[redacted]

d. Please provide the breakdown between futures and options (include quantities of each type on an annual and winter season basis).

[redacted]

e. How do you finance your swap/futures? Do you pay for them at the time of purchase or delivery?

[redacted]

f. What types of options do you use?

[redacted]

g. Describe how you decide which types of options to use.

[redacted]

h. How much and what percentage of total gas costs, booked to the GAC, do you spend on options?

[redacted]

i. How far out, when, and in what increments do you purchase futures?

[redacted]

j. How has your hedging strategy changed in the past year? Did your experience during the winter of 2017-18 lead to any changes? If so, what are the changes? If not, why not?

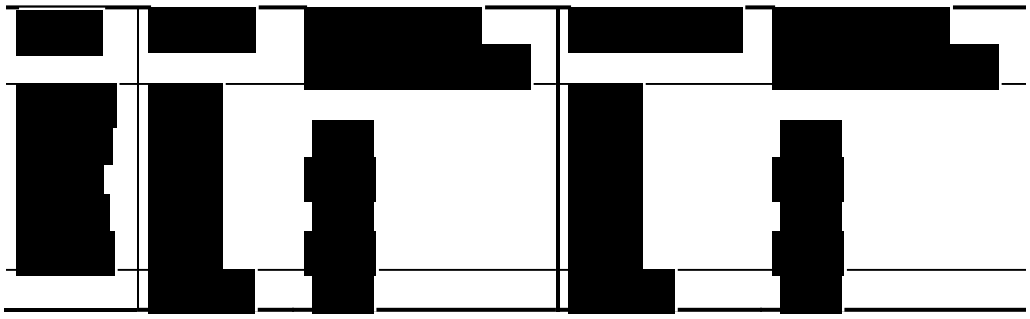
[redacted]

k. Describe any lessons learned in the past year.

[redacted]

l. Do you calculate gas price volatility, if so how, where and what time period do you use?

[redacted]



m. How do you determine the success or failure of your hedging program?

[redacted]

n. Please provide internal reporting, oversight, and audit structure of your hedging program.

The hedging program reporting and oversight is documented in the Company's US Energy Commodity Risk Management Policy.

National Grid's US based energy procurement activities are managed under the direction of National Grid Plc's Finance Committee. The Finance Committee authorized the Energy Procurement Risk Management Committee ("EPRMC") and delegated authority to focus on energy risk, metrics, energy strategies, financial impacts and other opportunities to the EPRMC. The EPRMC provides Vice President level review of Strategies, with a focus on market risks, inclusive of price, credit, operational, liquidity and reputational risks and Policy compliance. The EPRMC serves to administer a consistent and comprehensive sanctioning process for such Strategies and commitments across the organization. The EPRMC has delegated some of its authority to the Commodity Management Committee ("CMC"). The EPRMC appoints the CMC members and chairperson. The CMC provides detailed review of Strategies, products and new opportunities with a focus on energy risk, metrics, financial impacts and opportunities. The CMC conducts research to understand market rules, regulatory

requirements, customer needs, risks, barriers to entry, logistical requirements, competition, deal economics, risk measurement and valuation requirements, all related requirements and exposures (including credit, liquidity, tax and legal), and the impact of current and proposed Strategies on existing controls and limits, while administering a consistent and comprehensive sanctioning process for such Strategies and commitments across the organization.

The Company reviews the hedging program and highlights any changes with the NYPSC on an annual basis. The Company has a comprehensive internal approval process to authorize a hedge strategy and maintain oversight requirements. Prior to execution, a hedging strategy is documented with details that may include: risks to be hedged, volumetric targets, duration and cadence of hedge plan transactions, and types of instruments to be used. This documentation is then reviewed and approved by the CMC prior to going to the EPRMC for approval and authorization.

The Company employs the industry standard best practice of an independent three-office model:

- The Front Office develops and executes the hedge strategies hedges in accordance with the prevailing policies and strategies by entering into transactions with counterparties to mitigate natural gas price volatility;
- Risk Management is part of the Middle Office and maintains the overall control environment and assesses compliance with the Company's risk policy. The Middle Office reports through the Treasury organization and provides a significant level of control and independent policing of the Front Office's activities. The Middle Office confirms all trades with counterparties and monitors the risk exposures of the deals, as well as verifying approved strategies are executed in accordance with the plan; and
- The Accounting Department is part of the Back Office whose functions include processes in support of the Front Office, such as accounting, invoicing, check-outs, actualization, accounts receivable and payable, and financial reporting. The Back Office is also independent of both the Front and Middle Offices and reports up through the Finance organization.

Additionally, the Company's internal auditing department has performed audits to ensure compliance, for example, with Sarbanes-Oxley.

o. Table 7: Actual price hedging performance versus planned price hedging performance for last year, a summary of "lessons learned", and arrangements for this year. Include separate quantities for each hedging instrument.

Please see the following tables:

- Table 7a, 7b: Winter Supply Hedges Summary

31. How has your use of local production/landfill/renewable gas changed over the past year? Please provide the average daily volumes of local produced gas acquired for the

previous heating season and a forecast for the upcoming season. Also, include any plans to connect new or additional local production to your distribution systems. What percentage of your system throughput is local production? Include the total volume of locally produced gas that entered your system annually since 2008 until the present. How much of this gas is directly tied into your distribution system?

* Newtown Creek should be operational spring / summer of 2019.

Local production is currently less than 1% of the Company's system throughput. The Fresh Kills Landfill, located in Staten Island, is currently the only source of local production for all three of National Grid's NY LDCs. The landfill is owned by the City of New York. KEDNY has a supply agreement with the Landfill for up to 6,500 Dth/day, but there is no contractual minimum daily quantity. Landfill supply is directly fed into the distribution system.

The Company's use of local production has not changed over the past several years; the Company continues to purchase any supply the Landfill can provide. There is no forecast of Landfill production, but Gas System Operations communicates with the Landfill regularly and is notified of maintenance work and outages. When the Landfill is unable to sell supply, the Company replaces planned volumes with contingency supplies.

Please see the table below for Fresh Kills Landfill production through June 2018:

Year	Total (Dth)
2008	1,265,300
2009	1,088,325
2010	1,056,737
2011	949,997
2012	883,368
2013	775,931
2014	659,709
2015	611,122
2016	510,838
2017	406,245
2018	206,375

32. Identify when and how standards for connecting Renewable Natural Gas (RNG) projects will be filed with the Commission. If there are no plans to file RNG standards, please explain why.

National Grid recently filed the attached Engineering Service Agreement and Gas Sales (Interconnect) Agreement templates with the PSC for KEDNY, KEDLI, and NiMo.

These documents, once completed, define the responsibilities of the Developer as well as National Grid's responsibilities. They also list the billing and payment terms and conditions,

gas quality specifications, etc. for interconnecting to our system. The Interconnect Agreement is modeled after the existing agreement with the Freshkills Landfill.

For additional standards, National Grid was part of the development of the NYS Interconnect Guide for RNG document that was created by GTI, NGA, and utilities in NYS. The PSC has reviewed this document and supports it. It will be used as a guideline and shared with Developers. The document lays out the entire interconnect process for RNG projects and also contains in its appendices the Engineering Services Agreement and Gas Sales (Interconnect) Agreement templates discussed above for utilities to follow as an example of each document. The document is to reside within each utility and be distributed as needed for projects.

Please see the following attachments:

- Exhibit 30: Engineering Services Agreement for RNG Interconnections Template
- Exhibit 31: Gas Sales (Interconnect) for RNG Interconnections Template

Issue 5

Please provide the following information related to the changing market conditions:

33. A discussion of the impacts of the convergence of the gas and electric markets in your company's service territory, including:

(Business Confidential - Responses d and e)

a. Increase in summer load from last year.

There has been no new electric generation added to the Gas System between last summer and the current.

b. Changes in system operations from last year including how gas-fired electric generators' needs and behavior during last winter impacted your distribution system operations.

During this past winter season Gas Control initiated several Operational Flow Orders to control Generator behavior consistent their contractual tariffs. There were multiple instances where Generator behavior was inconsistent with the needs of our operational needs to serve our firm ratepayers and these have been addressed through penalty billing consistent with National Grids Tariff.

c. Need for distribution system facilities improvements.

There are no system improvements needed to support power generation based on the sourcing of gas and level of service contracted. Should power generators wish to modify their points of delivery or level of service, additional facility improvements may be required. At this point in time all of the gas system reinforcement projects identified are to support the needs of our firm gas ratepayer.

d. Available information on generators' upstream capacity arrangements.

[redacted]

e. Relationship between your company and its electric, steam operations or affiliate-owned generation, including any peaking service agreements.

[redacted]

f. Distributed Generation/CHP, including any micro-grid applications, and their impact on peak design-day forecasting. List all known planned projects and locations on the system.

In the majority of National Grid's New York service territory, NYSERDA is the service provider for the programs that offer customers incentives for CHP projects. LIPA (PSEG) customers for the past few years are now eligible for CHP programs, which are modeled after the NYSERDA programs. Various market factors such as spark spread, cost of land, electrically constrained areas, and interest in resilient generation has resulted in most of the CHP growth in the Downstate NY. Accordingly, the Company has a significant amount of interest and dedicated resources to promote the deployment of CHP in Downstate NY. In Upstate NY, the Company has fewer CHP projects and overall growth in DG capacity has been from solar.

The Company's current in-queue CHP projects are:

Subject	Size of Generator (kW)	Jurisdiction District	Submission Date	Estimated Connected Date	Status	Step
Albany VA Hospital CHP	1003	030 - Albany	6/1/2016 1:00	10/1/2018	Construction	In Progress
Turning Stone Casino	1500	020 - Oneida	3/21/2016 1:00	8/31/2018	Construction	In Progress
Intergrow Greenhouses (Albion, NY)	2384	006 - Albion/Medina	7/29/2014 1:00	12/10/2018	Construction	Not Started
Oneida County Sewer District	480	017 - Utica	4/22/2014 1:00	9/28/2018	Construction	In Progress
41 CROSS ST HOSPITALITY LLC	125	033 - Hudson	3/29/2017 1:00	9/5/2017	Agreement	-
NOTTINGHAM RETIREMENT COMM	10	011 - Syracuse	3/6/2012 0:00	-	Application	Hold
NYS SUNY COBLESKILL	130	037 - Cobleskill	8/29/2016 20:00	1/16/2019	Construction	In Progress
SARATOGA HOSPITAL	1000	039 - Saratoga	11/20/2016 19:00	2/1/2019	Agreement	Draft
Proctors Theatre	300	-	4/25/2012 1:00	-	Hold	-
New Application for Generator	300	006 - Albion/Medina	9/28/2017 9:38	-	Application	Hold

g. Provide a list of all electric generators in your service territory, and indicate whether or not they are currently attached to your distribution system. If so, indicate the maximum daily delivery quantity of the generating station.

Please see the following attachment:

- Exhibit 32: List of All Electric Generators

h. Outline typical communications between gas-fired generators in your service territory and your natural gas control center, and explain any improvements planned for those communications.

Formalized communications with gas fired generators takes place via the Company's Electric Bulletin Board (EBB/GTIS). Site specific communications takes place with individual generators from control room to control room.

34. A discussion of the realized and expected impacts on gas prices, gas price volatility, and required upstream assets from recent and proposed pipeline expansions and the development of Northeast (Marcellus/Utica) shale gas wells and related infrastructure.

(Business Confidential)

[redacted]



[redacted]

35. Please discuss any other major projects that will affect your purchasing strategy over the next five years and your anticipated responses to these changes.

(Business Confidential)

[redacted]

36. Currently, how much natural gas is being sold on an annual basis for use in natural gas vehicles? How has this changed over the past few years and how do you anticipate that this will change over the five year planning period in your service territory?

Region NGV Sales (MDth)

Company	2017	2016	2015	2014	2013	2012	2011
NYC	657	667	715	679	828	632	588
LI	747	840	959	741	760	777	796
NMPC	207	209	189	267	164	162	160
Total	1,611	1,716	1,863	1,688	1,753	1,571	1,544

After a decade of rapid growth from the inception of natural gas vehicle programs through the 1990s, steady growth followed, fueled by federal and state tax incentives for vehicle conversions and fueling stations. Natural gas sales for use in NGVs then leveled off due to the expiration of those incentives and the loss of key product offerings for high fuel-use vehicle platforms. More recently, the NGV industry continues to enjoy the release of new and upgraded offerings and new market entrants. As expected, natural gas sales for NGV use rose again due to the sustained low price of natural gas relative to conventional fuels and the reintroduction of state and federal tax credits for the use of natural gas fuel. However, recent dramatic reductions in petroleum prices have again slowed growth. The Company's forecast is for flat to slight declining volumes.

37. Have you been approached by or had any discussions with outside entities regarding the construction of compressed natural gas (CNG) or liquefied natural gas (LNG) fueling stations? If so, please explain. What impediments do you see for the expansion of CNG and LNG transportation?

Customers and other outside entities have approached the Company regarding the construction of CNG fueling stations for transportation uses and more recently in upstate New York and Long Island for the loading of CNG trailers to serve stationary facilities. These discussions typically involve requests by customers for technical advice on the installation of such facilities and/or the availability of natural gas at potential fueling station sites.

The major impediments for the expansion of both CNG and LNG for transportation use include the lack of existing fueling demand, high incremental costs for vehicles, the high cost of installing fueling stations on customer sites, the small number of publicly available fueling stations, the uncertainty of future petroleum and natural gas prices and the recent significant advances in electric vehicle market development and consumer awareness. Recent declines in gasoline and diesel pricing has slowed but not stopped CNG project development. For LNG as a transportation fuel, despite the effective repeal of the LNG moratorium outside New York City, the inability to obtain a permit for intrastate transportation of LNG combined with state and local permit requirements for storage under development effectively preclude significant growth. Finally, at this time the Company does not foresee the availability of gas as a significant impediment to expansion of these services, however there are areas of our distribution system where main reinforcements may be necessary in which case the Company works collaboratively with station developers to optimize the overall fueling station and manage costs.

38. Please list potential pipeline projects you are interested in, identifying the pipeline, delivery point, and daily delivery quantity you might take. Provide a list of all pipelines that you are communicating with on a regular basis regarding expansion projects on their pipeline systems that might be filed with the Federal Energy Regulatory Commission.

(Business Confidential)

[redacted]

Issue 6

Please provide the following information regarding bill impacts:

39. Table 8: Bill impact comparison of last winter versus the forecasted 2017-18 winter. Include the workpapers used to develop Table 8 (note: they should also tie to the numbers in Table 7).

Please see the following attachments:

- Table 8a: Bill Comparison - The Brooklyn Union Gas Company
- Table 8b: Bill Comparison – KeySpan Gas East Corporation
- Table 8c: Bill Comparison – Niagara Mohawk Power Corporation
- Exhibit 33: Table 8a Backup - The Brooklyn Union Gas Company
- Exhibit 34: Table 8b Backup KeySpan Gas East Corporation
- Exhibit 35: Table 8c Backup Niagara Mohawk Power Corporation

The 6/27/2018 NYMEX was used to create these tables.

Case 18-M-0272 - Winter Supply 2018-19 Forms

Table 1a - Total System Firm Peak Day Capacity (DT)

Company: **National Grid**
Gate Station or Service Area: **The Brooklyn Union Gas Company & KeySpan Gas East Corporation**
Submission Date: **7/16/2018**
Version #: 1 of 1

	2017-18 Winter	2018-19 Winter	Design Peak Day Demand
Flowing Supplies	693,357	915,326	
Storage Withdrawals	797,428	793,358	
Winter Peaking Service *	180,000	264,000	
Renewable Gas**	1,600	1,600	
LNG	394,500	394,500	
CNG	2,850	-	
Cogen Supplies	95,896	65,056	
Local Production*	-	-	
Recallable Capacity (AMAs)	204,776	-	
Marketer Provided Supplies***	416,266	402,891	
Peak Day Totals	**2,786,673**	**2,836,731**	**2,836,634**
		Peak Day Design Temp:	**0° F**

* City Gate Deliverd by Others and In-Territory Supplies (not LNG or CNG)
** Local Production, landfill gas, renewables, etc. delivered directly into the LDC distribution system.

*** Capacity released to or held by the marketers. Add additional rows for non-mandatory released capacity, grandfathered capacity and capacity associated with non-core customers if applicable.

Winter Peaking Service

Daily Price Indexed Contracts	264,000	Firm Primary Capacity Supported?
Non Daily Price Indexed Contracts	N/A	(storage, other?)

Marketer Provided Supplies

Retail Access Supplies	402,891	mandatory assignment of company capacity
Retail Access Supplies	0	marketer owned capacity (grandfathered capacity)
Other Supplies	0	marketer owned capacity

Case 18-M-0272 - Winter Supply 2018-19 Forms

Table 1b - Total System Firm Peak Day Capacity (DT)

Company: **National Grid**
Gate Station or Service Area: **Niagara Mohawk Power Corporation**
Submission Date: **7/16/2018**
Version #: 1 of 1

	2017-18 Winter	2018-19 Winter	Design Peak Day Demand
Flowing Supplies	336,742	388,161	
Storage Withdrawals	300,346	302,223	
Winter Peaking Service *			
Renewable Gas**			
LNG			
CNG			
Cogen Supplies	29,200	43,200	
Local Production*			
Recallable Capacity (AMAs)	50,000		
Marketer Provided Supplies***	250,408	247,112	
Peak Day Totals	**966,696**	**980,696**	**912,832**
		Peak Day Design Temp:	**75 HDD**

* City Gate Deliverd by Others and In-Territory Supplies (not LNG or CNG)
** Local Production, landfill gas, renewables, etc. delivered directly into the LDC distribution system.

*** Capacity released to or held by the marketers. Add additional rows for non-mandatory released capacity, grandfathered capacity and capacity associated with non-core customers if applicable.

Winter Peaking Service

Daily Price Indexed Contracts _____ Firm Primary Capacity Supported?
Non Daily Price Indexed Contracts _____ (storage, other?)

Case 18-M-0272 - Winter Supply 2018-19 Forms

Table 1c - Total System Firm Peak Day Capacity (DT)

Company: **National Grid**
Gate Station or Service Area: **Niagara Mohawk Power Corporation East Gate**
Submission Date: **7/16/2018**
Version #: 1 of 1

	2017-18 Winter	2018-19 Winter	Design Peak Day Demand
Flowing Supplies	156,530	206,371	
Storage Withdrawals	141,664	142,060	
Winter Peaking Service *			
Renewable Gas**			
LNG			
CNG			
Cogen Supplies	26,200	40,200	
Local Production*			
Recallable Capacity (AMAs)	50000		
Marketer Provided Supplies***	114,005	112,905	
Peak Day Totals	**488,399**	**501,536**	**449,285**
		Peak Day Design Temp:	**75 HDD**

* City Gate Deliverd by Others and In-Territory Supplies (not LNG or CNG)
** Local Production, landfill gas, renewables, etc. delivered directly into the LDC distribution system.

*** Capacity released to or held by the marketers. Add additional rows for non-mandatory released capacity, grandfathered capacity and capacity associated with non-core customers if applicable.

Winter Peaking Service

Daily Price Indexed Contracts _____ Firm Primary Capacity Supported?
Non Daily Price Indexed Contracts _____ (storage, other?)

Case 18-M-0272 - Winter Supply 2018-19 Forms

Table 1d - Total System Firm Peak Day Capacity (DT)

Company: **National Grid**
Gate Station or Service Area: **Niagara Mohawk Power Corporation West Gate**
Submission Date: **7/16/2018**
Version #: 1 of 1

	2017-18 Winter	2018-19 Winter	Design Peak Day Demand
Flowing Supplies	180,212	181,790	
Storage Withdrawals	158,682	160,163	
Winter Peaking Service *			
Renewable Gas**			
LNG			
CNG			
Cogen Supplies	3,000	3,000	
Local Production*			
Recallable Capacity (AMAs)			
Marketer Provided Supplies***	136,403	134,207	
Peak Day Totals	**478,297**	**479,160**	**463,547**
		Peak Day Design Temp:	**75 HDD**

* City Gate Deliverd by Others and In-Territory Supplies (not LNG or CNG)
** Local Production, landfill gas, renewables, etc. delivered directly into the LDC distribution system.

*** Capacity released to or held by the marketers. Add additional rows for non-mandatory released capacity, grandfathered capacity and capacity associated with non-core customers if applicable.

Winter Peaking Service

Daily Price Indexed Contracts _____ Firm Primary Capacity Supported?
Non Daily Price Indexed Contracts _____ (storage, other?)

Case 18-M-0272 - Winter Supply 2018-19 Forms

Table 2a - Estimated Design Weather Requirements (MDT)

Company: **National Grid**
Gate Station or Service Area: **The Brooklyn Union Gas Company**
Submission Date: **7/16/2018**
Version #: 1 of 1

	2017-18 Forecast	2017-18 Actual	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25	2025-26	2026-27	2027-28
Annual												
Firm												
Sales	112,348		114,248	117,013	117,691	119,342	121,902	125,073	126,858	129,692	132,628	136,650
Transportation	60,677		60,677	60,985	60,677	60,677	60,677	60,985	60,677	60,677	60,677	60,985
Temperature Controlled												
Sales	9,899		9,455	9,104	8,588	8,155	7,770	7,387	7,045	6,688	6,311	5,942
Transportation	5,628		5,451	5,260	5,064	4,859	4,652	4,456	4,259	4,054	3,837	3,623
Non-Firm												
Sales	-		-	-	-	-	-	-	-	-	-	-
Transportation	-		-	-	-	-	-	-	-	-	-	-
Total	**188,552**		**189,832**	**192,362**	**192,020**	**193,033**	**195,002**	**197,901**	**198,838**	**201,112**	**203,454**	**207,199**
Winter Season												
Firm												
Sales	82,997	72,241	84,311	86,608	86,936	87,807	89,751	92,174	93,151	95,162	97,214	100,375
Transportation	40,088	38,106	40,088	40,395	40,088	40,088	40,088	40,395	40,088	40,088	40,088	40,395
Temperature Controlled												
Sales	6,425	8,414	6,128	5,931	5,604	5,296	5,080	4,801	4,589	4,360	4,143	3,899
Transportation	3,320	1,269	3,224	3,112	3,003	2,863	2,773	2,640	2,531	2,406	2,288	2,155
Non-Firm												
Sales	-	-	-	-	-	-	-	-	-	-	-	-
Transportation	-	807	-	-	-	-	-	-	-	-	-	-
Total	**132,829**	**120,836**	**133,751**	**136,046**	**135,631**	**136,054**	**137,691**	**140,010**	**140,359**	**142,016**	**143,732**	**146,824**
Peak Day												
Firm												
Sales	1,160	930	1,178	1,198	1,213	1,225	1,251	1,272	1,297	1,324	1,352	1,382
Transportation	500	474	500	500	500	500	500	500	500	500	500	500
Temperature Controlled												
Sales	2	6	2	1	1	1	1	1	1	1	1	1
Transportation	-	7	-	-	-	-	-	-	-	-	-	-
Total	**1,662**	**1,417**	**1,680**	**1,699**	**1,715**	**1,727**	**1,753**	**1,774**	**1,799**	**1,826**	**1,853**	**1,883**

1/ Design Weather is defined as:

Annual	5,370	HDD
Winter	4,471	HDD
Peak Day	65	HDD

Case 18-M-0272 - Winter Supply 2018-19 Forms

Table 2b - Estimated Design Weather Requirements (MDT)

Company: **National Grid**
Gate Station or Service Area: **KeySpan Gas East Corporation**
Submission Date: **7/16/2018**
Version #: 1 of 1

	2017-18 Forecast	2017-18 Actual	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25	2025-26	2026-27	2027-28
Annual												
Firm												
Sales	91,156		92,519	94,358	94,629	95,557	97,108	99,120	99,660	100,486	101,416	103,701
Transportation	28,527		28,527	28,674	28,527	28,527	28,527	28,674	28,527	28,527	28,527	28,674
Temperature Controlled												
Sales	448		439	438	433	434	437	441	438	436	438	442
Transportation	157		147	141	135	130	126	122	117	112	109	106
Non-Firm												
Sales	466		437	418	396	385	369	364	350	339	335	332
Transportation	4		4	4	4	4	4	4	4	4	4	4
Total	**120,758**		**122,073**	**124,033**	**124,124**	**125,036**	**126,571**	**128,725**	**129,095**	**129,904**	**130,828**	**133,258**
Winter Season												
Firm												
Sales	65,475	57,598	66,469	68,010	68,095	68,548	69,705	71,336	71,594	72,280	72,621	74,759
Transportation	19,092	18,416	19,092	19,239	19,092	19,092	19,092	19,239	19,092	19,092	19,092	19,239
Temperature Controlled												
Sales	294	318	286	285	281	280	282	286	283	281	281	286
Transportation	93	38	86	82	79	75	73	71	68	65	63	62
Non-Firm												
Sales	335	288	314	300	285	278	266	263	254	246	243	242
Transportation	3	0	3	3	3	3	3	3	3	3	3	3
Total	**85,293**	**76,658**	**86,249**	**87,921**	**87,835**	**88,276**	**89,422**	**91,199**	**91,295**	**91,968**	**92,304**	**94,591**
Peak Day												
Firm												
Sales	902	782	916	928	938	944	960	973	985	994	999	1,018
Transportation	241	232	241	241	241	241	241	241	241	241	241	241
Temperature Controlled												
Sales	0	-	0	0	0	0	0	0	0	0	0	0
Transportation		-										
Total	**1,143**	**1,015**	**1,157**	**1,169**	**1,179**	**1,185**	**1,201**	**1,214**	**1,226**	**1,235**	**1,240**	**1,259**

1/ Design Weather is defined as:

Annual	5,370	HDD
Winter	4,471	HDD
Peak Day	65	HDD

Case 18-M-0272 - Winter Supply 2018-19 Forms

Table 2c - Estimated East Gate Design Weather Requirements (MDT)

Company: **National Grid**
Gate Station or Service Area: **Niagara Mohawk Power Corporation East Gate**
Submission Date: **7/16/2018**
Version #: 1 of 1

	2017-18 Forecast	2017-18 Actual	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25	2025-26	2026-27	2027-28
Annual												
Firm												
Sales	32,429		32,710	33,488	33,484	33,626	34,394	35,203	35,459	35,777	36,087	36,844
Transportation	14,567		14,716	14,834	15,095	15,371	15,487	15,711	15,752	15,929	16,079	16,437
Non-Firm												
Sales												
Transportation	61,712		62,252	63,305	63,627	64,058	65,201	66,173	66,569	67,103	67,677	68,922
Total	108,708		109,678	111,627	112,206	113,055	115,082	117,087	117,780	118,809	119,843	122,203
Winter Season												
Firm												
Sales	24,757	21,094	24,950	25,614	25,625	25,616	26,235	26,954	27,098	27,403	27,550	28,251
Transportation	10,469	8,440	10,555	10,619	10,802	10,992	11,061	11,217	11,227	11,335	11,421	11,701
Non-Firm												
Sales												
Transportation	41,596	12,762	41,949	42,803	43,029	43,098	43,972	44,762	44,968	45,415	45,679	46,679
Total	76,822	42,296	77,454	79,036	79,456	79,706	81,268	82,933	83,293	84,153	84,650	86,631
Peak Day												
Firm												
Sales	320	297	322	328	331	328	336	345	350	354	356	361
Transportation	130	100	131	130	133	139	140	138	139	140	140	144
Total	449	396	453	458	465	467	476	483	489	494	497	505

1/ Design Weather is defined as:

Annual	7,400	HDD
Winter	5,821	HDD
Peak Day	75	HDD

Case 18-M-0272 - Winter Supply 2018-19 Forms

Table 2d - Estimated West Gate Design Weather Requirements (MDT)

Company: **National Grid**
Gate Station or Service Area: **Niagara Mohawk Power Corporation West Gate**
Submission Date: **7/16/2018**
Version #: 1 of 1

	2017-18 Forecast	2017-18 Actual	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25	2025-26	2026-27	2027-28
Annual												
Firm												
Sales	33,537		33,949	34,819	34,726	34,981	35,694	36,477	36,777	37,148	37,077	37,296
Transportation	15,685		15,531	14,997	14,742	14,408	13,821	13,355	12,858	12,451	12,381	12,448
Non-Firm												
Sales												
Transportation	63,764		63,758	64,298	63,873	63,725	63,925	64,206	63,958	63,950	63,809	64,150
Total	79,449		79,289	79,295	78,615	78,133	77,746	77,561	76,816	76,401	76,190	76,598
Winter Season												
Firm												
Sales	25,212	25,112	25,376	26,128	26,032	26,153	26,675	27,295	27,474	27,786	27,715	27,934
Transportation	11,180	10,341	11,253	10,843	10,635	10,391	9,995	9,662	9,281	8,983	8,913	8,980
Non-Firm												
Sales		-										
Transportation	44,312	26,837	44,339	44,866	44,481	44,288	44,480	44,772	44,540	44,585	44,444	44,783
Total	80,704	62,290	80,968	81,837	81,148	80,832	81,150	81,729	81,295	81,354	81,072	81,697
Peak Day												
Firm												
Sales	323	320	325	332	333	335	341	345	351	355	353	353
Transportation	141	123	142	136	134	131	126	122	117	114	113	113
Total	464	442	467	467	467	465	467	467	468	468	467	467

1/ Design Weather is defined as:

Annual	7,462 HDD
Winter	5,812 HDD
Peak Day	75 HDD

Case 18-M-0272 - Winter Supply 2018-19 Forms

Table 3a - Estimated Normal Weather Requirements (MDT)

Company: **National Grid**
Gate Station or Service Area: **The Brooklyn Union Gas Company**
Submission Date: **7/16/2018**
Version #: 1 of 1

	2017-18 Forecast	2017-18 Actual	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25	2025-26	2026-27	2027-28
Annual												
Firm												
Sales	98,001		99,690	102,108	102,730	104,217	106,479	109,270	110,883	113,398	116,004	119,540
Transportation	55,466		55,466	55,741	55,466	55,466	55,466	55,741	55,466	55,466	55,466	55,741
Temperature Controlled												
Sales	9,963		9,507	9,172	8,638	8,201	7,817	7,447	7,085	6,728	6,351	5,991
Transportation	5,622		5,455	5,274	5,069	4,861	4,656	4,462	4,263	4,057	3,841	3,630
Non-Firm												
Sales	-		-	-	-	-	-	-	-	-	-	-
Transportation	-		-	-	-	-	-	-	-	-	-	-
Total	**169,053**		**170,118**	**172,295**	**171,903**	**172,746**	**174,419**	**176,919**	**177,698**	**179,649**	**181,662**	**184,902**
Winter Season												
Firm												
Sales	70,856	72,241	71,994	73,974	74,268	75,022	76,706	78,794	79,650	81,392	83,169	85,896
Transportation	35,748	38,106	35,748	36,022	35,748	35,748	35,748	36,022	35,748	35,748	35,748	36,022
Temperature Controlled												
Sales	6,652	8,414	6,337	6,149	5,795	5,477	5,254	4,983	4,746	4,510	4,285	4,045
Transportation	3,413	1,269	3,323	3,217	3,096	2,950	2,857	2,724	2,608	2,479	2,357	2,226
Non-Firm												
Sales	-	-	-	-	-	-	-	-	-	-	-	-
Transportation	-	807	-	-	-	-	-	-	-	-	-	-
Total	**116,669**	**120,836**	**117,402**	**119,361**	**118,906**	**119,197**	**120,564**	**122,522**	**122,752**	**124,128**	**125,559**	**128,188**
Peak Day												
Firm												
Sales	-	930	-	-	-	-	-	-	-	-	-	-
Transportation	-	474	-	-	-	-	-	-	-	-	-	-
Temperature Controlled	-		-	-	-	-	-	-	-	-	-	-
Sales	-	6	-	-	-	-	-	-	-	-	-	-
Transportation	-	7	-	-	-	-	-	-	-	-	-	-
Total	**-**	**1,417**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**

1/ Design Weather is defined as:

Annual	4,587	HDD
Winter	3,819	HDD
Peak Day	-	HDD

Case 18-M-0272 - Winter Supply 2018-19 Forms

Table 3b - Estimated Normal Weather Requirements (MDT)

Company: **National Grid**
Gate Station or Service Area: **KeySpan Gas East Corporation**
Submission Date: **7/16/2018**
Version #: 1 of 1

	2017-18 Forecast	2017-18 Actual	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25	2025-26	2026-27	2027-28
Annual												
Firm												
Sales	80,342		81,551	83,162	83,416	84,255	85,629	87,396	87,889	88,615	89,466	91,452
Transportation	25,990		25,990	26,121	25,990	25,990	25,990	26,121	25,990	25,990	25,990	26,121
Temperature Controlled												
Sales	450		440	440	434	435	438	443	439	437	439	444
Transportation	156		147	141	135	130	126	122	117	112	109	105
Non-Firm												
Sales												
Transportation												
Total	**106,939**		**108,128**	**109,864**	**109,975**	**110,809**	**112,183**	**114,082**	**114,435**	**115,154**	**116,003**	**118,122**
Winter Season												
Firm												
Sales	56,297	57,598	57,158	58,489	58,567	58,959	59,963	61,371	61,599	62,194	62,489	64,337
Transportation	16,980	18,416	16,980	17,111	16,980	16,980	16,980	17,111	16,980	16,980	16,980	17,111
Temperature Controlled												
Sales	304	318	294	295	290	288	291	295	292	290	290	296
Transportation	95	38	88	85	81	77	75	73	70	67	65	63
Non-Firm												
Sales	-	288	-	-	-	-	-	-	-	-	-	-
Transportation	-	0	-	-	-	-	-	-	-	-	-	-
Total	**73,676**	**76,658**	**74,520**	**75,980**	**75,918**	**76,305**	**77,308**	**78,850**	**78,941**	**79,530**	**79,823**	**81,806**
Peak Day												
Firm												
Sales	-	782	-	-	-	-	-	-	-	-	-	-
Transportation	-	232	-	-	-	-	-	-	-	-	-	-
Temperature Controlled												
Sales	-	-	-	-	-	-	-	-	-	-	-	-
Transportation	-	-	-	-	-	-	-	-	-	-	-	-
Total	**-**	**1,015**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**

1/ Design Weather is defined as:

Annual	4,587	HDD
Winter	3,819	HDD
Peak Day	-	HDD

Case 18-M-0272 - Winter Supply 2018-19 Forms

Table 3c - Estimated East Gate Normal Weather Requirements (MDT)

Company: **National Grid**
Gate Station or Service Area: **Niagara Mohawk Power Corporation East Gate**
Submission Date: **7/16/2018**
Version #: 1 of 1

	2017-18 Forecast	2017-18 Actual	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25	2025-26	2026-27	2027-28
Annual												
Firm												
Sales	28,402		28,652	29,320	29,331	29,462	30,140	30,837	31,081	31,360	31,638	32,287
Transportation	12,943		13,076	13,182	13,416	13,661	13,766	13,964	14,003	14,160	14,294	14,611
Non-Firm												
Sales												
Transportation	15,760		15,861	16,023	16,063	16,112	16,347	16,328	16,400	16,435	16,553	16,717
Total	28,703		28,937	29,205	29,479	29,773	30,113	30,292	30,403	30,595	30,847	31,328
Winter Season												
Firm												
Sales	21,396	21,094	21,565	22,130	22,158	22,150	22,693	23,305	23,449	23,716	23,845	24,443
Transportation	9,179	8,440	9,255	9,311	9,471	9,638	9,699	9,835	9,844	9,939	10,014	10,260
Non-Firm												
Sales		-										
Transportation	7,330	12,762	7,475	7,576	7,553	7,513	7,639	7,608	7,673	7,681	7,705	7,778
Total	37,905	42,296	38,295	39,017	39,182	39,301	40,031	40,748	40,966	41,336	41,564	42,481
Peak Day												
Firm												
Sales		297										
Transportation		100										
Total		396										

1/ Design Weather is defined as:

Annual	6,539	HDD
Winter	5,141	HDD
Peak Day		HDD

Case 18-M-0272 - Winter Supply 2018-19 Forms

Table 3d - Estimated West Gate Normal Weather Requirements (MDT)

Company: **National Grid**
Gate Station or Service Area: **Niagara Mohawk Power Corporation West Gate**
Submission Date: **7/16/2018**
Version #: 1 of 1

	2017-18 Forecast	2017-18 Actual	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25	2025-26	2026-27	2027-28
Annual												
Firm												
Sales	29,678		30,047	30,815	30,733	30,962	31,594	32,287	32,553	32,881	32,819	33,011
Transportation	13,925		13,782	13,308	13,084	12,787	12,266	11,852	11,413	11,052	10,991	11,050
Non-Firm												
Sales												
Transportation	13,926		13,697	13,875	13,807	13,751	13,813	13,777	13,723	13,743	13,744	13,795
Total	27,851		27,479	27,183	26,891	26,538	26,079	25,629	25,136	24,795	24,735	24,845
Winter Season												
Firm												
Sales	22,106	25,112	22,250	22,910	22,825	22,931	23,389	23,933	24,089	24,363	24,301	24,493
Transportation	9,804	10,341	9,868	9,509	9,326	9,112	8,765	8,473	8,139	7,877	7,816	7,875
Non-Firm												
Sales		-										
Transportation	7,323	26,837	7,139	7,284	7,193	7,143	7,188	7,215	7,190	7,200	7,199	7,249
Total	39,233	62,290	39,257	39,703	39,344	39,186	39,342	39,621	39,418	39,440	39,316	39,617
Peak Day												
Firm												
Sales		320										
Transportation		123										
Total		442										

1/ Design Weather is defined as:

Annual	6,588	HDD
Winter	5,128	HDD
Peak Day		HDD

Case 18-M-0272 - Winter Supply 2018-19 Forms

Table 4a - Firm Transportation Capacity*

(2018-19 Winter)

Company: **The Brooklyn Union Gas Company**
Submission Date: **7/16/2018**
Version #: 1 of 1

Pipeline Company Name	Rate Schedule	Daily Quanity (DT)	Winter Quanity (MDT)	Annual Quanity (MDT)	Expiration Date
Flowing Gas To Citygate					
Transco Year-Round	FT	245,955	37,139	89,774	5/31/2019
Transco Year-Round	FT	1,969	297	719	3/19/2019
Transco Seasonal - 90 Day	FT	4,244	382	382	7/31/2019
Texas Eastern	CDS	51,315	7,749	18,730	10/31/2020
Texas Eastern	CDS	5,403	816	1,972	10/31/2020
Tennessee	FT-A	30,180	4,557	11,016	5/31/2022
Iroquois	RTS	80,936	12,221	29,542	11/1/2022
~~Transco* 1/~~	~~FT (X-265)~~	~~3,500~~	~~529~~	~~1,278~~	~~1/1/2014~~
Transco	FT (X-266)	3,250	491	1,186	12/31/2019
Texas Eastern	FTS	2,560	387	934	10/31/2020
Texas Eastern *	FT-1	27,500	4,153	10,038	3/31/2023
Texas Eastern	FTS-4-2	5,000	755	1,825	12/1/2019
Texas Eastern	X-130	12,161	1,836	4,439	10/31/2019
Transco (avail Nov - Mar)	FT	13,945	2,106	2,106	3/31/2013
Transco	FDLS	353,700	53,409	129,101	5/14/2030
Upstream Pipeline Support [1]					
First Leg					
Transco	FT	10,688	1,614	3,901	10/31/2019
Texas Eastern	FT-1	20,604	3,111	7,520	10/31/2024
Dominion	FTNN	40,301	6,085	14,710	3/31/2023
Dominion - New Market Project 5/	FT	82,000	12,382	29,930	10/31/2031
TransCanada (Dawn to Wad)	FT	12,142	1,833	4,432	10/31/2022
TransCanada (Dawn to Wad)	FT	28,326	4,277	10,339	10/31/2022
Union (Dawn to Parkway)	M12	28,640	4,325	10,454	10/31/2020
Union (Dawn to Parkway)	M12	12,277	1,854	4,481	10/31/2020
Equitrans Keystone SS-3 Storage, Winter 2/	STS-1	16,193	2,445	2,445	4/1/2020
Equitrans Keystone SS-3 Storage, Summer 2/	STS-1	8,465	0	1,812	4/1/2020
Transco	FT	100,000	15,100	36,500	5/14/2030
Transco 5/	FT	115,000	17,365	41,975	11/1/2032

Pipeline Company Name	Rate Schedule	Daily Quanity (DT)	Winter Quanity (MDT)	Annual Quanity (MDT)	Expiration Date
Deliveries from Storage					
Dominion GSS Storage 1/	FT (X-285)	50,075	7,561	18,277	12/13/2019
Transco GSS Market Area Storage 3/	GSS	180,137	11,129	65,750	3/31/2023
Transco LSS Market Area Storage 3/	LSS	31,940	3,354	11,658	3/31/2023
Transco S-2 Market Area Storage 3/	S-2	22,838	2,053	8,336	4/15/2020
Equitrans Keystone SS-3 Storage	FTS-2	17,477	2,639	6,379	3/31/2020
Texas Eastern SS-1 Market Area Storage 3/	SS-1	114,190	7,370	41,679	4/30/2024
Dominion GSS-TE Storage	FTS-7	21,332	3,221	7,786	4/15/2020
Dominion GSS-TE Storage	FTS-8	10,340	1,561	3,774	3/31/2020
Tennessee FT-A Transport - Honeoye Storage 4/	FT-A	6,834	1,032	2,494	10/31/2019
Tennessee FT-A Transport - Tenn FS-MA Storag	FT-A	20,808	3,142	7,595	10/31/2019
Winter Peaking Service					
Total (Flowing Gas to City Gate, Deliveries from Storage, and Winter Peaking Service)					
		1,175,389	**148,416**	**424,866**	

* Please highlight any changes from the previous year's report.

[1] Capacity used to deliver gas to pipelines that deliver to the citygate.

* Contract does NOT have renewal rights.

1/ Effective 10/1/2016, contract consolidation
2/ Transportation associated with Keystone Storage.
3/ Bundled Transportation and Storage contracts.
4/ Transportation associated with Honeoye Storage does not allow max withdrawals from Honeoye Storage.
5/ New Project. Final expiration date determined upon actual in - service date of 10/8/2017.

Case 18-M-0272 - Winter Supply 2018-19 Forms

Table 4b - Firm Transportation Capacity*

(2018-19 Winter)

Company: **KeySpan Gas East Corporation**
Submission Date: **7/16/2018**
Version #: 1 of 1

Pipeline Company Name	Rate Schedule	Daily Quanity (DT)	Winter Quanity (MDT)	Annual Quanity (MDT)	Expiration Date
Flowing Gas To Citygate					
Transco Year-Round 2/	FT	154,287	18,722	45,254	5/31/2020
Transco Year-Round	FT	1,811	273	661	2/24/2020
Transco Seasonal - 90 Day	FT	1,863	168	168	7/31/2020
Texas Eastern	CDS	8,106	1,224	2,959	10/31/2019
Texas Eastern	CDS	25,001	3,775	9,125	10/31/2019
Tennessee	FT-A	2,518	380	919	5/31/2022
Iroquois 6/	RTS	87,760	13,252	32,032	11/1/2021
Iroquois	RTS	25,000	3,775	9,125	11/1/2021
Iroquois NE07	RTS	200,000	30,200	73,000	11/1/2023
Transco	FT (X-271)	2,100	317	767	1/31/2019
Transco 3/	FT (X-287)	536	81	196	10/31/2019
Texas Eastern	FTS	1,110	168	405	10/31/2019
Texas Eastern *	FT-1	22,500	3,398	8,213	3/31/2023
Transco (avail Nov - Mar)	FT	17,433	2,632	2,632	3/31/2020
Transco Leidy East	FT	25,000	3,775	9,125	3/31/2020
Transco Market Link	FT	25,000	3,775	9,125	11/30/2019
Transco	FDLS	293,300	26,397	89,163	5/14/2030
Iroquois 5/	RTS	7,000	1,057	2,555	11/1/2021
Upstream Pipeline Support [1]					
Texas Eastern	FT-1	12,578	1,899	4,591	10/31/2019
Dominion	FTNN	26,021	3,929	9,498	3/31/2023
TransCanada (Dawn to Wad)	FT	16,086	2,429	5,871	10/31/2024
TransCanada (Dawn to Wad)	FT	21,347	3,223	7,792	10/31/2024
Union (Dawn to Parkway)	M12	21,584	3,259	7,878	10/31/2020
Union (Dawn to Parkway)	M12	16,266	2,456	5,937	10/31/2020
Transcanada*	FT	130,000	19,630	47,450	10/31/2018
Empire*	FT	150,750	22,763	55,024	12/21/2018
Millennium	FT-1	150,000	22,650	54,750	12/31/2023
Millennium	FT-1	50,000	7,550	18,250	12/31/2023
Millennium	FT-1	50,000	7,550	18,250	12/31/2023
Algonquin	AFT-1	196,000	29,596	71,540	10/31/2023

Pipeline Company Name	Rate Schedule	Daily Quanity (DT)	Winter Quanity (MDT)	Annual Quanity (MDT)	Expiration Date
Deliveries from Storage					
Dominion GSS Storage 3/	FT (X-287)	35,689	5,389	13,026	10/31/2007
Leidy to Long Island for Dominion DTI GSS	FT	50,000	7,550	18,250	12/12/2027
Leidy to Long Island for Dominion DTI GSS	FT	50,000	7,550	18,250	12/12/2027
Transco GSS Market Area Storage 4/	GSS	112,484	6,669	41,057	3/31/2023
Transco LSS Market Area Storage 4/	LSS	19,807	2,100	7,230	3/31/2023
Transco SS-2 Market Area Storage 4/	SS-2	23,184	2,550	8,462	3/31/2020
Texas Eastern SS-1 Market Area Storage 4	SS-1	2,076	187	758	4/30/2020
Texas Eastern SS-1 Market Area Storage 4	SS-1	15,572	934	5,684	4/30/2020
Dominion GSS N. Summit	FTS-5	20,000	3,020	7,300	3/31/2020
Dominion GSS N. Summit	FTS-5	15,000	2,265	5,475	3/31/2020
Dominion GSS Apec	FTS-5	15,000	2,265	5,475	3/31/2020
Dominion GSS-TE Storage	FTS-8	14,771	2,230	5,391	3/31/2020
Tenn FS-MA Storage	FT-A	5,202	786	1,899	10/31/2019
GSS "Apec"	FT-GSS	15,000	2,265	5,475	3/31/2022
DTI GSS	FT-GSS	100,000	15,100	15,100	3/31/2022
Winter Peaking Service					
Total (Flowing Gas to City Gate, Deliveries from Storage, and Winter Peaking Service)					
		985,810	**130,466**	**344,517**	

* Please highlight any changes from the previous year's report.

[1] Capacity used to deliver gas to pipelines that deliver to the citygate.

1/ Capacity used to deliver gas to pipelines that deliver to the citygate.
2/ The actual max daily contract volume is 154,287 dth per day, 30,303 dt per day is released to the Brooklyn Navy Yard.
3/ MDQ is 36,225 Dth/day
4/ Bundled Transportation and Storage contracts.
5/ New South Commack Capacity acquired 11/15/2016.
6/ Effective 11/1/2017, contract consolidation

Case 18-M-0272 - Winter Supply 2018-19 Forms

Table 4c - Firm Transportation Capacity*

(2018-19 Winter)

Company: **Niagara Mohawk Power Corporation**
Submission Date: **7/16/2018**
Version #: 1 of 1

Pipeline Company Name	Rate Schedule	Daily Quanity (DT)	Winter Quanity (MDT)	Annual Quanity (MDT)	Expiration Date
Flowing Gas To Citygate					
Dominion Transmission Inc.	FTNN	340,122	51,358	117,975	3/31/2021
Iroquois Gas Transmission	RTS	51,596	7,791	18,833	11/1/2021
Dominion Transmission Inc.	FT	10,000	1,510	2,712	3/31/2021
Dominion Transmission Inc.	FT	17,700	2,673	6,461	10/31/2025
Dominion - New Market Project	FT	30,000	4,530	10,950	10/31/2032
Tennessee	FT-A	20,000	3,020	7,300	10/31/2038
Tennessee	FT-A	30,000	4,530	10,950	10/31/2037
Upstream Pipeline Support [1]					
~~Transcontinental Gas Pipeline~~	~~FT~~	~~8,306~~	~~1,254~~	~~3,032~~	~~10/31/2017~~
Tennessee	FT-A	14,000	2,114	5,110	10/31/2019
Union (Dawn to Parkway)	M12	52,247	7,889	19,070	10/31/2020
TransCanada (Parkway to Waddington)	FT	51,596	7,791	18,833	10/31/2024
Deliveries from Storage					
Dominion Transmission Inc.	FTNN-GSS	434,078	22,309	65,546	3/31/2021
Dominion Transmission Inc.	FT	4,000	604	604	3/31/2021
Winter Peaking Service					
Total (Flowing Gas to City Gate, Deliveries from Storage, and Winter Peaking Service)					
		937,496	**98,325**	**241,330**	

* Please highlight any changes from the previous year's report.

[1] Capacity used to deliver gas to pipelines that deliver to the citygate.

* Contract does NOT have renewal rights

Case 18-M-0272 - Winter Supply 2018-19 Forms

Table 5a - Firm Storage Capacity*

(2018-19 Winter)

Company: **The Brooklyn Union Gas Company**
Submission Date: **7/16/2018**
Version #: 1 of 1

Storage Company Name	Rate Schedule	Daily Winter Quantity (DT)	Quantity (MDT)	Expiration Date
Market Area Storage				
Transco	GSS	180,137	11,129	3/31/2023
Transco	LSS (1)	31,940	3,354	3/31/2023
Transco	S-2	22,838	2,053	4/16/2020
Texas Eastern	SS-1	114,190	7,370	4/30/2024
Equitrans-Keystone	SS-3/STS-1	16,193	1,693	4/1/2020
Tennessee	FS-MA (5)	20,808	2,497	10/31/2019
Honeoye	SS-NY (7)	10,220	1,226	4/1/2020
Dominion	GSS (3)	46,351	2,874	3/31/2023
Dominion	GSS-TE (4)	32,267	3,098	3/31/2021
Total		**474,944**	**35,294**	
Production Area Storage				
Transco	WSS (2)	162,680	15,455	8/31/2019
Transco	ESS (6)(8)	32,884	277	10/31/2018
Transco	ESS (6)	54,855	458	4/11/2019
Total		**250,419**	**16,190**	

* Please highlight any changes from the previous year's report.

** Deliveries do not reflect fuel losses at the Citygate

(1) Extended term of LSS from March 31, 1994 to March 31, 2013 by amendment dated March 31, 2008.
(2) Quantity reduced to 162,680 from 181,819 by amendment dated 5/1/2011.
(3) Extended term of GSS to March 31, 2013 by amendment dated July 20, 2006.
(4) Extended term of GSS-TE from March 31, 2006 to March 31, 2013 by amendment dated July 20, 2006.
(5) Extended term of FS-MA to October 31, 2014 by amendment dated August 1, 2008.
(6) Contract volumes reduced to reflect the abandonment of ESS caverns 1-4
(7) The Company cannot withdraw maximum amount from Honeoye Storage due to transportation MDQ
(8) Notice of termination provided by the Company

Case 18-M-0272 - Winter Supply 2018-19 Forms

Table 5b - Firm Storage Capacity*

(2018-19 Winter)

Company: **KeySpan Gas East Corporation**
Submission Date: **7/16/2018**
Version #: 1 of 1

Storage Company Name	Rate Schedule	Daily Winter Quantity (DT)	Quantity (MDT)	Expiration Date
Market Area Storage				
Transco	GSS	112,484	6,669	3/31/2023
Transco	LSS (1)	19,807	2,100	3/31/2023
Transco	SS-2	23,184	2,550	3/31/2028
Texas Eastern	SS-1	15,572	934	4/30/2024
Texas Eastern	SS-1	2,076	187	4/30/2020
Tennessee	FS-MA	5,202	468	10/31/2019
Dominion	GSS	35,814	2,164	3/31/2023
Dominion	GSS-TE (3)	15,000	1,443	3/31/2021
Dominion	GSS-N Summit	35,000	3,500	3/31/2022
Dominion	GSS-APEC	15,000	1,500	3/31/2022
Dominion	GSS	100,000	6,000	3/31/2022
Total		**379,139**	**27,515**	
Production Area Storage				
Transco	WSS (2)	46,939	4,459	8/31/2019
Transco	ESS (4)(5)	35,934	303	10/31/2018
Transco	ESS (4)	33,074	276	3/4/2019
Total		**115,947**	**5,039**	

* Please highlight any changes from the previous year's report.

** Deliveries do not reflect fuel losses at the Citygate

(1) Extended term of LSS from March 31, 1994 to March 31, 2013 by amendment dated March 31, 2008.
(2) Quantity reduced to 46,939 from 52,461 by amendment dated 5/1/2011.
(3) Extended term of GSS-TE from March 31, 2006 to March 31, 2011 by amendment dated August 20, 2004.
(4) Contract volumes reduced to reflect the abandonment of ESS caverns 1-4
(5) Notice of termination provided by the Company

Case 18-M-0272 - Winter Supply 2018-19 Forms

Table 5c - Firm Storage Capacity*

(2018-19 Winter)

Company: **Niagara Mohawk Power Corporation**
Submission Date: **7/16/2018**
Version #: 1 of 1

Storage Company Name	Rate Schedule	Daily Winter Quantity (DT)	Quantity (MDT)	Expiration Date
Market Area Storage				
Dominion Transmission Inc.	GSS	438,078	22,917	3/31/2021
Total		**438,078**	**22,917**	
Production Area Storage				
Total		**0**	**0**	

* Please highlight any changes from the previous year's report.

Case 18-M-0272 - Winter Supply 2018-19 Forms

Table 6a - Gas Supply Contracts

(2018-19)

Company: The Brooklyn Union Gas Company
Submission Date: 7/16/2018
Version #: 1 of 1

(Business Confidential)

Transporter	Supplier	Receipt Point	Contract Number	Daily^ Quantity (DT)	Winter Quantity (MDT)	Annual Quantity (MDT)	Pricing Terms *	Flexibility	Expiration Date	Supply Basin **
Long Term (>5 years) by FT										
Intermediate Term (1 to 5 years) by FT										
[redacted]										
Short Term (<1 year) by FT										
Purchases other than by FT										
[redacted]										
Total										

* Fixed Price, Monthly Index (Identify Index), First of Month (FOM), Daily Indexed (Identify Index), etc.

** Mark "M" for Marcellus, "C" for Canadian, "G" for Gulf, or "W" for West/Mid-Continent supplies or leave blank for others if not known.

^ If daily amounts differ by winter month, please show on separate form.

Case 18-M-0272 - Winter Supply 2018-19 Forms

Table 6b - Gas Supply Contracts

(2018-19)

Company: KeySpan Gas East Corporation
Submission Date: 7/16/2018
Version #: 1 of 1

(Business Confidential)

Transporter	Supplier	Receipt Point	Contract Number	Daily^ Quantity (DT)	Winter Quantity (MDT)	Annual Quantity (MDT)	Pricing Terms *	Flexibility	Expiration Date	Supply Basin **
Long Term (>5 years) by FT										
Intermediate Term (1 to 5 years) by FT										
[redacted]										
Short Term (<1 year) by FT										
Purchases other than by FT										
[redacted]										
Total										

* Fixed Price, Monthly Index (Identify Index), First of Month (FOM), Daily Indexed (Identify Index), etc.

** Mark "M" for Marcellus, "C" for Canadian, "G" for Gulf, or "W" for West/Mid-Continent supplies or leave blank for others if not known.

^ If daily amounts differ by winter month, please show on separate form.

Case 18-M-0272 - Winter Supply 2018-19 Forms

Table 6c - Gas Supply Contracts

(2018-19)

Company: Niagara Mohawk Power Corporation
Submission Date: 7/16/2018
Version #: 1 of 1

(Business Confidential)

Transporter	Supplier	Receipt Point	Contract Number	Daily^ Quantity (DT)	Winter Quantity (MDT)	Annual Quantity (MDT)	Pricing Terms *	Flexibility	Expiration Date	Supply Basin **
Long Term (>5 years) by FT										
Intermediate Term (1 to 5 years) by FT										
Short Term (<1 year) by FT										
Purchases other than by FT										
[redacted]										
[redacted]										
[redacted]

Case 14-G-0185 - Winter Supply 2015-16 Forms

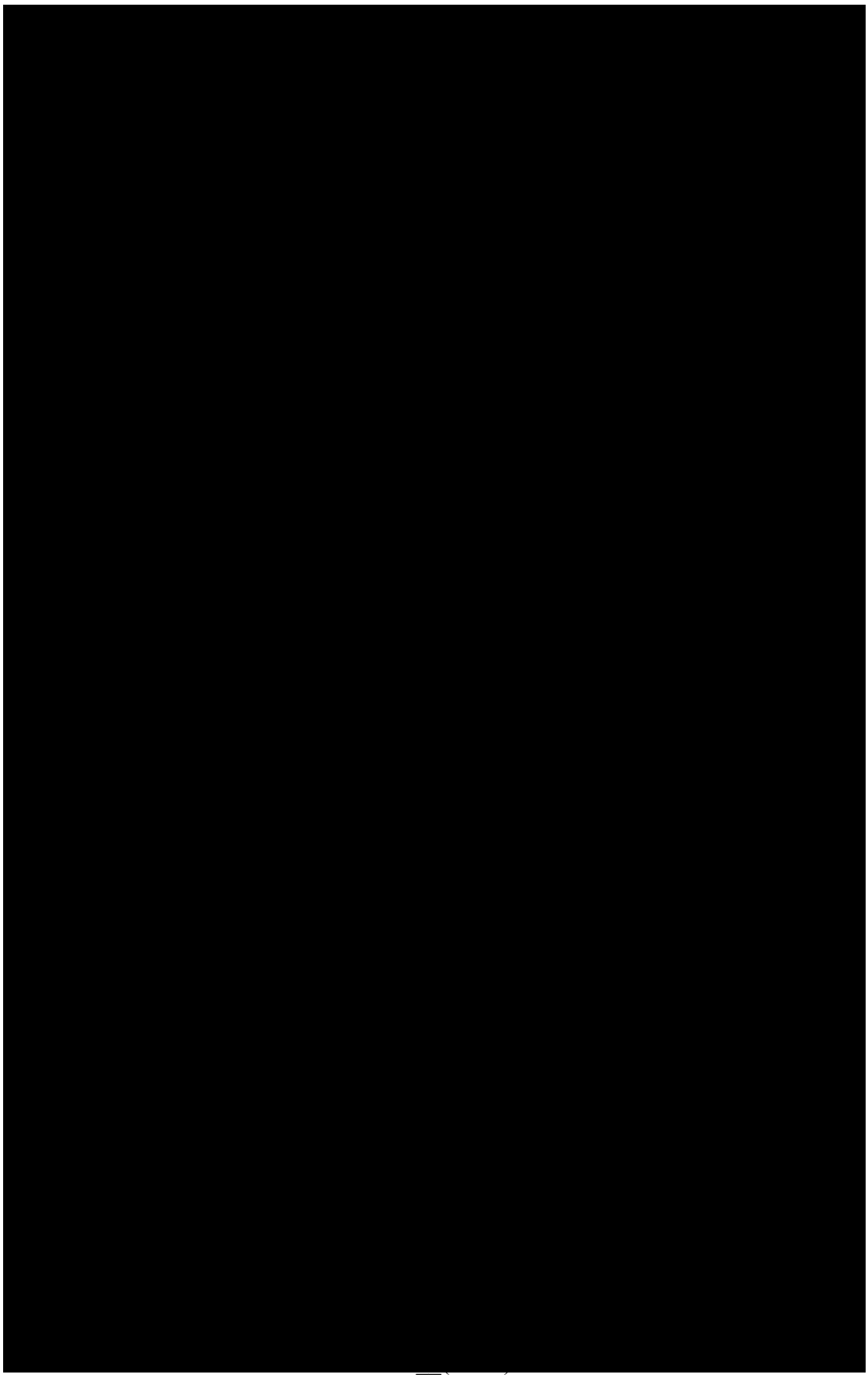
Table 7a - Winter Supply Hedges Summary

Company: The Brooklyn Union Gas Company & KeySpan Gas East Corporation

Submission Date: 7/16/2018

Version #: 1

Winter 2017-18 Purchasing Plan & Projected Prices



Case 14-G-0185 - Winter Supply 2015-16 Forms

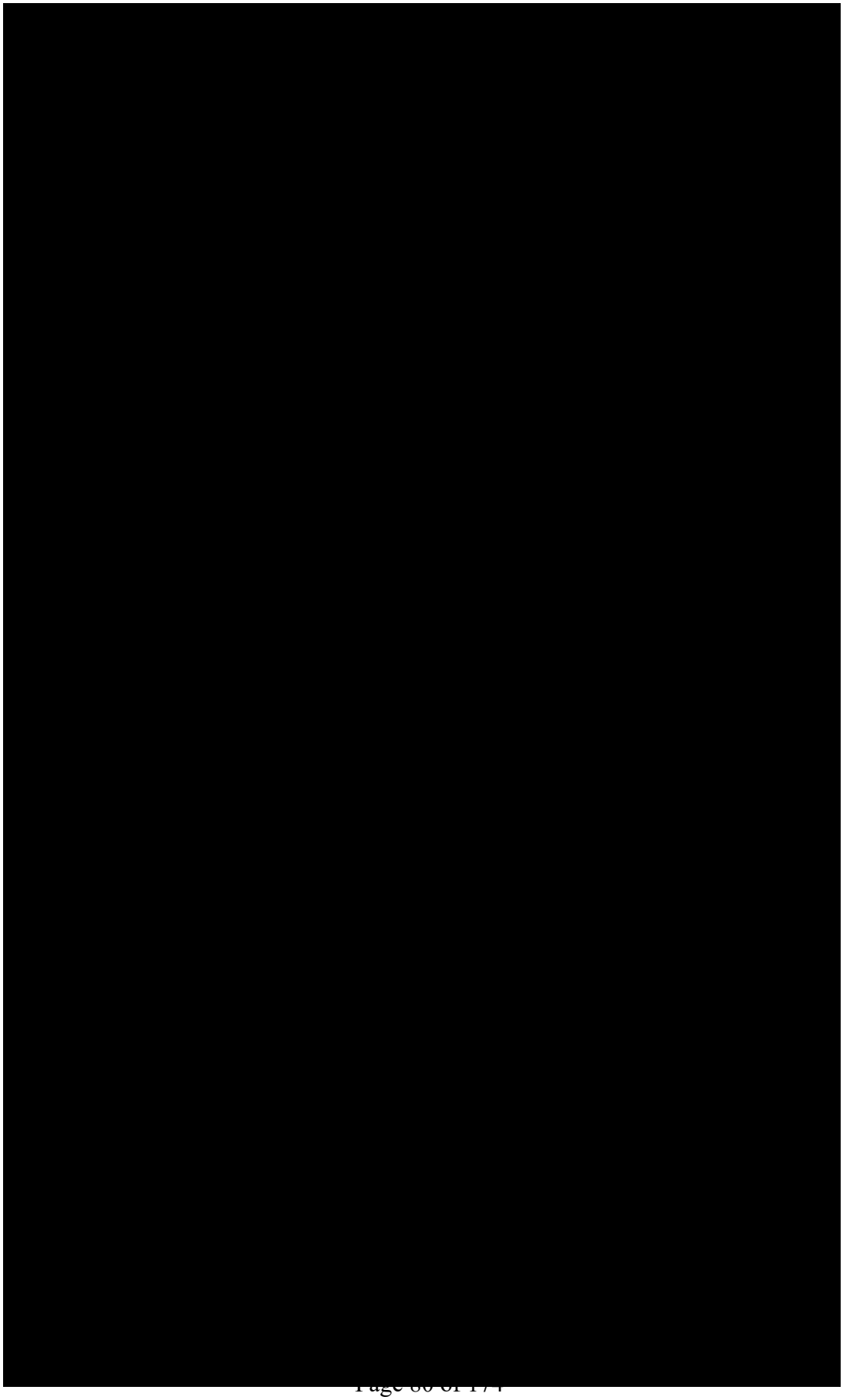
Table 7b Winter Supply Hedges Summary

Company: Niagara Mohawk Power Corporation

Submission Date: 7/16/2018

Version #: 1

Winter 2017-18 Purchasing Plan & Projected Prices



Case 18-M-0272 - Winter Supply 2018-19 Forms

Table 8a - Bill Comparison (Excluding Taxes)

Winter 2017-18 to Winter 2018-189

Company(s): **The Brooklyn Union Gas Company**
Submission Date: **7/16/2018**
Version #:

1	2	3	4	5	6
Company	2017-18 Average Residential Heating Customer Winter Bill	2017-18 Average Residential Heating Customer Winter Bill	2018-19 Forecasted Residential Heating Customer Winter Bill	2018-19 Percentage Change from Last Winter (Normalized)	2018-19 Percent Change from Last Winter (Actual)
	Actual	Normalized	Normalized	(column 4 - column 3) / column 3	(column 4 - column 2) / column 2
Total Delivery Costs	$501.00	$503.00	$583.00	$0.16	16.2%
GAC Costs (Commodity)			$350		
1 GAC Reconciliation			$1		
2 GAC Surcharges and Refunds			$14		
Total Commodity Costs	$408	$403	$365	-9.3%	-10.5%
Total Winter Bill	$909.00	$906.00	$948.00	$0.05	4.2%

Assumptions:

Normal for 2018-2019 = 1,000 Therms/year with 720 Therms of winter use
Normal for 2017-2018 = 1,000 Therms/year with 720 Therms of winter use

Actual Nov 17 - Mar 18 = 733 Therms of winter use

Notes:

[1] Identify the impact of any GAC reconcilliation surcharge or refund mechanism.

[2] Identify the impact of any other surcharges or refunds included in bills.

[3] Normalized winter bill impact covers five months, November to March.

[4] The Gas Cost Factor and Merchant Function Charge are embedded in the Monthly Rate (GAC) Adjustment.

[5] Delivery and Commodity Costs includ GRT and MTA

[6] Delivery costs are calculated based on the tariff rates (Case 16-G-0059).

[3 4] **Make up of Winter Bill Impact -**	**$948.00**
1 Delivery	57.45%
2 GRT	3.75%
3 Monthly Rate (GAC) Adjustment	37.53%
4 Gas Cost Factor	NA
5 Systems Benefits Charge	1.16%
6 Merchant Function Charge	NA
7 Revenue Decoupling Mechanism (RDM)	-0.52%
8 Temporary NY State Surcharge	0.00%
9 Other - Delivery Rate Surcharge	0.63%
	100.00%

Case 18-M-0272 - Winter Supply 2018-19 Forms

Table 8b - Bill Comparison (Excluding Taxes)

Winter 2017-18 to Winter 2018-19

Company(s): **KeySpan Gas East Corporation**
Submission Date: **7/16/2018**
Version #: 1 of 1

1	2	3	4	5	6
Company	2017-18 Average Residential Heating Customer Winter Bill	2017-18 Average Residential Heating Customer Winter Bill	2018-19 Forecasted Residential Heating Customer Winter Bill	2018-19 Percentage Change from Last Winter (Normalized)	2018-19 Percent Change from Last Winter (Actual)
	Actual	Normalized	Normalized	(column 4 - column 3) / column 3	(column 4 - column 2) / column 2
Total Delivery Costs	$544.35	$525.29	$557.94	$0.06	2.5%
GAC Costs (Commodity)			$335		
[1] GAC Reconciliation			$2		
[2] GAC Surcharges and Refunds			$11		
Total Commodity Costs	$368	$364	$348	-4.4%	-5.3%
Total Winter Bill	$912.06	$889.42	$906.06	$0.02	-0.7%

Assumptions:

Normal for 2018-19 = 1000 Therms/year with 720 Therms of winter use
Normal for 2017-18 = 1000 Therms/year with 720 Therms of winter use

Actual Nov 17 - Mar 18 = 732 Therms of winter use

Notes:

[1] Identify the impact of any GAC reconcilliation surcharge or refund mechanism.
[2] Identify the impact of any other surcharges or refunds included in bills.

3 4 Make up of Winter Bill Impact -	$906.06
1 Delivery	59.74%
2 GRT and MTA	2.36%
3 Monthly Rate (GAC) Adjustment	37.99%
4 Gas Cost Factor	NA
5 Systems Benefits Charge	1.08%
6 Merchant Function Charge	NA
7 Revenue Decoupling Mechanism (RDM)	-1.20%
8 Temporary NY State Surcharge	0.00%
9 Other	0.03%
	100.00%

Case 18-M-0272 - Winter Supply 2018-19 Forms

Table 8c - Bill Comparison (Excluding Taxes)

Winter 2017-18 to Winter 2018-19

Company(s): **Niagara Mohawk Power Corporation**
Submission Date: **7/16/2018**
Version #: 1 of 1

1	2	3	4	5	6
Company	2017-18 Average Residential Heating Customer Winter Bill	2017-18 Average Residential Heating Customer Winter Bill	2018-19 Forecasted Residential Heating Customer Winter Bill	2018-19 Percentage Change from Last Winter (Normalized)	2018-19 Percent Change from Last Winter (Actual)
	Actual	Normalized	Normalized	(column 4 - column 3) / column 3	(column 4 - column 2) / column 2
National Grid (Niagara Mohawk)	$548	$538	$517	-4.0%	-5.7%
GAC Reconciliation Refund	-$2	-$2	-$2		

* Total bill includes filed GAC imbalance effective 1/1/2018

Assumptions:

Normal for 2018-19 = 1000 Therms/year with 713 Therms of winter use
Normal for 2017-18 = 1000 Therms/year with 713 Therms of winter use

Actual Nov 17 - Mar 18 = 730 Therms of winter use

Notes:

[1] Identify the impact of any GAC reconcilliation surcharge or refund mechanism.

[2] Identify the impact of any other surcharges or refunds included in bills.

	Make up of Monthly Bill Impact -	$516.52
1	Delivery	45.62%
2	GRT	0.99%
3	Monthly Rate (GAC) Adjustment	0.00%
4	Gas Cost Factor	47.68%
5	Systems Benefits Charge	1.56%
6	Merchant Function Charge	2.81%
7	Revenue Decoupling Mechanism (RDM)	1.33%
8	Temporary NY State Surcharge	0.00%
9	Net Revenue Sharing	0.01%
		100.00%

Case 18-M-0272

Exhibit 1
2017/18 Table 1a

Case 17-M-0280 - Winter Supply 2017-18 Forms

Total System Firm Peak Day Capacity (DT)

Company: **National Grid**
Gate Station or Service Area: **The Brooklyn Union Gas Company & KeySpan Gas East Corporation**
Submission Date: **11/9/2017**
Version #: 4 of 4

	2016-17 Winter	2017-18 Winter	Design Peak Day Demand
Flowing Supplies	795,769	693,357	
Storage Withdrawals	795,919	797,428	
Winter Peaking Service	292,500	180,000	
LNG	394,500	394,500	
CNG		2,850	
Cogen Supplies	95,896	95,896	
Local Production *	1,600	1,600	
Recallable Capacity (AMAs, etc.)		204,776	
Marketer Provided Supplies:	402,450	416,266	
Mandatory Retail Access Capacity			
Grandfathered Retail Access Capacity			
Non- Mandatory Firm Capacity			
Other Customer Capacity			
Peak Day Totals	**2,778,634**	**2,786,673**	**2,762,225**
		Peak Day Design Temp:	**0° F**

\+ From Table 2 data
* Local Production, landfill gas, renewables, etc. delivered directly into the LDC distribution system.

Winter Peaking Service

Daily Price Indexed Contracts	180,000	Firm Primary Capacity Supported?
Non Daily Price Indexed Contracts	N/A	(storage, other?)

Marketer Provided Supplies

Retail Access Supplies	416,266	mandatory assignment of company capacity
Retail Access Supplies	0	marketer owned capacity (grandfathered capacity)
Other Supplies	0	marketer owned capacity

Case 18-M-0272

Exhibit 2
2017/18 Table 1b

Case 17-M-0280 - Winter Supply 2017-18 Forms
Total System Firm Peak Day Capacity (DT)

Company: **National Grid**
Gate Station or Service Area: **Niagara Mohawk Power Corporation**
Submission Date: **11/9/2017**

Version #: 4 of 4

	2016-17 Winter	2017-18 Winter	Design Peak Day Demand
Flowing Supplies	322,031	336,742	
Storage Withdrawals	294,766	300,346	
Winter Peaking Service	30,000		
LNG			
CNG			
Cogen Supplies	43,200	29,200	
Local Production *			
Recallable Capacity (AMAs, etc.)		50,000	
Marketer Provided Supplies:			
Mandatory Retail Access Capacity	260,699	250,408	
Grandfathered Retail Access Capacity			
Non- Mandatory Firm Capacity			
Other Customer Capacity			
Peak Day Totals	950,696	966,696	897,107
		Peak Day Design Temp:	75 HDD

\+ From Table 2 data
* Local Production, landfill gas, renewables, etc. delivered directly into the LDC distribution system.

Winter Peaking Service

Daily Price Indexed Contracts _____ Firm Primary Capacity Supported?
Non Daily Price Indexed Contracts _____ (storage, other?)

Case 18-M-0272

Exhibit 3
2017/18 Table 1c

Case 17-M-0280 - Winter Supply 2017-18 Forms
Total System Firm Peak Day Capacity (DT)

Company: **National Grid**
Gate Station or Service Area: **Niagara Mohawk Power Corporation East Gate**
Submission Date: **11/9/2017**
Version #: 4 of 4

	2016-17 Winter	2017-18 Winter	Design Peak Day Demand
Flowing Supplies[1]	143,647	156,530	
Storage Withdrawals[1]	138,563	141,664	
Winter Peaking Service	30,000		
LNG			
CNG			
Cogen Supplies	40,200	26,200	
Local Production *			
Recallable Capacity (AMAs, etc.)		50,000	
Marketer Provided Supplies:			
Mandatory Retail Access Capacity	119,841	114,005	
Grandfathered Retail Access Capacity			
Non- Mandatory Firm Capacity			
Other Customer Capacity			
Peak Day Totals	472,251	488,399	443,184
		Peak Day Design Temp:	75 HDD

\+ From Table 2 data

* Local Production, landfill gas, renewables, etc. delivered directly into the LDC distribution system.

1 Contracts 100001 and 700001 are split by design day allocation %;East Gate- 47.08%;West Gate-52.92%

Winter Peaking Service

Daily Price Indexed Contracts _____ Firm Primary Capacity Supported?
Non Daily Price Indexed Contracts _____ (storage, other?)

Case 18-M-0272

Exhibit 4
2017/18 Table 1c

Case 17-M-0280 - Winter Supply 2017-18 Forms
Table 1d - Total System Firm Peak Day Capacity (DT)

Company: **National Grid**
Gate Station or Service Area: **Niagara Mohawk Power Corporation West Gate**
Submission Date: **11/9/2017**
Version #: 2 of 2

	2016-17 Winter	2017-18 Winter	Design Peak Day Demand
Flowing Supplies[1]	178,384	180,212	
Storage Withdrawals[1]	156,203	158,682	
Winter Peaking Service	-		
LNG			
CNG			
Cogen Supplies	3,000	3,000	
Local Production *			
Recallable Capacity (AMAs, etc.)			
Marketer Provided Supplies:			
Mandatory Retail Access Capacity	140,858	136,403	
Grandfathered Retail Access Capacity			
Non- Mandatory Firm Capacity			
Other Customer Capacity			
Peak Day Totals	478,445	478,297	453,923
		Peak Day Design Temp:	75 HDD

\+ From Table 2 data

* Local Production, landfill gas, renewables, etc. delivered directly into the LDC distribution system.

1 Contracts 100001 and 700001 are split by design day allocation %;East Gate- 47.08%;West Gate-52.92%

Winter Peaking Service

Daily Price Indexed Contracts _____ Firm Primary Capacity Supported?
Non Daily Price Indexed Contracts _____ (storage, other?)

Exhibit 5
2017/18 Table 3a

Case 17-M-0280 - Winter Supply 2017-18 Forms

Table 3a - Estimated Normal Weather Requirements (MDT)

Company: **National Grid**
Gate Station or Service Area: **The Brooklyn Union Gas Company**
Submission Date: **7/14/2017**
Version #: **1**

	2016-17 Forecast	2016-17 Actual	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25	2025-26	2026-27
Annual												
Firm												
Sales	93,292		94,885	96,372	98,486	99,292	101,027	102,863	105,293	106,090	107,705	57,339
Transportation	56,946		56,946	56,946	57,201	56,946	56,946	56,946	57,201	56,946	56,946	56,946
Temperature Controlled												
Sales	11,679		11,292	11,118	10,991	10,660	10,384	10,128	9,928	9,598	9,308	6,326
Transportation	6,583		6,425	6,360	6,334	6,215	6,114	6,014	5,951	5,815	5,698	3,601
Non-Firm												
Sales	-		-	-	-	-	-	-	-	-	-	-
Transportation	-		-	-	-	-	-	-	-	-	-	-
Total	**168,500**		**169,548**	**170,796**	**173,011**	**173,112**	**174,471**	**175,951**	**178,373**	**178,449**	**179,657**	**124,213**
Winter Season												
Firm												
Sales	66,479	64,689	67,595	68,654	70,318	70,605	71,758	73,051	75,010	75,303	76,423	77,520
Transportation	36,766	35,423	36,766	36,766	37,021	36,766	36,766	36,766	37,021	36,766	36,766	36,766
Temperature Controlled												
Sales	8,184	9,347	7,859	7,741	7,687	7,448	7,253	7,081	6,955	6,714	6,508	6,326
Transportation	4,254	1,590	4,112	4,071	4,073	3,991	3,921	3,862	3,834	3,748	3,669	3,601
Non-Firm												
Sales	-	-	-	-	-	-	-	-	-	-	-	-
Transportation	-	838	-	-	-	-	-	-	-	-	-	-
Total	**115,683**	**111,887**	**116,332**	**117,232**	**119,098**	**118,809**	**119,697**	**120,760**	**122,820**	**122,530**	**123,366**	**124,213**
Peak Day												
Firm												
Sales	-	906	-	-	-	-	-	-	-	-	-	-
Transportation	-	326	-	-	-	-	-	-	-	-	-	-
Temperature Controlled												
Sales	-	0	-	-	-	-	-	-	-	-	-	-
Transportation	-	6	-	-	-	-	-	-	-	-	-	-
Total	**-**	**1,238**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**

1/ Normal Weather is defined as:

Annual	4,587	HDD
Winter	3,819	HDD
Peak Day	-	HDD

Exhibit 6
2017/18 Table 3b

Case 17-M-0280 - Winter Supply 2017-18 Forms

Table 3b - Estimated Normal Weather Requirements (MDT)

Company: **National Grid**
Gate Station or Service Area: **KeySpan Gas East Corporation**
Submission Date: **7/14/2017**
Version #: **1**

	2016-17 Forecast	2016-17 Actual	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25	2025-26	2026-27
Annual												
Firm												
Sales	77,153		78,117	80,055	82,265	83,498	84,982	86,894	89,235	90,731	92,096	55,017
Transportation	27,489		27,489	27,489	27,615	27,489	27,489	27,489	27,615	27,489	27,489	27,489
Temperature Controlled												
Sales	496.68		499	529	555	571	591	616	643	660	682	482
Transportation	187.84		190	199	206	210	214	220	227	230	235	147
Non-Firm												
Sales	-		-	-	-	-	-	-	-	-	-	-
Transportation	-		-	-	-	-	-	-	-	-	-	-
Total	**105,327**		**106,294**	**108,271**	**110,641**	**111,768**	**113,276**	**115,219**	**117,719**	**119,110**	**120,502**	**83,134**
Winter Season												
Firm												
Sales	53,591	50,725	53,880	55,355	56,921	57,778	58,601	60,049	61,660	62,662	63,741	64,367
Transportation	18,139	18,278	18,139	18,139	18,265	18,139	18,139	18,139	18,265	18,139	18,139	18,139
Temperature Controlled												
Sales	348	356	339	362	382	393	404	423	442	453	469	482
Transportation	117	19	116	122	127	130	132	136	140	142	145	147
Non-Firm												
Sales	-	281	-	-	-	-	-	-	-	-	-	-
Transportation	-	0	-	-	-	-	-	-	-	-	-	-
Total	**72,195**	**69,659**	**72,474**	**73,978**	**75,695**	**76,440**	**77,276**	**78,746**	**80,507**	**81,396**	**82,494**	**83,134**
Peak Day												
Firm												
Sales	-	620	-	-	-	-	-	-	-	-	-	-
Transportation	-	173	-	-	-	-	-	-	-	-	-	-
Temperature Controlled												
Sales	-		-	-	-	-	-	-	-	-	-	-
Transportation	-		-	-	-	-	-	-	-	-	-	-
Total	**-**	**793**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**

1/ Normal Weather is defined as:

Annual	4,587	HDD
Winter	3,819	HDD
Peak Day	-	HDD

Case 18-M-0272

Exhibit 7
2017/18 Table 3c

Case 17-M-0280 - Winter Supply 2017-18 Forms
Table 3c - Estimated East Gate Normal Weather Requirements (MDT)

Company: **National Grid**
Company: **Niagara Mohawk Power Corporation East Gate**
Submission Date: **11/9/2017**
Version #: **2**

		2016-17 Forecast	2016-17 Actual	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25	2025-26	2026-27
Annual													
Firm													
	Sales	27,614		28,299	28,617	29,212	29,403	29,684	30,315	31,079	31,265	31,656	
	Transportation	13,243		13,594	13,689	13,845	13,716	13,781	13,901	14,096	14,022	14,102	
Non-Firm													
	Sales												
	Transportation	16,882		17,509	17,598	17,616	17,782	17,870	18,184	18,224	18,373	18,451	18,467
Total		57,739		59,402	59,904	60,673	60,901	61,335	62,400	63,399	63,661	64,209	18,467
Winter Season													
Firm													
	Sales	20,811	18,779	21,445	21,696	22,198	22,331	22,472	22,961	23,661	23,723	24,111	24,151
	Transportation	9,415	8,212	9,716	9,751	9,896	9,771	9,784	9,845	10,014	9,904	9,969	9,981
Non-Firm													
	Sales		-										
	Transportation	7,776	12,351	8,167	8,310	8,311	8,379	8,339	8,505	8,523	8,638	8,679	8,696
Total		38,002	39,342	39,328	39,757	40,405	40,481	40,595	41,311	42,197	42,265	42,760	42,827
Peak Day													
Firm													
	Sales		226										
	Transportation		79										
Total			305										

1/ Design Weather is defined as:

Annual	6,561	HDD
Winter	5,151	HDD
Peak Day		HDD

Case 18-M-0272

Exhibit 8
2017/18 Table 3d

Case 17-M-0280 - Winter Supply 2017-18 Forms

Table 3d - Estimated West Gate Normal Weather Requirements (MDT)

Company: **National Grid**
Company: **Niagara Mohawk Power Corporation West Gate**
Submission Date: **11/9/2017**
Version #: **2**

		2016-17 Forecast	2016-17 Actual	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25	2025-26	2026-27
Annual													
Firm													
	Sales	29,640		29,409	29,523	29,767	29,656	29,689	29,822	30,089	30,074	30,125	
	Transportation	14,958		14,855	14,820	14,763	14,586	14,395	14,361	14,351	14,270	14,104	
Non-Firm													
	Sales												
	Transportation	13,999		14,651	14,300	14,033	13,895	13,738	13,655	13,433	13,202	13,101	13,146
Total		58,597		58,914	58,643	58,563	58,137	57,821	57,837	57,872	57,546	57,330	13,146
Winter Season													
Firm													
	Sales	22,044	22,873	21,790	21,896	22,082	22,000	21,976	22,123	22,306	22,283	22,337	22,276
	Transportation	10,735	9,870	10,611	10,627	10,603	10,473	10,290	10,275	10,278	10,220	10,125	9,983
Non-Firm													
	Sales		-										
	Transportation	6,903	23,381	7,593	7,511	7,391	7,312	7,221	7,200	7,092	7,036	6,998	7,042
Total		39,681	56,124	39,994	40,034	40,076	39,785	39,487	39,598	39,676	39,539	39,460	39,302
Peak Day													
Firm													
	Sales		259										
	Transportation		101										
Total			360										

1/ Design Weather is defined as:

Annual	6,561	HDD
Winter	5,151	HDD
Peak Day		HDD

Exhibit 9
Downstate New York 2018/19 Load Duration Curve

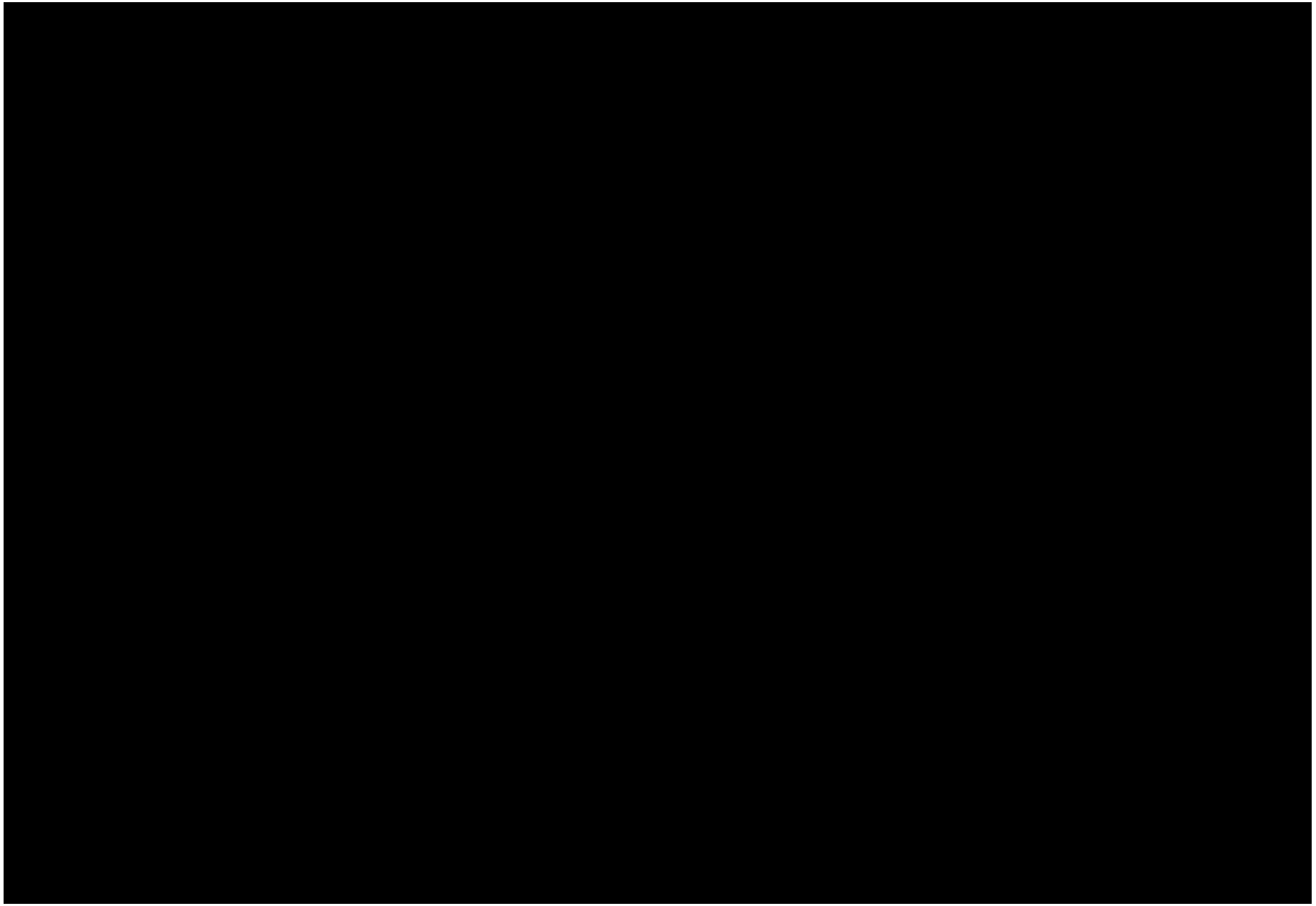


Exhibit 10
Upstate New York 2018/19 Load Duration Curve

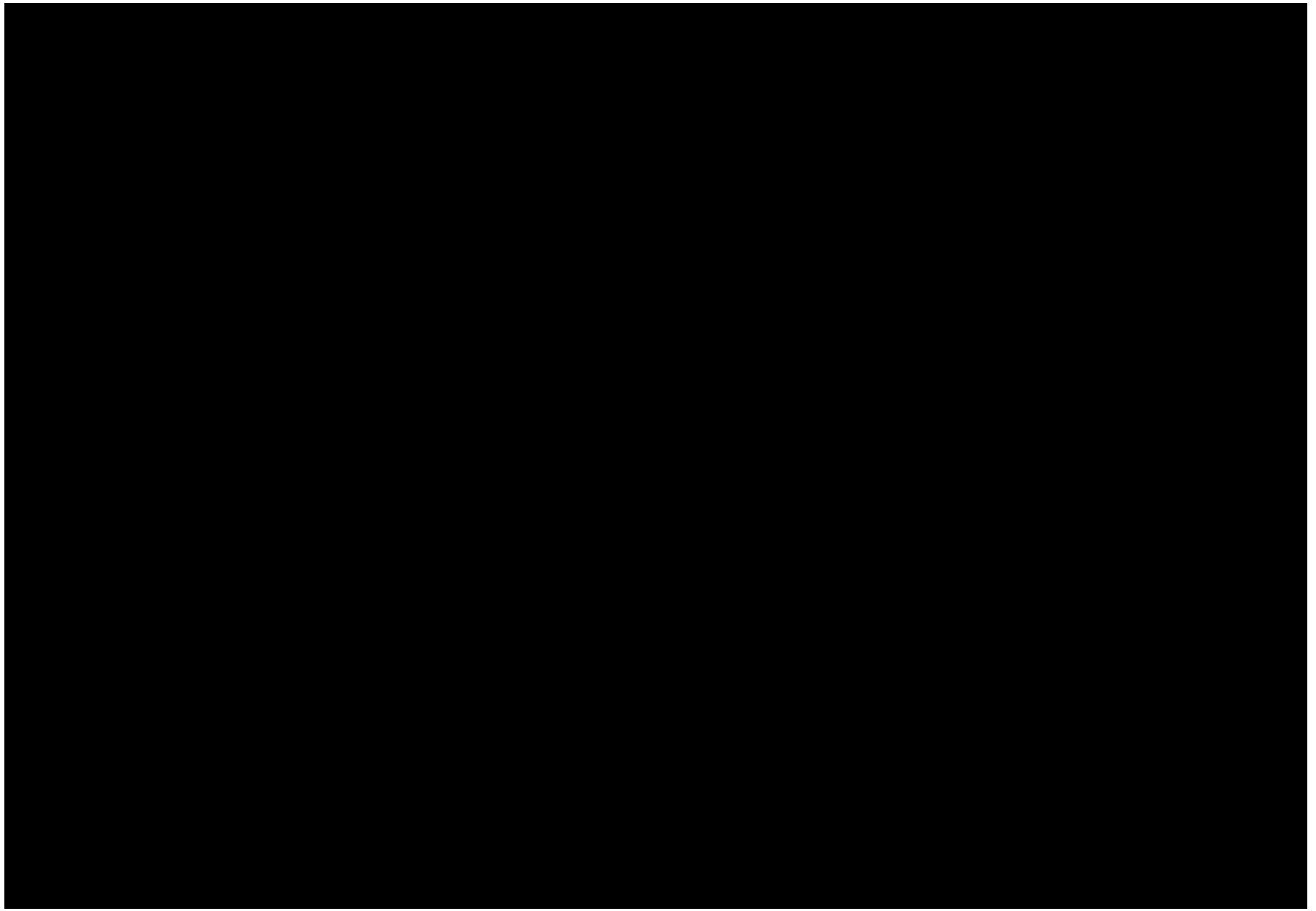


Exhibit 11
Upstate New York 2018/19 East Gate Load Duration Curve

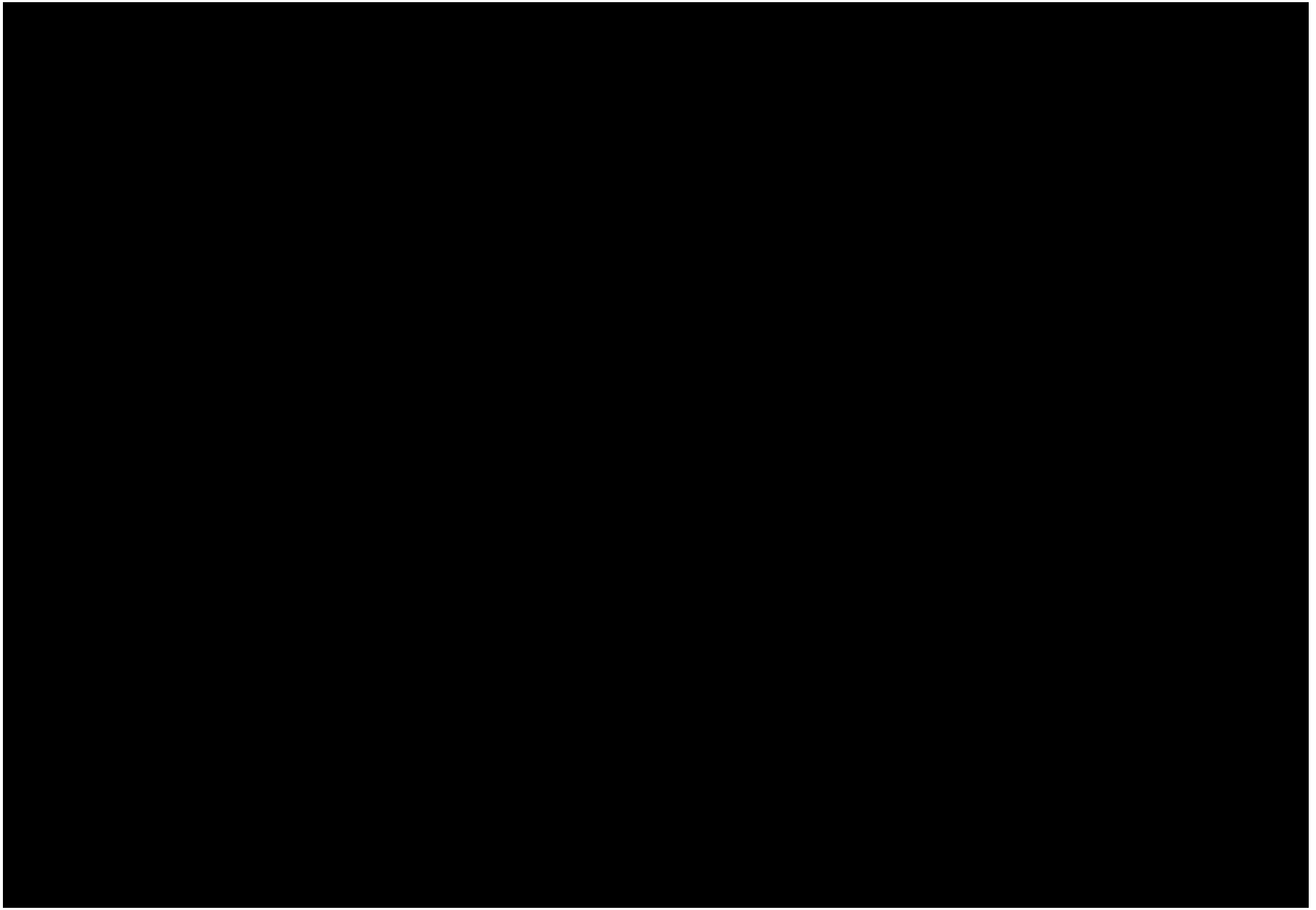
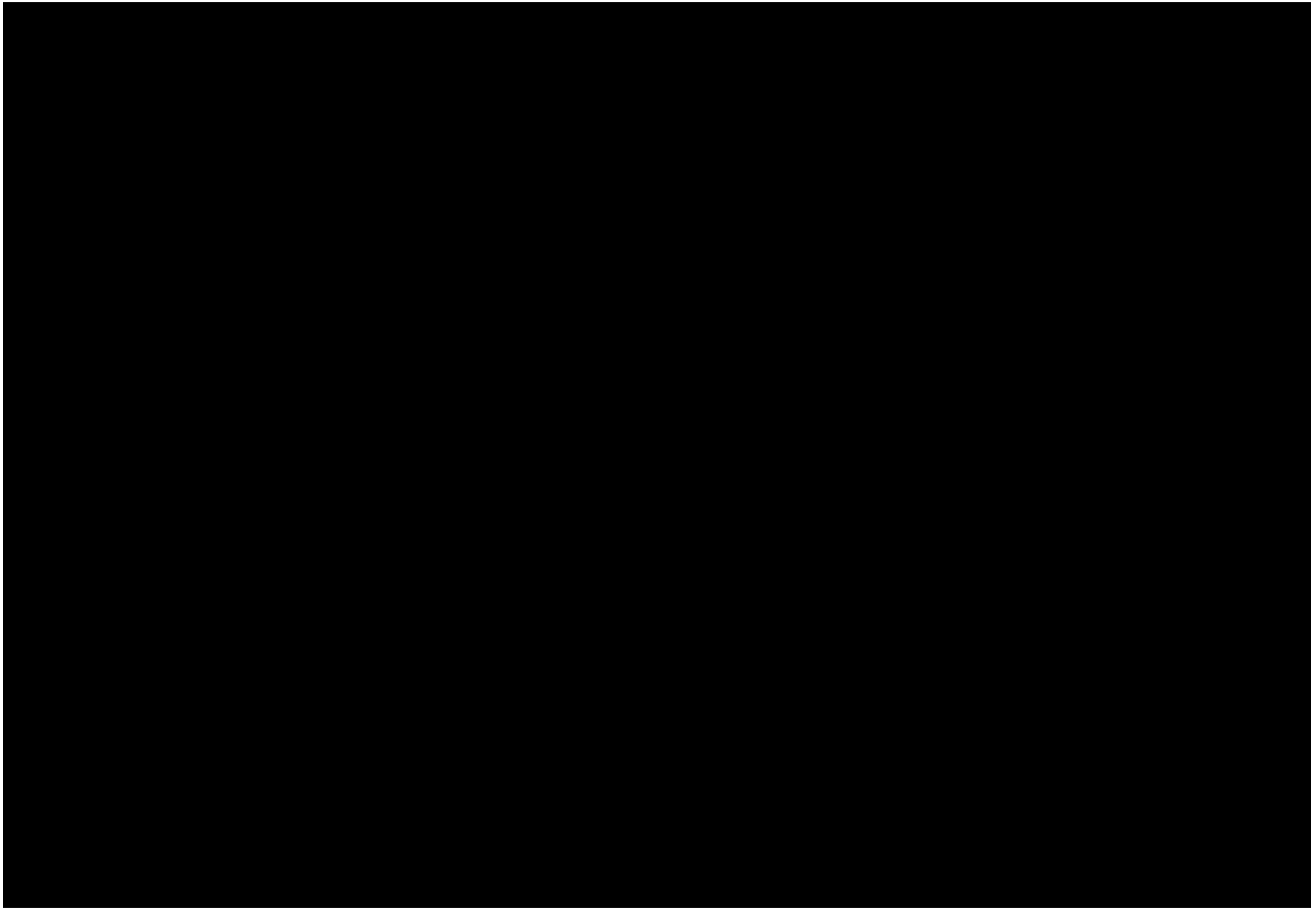


Exhibit 12
Upstate New York 2018/19 West Gate Load Duration Curve



Case 18-M-0272

Exhibit 13
Equations For Predicting November 2018 Firm & TC Sendout (Dth/day)

Coefficients

		BUG	KGE	NMPC East	NMPC West	
Constant		1,091,260	989,632	319,353	367,396	
Temperature (d)	x	26,991	-3,819	-3902	-4,432	+
Temperature (d-1)	x	-5,163	-3718	-1,000	-1,066	+
Temperature (d-2)	x	-666	114	-321	-464	+
$Temperature^2$ (d)	x	-1,167	-309	0	0	+
$Temperature^3$ (d)	x	10	3	0	0	+
Cloud Cover (d)	x			98	162	+
$Wind_speed^2$ (d)	x	-162	-27	0	0	+
Wind_speed (d)	x	8,319	1,798	1,500	1,774	+
Day Type	x	979	12,939	3,245	784	+
DOM	x	718	261	299	196	+

Notes:
Temperature (d) = current day average temperature (°F)
Temperature (d-1) = prior day average temperature (°F)
Temperature (d-2) = average temperature two days prior (°F)
Cloud Cover (d) = current day cover (oktas)
Wind_speed (d) = current day average wind speed (mph)
Day Type = weekend/holiday (2) or weekday (3)
DOM = day of the month (1-31)

Exhibit 14

Brooklyn Union Gas & KeySpan Gas East
Combined Flow Diagram

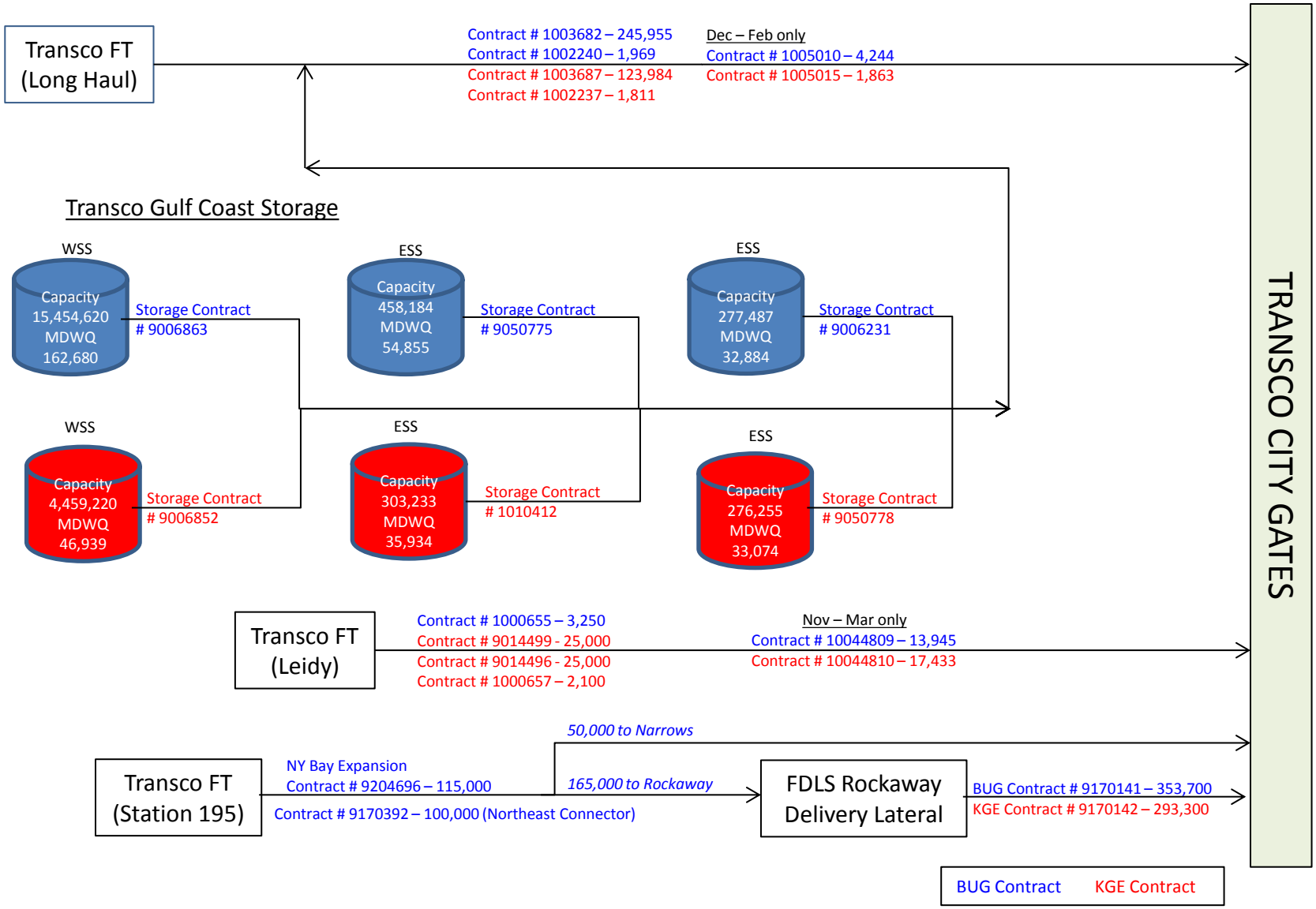


Exhibit 14

Brooklyn Union Gas & KeySpan Gas East
Combined Flow Diagram

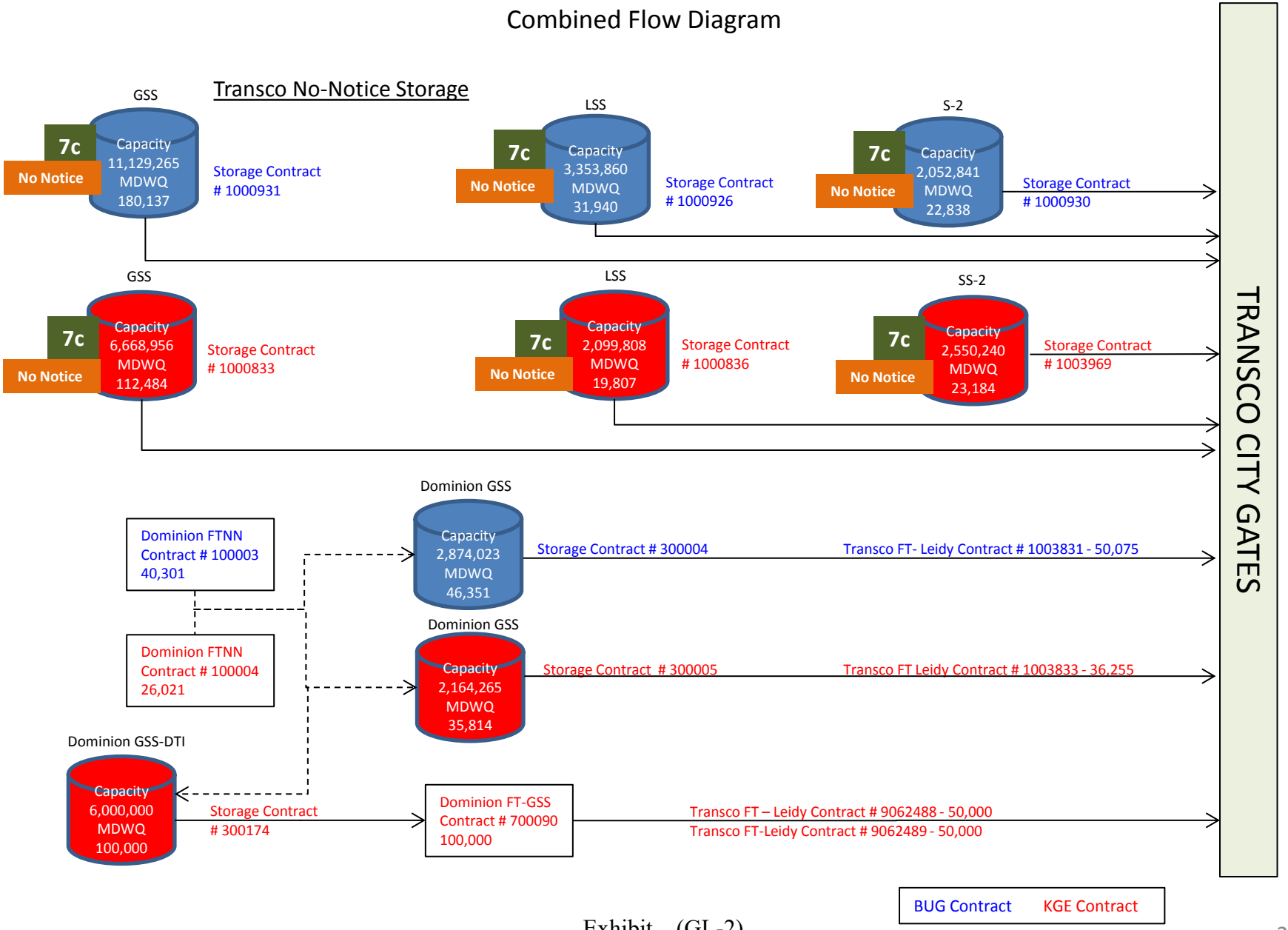


Exhibit 14

Brooklyn Union Gas & KeySpan Gas East
Combined Flow Diagram

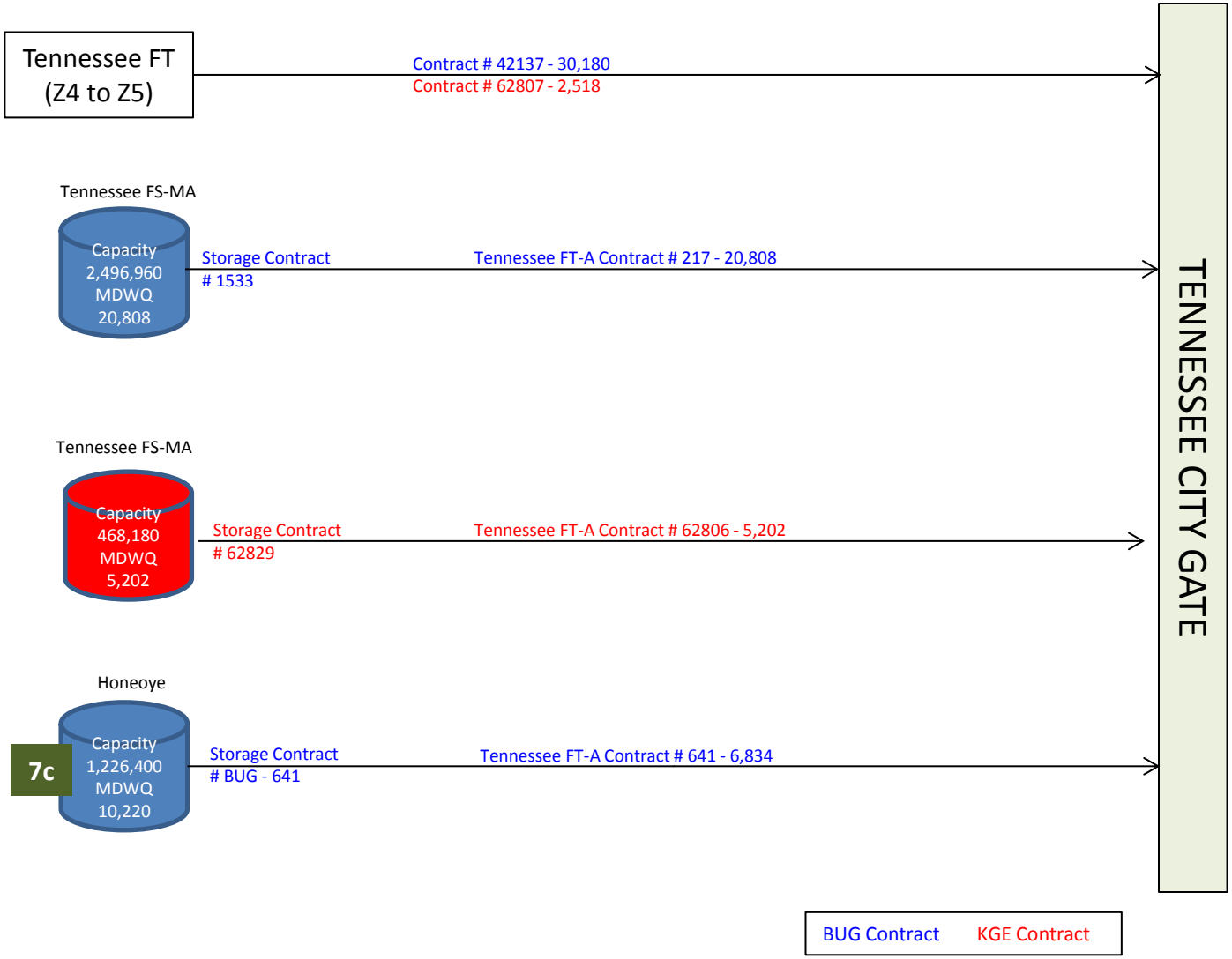


Exhibit 14

Brooklyn Union Gas & KeySpan Gas East
Combined Flow Diagram

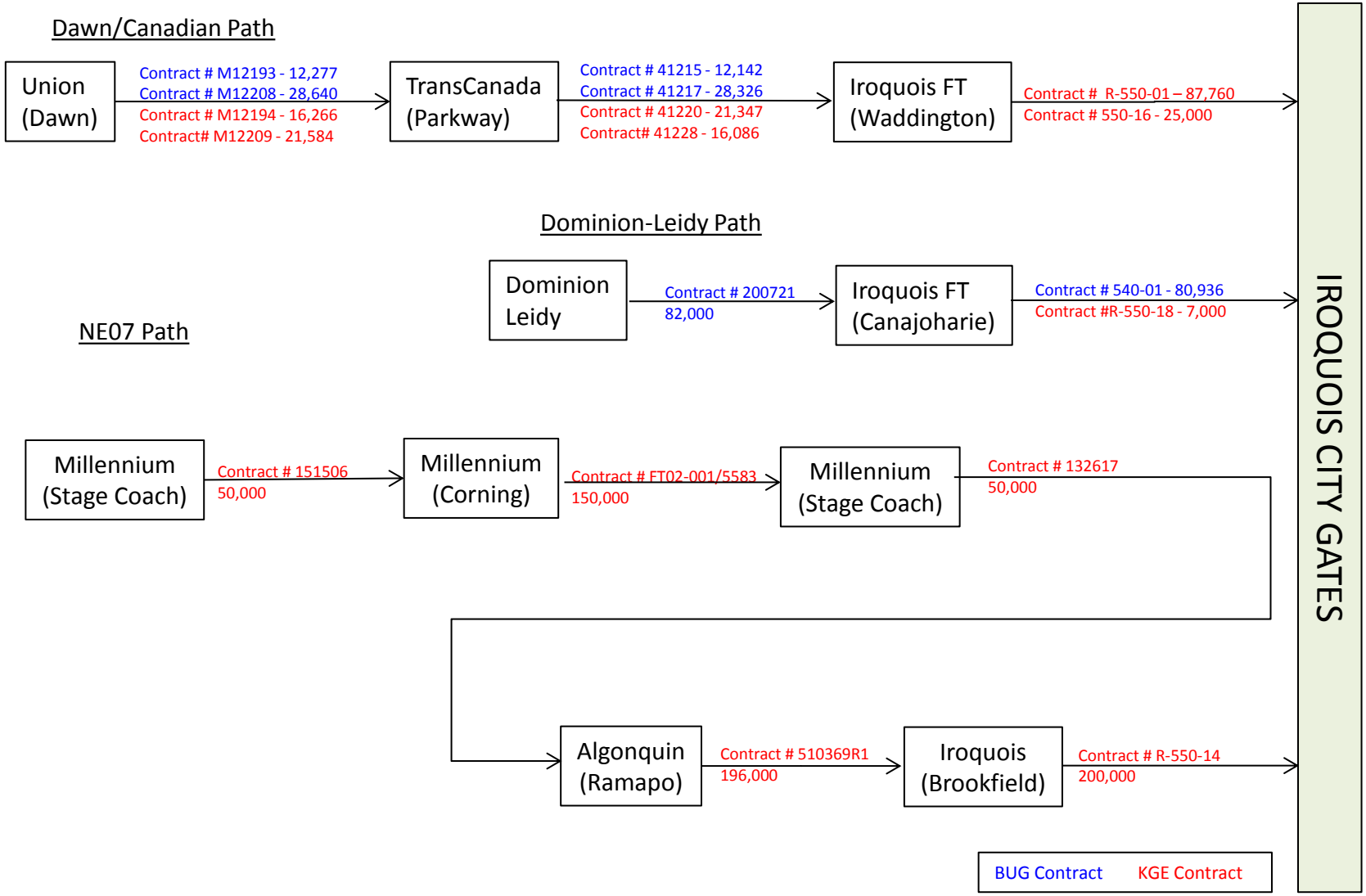


Exhibit 14
Brooklyn Union Gas & KeySpan Gas East
Combined Flow Diagram

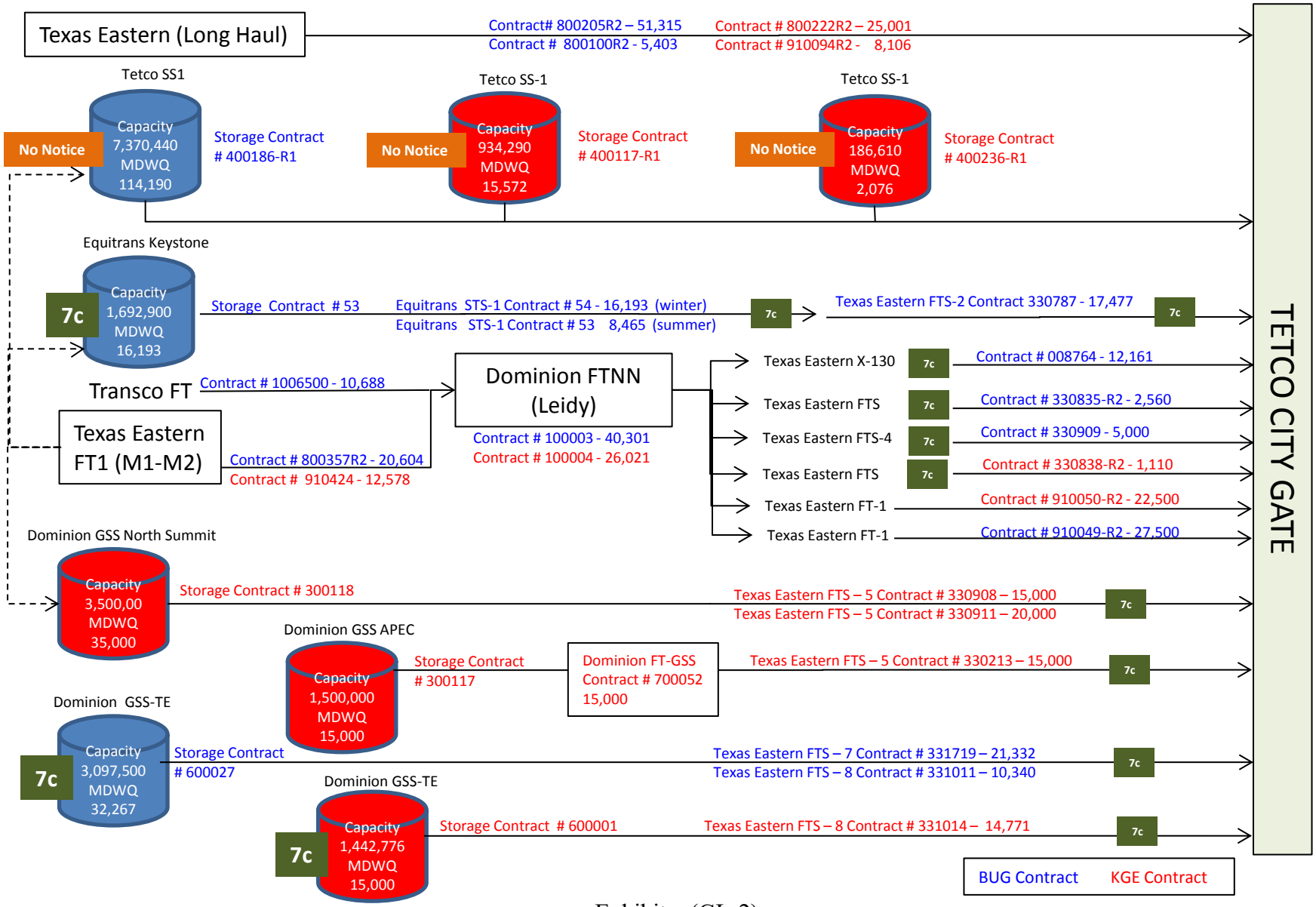
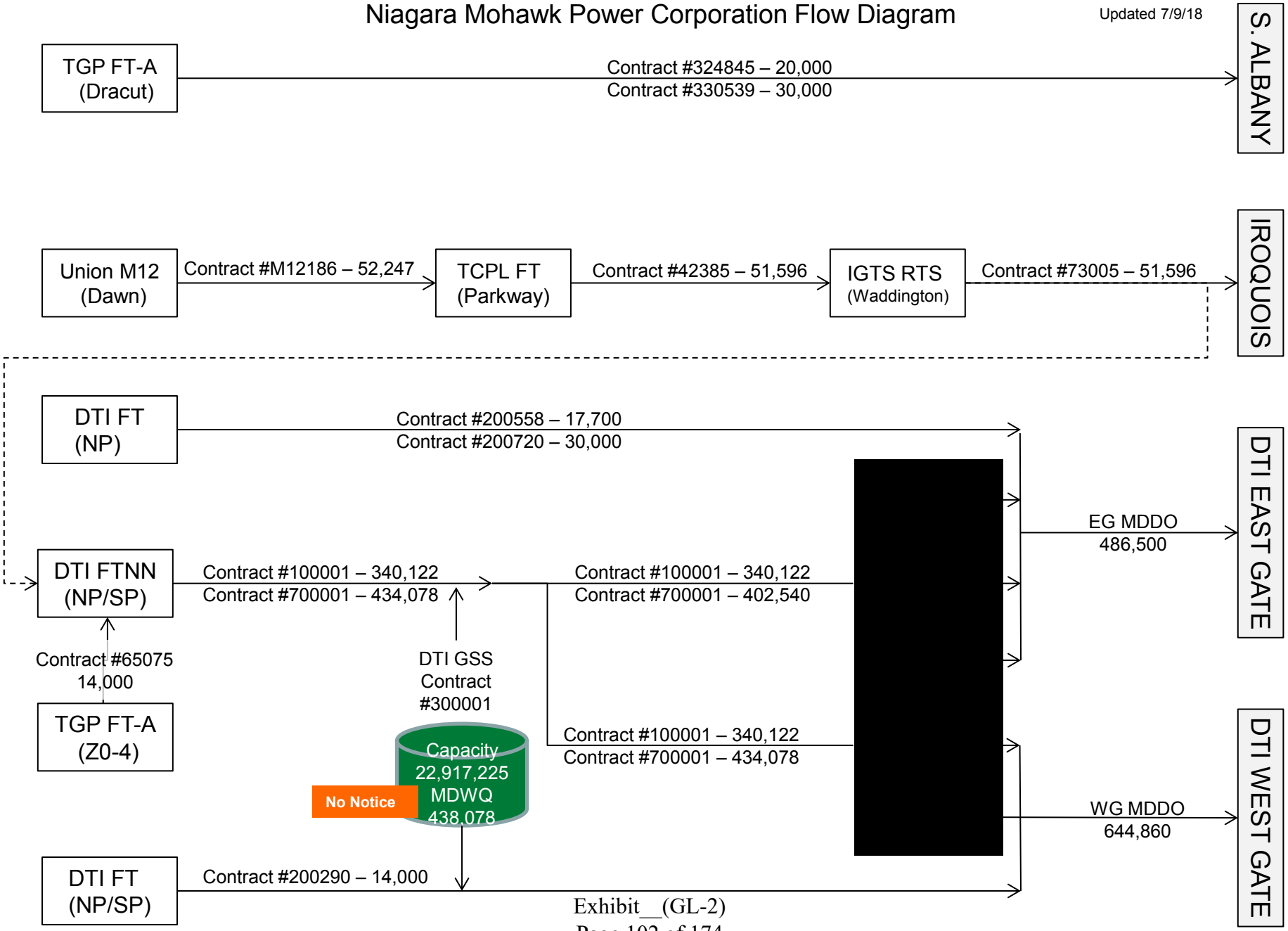


Exhibit 15

(Business Confidential)

Niagara Mohawk Power Corporation Flow Diagram

Updated 7/9/18



Case 18-M-0272

Exhibit 16
Downstate OFOs, Interruptions & Curtailments

BALANCING OFO					
Placed DATE	**START**	**LIFTED DATE**	**END**	**DURATION**	**NOTES**
12/14/2017	6:00 AM	12/14/2017	10:00 AM	4 hours	Hourly 1/24th Power Gen
12/14/2017	10:00 AM	12/16/2017	10:00 AM	2 Days	Daily 5%
12/26/2017	10:00 AM	1/9/2018	10:00 AM	14 days	Daily 2%
12/28/2017	10:00 AM	1/8/2018	10:00 AM	11 days	Hourly 1/24th Power Gen
1/14/2018	12:00 AM	1/16/2018	10:00 AM	2-1/2 days	Hourly 1/24th Power Gen
1/14/2018	10:00 AM	2/4/2018	10:00 AM	7 days	Daily 2%

INTERRUPTIONS/CURTAILMENTS					
Placed DATE	**START**	**LIFTED DATE**	**END**	**DURATION**	**NOTES**
11/14/2017	8:00 AM	11/14/2017	4:00 PM	8 hours	Planned LI TC Equip test
11/16/2017	6:00 AM	11/16/2018	2:00 PM	8 hours	Planned NY TC Equip test
12/28/2018	10:00 AM	1/3/2018	10:00 AM	6 days	DNY - Planned TC & IT
[redacted]					
1/4/2018	10:00 PM	1/8/2018	10:00 AM	3-1/2 days	DNY - Planned TC & IT
[redacted]					
1/14/2018	10:00 AM	1/14/2018	2:00 PM	4 hours	LI TC - Unplanned due to temp
2/3/2018	12:30 AM	2/3/2018	9:26 AM	9 hours	LI TC - Unplanned due to temp
[redacted]					

Notes:

1) No curtailments issued in UNY or DNY for 2017-18

[redacted]

Case 18-M-0272

Exhibit 17
Upstate OFOs, Interruptions & Curtailments

BALANCING OFO					
Placed DATE:	**START**	**LIFTED DATE:**	**END**	**DURATION**	**NOTES**
11/10/2017	10:00 AM	11/11/2017	10:00 AM	1 day	Daily 5% (East & West gate)
12/26/2017	10:00 AM	1/9/2018	10:00 AM	14 days	Daily 5% (East & West gate)
1/13/2018	10:00 AM	1/16/2018	10:00 AM	3 days	Daily 5% (East & West gate)

INTERRUPTIONS/CURTAILMENTS					
Placed DATE:	**START**	**LIFTED DATE:**	**END**	**DURATION**	**NOTES**
11/14/2017	8:00 AM	11/14/2017	10:00 AM	2 hours	East gate only (interruptibles)
12/28/2017	10:00 AM	12/29/2017	10:00 AM	1 day	East gate only (interruptibles)
[redacted]					
12/31/2017	10:00 AM	1/2/2018	10:00 AM	2 days	East gate only (interruptibles)
[redacted]					
1/5/2018	10:00 AM	1/7/2018	10:00 AM	2 days	East gate only (interruptibles)

Notes:

1) No curtailments issued in UNY for 2017-18

[redacted]

Case 18-M-0272

Exhibit 18

National Grid
2017/18 Interruption-Curtailment Summary

LDC	Start	End	Duration	Customer	Delivery Pipe	Type	Comments
NMPC	11/14/17 8:00	11/14/17 10:00	2 hrs	Interruptibles	DTI - East gate	Interruption	East gate only interruption
	12/28/17 10:00	12/29/17 10:00	1 day	Interruptibles	DTI - East gate	Interruption	East gate only interruption
	[redacted]						
	12/31/17 10:00	1/2/18 10:00	2 days	Interruptibles	DTI - East gate	Interruption	East gate only interruption
	[redacted]						
	1/5/18 10:00	1/7/18 10:00	2 days	Interruptibles	DTI - East gate	Interruption	East gate only interruption
DNY	12/28/17 10:00	1/3/18 10:00	6 days	Interruptibles & TC	Various	Interruption	Planned interruption
	1/4/18 22:00	1/8/18 10:00	3.5 days	Interruptibles & TC	Various	Interruption	Planned interruption
KEDLI	11/14/17 8:00	11/14/17 16:00	8 hrs	TC	Various	Interruption	TC Equipment test
	[redacted]						
	1/14/18 10:00	1/14/18 14:00	4 hrs	TC	Various	Interruption	Unplanned interruption due to temperature
	2/3/18 0:30	2/3/18 9:26	9 hrs	TC	Various	Interruption	Unplanned interruption due to temperature
	[redacted]						
KEDNY	11/16/17 6:00	11/14/17 14:00	8 hrs	TC	Various	Interruption	TC Equipment test
	[redacted]						

Notes:

1) No curtailments issued in UNY or DNY for 2017-18

[redacted]

Exhibit 19
Downstate NY Interruptibilty Matrix

Effective 12-1-2016

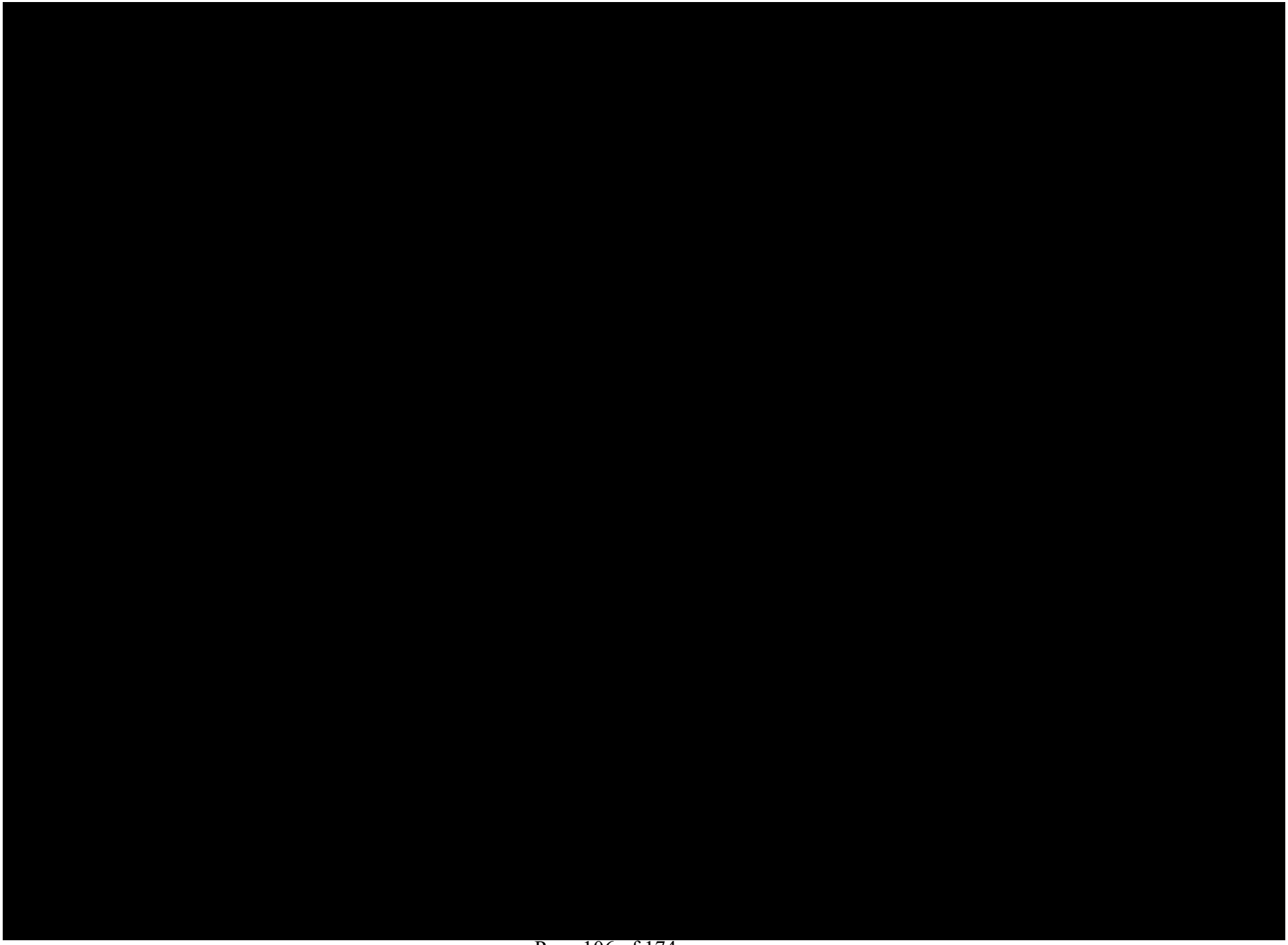


Exhibit 20
Matching Days Selection



Dear Customer:

The purpose of this letter is to notify you that on November 16, 2017 National Grid ("Company") will perform a test to confirm your ability to comply with the interruption provisions of the Company's Temperature Controlled ("TC") tariff.

1. The annual gas interruption test is scheduled for November 16, 2017, beginning at 6:00 a.m. For customers with remote switch over capability, National Grid will remotely switch your dual-fuel equipment for up to eight (8) hours. All other customers must remain off natural gas from the time they receive an alarm to switch over from gas until such time they receive an alarm notifying them that they can switch back to natural gas. It is imperative that your dual-fuel system be in good working order and that you have adequate alternate fuel on hand to sustain this interruption. If you have any equipment issues, it is imperative that you contact the Company within one (1) hour of when you or your operating personnel become aware of any equipment issues.

2. Please promptly notify National Grid at 1-718-403-3134 of any condition that may prevent your equipment from switching to an alternate fuel.

National Grid would like to take this opportunity to remind you of the tariff provisions for TC service:

1. PLEASE NOTE National Grid reserves the right to raise the designated interruption temperature (for example, from 15°F to 20°F or higher) at any time as may be required, in National Grid's sole judgment, to maintain system integrity or for other operational consideration.

2. As a TC customer, you are required to stop burning natural gas when the outdoor temperature reaches your designated interruption temperature. If you have automatic switchover capability, your equipment will switch automatically from natural gas to an alternate fuel when the outdoor temperature falls to your thermostat's designated temperature (15° F). If you do not have automatic switchover capability, you must manually switch your equipment to an alternate fuel when the outdoor temperature falls to the designated temperature for manual equipment, (15° F). In addition, please be aware that the Company will initiate a remote switchover and send an alarm that manual equipment must be switched any time National Grid receives a report that the temperature has reached 15° F at LaGuardia Airport. National Grid may also initiate a remote switchover or send an alarm requesting that you switch your equipment manually in the event of a system reliability or supply issue regardless of temperature. However, customers in the Northern Queens area may be interrupted at temperatures higher than 15° F. After this remote switchover, when National Grid receives a report that the temperature has reached 20° F, customers with automatic switchover equipment will be switched back to natural gas and all others will receive an alarm indicating that they can manually switch their equipment back to burn natural gas.

3. Your dual-fuel equipment and controls must be in working order. In addition, you must have a minimum ten (10) day supply of alternate fuel on hand. The "on hand" requirement can be satisfied by either storing the full 10 day fuel requirement onsite, or through a combination of onsite storage and a delivery contract with a fuel supplier for the difference. Whenever National Grid becomes aware that a Customer's equipment is incapable of switching over to an alternate fuel and/or that the Customer does not have an adequate supply of alternate fuel on hand, the Company will notify the Customer that the violation must be corrected within ten (10) calendar days. It will be the responsibility of the Customer to contact the Company and provide proof that the violation has been corrected. With regard to an inadequate fuel situation and/or equipment failure, Customers who had not supplied the Company with such proof will be subject to a non-compliance charge as stated in the Company's TC tariff. The non-compliance charge will cease the day the Company receives satisfactory proof that the violation has been corrected.

4. National Grid must be permitted to install, access, inspect and repair remote meter reading and fuel switching devices and associated controls located on your premises as required to assure proper operating and monitoring of your system. It is the Customer's responsibility to avoid damage to or disconnection of the Company's equipment. Any such event must be reported to the Company immediately.

5. National Grid has a communications system that will enable you to receive information regarding your account. This system provides real-time communications to interruptible customers regarding their natural gas service, including service interruption notifications. In addition to receiving traditional automated telephone calls, the system will allow you to receive messages via email, text and facsimile. You can register by visiting https://ngrid.myenergysites.com/nonfirm-communication.

6. During each winter season, any Customer who does not interrupt gas usage when required to do so, including during system-wide tests, will be notified that they have violated the requirements of the tariff. Please provide prompt access to National Grid representatives when they come to inspect your equipment.

7. Any failure to interrupt gas usage when required to do so will result in penalties and/or changes in service. Any gas consumed (above the minimum tolerance level established in the tariff) will result in unauthorized use charges as provided for in the Company's TC tariff. In addition, if proof is not provided within ten (10) calendar days that the violation has been remedied, your account will also incur non-compliance charges.
   a. Customers who receive two (2) violations within one (1) winter season (November through March), will have their account transferred, effective no later than the second billing period following the second notice of violation, to the appropriate firm service classification unless (i) the Company has been notified in writing that the Customer has chosen to terminate gas service or (ii) the Customer has been notified that the Company has determined, in its sole discretion that it cannot provide firm service to the Customer and service would be terminated in thirty (30) days. If a customer fails to switch to its alternate fuel during a TC interruption that occurs within 48 hours of a previous interruption, the customer will not earn a second strike.

8. Customers are responsible for the maintenance and repair of their dual-fuel equipment and all associated non-National Grid controls. You are required to manually switch to your alternate fuel if any control device fails to switch the equipment over to the alternate fuel when the outside temperature falls below the designated temperature setting. Please notify the Company at 1-718-643-4050 if you experience any equipment difficulty or alternate fuel issues.

9. Customers must have qualified personnel available to curtail the use of natural gas 24 hours a day, 7 days a week.

If you have any questions regarding your TC service or require a copy of the Company's TC tariff, please contact your account executive or Account Services at 1-718-270-5990.

Sincerely,

National Grid

Dear Customer:

The purpose of this letter is to notify you that on November 14, 2017 National Grid ("Company") will perform a test to confirm your ability to comply with the interruption provisions of the Company's Temperature Controlled ("TC") tariff.

1. The annual gas interruption test is scheduled for November 14, 2017, beginning at 8:00 a.m. For customers with remote switch over capability, National Grid will remotely switch your dual-fuel equipment for up to eight (8) hours. All other customers must remain off natural gas from the time they receive an alarm to switch over from gas until such time they receive an alarm notifying them that they can switch back to natural gas. It is imperative that your dual-fuel system be in good working order and that you have adequate alternate fuel on hand to sustain this interruption. If you have any equipment issues, it is imperative that you contact the Company within one (1) hour of when you or your operating personnel become aware of any equipment issues.

2. Please promptly notify National Grid at 1-800-930-5003 of any condition that may prevent your equipment from switching to an alternate fuel.

National Grid would like to take this opportunity to remind you of the tariff provisions for TC service:

1. PLEASE NOTE: National Grid reserves the right to raise the designated interruption temperature (for example, from 15°F to 20°F or higher) at any time as may be required, in National Grid's sole judgment, to maintain system integrity or for other operational consideration.

2. As a TC customer, you are required to stop burning natural gas when the outdoor temperature reaches your designated interruption temperature. If you have automatic switchover capability, your equipment will switch automatically from natural gas to an alternate fuel when the outdoor temperature falls to your thermostat's designated temperature (15° F). If you do not have automatic switchover capability, you must manually switch your equipment to an alternate fuel when the outdoor temperature falls to the designated temperature for manual equipment, (15° F). In addition, please be aware that the Company will initiate a remote switchover and send an alarm that manual equipment must be switched any time National Grid receives a report that the temperature has reached 15° F at LaGuardia Airport. National Grid may also initiate a remote switchover or send an alarm requesting that you switch your equipment manually in the event of a system reliability or supply issue regardless of temperature. After this remote switchover, when National Grid receives a report that the temperature has reached 20° F, customers with automatic switchover equipment will be switched back to natural gas and all others will receive an alarm indicating that they can manually switch their equipment back to burn natural gas.

3. Your dual-fuel equipment and controls must be in working order. In addition, you must have a minimum ten (10) day supply of alternate fuel on hand. The "on hand" requirement can be satisfied by either storing the full 10 day fuel requirement onsite, or through a combination of onsite storage and a delivery contract with a fuel supplier for the difference. Whenever National Grid becomes aware that a Customer's equipment is incapable of switching over to an alternate fuel and/or that the Customer does not have an adequate supply of alternate fuel on hand, the Company will notify the Customer that the violation must be corrected within ten (10) calendar days. It will be the responsibility of the Customer to contact the Company and provide proof that the violation has been corrected. With regard to an inadequate fuel situation and/or equipment failure, Customers who had not supplied the Company with such proof will be subject to a non-compliance charge as stated in the Company's TC tariff. The non-compliance charge will cease the day the Company receives satisfactory proof that the violation has been corrected.

4. National Grid must be permitted to install, access, inspect and repair remote meter reading and fuel switching devices and associated controls located on your premises as required to assure proper operating and monitoring of your system. It is the Customer's responsibility to avoid damage to or disconnection of the Company's equipment. Any such event must be reported to the Company immediately.

5. National Grid has a communications system that will enable you to receive information regarding your account. This system provides real-time communications to interruptible customers regarding their natural gas service, including service interruptions notifications. In addition to receiving traditional automated telephone calls, the system will allow you to receive messages via email, text and facsimile. You can register by visiting https://ngrid.myenergysites.com/nonfirm-communication.

6. During each winter season, any Customer who does not interrupt gas usage when required to do so, including during system-wide tests, will be notified that they have violated the requirements of the tariff. Please provide prompt access to National Grid representatives when they come to inspect your equipment.

7. Any failure to interrupt gas usage when required to do so will result in penalties and/or changes in service. Any gas consumed (above the minimum tolerance level established in the tariff) will result in unauthorized use charges as provided for in the Company's TC tariff. In addition, if proof is not provided within ten (10) calendar days that the violation has been remedied, your account will also incur non-compliance charges.
    a. Customers who receive two (2) violations within one (1) winter season (November through March), will have their account transferred, effective no later than the second billing period following the second notice of violation, to the appropriate firm service classification unless (i) the Company has been notified in writing that the Customer has chosen to terminate gas service or (ii) the Customer has been notified that the Company has determined, in its sole discretion that it cannot provide firm service to the Customer and service would be terminated in thirty (30) days. If a customer fails to switch to its alternate fuel during a TC interruption that occurs within 48 hours of a previous interruption, the customer will not earn a second strike.

8. Customers are responsible for the maintenance and repair of their dual-fuel equipment and all associated non-National Grid controls. You are required to manually switch to your alternate fuel if any control device fails to switch the equipment over to the alternate fuel when the outside temperature falls below the designated temperature setting. Please notify the Company at 1-800-930-5003 if you experience any equipment difficulty or alternate fuel issues.

9. Customers must have qualified personnel available to curtail the use of natural gas 24 hours a day, 7 days a week.

If you have any questions regarding your TC service or require a copy of the Company's TC tariff, please contact your account executive or Account Services at 1-718-270-5990.

Sincerely,

National Grid

Dear Customer:

This purpose of this letter is to notify you that on November 14, 2017 National Grid ("National Grid" or "Company") will perform a test to confirm your ability to comply with the interruption provisions of the Company's interruptible gas tariff, specifically the tariff's requirement that interruptible customers maintain alternate fuel capabilities that will be available in the event of a system interruption. Please take the time to share this information with all individuals responsible for operating and maintaining the heating equipment at your premises. Please refer to the tariff for additional provisions.

1. National Grid will attempt to notify you of the upcoming test or an interruption by calling the telephone number that is currently on file for your account. If this number is no longer valid, please call us at **1-800-930-5003** so that we can update our records with the correct number.

2. **There is a test scheduled for November 14, 2017 beginning at 8:00 am.** National Grid will call you instructing you to switch over your dual-fuel equipment for up to eight (8) hours. If you have any equipment issues, it is imperative that you contact the Company within one hour of when you or your operating personnel become aware of the equipment issues.

3. Please promptly notify National Grid at 1-800-930-5003 of any condition that may prevent your equipment from switching to an alternate fuel.

Tariff Provisions: National Grid would like to remind you of the tariff provisions for Interruptible ("IT") service under either Service Classification No. 4 (Sales) or Service Classification No. 7 (Transportation).

1. If at any time, the Company needs to call on its peaking supplies to serve firm customers, or if the Company, in its sole discretion, determines that it is necessary to protect the reliability and safety of the system for firm customers, the Company may interrupt service to all customers served under the interruptible service classes. The Company will endeavor to provide notification (within two hours) of this scheduled interruption. A customer that continues to take gas after receiving notice of interruption by the Company may incur additional charges as provided in the interruptible tariff.

2. Your dual-fuel equipment and controls must be in good working order. In addition, you must have a minimum ten (10) day supply of alternate fuel on hand. The "on hand" requirement can be satisfied by either storing the full ten-day fuel requirement onsite, or through a combination of onsite storage and a delivery contract with a fuel supplier for the difference. Whenever National Grid becomes aware that a Customer's equipment is incapable of switching over to alternate fuel and/or that the customer does not have an adequate supply of alternate fuel on hand, the Company shall notify the Customer that the violation must be corrected within ten (10) calendar days. It shall be the responsibility of the Customer to contact the Company and provide proof that the violation has been corrected. With regard to an inadequate fuel situation and/or equipment failure, customers who have not supplied the Company with such proof shall be subject to a variable rate (**non-compliance charge**) that is higher than the monthly posted rate as provided for in the Company's IT tariff. The non-compliance charge will cease the day the Company receives satisfactory proof that the violation has been corrected.

3. National Grid must be permitted to install, access, inspect and repair remote meter reading and fuel switching devices and associated controls located on your premises as required to assure proper operation and monitoring of your system. It is the Customer's responsibility to avoid damage to or disconnection of the Company's equipment. Any such event must be reported to the Company immediately.

4. During each winter season, any Customer who does not interrupt gas usage when required to do so, including during any system-wide tests, will be notified that they have violated the requirements of the tariff. Please provide National Grid representatives with prompt access when they come to inspect your equipment. Although they may arrive at any time and without prior notice, the Company will always try to minimize any inconvenience this may cause.

5. Any failure to interrupt gas usage when required to do so will result in penalties and potentially changes in service. Any gas consumed (above the minimum tolerance level established in the tariff) when there is an interruption will result in unauthorized use charges as provided for in the Company's IT tariff. In addition, if proof is not provided within ten calendar (10) days that the violation has been remedied, your account will also incur a non-compliance charge as mentioned in item (2) above. If a customer fails to switch to its alternate fuel during an interruption that occurs within 48 hours of a previous interruption, the customer will not earn a second strike.

6. Customers who receive two violations within one winter season (November through March) will have their account transferred, effective with their next billing period following the second notice of violation, to the appropriate firm service classification, unless (i) the Company has been notified in writing that the Customer has chosen to terminate gas service or (ii) the Customer has been notified that the Company has determined in its sole discretion that it cannot provide firm service to the Customer and service would be terminated in thirty (30) days.

7. Customers are responsible for the maintenance and repair of their dual-fuel equipment and all associated non-National Grid controls.

8. Customers must have qualified personnel available to cease the use of natural gas 24 hours a day, 7 days a week.

9. National Grid has a communications system that will enable you to receive information regarding your account. This system is intended to enhance real-time communications with interruptible customers regarding their natural gas service, including notifications regarding service interruptions. In addition, to receiving traditional automated telephone calls, this system will allow you to receive messages via different forms of communications, including emails, texts and facsimile. You can register by visiting https://ngrid.myenergysites.com/nonfirm-communication. Please take a few moments to register on the website and fill out the requested information. If you have any questions, please feel free to contact your Account Manager or contact us at 718-270-5990.

If you have any questions, regarding your Interruptible Service or require a copy of the Company's IT tariff, please contact your account executive or Account Services at 718-270-5990.

Sincerely,

National Grid

<insert company header>

<Date>

<Customer Name & Address>

Dear <Mr./Ms. XXXXX>:

On August 24, 2000, the NYS Public Service Commission issued an Order that established the Criteria for Interruptible Gas Service. The Order requires National Grid completion of the following items on a yearly basis:

- Verification a customer's alternative fuel source,
- Establishment of an alternative fuel source as specified under Rule 3 of PSC Tariff No. 219 – Gas for eligible customers, and
- If applicable, performance of an on-site audit to verify compliance with the alternative fuel source requirements.

Customers meeting the following criteria are subject to the alternative fuel source requirements:

- Customer receiving gas service from National Grid under S.C. Nos. 6, 9, or 14, and having an alternative fuel source of either Fuel Oil No. 2, diesel, or kerosene, or

- Customer receiving gas service from National Grid under S.C. Nos. 6, 9, or 14, and providing gas service to a Human Needs facility (as defined in PSC Tariff No. 219).

Customers subject to these requirements that receive gas service from National Grid under S.C. No. 6 are required to have ten (10) days of alternative fuel source requirements. Customers subject to these requirements receiving gas service from National Grid under S.C. Nos. 9 or 14 are required to have five (5) days of alternative fuel source requirements.

National Grid requests that you complete and return the following:

- **Year 2017 Alternative Fuel Source Verification form no later than September 1, 2017**

- **Year 2017 Alternative Fuel Source Requirements form no later than September 1, 2017 (if applicable).**

National Grid has calculated the Required Number of Gallons of Alternative Fuel Source for the winter season 2017-2018 in anticipation that you would be subject to Rule 3 and be required to complete the Year 2009 Alternative Fuel Source Requirements form. I will be contacting you in the fall of 2017 to arrange for an on-site audit to verify compliance with the Alternative Fuel Source Requirements. An on-site audit may be required to verify your Alternative Fuel Requirements.

If it is determined that you are not subject to Rule 3, based on our review of your completed Year 2017 Alternative Fuel Source Verification form, you will be notified that the completion of the **Year 2017 Alternate Fuel Source Requirement** form will not be necessary.

National Grid thanks you for your cooperation and I encourage you to contact me at (XXX) ###-#### should you have any questions.

Sincerely yours,

<Energy Consultant Name>
<Energy Consultant Title>

Encl

Alternative Fuel Source Verification

nationalgrid

National Grid (the Company), pursuant to PSC Tariff No. 219, General Information Rule No. 3, requires interruptible sales or transportation customers to identify (a) your facility's primary alternative fuel source that would allow you to operate your facility in the event that the Company was required to interrupt sales or transportation service to your facility and (b) your facility's total combined estimated load in Therms. You are required to complete the required data fields identified below. A customer's failure to return this form back to the Company by October 1st will result in the Company classifying the customer as utilizing Fuel Oil #2 as the primary alternative fuel source. Please return or fax this form to the address identified below:

National Grid
Manager Gas Rates
300 Erie Boulevard West, B-1
Syracuse, New York 13202
Fax: 315.401.7891

Required Data to be completed by the Customer

Customer Name: _____
Customer Facility: _____
Address (Street / PO Box):_____
State / Zip Code: _____
Indicate the Alternate Fuel Source (Please place an "X" in the appropriate ballot box): _____ Fuel Oil No. 2 _____ Fuel Oil No. 4 _____ Fuel Oil No. 6 _____ Diesel _____ Propane _____ Kerosene _____Other (please specify)_____
Indicate the facility's total combined annual estimate load (natural gas and other alternative fuels in Therms: _____ 0 to 3,000,000 Therms _____3,000,000 to 12,000,000 Therms _____ 12,000,000 to 20,000,000 Therms _____ Greater than 20,000,000 Therms

The Customer, if requested by the Company, shall be prepared to provide documentation acceptable to the Company validating their selection of alternative fuel source.

Customer Name and Title:
Customer Signature: _____
Date: _____

Alternative Fuel Source Requirements

nationalgrid

National Grid (the Company) has determined that your facility identified below meets the Alternative Fuel Source requirements outlined in PSC Tariff No. 219, Rule No. 3.4. You are required to complete the required fields below. Failure to return this form back to the Company by September 1st may subject your facility to the penalty provisions identified within Rule 3.4. Please return or fax this form to: National Grid, Manager Gas Rates, 300 Erie Boulevard West, B-1, Syracuse, New York 13202. Fax: 315.401.7891.

Customer Facility Name:
Customer Facility Address:

National Grid Assigned Alternative Fuel Source Storage Capacity Requirements

The Company has assigned below your facility's alternative fuel source storage capacity requirements. Your facility has the opportunity to contest the alternative fuel source calculation, as defined in Rule 3 of PSC Tariff No. 219, Alternative Fuel Source Requirements, by providing documentation to your National Grid, within ten (10) days of receipt of letter, as to why you believe the alternative fuel source requirement should be changed. The Company has five (5) days to respond, and the Company shall have the sole authority to accept or reject a customer's request for an alternative fuel source value.

Required number of gallons of alternate fuel source	**Gallons**

Required Data to be completed by Customer

Customer's total capacity (in Gallons) for on-site alternative fuel source storage capacity	**Gallons**

Customer Verification of Alternative Fuel Source Requirements: (place an "X" in the appropriate ballot box):

☐ **Option A:** I certify that my facility, prior to November 1 of the current winter heating season, has sufficient alternative fuel source storage capacity and/or has entered into a relationship with an alternative fuel source provider to supply the difference between on-site storage capacity and the Company's determined alternative fuel source requirements specified above and as defined in Rule 3 of the Tariff. If requested by the Company I shall provide documentation to substantiate the storage capacity and/or relationship with an alternative fuel source provider as defined in Rule 3 of the Tariff.

☐ **Option B:** I elect, prior to November 1 of the current winter heating season, to curtail my facility's operations when requested rather than be subject to the requirements outlined in Rule 3.4. Failure to comply completely with an interruption request will subject the customer to penalty charges outlined in Rule 3.4.4. Curtailment of a customer's operation shall be defined to mean that zero gas consumption is recorded on the customer's meter for the applicable interruption period. For customers with both interruptible and firm loads, curtailment of the customer's operations shall be defined to mean that zero gas is consumed for the interruptible portion of the customer's load. The customer must accomplish this zero gas consumption for the duration of the interruption period by curtailing the operation of its facility rather than switching to its primary alternative fuel source.

☐ **Option C:** I certify that my facility's available alternative fuel storage capacity is less than the required alternative fuel storage capacity at full operational levels and I elect, prior to November 1 of the current winter heating season, to curtail my facility's operations to a reduced level to the degree necessary to supplement its primary alternative fuel source so that it can completely interrupt gas consumption for ten (10) full days (or five (5) full days for customers taking transportation service under SC-9 or SC-14) rather than be subject to the requirements outlined in the Rule 3.4. Failure to comply completely with an interruption request will subject the customer to penalty charges outlined in Rule 3.4.4. Curtailment of a customer's operation shall be defined to mean that zero gas consumption is recorded on the customer's meter for the applicable interruption period. The customer must accomplish this zero gas consumption for the duration of the interruption period by reducing/curtailing the operation of its facility for the portion of their requirements that exceeds the customer's available alternative fuel source storage capacity rather than switch to its primary alternative fuel source.

Customer Name:	Customer Title:
Customer Signature:	Date:

Universal NY Replenishment Letter

Date

Customer
Address
City, State, Zip

Re: Alternate Fuel Requirements

Dear:

Currently this winter has resulted in at least five days of natural gas service interruptions. As a requirement under your service classification, you are required to, at all times, maintain dual-fuel equipment and have a sufficient standby alternate fuel supply throughout the heating season in the event that transfer off of natural gas is required.

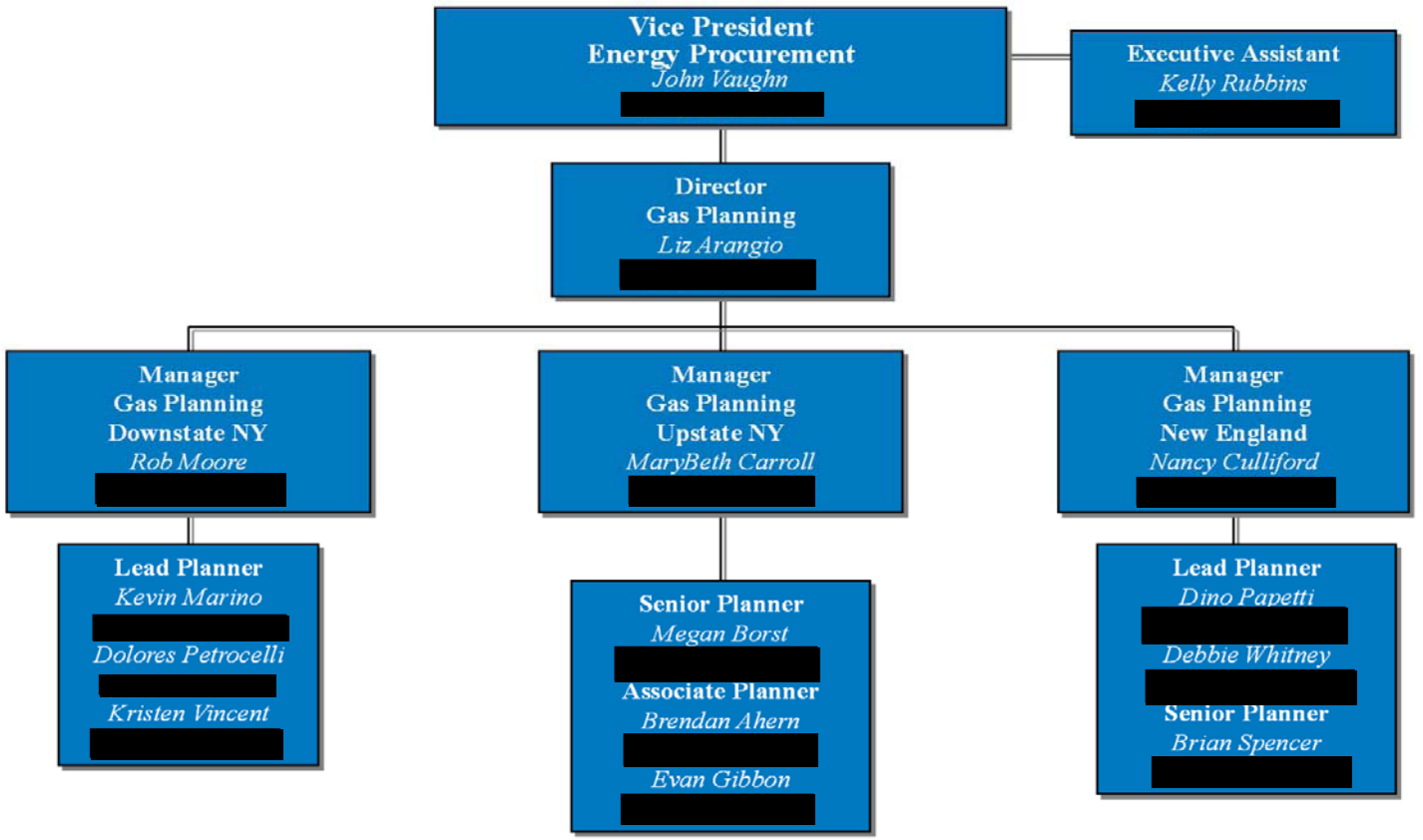
Considering the interruptions that have already occurred, we are asking that you please replenish the amount of your alternate fuel on hand. By doing so, you will be able to avoid any penalties/non-compliance charges should there be further interruptions of service.

If you have any questions, please call (XXX) XXX-XXXX or your account manager.

Sincerely,

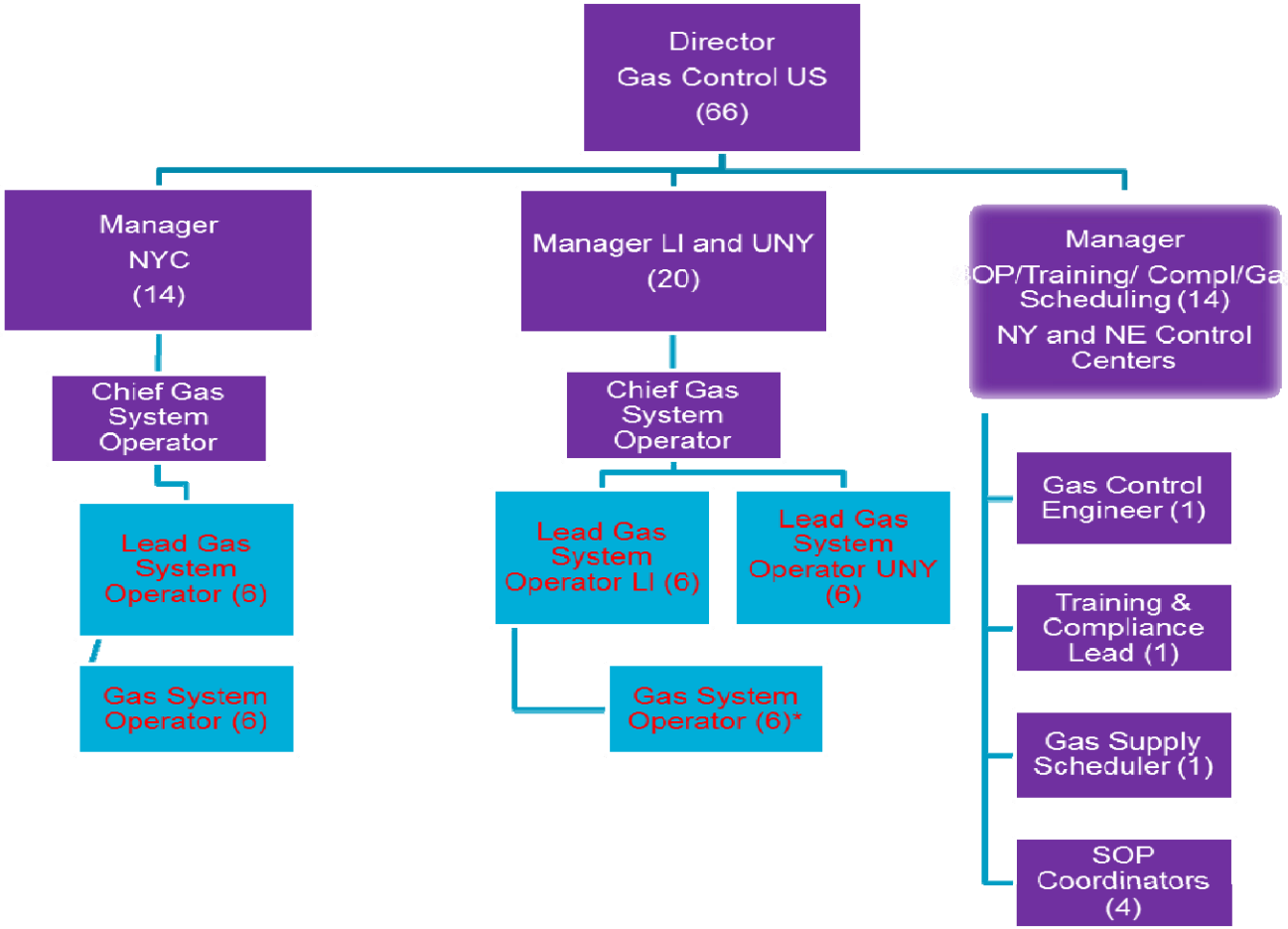
Exhibit 26
Gas Supply Planning Organization Chart

nationalgrid
HERE WITH YOU. HERE FOR YOU.



(Business Confidential)

Exhibit 27
Gas System Control Organzation Chart



Contact Information:
Lead System Operator KEDLI: [redacted]
Lead System Operator KEDNY:
Lead System Operator NIMO:

Exhibit 28
Air Permit Holders

Facility Name	Address	Town	State	Zip	NATLGRID_REGION	NATLGRID_SERVICE_FLAG	cust_stat	Gas Available
3M Tonawanda	305 Sawyer Ave	Tonawanda	NY	14150	UNY	ELEC ONLY	UKN	No
AES SOMERSET LLC	7725 LAKE RD	Barker	NY	14012	UNY	ELEC ONLY	UKN	No
AL TURI LANDFILL & LFGTE FACILITY	73 Hartley RD	Goshen	NY	10924	UNY	ELEC ONLY		No
API AIRTECH	91 NORTH ST	Arcade	NY	14009	UNY	ELEC ONLY	UKN	No
ARCONIC INC MASSENA OPERATIONS	PARK AVE E	Massena	NY	13662	UNY	ELEC ONLY		No
ARKEMA INC	3289 GENESEE ST	Piffard	NY	14533	UNY	ELEC ONLY	UKN	No
AURUBIS BUFFALO INC	70 SAYRE ST	Buffalo	NY	14207	UNY	ELEC ONLY	UKN	No
BATAVIA POWER PLANT	163 CEDAR ST	Batavia	NY	14020	UNY	ELEC ONLY	UKN	No
BEAVER FALLS LLC	9644 MAIN ST	Beaver Falls	NY	13305	UNY	ELEC ONLY		No
BEECH HILL COMPRESSOR STATION	PEET RD BETWEEN GRAVES & BEECH HILL RDS	Wellsville	NY	14895	UNY	ELEC ONLY	UKN	No
BIRD ISLAND STP	90 WEST FERRY ST	Buffalo	NY	14213	UNY	ELEC ONLY	UKN	No
BUFFALO TERMINAL	625 ELK ST	Buffalo	NY	14210	UNY	ELEC ONLY	UKN	No
CELLO-PACK CORPORATION OF BUFFALO	55 INNSBRUCK DR	Buffalo	NY	14227	UNY	ELEC ONLY	UKN	No
CFSWMA REGIONAL SOLID WASTE DISPOSAL FAC	828 CO RTE 20	Constable	NY	12926	UNY	ELEC ONLY	UKN	No
CHAFFEE LANDFILL	10860 Olean Rd	Chaffee	NY	14030-9799	UNY	ELEC ONLY		No
CHEMOURS NIAGARA	BUFFALO AVE & 26TH ST	Niagara Falls	NY	14302-0787	UNY	ELEC ONLY		No
CHEMUNG COUNTY LANDFILL	1488 CO RTE 60	Elmira	NY	14901	UNY	ELEC ONLY		No
COMMODORE TECHNOLOGY/PLASTICS LLC	26 MAPLE AVE	Bloomfield	NY	14469	UNY	ELEC ONLY	UKN	No
COMMONWEALTH PLYWOOD INC	10068 US RTE 4	Whitehall	NY	12887	UNY	ELEC ONLY	UKN	No
COMPRESSOR STATION 249	2840 US RTE 20	Carlisle	NY	12031	UNY	ELEC ONLY	UKN	No
COMPRESSOR STATION 254	ST RTE 66 - E SIDE - S OF COUNTY LINE	Nassau	NY	12123	UNY	ELEC ONLY	UKN	No
CONCORD COMPRESSOR STATION	5510 GENESEE RD	Springville	NY	14141	UNY	ELEC ONLY	UKN	No
CORNING DIESEL MANUFACTURING FACILITY	890 ADDISON RD (ST RTE 417)	Painted Post	NY	14870	UNY	ELEC ONLY	UKN	No
CORNING INC CANTON PLANT	MCADOO RD|334 CO RTE 16	Canton	NY	13617	UNY	ELEC ONLY	UKN	No
COVANTA NIAGARA LP	100 ENERGY BLVD AT 56TH ST	Niagara Falls	NY	14304	UNY	ELEC ONLY	UKN	No
CRICKET VALLEY ENERGY CENTER	2241 ST RTE 22	Dover Plains	NY	12522	UNY	ELEC ONLY		No
CUMMINS INC - JAMESTOWN ENGINE PLANT	4720 BAKER ST EXT	Lakewood	NY	14750	UNY	ELEC ONLY	UKN	No
DANC SOLID WASTE MANAGEMENT FACILITY	23400 ST RTE 177	Rodman	NY	13682	UNY	ELEC ONLY	UKN	No
DUNKIRK STEAM GENERATING STATION	106 POINT DR NORTH	Dunkirk	NY	14048	UNY	ELEC ONLY	UKN	No
DUPONT COMPANY	BUFFALO AVE & 26TH ST	Niagara Falls	NY	14302	UNY	ELEC ONLY	UKN	No
Durez Niagara	5000 Packard Rd	Niagara Falls	NY	14304	UNY	ELEC ONLY		No
E I DUPONT YERKES PLANT	3115 RIVER RD	Buffalo	NY	14207	UNY	ELEC ONLY		No
ENGINEERED COMPOSITES INC	55 ROBERTS AVE	Buffalo	NY	14206	UNY	ELEC ONLY		No
ERWIN MANUFACTURING COMPLEX	792 ADDISON RD - S HAMILTON RD	Painted Post	NY	14870	UNY	ELEC ONLY	UKN	No
FCINA - DELHI	40196 ST HWY 10	Delhi	NY	13753	UNY	ELEC ONLY		No
FLEXO-TRANSPARENT INC	28 WASSON ST	Buffalo	NY	14210	UNY	ELEC ONLY	UKN	No
FORTISTAR NORTH TONAWANDA INC	1070 ERIE AVE	North Tonawanda	NY	14120	UNY	ELEC ONLY	UKN	No
FRIESLAND CAMPINA DOMO	40196 ST HWY 10	Delhi	NY	13753	UNY	ELEC ONLY	UKN	No
GENERAL MILLS OPERATIONS LLC	54 S MICHIGAN AVE	Buffalo	NY	14203	UNY	ELEC ONLY	UKN	No
GENERAL MILLS OPERATIONS LLC	54 S MICHIGAN AVE	Buffalo	NY	14203	UNY	ELEC ONLY	UKN	No
GENERAL MILLS OPERATIONS LLC	54 S MICHIGAN AVE	Buffalo	NY	14203	UNY	ELEC ONLY	UKN	No
GENPAK LLC MIDDLETOWN MAIN PLANT	26 REPUBLIC PLZ	Middletown	NY	10940	UNY	ELEC ONLY		No
GLOBAL COMPANIES - CARGO TERMINAL	1096 RIVER RD	New Windsor	NY	12553	UNY	ELEC ONLY		No
GLOBAL COMPANIES - NORTH TERMINAL	1254 RIVER RD|HUDSON RIVER	New Windsor	NY	12553	UNY	ELEC ONLY		No
GLOBAL COMPANIES - SOUTH TERMINAL	1184 RIVER RD	New Windsor	NY	12553	UNY	ELEC ONLY		No
GLOBAL COMPANIES LLC - NEWBURGH TERMINAL	1281 RIVER RD	New Windsor	NY	12553	UNY	ELEC ONLY		No
GLOBE METALLURGICAL INC	3807 HIGHLAND AVE	Niagara Falls	NY	14305	UNY	ELEC ONLY	UKN	No
GOOD SAMARITAN HOSPITAL	255 LAFAYETTE AVE	Suffern	NY	10901	UNY	ELEC ONLY		No
GOODYEAR DUNLOP TIRES NORTH AMERICA LTD	3333 RIVER RD|10 SHERIDAN DR	Tonawanda	NY	14150	UNY	ELEC ONLY	UKN	No
HAMMER PACKAGING CORP	200 LUCIUS GORDON DR	West Henrietta	NY	14586	UNY	ELEC ONLY	UKN	No
HARDEN FURNITURE INC	8550 MILL POND WAY	Mc Connellsville	NY	13401	UNY	ELEC ONLY	UKN	No
HARDEN FURNITURE, LLC	8550 MILL POND WAY	McConnellsville	NY	13401	UNY	ELEC ONLY		No
HUNTLEY STEAM GENERATING STATION	3500 RIVER RD	Tonawanda	NY	14150	UNY	ELEC ONLY	UKN	No
INDECK OLEAN ENERGY CENTER	140 MOORE AVE	Olean	NY	14760	UNY	ELEC ONLY	UKN	No
INDECK-YERKES ENERGY SERVICES	1 SHERIDAN DRIVE	Tonawanda	NY	14150	UNY	ELEC ONLY	UKN	No
INDEPENDENCE STATION	CO RTE 22 BETW FULMER VALLEY RD & CO RTE 22A	Andover	NY	14806	UNY	ELEC ONLY	UKN	No
INNOVATIVE DANC	23400 ST RTE 177	Rodman	NY	13682	UNY	ELEC ONLY	UKN	No
INTERNATIONAL IMAGING	310 COMMERCE DR	Buffalo	NY	14228	UNY	ELEC ONLY	UKN	No
INTERNATIONAL PAPER TICONDEROGA MILL	568 SHORE AIRPORT RD	Ticonderoga	NY	12883	UNY	ELEC ONLY	UKN	No
KEYWELL LLC	300 FALCONER STREET	Frewsburg	NY	14738	UNY	ELEC ONLY	UKN	No
LACKAWANNA PLANT- REPUBLIC STEEL	3049 LAKESHORE RD	Buffalo	NY	14219	UNY	ELEC ONLY	UKN	No
LAIDLAW ENERGY AND ENVIRONMENTAL INC	6662 ROUTE 219	Ellicottville	NY	14731	UNY	ELEC ONLY	UKN	No
LAKESIDE BEAVER FALLS LLC	9644 MAIN ST	Beaver Falls	NY	13305	UNY	ELEC ONLY	UKN	No
LCOA MASSENA OPERATIONS (WEST PLANT)	PARK AVE E	Massena	NY	13662	UNY	ELEC ONLY	UKN	No
LEHIGH NORTHEAST CEMENT CO - CEMENTON TERMINAL	120 ALPHA BLVD US RTE 9W	Cementon	NY	12415	UNY	ELEC ONLY		No
LOCKPORT COGENERATION FACILITY	5087 JUNCTION RD	Lockport	NY	14094	UNY	ELEC ONLY	UKN	No
LYONSDALE BIOMASS LLC	MARMON LOWDALE & LYONSDALE RDS	Port Leyden	NY	13433	UNY	ELEC ONLY	UKN	No
MANCHESTER WOOD INC	NORTH ST	Granville	NY	12832	UNY	ELEC ONLY	UKN	No
MASSENA ENERGY FACILITY	ALCOA POWER CANAL RD|N MAIN ST GATE	Massena	NY	13662	UNY	ELEC ONLY	UKN	No
METAL CONTAINER CORP	130 BRUENIG RD	New Windsor	NY	12553	UNY	ELEC ONLY		No
METAULLICS SYSTEMS DIVISION OF PYROTEK INC	2040 CORY RD	Sanborn	NY	14132	UNY	ELEC ONLY	UKN	No
MILL SEAT LANDFILL	303 BREW RD	Bergen	NY	14416	UNY	ELEC ONLY		No
MILL SERVICES INC	128 MACARTHUR AVE	Cobleskill	NY	12043	UNY	ELEC ONLY	UKN	No
MODEL CITY ENERGY FACILITY	1447 PLETCHER RD	Youngstown	NY	14174	UNY	ELEC ONLY	UKN	No
MODERN LANDFILL INC	PLETCHER RD	Model City	NY	14107	UNY	ELEC ONLY	UKN	No
MONROE LIVINGSTON SANITARY LANDFILL	1241 SOUTH RD	Scottsville	NY	14546	UNY	ELEC ONLY	UKN	No
NASHVILLE COMPRESSOR STATION	11413 ALLEGANY ROAD	Forestville	NY	14062	UNY	ELEC ONLY	UKN	No
NEENAH NORTHEAST - LOWVILLE	5492 BOSTWICK ST	Lowville	NY	13367	UNY	ELEC ONLY		No
NEW BATH LANDFILL	TURNPIKE RD	Bath	NY	14810	UNY	ELEC ONLY		No
NIAGARA GENERATING FACILITY	5300 FRONTIER AVE	Niagara Falls	NY	14304	UNY	ELEC ONLY	UKN	No
NOCO ENERGY CORP	700 GRAND ISLAND BLVD	Tonawanda	NY	14150	UNY	ELEC ONLY	UKN	No

Facility Name	Address	Town	State	Zip	NATLGRID_REGION	NATLGRID_SERVICE_FLAG	cust_stat	Gas Available
NOCO ENERGY CORP - TONAWANDA	700 GRAND ISLAND BLVD	Tonawanda	NY	14150	UNY	ELEC ONLY		No
NORTH ALBANY TERMINAL CO - GLENMONT TERMINAL	552 RIVER RD	Glenmont	NY	12077	UNY	ELEC ONLY		No
OCCIDENTAL CHEMICAL CORP - NIAGARA PLANT	BUFFALO AVE AND 47TH ST	Niagara Falls	NY	14302	UNY	ELEC ONLY	UKN	No
OCCIDENTAL CHEMICAL CORP - NIAGARA PLANT	BUFFALO AVE AND 47TH ST	Niagara Falls	NY	14302	UNY	ELEC ONLY	UKN	No
OGDENSBURG ENERGY FACILITY	21 ENTRANCE AVE	Ogdensburg	NY	13669	UNY	ELEC ONLY	UKN	No
PRESTOLITE ELECTRIC NY INC	400 MAIN ST	Arcade	NY	14009	UNY	ELEC ONLY	UKN	No
PVS CHEMICAL SOLUTIONS INC	55 LEE ST	Buffalo	NY	14210	UNY	ELEC ONLY		No
PVS CHEMICALS	55 LEE ST	Buffalo	NY	14210	UNY	ELEC ONLY	UKN	No
REENERGY BLACK RIVER	EUPHRATES RIVER VALLEY RD & ONEIDA AVE	Fort Drum	NY	13602	UNY	ELEC ONLY		No
REENERGY BLACK RIVER LLC	EUPHRATES RIVER VALLEY RD & ONEIDA AVE	Fort Drum	NY	13602	UNY	ELEC ONLY	UKN	No
REYNOLDS METALS ST LAWRENCE REDUCTION PLANT	194 CO RTE 45	Massena	NY	13662	UNY	ELEC ONLY	UKN	No
RG&E ALLEGANY STATION #133	11537 ST RTE 19A	Fillmore	NY	14735	UNY	ELEC ONLY	UKN	No
RIGA/MILL SEAT LANDFILL	303 BREW RD	Bergen	NY	14416	UNY	ELEC ONLY	UKN	No
SAINT GOBAIN PERFORMANCE PLASTICS	1 LIBERTY ST	Hoosick Falls	NY	12090	UNY	ELEC ONLY	UKN	No
SAINT-GOBAIN ADFORS AMERICA INC	14770 EAST AVE	Albion	NY	14411	UNY	ELEC ONLY	UKN	No
SAMUEL A CARLSON GENERATING STATION	136 STEELE ST	Jamestown	NY	14701	UNY	ELEC ONLY	UKN	No
SARANAC POWER PARTNERS COGENERATION FAC	99 WEED ST EXT	Plattsburgh	NY	12901	UNY	ELEC ONLY		No
SENECA FOODS MARION PLANT	3709 MILL ST	Marion	NY	14505	UNY	ELEC ONLY		No
SENECA MEADOWS SWMF	1786 SALCMAN RD	Waterloo	NY	13165	UNY	ELEC ONLY		No
SILGAN CONTAINERS CORP - LYONS PLANT	8673 LYONS MARENGO RD	Lyons	NY	14489-9726	UNY	ELEC ONLY		No
SKF AEROENGINE NORTH AMERICA	ONE MAROCO RD|PO BOX 263	Falconer	NY	14733	UNY	ELEC ONLY	UKN	No
SOMERSET OPERATING COMPANY LLC	7725 LAKE RD	Barker	NY	14012	UNY	ELEC ONLY		No
SOUTH OSWEGO TERMINAL	RIDGE RD & CO RTE 20	Oswego	NY	13126	UNY	ELEC ONLY		No
SUMITOMO RUBBER USA LLC	10 Sheridan Dr	Tonawanda	NY	14150	UNY	ELEC ONLY		No
SUNOCO TONAWANDA TERMINAL	3733 RIVER RD	Tonawanda	NY	14150	UNY	ELEC ONLY	UKN	No
TAM CERAMICS LLC	4511 HYDE PARK BLVD	Niagara Falls	NY	14305	UNY	ELEC ONLY	UKN	No
TENNESSEE GAS PIPELINE CO COMPRESSOR STATION 224	9766 RAVLIN HILL RD	Clymer	NY	14724	UNY	ELEC ONLY	UKN	No
TENNESSEE GAS PIPELINE CO COMPRESSOR STATION 229	7586 E EDEN RD	Eden	NY	14057	UNY	ELEC ONLY	UKN	No
TENNESSEE GAS PIPELINE CO COMPRESSOR STATION 230-C	5186 LOCKPORT JUNCTION RD	Lockport	NY	14094	UNY	ELEC ONLY	UKN	No
TGP COMPRESSOR STATION 233	2251 DOW RD	Piffard	NY	14533	UNY	ELEC ONLY	UKN	No
TGP COMPRESSOR STATION 237	2001 ARCHER RD	Clifton Springs	NY	14432-9349	UNY	ELEC ONLY		No
TGP COMPRESSOR STATION 245	457 BURROWS RD	West Winfield	NY	13491	UNY	ELEC ONLY	UKN	No
THOMAS J WATSON RESEARCH CENTER	1101 KITCHAWAN RD/ST RTE 134	Yorktown Heights	NY	10598	UNY	ELEC ONLY		No
TONAWANDA COKE CORP	3875 RIVER RD	Tonawanda	NY	14150	UNY	ELEC ONLY	UKN	No
TONAWANDA ENGINE PLANT-GENERAL MOTORS POWERTRAIN	2995 RIVER RD	Buffalo	NY	14207-1099	UNY	ELEC ONLY		No
TONAWANDA POWERTRAIN ENGINE PLANT	2995 RIVER RD	Buffalo	NY	14207	UNY	ELEC ONLY	UKN	No
TRINITY PACKAGING - A DIVISION OF PROAMPAC	55 INNSBRUCK DR	Cheektowaga	NY	14227	UNY	ELEC ONLY		No
U S GYPSUM CO OAKFIELD PLANT	2750 MAPLE AVE	Oakfield	NY	14125	UNY	ELEC ONLY	UKN	No
UNICELL BODY COMPANY INC	575 HOWARD ST	Buffalo	NY	14206	UNY	ELEC ONLY	UKN	No
VANDEMARK CHEMICAL INC	1 NORTH TRANSIT RD	Lockport	NY	14094	UNY	ELEC ONLY	UKN	No
WASHINGTON MILLS ELECTRO MINERALS	1801 BUFFALO AVE	Niagara Falls	NY	14302	UNY	ELEC ONLY	UKN	No
WHITING DOOR MANUFACTURING CORP	113 CEDAR ST	Akron	NY	14001	UNY	ELEC ONLY	UKN	No
CHAUTAUQUA COUNTY LANDFILL	3889 TOWERVILLE RD	JAMESTOWN	NY	14701	UNY	ELEC ONLY		No
CENEX TERMINALS	54 RIVERSIDE AVE	Rensselaer	NY	12144	UNY	GAS & ELEC		Yes
CRUCIBLE INDUSTRIES	575 STATE FAIR BLVD	Geddes	NY	13209	UNY	GAS & ELEC		Yes
GENERAL ELECTRIC STEAM TURBINE GENERATOR GLOBAL	1 RIVER RD	Schenectady	NY	12345	UNY	GAS & ELEC	UKN	Yes
GENERAL ELECTRIC STEAM TURBINE GENERATOR GLOBAL	1 RIVER RD	Schenectady	NY	12345	UNY	GAS & ELEC	UKN	Yes
GENERAL ELECTRIC STEAM TURBINE GENERATOR GLOBAL	1 RIVER RD	Schenectady	NY	12345	UNY	GAS & ELEC	UKN	Yes
INNOVATIVE / COLONIE	16 ARROWHEAD LN	Cohoes	NY	12047	UNY	GAS & ELEC		Yes
ALBANY LANDFILL	525 RAPP RD	Albany	NY	12205	UNY	GAS & ELEC	UKN	Yes
ALBANY LANDFILL GAS TO ENERGY FACILITY	RAPP RD	Albany	NY	12205	UNY	GAS & ELEC	UKN	Yes
ALCOA USA CORP	PARK AVE E	Massena	NY	13662	UNY	GAS & ELEC		Yes
ALGONQUIN GAS SOUTHEAST COMPRESSOR STATION	142 Tulip Rd	Brewster	NY	10509	UNY	GAS & ELEC		Yes
ALGONQUIN GAS: STONY POINT COMPRESSOR STA	Lindberg Rd	Stony Point	NY	10980	UNY	GAS & ELEC		Yes
ANHEUSER BUSCH BALDWINSVILLE BREWERY	2885 BELGIUM RD	Baldwinsville	NY	13027	UNY	GAS & ELEC	UNY	Yes
AUBURN LANDFILL GAS ELECTRIC GENERATORS FACILITY	43 ALLEN ST	Auburn	NY	13021	UNY	GAS & ELEC	UKN	Yes
AUBURN SANITARY LANDFILL NO 2	311 N DIVISION ST	Auburn	NY	13021	UNY	GAS & ELEC	UKN	Yes
AVA LANDFILL	7044 ST RTE 294	Boonville	NY	13309	UNY	GAS & ELEC	UKN	Yes
BALL METAL BEVERAGE CONTAINER CORP	11 ADAMS RD|CADY HILL INDUSTRIAL PARK	Saratoga Springs	NY	12866	UNY	GAS & ELEC	UKN	Yes
BETHLEHEM ENERGY CENTER	380 RIVER RD	Glenmont	NY	12077	UNY	GAS & ELEC	UKN	Yes
BINGHAMTON CO-GENERATION PLANT	22 Charles St	Binghamton	NY	13905	UNY	GAS & ELEC		Yes
BRISTOL-MYERS SQUIBB COMPANY	6000 THOMPSON RD	Syracuse	NY	13211	UNY	GAS & ELEC	UKN	Yes
BUCKEYE ALBANY TERMINAL LLC	301 NORMANSKILL ST|PORT OF ALBANY	Albany	NY	12202	UNY	GAS & ELEC	UNY	Yes
BUCKEYE PIPE LINE CO	3186 GATES RD	Auburn	NY	13021	UNY	GAS & ELEC	UKN	Yes
BUCKEYE PIPE LINE CO - AUBURN	3186 GATES RD	Auburn	NY	13021	UNY	GAS & ELEC		Yes
BUCKEYE ROCHESTER SOUTH TERMINAL	675 Brooks Ave	Rochester	NY	14619	UNY	GAS & ELEC		Yes
CARESTREAM HEALTH AT EASTMAN BUSINESS PARK	1049 W Ridge Rd	Rochester	NY	14619	UNY	GAS & ELEC		Yes
CARR STREET GENERATING STATION	64 CARR ST	East Syracuse	NY	13057	UNY	GAS & ELEC	UNY	Yes
CARTHAGE ENERGY COGEN FACILITY	701 WESTEND AVE	Carthage	NY	13619	UNY	GAS & ELEC	UNY	Yes
CASTLETON ENERGY CENTER	1902 RIVER RD|ST RTE 9J	1902 RIVER RD|ST RTE 9J	NY	12033	UNY	GAS & ELEC		Yes
CHEMPRENE INC	483 Fishkill Ave	Beacon	NY	12508-1200	UNY	GAS & ELEC		Yes
CITGO PETROLEUM GLENMONT TERMINAL	495 RIVER RD	Glenmont	NY	12077	UNY	GAS & ELEC	UKN	Yes
CLINTON COUNTY REGIONAL LANDFILL	286 Sand Rd	Morrisonville	NY	12962	UNY	GAS & ELEC		Yes
CORNELL UNIVERSITY CENTRAL ENERGY PLANT	COLLEGE AVE	Ithaca	NY	14853	UNY	GAS & ELEC		Yes
DANSKAMMER GENERATING STATION	994 RIVER RD	Newburgh	NY	12550	UNY	GAS & ELEC		Yes
DOMINION TRANSMISSION INC - UTICA STATION	HIGBY RD	Frankfort	NY	13340	UNY	GAS & ELEC	UKN	Yes
DUTCHESS CO RESOURCE RECOVERY FACILITY	96 SAND DOCK RD	Poughkeepsie	NY	12601	UNY	GAS & ELEC		Yes
EASTMAN BUSINESS PARK	1999 LAKE AVE	Rochester	NY	14619	UNY	GAS & ELEC		Yes
EMPIRE POWER PLANT	75 RIVERSIDE AVE	Rensselaer	NY	12144	UNY	GAS & ELEC		Yes
EPCOR POWER CASTLETON	1902 RIVER RD|ST RTE 9J	Castleton On Hudson	NY	12033	UNY	GAS & ELEC	UNY	Yes
FAB 8	400 STONE BREAK ROAD	Ballston Spa	NY	12020	UNY	GAS & ELEC	UNY	Yes
FAB 8	400 STONE BREAK ROAD	Ballston Spa	NY	12020	UNY	GAS & ELEC	UNY	Yes
FIBERMARK NORTH AMERICA INC	5492 BOSTWICK ST	Lowville	NY	13367	UNY	GAS & ELEC	UKN	Yes
FINCH PAPER LLC	1 GLEN ST	Glens Falls	NY	12801	UNY	GAS & ELEC	UKN	Yes
FULTON COUNTY MUD RD SANITARY LANDFILL	847 MUD RD (ST RTE 67)	Johnstown	NY	12095	UNY	GAS & ELEC	UKN	Yes
GE GLOBAL RESEARCH CENTER	1 RESEARCH CIR	Schenectady	NY	12309	UNY	GAS & ELEC	UKN	Yes

Facility Name	Address	Town	State	Zip	NATLGRID_REGION	NATLGRID_SERVICE_FLAG	cust_stat	Gas Available
GENERAL CHEMICAL LLC	1421 WILLIS AVE	Syracuse	NY	13204	UNY	GAS & ELEC	UNY	Yes
GLOBAL COMPANIES LLC - ALBANY TERMINAL	50 CHURCH ST - PORT OF ALBANY	Albany	NY	12202	UNY	GAS & ELEC	UNY	Yes
GREEN RIDGE RDF	PETERS RD & KOBOR RD	Gansevoort	NY	12831	UNY	GAS & ELEC		Yes
GREENIDGE STATION	590 PLANT RD	Dresden	NY	14441	UNY	GAS & ELEC		Yes
GUARDIAN INDUSTRIES CORP	50 FORGE AVE	Geneva	NY	14456	UNY	GAS & ELEC		Yes
GUNLOCKE CO	1 GUNLOCKE DR	Wayland	NY	14572	UNY	GAS & ELEC		Yes
HESS/SUNOCO SYRACUSE TED PARK	2951 ENERGY DR	Warners	NY	13164	UNY	GAS & ELEC	UKN	Yes
HILLBURN SUBSTATION	6 4TH ST	Hillburn	NY	10931	UNY	GAS & ELEC		Yes
HOLLINGSWORTH & VOSE GREENWICH MILL	2322 ST RTE 29	Greenwich	NY	12834	UNY	GAS & ELEC	UKN	Yes
HOLLINGSWORTH & VOSE-EASTON MILL	3235 CO RTE 113	Greenwich	NY	12834	UNY	GAS & ELEC	UKN	Yes
HURON CAMPUS	1701 NORTH ST - BLDG 39	Endicott	NY	13760	UNY	GAS & ELEC		Yes
HYLAND LANDFILL	6653 HERDMAN RD	Angelica	NY	14709	UNY	GAS & ELEC		Yes
i3 Electronics Huron Campus Industrial Waste Treatment Plant	1093 Clarks St	Endicott	NY	13760	UNY	GAS & ELEC		Yes
INDECK-CORINTH ENERGY CENTER	24 WHITE ST	Corinth	NY	12822	UNY	GAS & ELEC	UNY	Yes
INDECK-OSWEGO ENERGY CENTER	105 MITCHELL ST	Oswego	NY	13126	UNY	GAS & ELEC		Yes
INDECK-OSWEGO LIMITED PARTNERSHIP	105 MITCHELL ST	Oswego	NY	13126	UNY	GAS & ELEC	UNY	Yes
INDECK-SILVER SPRINGS ENERGY CENTER	1 INDECK DR	Silver Springs	NY	14550	UNY	GAS & ELEC		Yes
INDEPENDENCE STATION	76 INDEPENDENCE WAY	Oswego	NY	13126	UNY	GAS & ELEC	UKN	Yes
INNOVATIVE / FULTON	847 MUD RD	Johnstown	NY	12095	UNY	GAS & ELEC	UKN	Yes
INNOVATIVE/COLONIE	1319 NEW LOUDON RD	Cohoes	NY	12047	UNY	GAS & ELEC	UKN	Yes
INTERFACE PERFORMANCE MATERIALS	2885 ST RTE 481	Fulton	NY	13069	UNY	GAS & ELEC		Yes
INTERFACE SOLUTIONS INC	2885 ST RTE 481	Fulton	NY	13069	UNY	GAS & ELEC	UNY	Yes
INTERFACE SOLUTIONS INC	2885 ST RTE 481	Fulton	NY	13069	UNY	GAS & ELEC	UNY	Yes
KENNEDY VALVE DIV MC WANE INC	1021 E WATER ST	Elmira	NY	14901	UNY	GAS & ELEC		Yes
KERRY	158 ST RTE 320	Norwich	NY	13815	UNY	GAS & ELEC		Yes
KEYMARK CORP PLANT	1188 CAYADUTTA ST	Fonda	NY	12068	UNY	GAS & ELEC	UKN	Yes
KNOWLTON TECHNOLOGIES LLC	213 FACTORY ST	Watertown	NY	13601	UNY	GAS & ELEC	UNY	Yes
L & J G STICKLEY INC	1 STICKLEY DR	Manlius	NY	13104	UNY	GAS & ELEC	UNY	Yes
LAFARGE BUILDING MATERIALS INC	1916 US RTE 9W	Ravena	NY	12143	UNY	GAS & ELEC	UKN	Yes
LAFARGE RAVENA PLANT	1916 US RTE 9W	Ravena	NY	12143	UNY	GAS & ELEC		Yes
LEHIGH NORTHEAST CEMENT COMPANY	313 WARREN ST	Glens Falls	NY	12801	UNY	GAS & ELEC	UNY	Yes
MADISON COUNTY LANDFILL	BUYEA RD - W SIDE - N OF EDDY RD	Canastota	NY	13032	UNY	GAS & ELEC	UKN	Yes
MOMENTIVE PERFORMANCE MATERIALS	260 HUDSON RIVER RD	Waterford	NY	12188	UNY	GAS & ELEC	UNY	Yes
MOOG INC	SENECA ST, JAMISON RD & CONLEY RD	Elma	NY	14059-0018	UNY	GAS & ELEC		Yes
NORLITE CORP	628 S SARATOGA ST	Cohoes	NY	12047	UNY	GAS & ELEC	UNY	Yes
NORLITE LLC	628 S SARATOGA ST	Cohoes	NY	12047	UNY	GAS & ELEC		Yes
NORTHEAST SOLITE CORPORATION	962 OLD KINGS HWY	MT Marion	NY	12456	UNY	GAS & ELEC		Yes
NOVELIS CORPORATION	448 CO RTE 1A	Oswego	NY	13126	UNY	GAS & ELEC	UKN	Yes
NOVELIS CORPORATION	448 CO RTE 1A	Oswego	NY	13126	UNY	GAS & ELEC	UKN	Yes
NUCOR STEEL AUBURN INC	25 QUARRY ROAD	Auburn	NY	13021	UNY	GAS & ELEC	UKN	Yes
NYS OGS SHERIDAN STEAM PLANT	79 SHERIDAN AVE	Albany	NY	12210	UNY	GAS & ELEC	UNY	Yes
ONONDAGA CO RESOURCE RECOVERY FACILITY	5801 ROCK CUT RD	Jamesville	NY	13078	UNY	GAS & ELEC	UNY	Yes
ORGANICHEM CORP	33 RIVERSIDE AVE	Rensselaer	NY	12144	UNY	GAS & ELEC	UNY	Yes
OSWEGO CO ENERGY RECOVERY FAC	2801 ST RTE 481	Fulton	NY	13069	UNY	GAS & ELEC	UNY	Yes
OSWEGO HARBOR POWER	261 WASHINGTON BLVD	Oswego	NY	13126	UNY	GAS & ELEC	UNY	Yes
OWENS-BROCKWAY GLASS CONTAINER INC	7134 COUNTY HOUSE RD	Auburn	NY	13021	UNY	GAS & ELEC	UKN	Yes
OWENS-CORNING INSULATING SYSTEMS- FEURA BUSH	1277 FEURA BUSH RD	Feura Bush	NY	12067	UNY	GAS & ELEC	UKN	Yes
PACTIV LLC	5250 NORTH ST	Canadaigua	NY	14424-1095	UNY	GAS & ELEC		Yes
PACTIV TECHNOLOGY CENTER	2651 BRICKYARD RD	Canadaigua	NY	14424	UNY	GAS & ELEC		Yes
PALL TRINITY MICRO	3643 ST RTE 281 - NW CORNER @ MCLEAN RD	Cortland	NY	13045	UNY	GAS & ELEC	UKN	Yes
PAR PHARMACEUTICAL INC	1 RAM RIDGE RD	Chestnut Ridge	NY	10977	UNY	GAS & ELEC		Yes
PEARL LEATHER FINISHERS INC	11-21 INDUSTRIAL PARK	Johnstown	NY	12095	UNY	GAS & ELEC	UKN	Yes
PEARL RIVER CAMPUS LLC	401 N MIDDLETOWN RD	Pearl River	NY	10965-1299	UNY	GAS & ELEC		Yes
PETROLEUM FUEL & TERMINAL CO-RENSSELAER	54 RIVERSIDE AVE	Rensselaer	NY	12144	UNY	GAS & ELEC	UKN	Yes
PLIANT LLC (DBA BERRY PLASTICS CORPORATION)	200 EAST MAIN ST	Macedon	NY	14502	UNY	GAS & ELEC		Yes
QUADGRAPHICS	56 DUPLAINVILLE RD|GRANDE INDUSTRIAL PARK	Saratoga Springs	NY	12866	UNY	GAS & ELEC	UKN	Yes
RED-ROCHESTER LLC AT EASTMAN BUSINESS PARK	EASTMAN BUSINESS PARK UTILITIES WATER, POWER, WASTEWATER|1669 LAKE AVE	Rochester	NY	14619	UNY	GAS & ELEC		Yes
RENSSELAER COGEN FACILITY	39 RIVERSIDE AVE	Rensselaer	NY	12144	UNY	GAS & ELEC	UNY	Yes
REVERE SMELTING & REFINING CORP	65 BALLARD RD	Middletown	NY	10941	UNY	GAS & ELEC		Yes
ROCKLAND PSYCHIATRIC CENTER	140 OLD ORANGEBURG RD	Orangeburg	NY	10962	UNY	GAS & ELEC		Yes
ROSETON GENERATING STATION	992 RIVER RD	Newburgh	NY	12550	UNY	GAS & ELEC		Yes
SABIC INNOVATIVE PLASTICS US LLC	1 NORYL AVE	Selkirk	NY	12158	UNY	GAS & ELEC	UNY	Yes
SABIC INNOVATIVE PLASTICS US LLC	1 NORYL AVE	Selkirk	NY	12158	UNY	GAS & ELEC	UNY	Yes
SARATOGA CO SEWER DIST #1	1002 HUDSON RIVER RD	Mechanicville	NY	12020-1907	UNY	GAS & ELEC		Yes
SCHENECTADY INTERNATIONAL/ROTT JCT FAC	1000 MAIN ST RT 5S	Schenectady	NY	12303	UNY	GAS & ELEC	UKN	Yes
SELKIRK COGENERATION PROJECT	24 POWER PARK DR ON GE PROPERTY	Selkirk	NY	12158	UNY	GAS & ELEC	UKN	Yes
SEMICONDUCTOR COMPONENTS INDUSTRIES, LLC	1964 LAKE AVE|	Rochester	NY	14619	UNY	GAS & ELEC		Yes
SI GROUP INC - ROTTERDAM JUNCTION FACILITY	1000 MAIN ST (ST RTE 5S)	Rotterdam	NY	12150	UNY	GAS & ELEC		Yes
SOLID WASTE MANAGEMENT FACILITY	23400 ST RTE 177	Rodman	NY	13682	UNY	GAS & ELEC		Yes
SPEAR USA	6 MORRILL PL	Fulton	NY	13069	UNY	GAS & ELEC		Yes
SPRAGUE ENERGY CORP	58 RIVERSIDE AVE	Rensselaer	NY	12144	UNY	GAS & ELEC	UNY	Yes
SPRAGUE OPERATING RESOURCES LLC	58 RIVERSIDE AVE	Rensselaer	NY	12144	UNY	GAS & ELEC		Yes
SREC BATH LFGTE FACILITY	TURNPIKE RD - S SIDE - W OF CO RTE 15	Bath	NY	14810	UNY	GAS & ELEC		Yes
STERLING ENERGY FACILITY	110 E SENECA ST	Sherrill	NY	13461	UNY	GAS & ELEC	UNY	Yes
STONY POINT FACILITY	70 GRASSY POINT RD	Stony Point	NY	10980	UNY	GAS & ELEC		Yes
SULLIVAN COUNTY LANDFILL	91 LANDFILL DR	Monticello	NY	12710-3835	UNY	GAS & ELEC		Yes
SUNOCO PARTNERS M & T ROCHESTER TERMINAL	1840 LYELL AVE	Rochester	NY	14619	UNY	GAS & ELEC		Yes
SUNOCO SYRACUSE TED PARK TERMINAL	2951 ENERGY DR	Warners	NY	13164	UNY	GAS & ELEC		Yes
SUNY AT BINGHAMTON	4400 VESTAL PKWY E	Binghamton	NY	13905	UNY	GAS & ELEC		Yes
SYRACUSE LLC	300 BELLE ISLE RD	Solvay	NY	13209	UNY	GAS & ELEC		Yes
SYRACUSE UNIVERSITY - STEAM STATION	500 EAST TAYLOR ST	Syracuse	NY	13244-6500	UNY	GAS & ELEC		Yes
TALLMAN ISLAND WASTEWATER TREATMENT PLANT	127-01 POWELLS COVE BLVD	College Point	NY	11356	NY	GAS & ELEC		Yes
TENNESSEE GAS PIPELINE CO - COMP STA 241	3447 SENTINEL HEIGHTS RD	La Fayette	NY	13084	UNY	GAS & ELEC	UKN	Yes
TOWN OF COLONIE LANDFILL FACILITY	1319 LOUDON RD	Cohoes	NY	12047	UNY	GAS & ELEC	UKN	Yes
TOWN OF COLONIE LANDFILL FACILITY	1319 LOUDON RD	Cohoes	NY	12047	UNY	GAS & ELEC	UKN	Yes
U S SALT - WATKINS GLEN REFINERY	3580 SALT POINT RD	Watkins Glen	NY	14891	UNY	GAS & ELEC		Yes

Facility Name	Address	Town	State	Zip	NATLGRID_REGION	NATLGRID_SERVICE_FLAG	cust_stat	Gas Available
US ARMY GARRISON	IMNE-MIL-PWE-M|BLDG 667A RUGER RD	West Point	NY	10996	UNY	GAS & ELEC		Yes
VALHALLA CAMPUS (GRASSLANDS)	35 WOODS RD|OPERATIONS BUILDING	Valhalla	NY	10595	UNY	GAS & ELEC		Yes
VARFLEX CORPORATION	512 W COURT ST	Rome	NY	13440	UNY	GAS & ELEC	UKN	Yes
VARFLEX CORPORATION	512 W COURT ST	Rome	NY	13440	UNY	GAS & ELEC	UKN	Yes
VON ROLL USA INC	200 VON ROLL DR	Schenectady	NY	12306	UNY	GAS & ELEC	UKN	Yes
VULCRAFT OF NEW YORK INC	5362 RAILROAD ST	Chemung	NY	14825	UNY	GAS & ELEC		Yes
WHEELABRATOR HUDSON FALLS	93 RIVER ST	Hudson Falls	NY	12839	UNY	GAS & ELEC	UKN	Yes
WM ONEIDA HERKIMER RENEWABLE ENERGY FACILITY	7044 ST RTE 294	Boonville	NY	13309	UNY	GAS & ELEC	UKN	Yes
WOODHULL STATION & POOL	974 CO RTE 99	Woodhull	NY	14898	UNY	GAS & ELEC		Yes
WPS SYRACUSE GENERATION LLC	300 BELLE ISLE RD	Syracuse	NY	13209	UNY	GAS & ELEC	UNY	Yes
WPS SYRACUSE GENERATION LLC	300 BELLE ISLE RD	Syracuse	NY	13209	UNY	GAS & ELEC	UNY	Yes
XEROX CORPORATION - WILSON CENTER	800 PHILLIPS RD	Webster	NY	14580	UNY	GAS & ELEC		Yes
NYCDOS - FRESH KILLS LANDFILL	310 W SERVICE RD	Staten Island	NY	10314	NY	GAS ONLY		Yes
WHEELABRATOR WESTCHESTER LP	1 CHARLES PT AVE	Peekskill	NY	12839-1354	UNY	GAS ONLY		Yes
ADCHEM CORPORATION	1852 Old Country Rd	Riverhead	NY	11901	LI	GAS ONLY	UNY	Yes
ALADDIN PACKAGING	40 RANICK RD	Hauppauge	NY	11788	LI	GAS ONLY	UNY	Yes
ALLIANCE ENERGY - SHOEMAKER GAS TURBINE FACILITY	110 E SENECA ST	Sherrill	NY	10940	UNY	GAS ONLY		Yes
AMERICAN TECHNICAL CERAMICS	ONE NORDEN PL AND 10 & 15 STEPAR PL	Huntington Station	NY	11746	LI	GAS ONLY	UKN	Yes
ARTHUR KILL GENERATING STATION	4401 VICTORY BLVD	Staten Island	NY	10314	NY	GAS ONLY	DNY	Yes
BABYLON RESOURCE RECOVERY FACILITY	125 GLEAM ST	West Babylon	NY	11704	LI	GAS ONLY	UKN	Yes
BARKER BROS - RIDGEWOOD	16-66 SUMMERFIELD ST	Ridgewood	NY	11385	NY	GAS ONLY	DNY	Yes
BAYSWATER / JAMAICA BAY PEAKING FACILITY	14-25 BAY 24TH ST	Far Rockaway	NY	11691	LI	GAS ONLY	UKN	Yes
BAYSWATER / JAMAICA BAY PEAKING FACILITY	14-25 BAY 24TH ST	Far Rockaway	NY	11691	LI	GAS ONLY	UKN	Yes
BEMIS COMPANY INC	100 WILSHIRE BLVD	Brentwood	NY	11717	LI	GAS ONLY	UNY	Yes
BERGEN POINT STP & BERGEN AVE. DOCK	600 BERGEN AVE	West Babylon	NY	11704	LI	GAS ONLY	UNY	Yes
BERGEN POINT WWTP	600 BERGEN AVE	W Babylon	NY	11704	LI	GAS ONLY		Yes
BIG SIX TOWERS INC	59-55 47TH AVE	Woodside	NY	11377	NY	GAS ONLY	DNY	Yes
BLYDENBURGH ROAD LANDFILL	440 BLYDENBURGH RD	Islip	NY	11751	LI	GAS ONLY	UKN	Yes
BOWLINE GENERATING STATION	140 Samsondale Ave	West Haverstraw	NY	10993	UNY	GAS ONLY		Yes
BP PRODUCTS N AMERICA BROOKLYN TERMINAL	125 APOLLO ST	Brooklyn	NY	11222	NY	GAS ONLY	UKN	Yes
BRENTWOOD PLANT	WITHIN PILGRIM STATE PROPERTY OFF SAGTIKOS STATE PKWY	Islip	NY	11571	LI	GAS ONLY	UKN	Yes
BROOKDALE MED HOSP CTR-1275 LINDEN BLVD	1275 LINDEN BLVD	Brooklyn	NY	11212	NY	GAS ONLY	UKN	Yes
BROOKHAVEN LANDFILL & RECYCLING AREA	350 HORSEBLOCK RD	Yaphank	NY	11980	LI	GAS ONLY	UKN	Yes
BROOKHAVEN LANDFILL & RECYCLING AREA	350 HORSEBLOCK RD	Yaphank	NY	11980	LI	GAS ONLY	UKN	Yes
BROOKHAVEN LANDFILL & RECYCLING AREA	350 HORSEBLOCK RD	Yaphank	NY	11980	LI	GAS ONLY	UKN	Yes
BROOKHAVEN LANDFILL & RECYCLING AREA	350 HORSEBLOCK RD	Yaphank	NY	11980	LI	GAS ONLY	UKN	Yes
BROOKHAVEN LANDFILL CELL 6	350 HORSEBLOCK RD	Yaphank	NY	11980	LI	GAS ONLY		Yes
BROOKHAVEN LANDFILL GAS RECOVERY FACILITY	350 HORSEBLOCK RD	Yaphank	NY	11980	LI	GAS ONLY		Yes
BROOKHAVEN NATIONAL LABORATORY	53 BELL AVE	Upton	NY	11973	LI	GAS ONLY	UKN	Yes
BROOKLYN NAVY YARD COGENERATION PLANT	BROOKLYN NAVY YARD, 63 FLUSHING AVE|BLDG 41 UNIT #234	Brooklyn	NY	11205	NY	GAS ONLY	UKN	Yes
BROOME COUNTY LANDFILL	286 Knapp Rd	Binghamton	NY	13905	UNY	GAS ONLY		Yes
BROOME COUNTY LFG RECOVERY FACILITY	286 Knapp Rd	Binghamton	NY	13905	UNY	GAS ONlY		Yes
CAITHNESS LONG ISLAND ENERGY CENTER	50 ZORN BLVD	Yaphank	NY	11980	LI	GAS ONLY	UNY	Yes
CARBO INDUSTRIES PROPERTY	TERMINAL 7 PARKING	Lawrence	NY	11559	LI	GAS ONLY	UKN	Yes
CEDAR CREEK WPCP	3340 Merrick Rd & Cedar Creek PK	Wantagh	NY	11793	LI	GAS ONLY		Yes
CEDAR GRAPHICS IGI	1700 OCEAN AVE	Ronkonkoma	NY	11779	LI	GAS ONLY	UNY	Yes
COGEN CORP-111 LIVINGSTON ST	111 LIVINGSTON ST	Brooklyn	NY	11201	NY	GAS ONLY	DNY	Yes
CON EDISON - HUDSON AVE STATION	1 HUDSON AVE	Brooklyn	NY	11201	Not In territory	GAS ONLY	UKN	Yes
CONTINENTAL BUCHANAN LLC	350 BROADWAY	Brooklyn	NY	10511-1010	NY	GAS ONLY		Yes
DISC GRAPHICS FACILITY	10 GILPIN AVE	Hauppauge	NY	11788	LI	GAS ONLY	UNY	Yes
DOWNSTATE MEDICAL CENTER	450 CLARKSON AVE	Brooklyn	NY	11203	NY	GAS ONLY	DNY	Yes
EAST HAMPTON GT FACILITY	BUELL LN WEST OF COVE HOLLOW LN	East Hampton	NY	11937	LI	GAS ONLY	UKN	Yes
EDGEWOOD ENERGY LLC	998 CROOKED HILL RD	Brentwood	NY	11717	LI	GAS ONLY	UNY	Yes
EDGEWOOD ENERGY LLC	998 CROOKED HILL RD	Brentwood	NY	11717	LI	GAS ONLY	UNY	Yes
EF BARRETT POWER STATION	1 MCCARTHY RD	Island Park	NY	11558	LI	GAS ONLY	UKN	Yes
ELMHURST HOSP-79-01 BROADWAY	79-01 BROADWAY	Elmhurst	NY	11373	NY	GAS ONLY	DNY	Yes
ENTENMANN'S BAKERY	1724 FIFTH AVE	Bay Shore	NY	11706	LI	GAS ONLY	UNY	Yes
EPIC PHARMA LLC	227-15 N CONDUIT AVE	Springfield Gardens	NY	11413	NY	GAS ONLY	DNY	Yes
EQUUS FREEPORT POWER GENERATING STATION	289 BUFFALO AVE	Freeport	NY	11520	LI	GAS ONLY	UNY	Yes
FREEPORT POWER PLANT #1	220 W SUNRISE HWY	Freeport	NY	11520	LI	GAS ONLY		Yes
FREEPORT POWER PLANT #2	289 BUFFALO AVE	Freeport	NY	11520	LI	GAS ONLY		Yes
GERSHOW RECYCLING CORP	71 PECONIC AVE	Medford	NY	11763	LI	GAS ONLY	UNY	Yes
GLENWOOD COMBUSTION TURBINE FACILITY	SHORE RD	Glenwood Landing	NY	11547	LI	GAS ONLY		Yes
GLENWOOD LANDING ENERGY CENTER	SHORE RD	Glenwood Landing	NY	11547	LI	GAS ONLY	UKN	Yes
GLENWOOD MAIN POWER STATION	SHORE RD	Glenwood Landing	NY	11547	LI	GAS ONLY	UKN	Yes
GLOBAL COMPANIES - COMMANDER TERMINAL	1 COMMANDER SQ	Oyster Bay	NY	11771	LI	GAS ONLY	UKN	Yes
GLOBAL COMPANIES LLC - GLENWOOD TERMINAL	SHORE & GLENWOOD RDS	Glenwood Landing	NY	11547	LI	GAS ONLY	UKN	Yes
GLOBAL COMPANIES LLC - INWOOD TERMINAL	464 DOUGHTY BLVD	Inwood	NY	11096	LI	GAS ONLY	UNY	Yes
GOWANUS GENERATING STATION	29TH ST & 2ND AVE	Brooklyn	NY	11232	NY	GAS ONLY	UKN	Yes
GREENPOINT ENERGY CTR	287 MASPETH AVE	Brooklyn	NY	11211	NY	GAS ONLY	DNY	Yes
GREENPORT GENERATING FACILITY	1885 MOORES LN|SCTM # 1001-40-4-1.1	Greenport	NY	11944	LI	GAS ONLY	UKN	Yes
HEMPSTEAD RESOURCE RECOVERY FACILITY	600 MERCHANTS CONCOURSE	Westbury	NY	11590	LI	GAS ONLY	UKN	Yes
HOFSTRA UNIVERSITY	1000 FULTON AVE	Hempstead	NY	11550	LI	GAS ONLY	UNY	Yes
HOFSTRA UNIVERSITY	1000 FULTON AVE	Hempstead	NY	11550	LI	GAS ONLY	UNY	Yes
HOFSTRA UNIVERSITY	1000 FULTON AVE	Hempstead	NY	11550	LI	GAS ONLY	UNY	Yes
HOLTSVILLE GT & LNG FACILITY	180-A MORRIS AVE	Holtsville	NY	11742	LI	GAS ONLY	UKN	Yes
HOLTSVILLE GT & LNG FACILITY	180-A MORRIS AVE	Holtsville	NY	11742	LI	GAS ONLY	UKN	Yes
HOLTSVILLE GT & LNG FACILITY	180-A MORRIS AVE	Holtsville	NY	11742	LI	GAS ONLY	UKN	Yes
HOLTSVILLE GT & LNG FACILITY	180-A MORRIS AVE	Holtsville	NY	11742	LI	GAS ONLY	UKN	Yes
HOLTSVILLE TERMINAL, NORTHVILLE IND CORP	586 UNION AVE	Holtsville	NY	11742	LI	GAS ONLY	UNY	Yes
HUNTINGTON RESOURCE RECOVERY FACILITY	99 TOWN LINE RD	East Northport	NY	11731	LI	GAS ONLY	UKN	Yes
HUNTINGTON RESOURCE RECOVERY FACILITY	99 TOWN LINE RD	East Northport	NY	11731	LI	GAS ONLY	UKN	Yes
INTERSTATE BRANDS CORPORATION	168-23 DOUGLAS AVE	Jamaica	NY	11433	NY	GAS ONLY	DNY	Yes
ISLIP MCARTHUR RESOURCE RECOVERY FACILITY	4001 VETERANS MEMORIAL HWY	Ronkonkoma	NY	11779	LI	GAS ONLY	UNY	Yes
KIAC COGENERATION PLANT-JFK AIRPORT	JFK AIRPORT BLDG 49	Jamaica	NY	11430	NY	GAS ONLY	UKN	Yes
KINDER MORGAN LIQUIDS TERMINALS LLC	4101 ARTHUR KILL RD	Staten Island	NY	10309	NY	GAS ONLY	DNY	Yes

Facility Name	Address	Town	State	Zip	NATLGRID_REGION	NATLGRID_SERVICE_FLAG	cust_stat	Gas Available
KINGS PLAZA TOTAL ENERGY PLANT	5100 KINGS PLZ	Brooklyn	NY	11234	NY	GAS ONLY	UKN	Yes
LAWRENCE RIPAK FACILITY	165 FIELD ST	West Babylon	NY	11704	LI	GAS ONLY	UNY	Yes
LAWRENCE RIPAK FACILITY	165 FIELD ST	West Babylon	NY	11704	LI	GAS ONLY	UNY	Yes
LONG ISLAND JEWISH MEDICAL CENTER-QUEENS	270-05 76TH AVE	New Hyde Park	NY	11040	LI	GAS ONLY	UKN	Yes
LONG ISLAND JEWISH MEDICAL CENTER-QUEENS	270-05 76TH AVE	New Hyde Park	NY	11040	LI	GAS ONLY	UKN	Yes
MAGELLAN AEROSPACE PROCESSING LONG ISLAND INC	165 FIELD ST	West Babylon	NY	11704-1416	LI	GAS ONLY		Yes
MAIMONIDES MEDICAL CENTER	4802 TENTH AVE	Brooklyn	NY	11219	NY	GAS ONLY	DNY	Yes
MASON INDUSTRIES PROPERTY	350 RABRO DR	Hauppauge	NY	11788	LI	GAS ONLY	UNY	Yes
MONTAUK GENERATING FACILITY	NAVY RD	Montauk	NY	11954	LI	GAS ONLY	UKN	Yes
MOTIVA ENTERPRISES LLC	25 PAIDGE AVE	Brooklyn	NY	11222	NY	GAS ONLY	DNY	Yes
MOTIVA ENTERPRISES LLC PROPERTY	74 EAST AVE	Lawrence	NY	11559	LI	GAS ONLY	UNY	Yes
N 1ST STREET PLANT	N 1ST ST & RIVER ST	Brooklyn	NY		NY	GAS ONLY	UKN	Yes
N SHORE TOWERS APT TOTAL ENERGY PLANT	272-40 GRAND CENTRAL PARKWAY	Floral Park	NY	11005	LI	GAS ONLY	UKN	Yes
NAP - KENT AVENUE FACILITY	667 KENT AVE	Brooklyn	NY	11249	NY	GAS ONLY	UKN	Yes
NARROWS GENERATING STATION	53RD ST & FIRST AVE|4 WHALE SQUARE	Brooklyn	NY	11232	NY	GAS ONLY	UKN	Yes
NASSAU ENERGY CORPORATION	185 CHARLES LINDBERGH BLVD	Garden City	NY	11530	LI	GAS ONLY	UNY	Yes
NASSAU ENERGY LLC	185 CHARLES LINDBERGH BLVD	Garden City	NY	11530	LI	GAS ONLY		Yes
NEW YORK METHODIST HOSPITAL	506 6TH ST	Brooklyn	NY	11215	NY	GAS ONLY	DNY	Yes
NEWTOWN CREEK WASTEWATER TREATMENT PLANT	329-69 GREENPOINT AVE	Brooklyn	NY	11222	NY	GAS ONLY	UKN	Yes
NISSEQUOGUE COGEN PARTNERS PLANT	2099 SUNY@ STONY BROOK	Stony Brook	NY	11794	LI	GAS ONLY	UKN	Yes
NORTH SHORE UNIVERSITY HOSPITAL	300 COMMUNITY DR	Manhasset	NY	11030	LI	GAS ONLY	UNY	Yes
NORTH SHORE UNIVERSITY HOSPITAL	300 COMMUNITY DR	Manhasset	NY	11030	LI	GAS ONLY	UNY	Yes
NORTHPORT POWER STATION	WATERSIDE AVE & EATONS NECK RD	Northport	NY	11768	LI	GAS ONLY	UKN	Yes
NYC-DEP CONEY ISLAND WPCP	2591 KNAPP ST	Brooklyn	NY	11235	NY	GAS ONLY	UKN	Yes
NYC-DOC - RIKERS ISLAND	17-25 HAZEN ST	East Elmhurst	NY	11370	NY	GAS ONLY	UKN	Yes
NYC-HH - CONEY ISLAND HOSPITAL	2601 OCEAN PKWY @ AVE Z	Brooklyn	NY	11235	NY	GAS ONLY	DNY	Yes
NYC-HH - KINGS COUNTY HOSPITAL CENTER	451 CLARKSON AVE	Brooklyn	NY	11203	NY	GAS ONLY	DNY	Yes
NYC-HH - WOODHULL HOSPITAL	760 BROADWAY	Brooklyn	NY	11206	NY	GAS ONLY	DNY	Yes
NYC-TA CONEY ISLAND YARD	2556 MCDONALD AVE @ AVE X	Brooklyn	NY	11223	NY	GAS ONLY	DNY	Yes
NYPA JOSEPH J SEYMOUR - 23RD ST & 3RD AVE	730 3RD AVE	Brooklyn	NY	11232	NY	GAS ONLY	UKN	Yes
ONEIDA CO WATER POLLUTION CONTROL PLANT	51 LELAND AVE	Utica	NY	13502	UNY	GAS ONLY		Yes
OWLS HEAD WASTEWATER TREATMENT PLANT	6700 SHORE RD	Brooklyn	NY	11220	NY	GAS ONLY	UKN	Yes
PARKER TOWERS	104-40 QUEENS BLVD	Forest Hills	NY	11375	NY	GAS ONLY	DNY	Yes
PE - BAYSHORE LLC	55 PARADISE LN	Bay Shore	NY	11706	LI	GAS ONLY	UNY	Yes
PINELAWN POWER	380 PATTON AVE|NEXT TO BABYLON RESOURCE RECOVERY	West Babylon	NY	11704	LI	GAS ONLY	UKN	Yes
POLY-PAK MFG PLANT	125 SPAGNOLI RD	Melville	NY	11747	LI	GAS ONLY	UNY	Yes
PORT JEFFERSON ENERGY CENTER	BEACH ST	Port Jefferson	NY	11777	LI	GAS ONLY	UKN	Yes
PORT JEFFERSON POWER STATION	BEACH ST	Port Jefferson	NY	11777	LI	GAS ONLY	UKN	Yes
POUCH TERMINAL	LYNHURST AVE & EDGEWATER	Staten Island	NY	10305	NY	GAS ONLY	UKN	Yes
PRATT PAPER (NY), INC	4435 VICTORY BLVD	Staten Island	NY	10314	NY	GAS ONLY	DNY	Yes
RICHARD M FLYNN POWER PLANT	607 UNION AVE	Holtsville	NY	11742	LI	GAS ONLY	UNY	Yes
RICHARD M FLYNN POWER PLANT	607 UNION AVE	Holtsville	NY	11742	LI	GAS ONLY	UNY	Yes
RICHARD M FLYNN POWER PLANT	607 UNION AVE	Holtsville	NY	11742	LI	GAS ONLY	UNY	Yes
ROCHDALE VILLAGE	132-11 BEDELL ST	Jamaica	NY	11434	NY	GAS ONLY	UKN	Yes
ROCKVILLE CENTRE POWER PLANT	110 MAPLE AVE	Rockville Centre	NY	11571	LI	GAS ONLY	UNY	Yes
ROCKVILLE CENTRE POWER PLANT	110 MAPLE AVE	Rockville Centre	NY	11571	LI	GAS ONLY	UNY	Yes
SHOREHAM ENERGY LLC	227 N COUNTRY RD	Shoreham	NY	11786	LI	GAS ONLY	UKN	Yes
SOUTHAMPTON GT FACILITY	DAVID WHITES LN	Southampton	NY	11968	LI	GAS ONLY	UKN	Yes
SOUTHOLD GT FACILITY	ST RTE 25 - W OF CHAPEL LN	Greenport	NY	11944	LI	GAS ONLY	UKN	Yes
SPRAGUE OPERATING RESOURCES FACILITY	1 BAY BLVD	Lawrence	NY	11559	LI	GAS ONLY		Yes
ST JOHNS UNIVERSITY	81-50 UTOPIA PKWY	Jamaica	NY	11432	NY	GAS ONLY	UKN	Yes
STARRETT CITY POWER PLANT	165 ELMIRA LOOP	Brooklyn	NY	11239	NY	GAS ONLY	UKN	Yes
STATEN ISLAND LANDFILL	1010 MULDOON AVE	Staten Island	NY	10312	NY	GAS ONLY	UKN	Yes
TBG COGEN FACILITY	939 S BROADWAY	Hicksville	NY	11801	LI	GAS ONLY	UNY	Yes
TRANSCONTINENTAL ULTRA FLEX INC.	975 ESSEX ST	Brooklyn	NY	11208-5419	NY	GAS ONLY		Yes
ULTRA FLEX PKG CORP-975 ESSEX ST	975 ESSEX STREET	Brooklyn	NY	11208	NY	GAS ONLY	DNY	Yes
UNITED RIVERHEAD TERMINAL	212 SOUND SHORE RD	Riverhead	NY	11901	LI	GAS ONLY	UKN	Yes
WADING RIVER GT FACILITY	NORTH COUNTRY RD	Shoreham	NY	11786	LI	GAS ONLY	UKN	Yes
WALNUT PACKAGING PROPERTY	450 SMITH ST	Farmingdale	NY	11735	LI	GAS ONLY	UNY	Yes
WEST BABYLON GT FACILITY	RAILROAD AVE	West Babylon	NY	11704	LI	GAS ONLY	UKN	Yes
WYCKOFF HEIGHTS MEDICAL CENTER	374 STOCKHOLM ST	Brooklyn	NY	11237	NY	GAS ONLY	UKN	Yes
KINGS PLAZA ENERGY LLC	5100 KINGS PLZ	Brooklyn	NY	11234	NY	GAS ONLY		Yes
BUCKEYE RENSSELAER TERMINAL	367 American Oil Rd	Rensselaer	NY	12144	UNY	GAS ONLY		Yes
ZENITH ENERGY TERMINALS BROOKLYN	25 PAIDGE AVE	Brooklyn	NY	11222-1281	NY	GAS ONLY		Yes
CAYUGA OPERATING COMPANY, LLC	228 Cayuga Dr	Lansing	NY	14882	UNY	Not in Territory		
CONEY ISLAND WASTEWATER TREATMENT PLANT	2591 KNAPP ST	Brooklyn	NY	11235	NY	No Service		Yes
KM PHOENIX HOLDINGS LLC - BROOKLYN TERMINAL	125 APOLLO ST	Brooklyn	NY	11222	NY	No service		Yes
CON ED - ASTORIA FACILITY	31-01 20th Ave	Astoria	NY	11105	NY	Not In Territory		
CON ED-59TH ST STATION	850 12th Ave	New York	NY	10019	NY	Not In Territory		
CON ED-74TH STREET STA	506 E 75th St	New York	NY	10021	NY	Not In Territory		
CON ED-EAST 60TH STREET STEAM PLANT	514 E 60Th St	New York	NY	10022	NY	Not In Territory		
CON ED-EAST RIVER GENERATING STATION	701-827 E 14th St	New York	NY	10009	NY	Not In Territory		
FORDHAM UNIVERSITY	441 E FORDHAM RD	Bronx	NY	10458	NY	Not In Territory		
MONTEFIORE MEDICAL CTR-111 E 210TH ST	111 E 210TH ST	Bronx	NY	10458	NY	Not in Territory		
MOUNT SINAI HOSPITAL	1 GUSTAVE L LEVY PL|1450 MADISON AVE	New York	NY	10029	NY	Not in Territory		
MUTUAL REDEVELOPMENT HOUSES	315 W 25TH ST	New York	NY	10001	NY	Not in Territory		
NEW YORK PRESBYTERIAN HOSPITAL	622 W 168TH ST	New York	NY	10032-3702	NY	Not in Territory		
NORTH RIVER WASTEWATER TREATMENT PLANT	725 W 135TH ST	New York	NY	10031	NY	Not in Territory		
NY - PRESBYTERIAN HOSPITAL-525 E 68TH ST	525 EAST 68TH ST	New York	NY	10021	NY	Not in Territory		
PARKCHESTER SOUTH CONDOMINIUM	2020 E TREMONT AVE	Bronx	NY	10458	NY	Not In Territory		
POLETTI POWER PROJECT	31-03 20TH AVE	Astoria	NY	11105	NY	Not In Territory		
QUEENS FRESH MEADOWS FACILITY	67-10 192ND ST	Flushing	NY	11365	NY	Not in Territory		
RAVENSWOOD GENERATING STATION	38-54 VERNON BLVD	Queens	NY	11101	NY	Not in Territory		
RAVENSWOOD GENERATING STATION	38-54 VERNON BLVD	Queens	NY	11101	NY	Not in Territory		
RIVERBAY CORP-CO-OP CITY	2049 BARTOW AVE	Bronx	NY	10458	NY	Not In Territory		
ROCKEFELLER UNIVERSITY	1188 York Ave	New York	NY	10065	NY	Not In Territory		

Facility Name	Address	Town	State	Zip	NATLGRID_REGION	NATLGRID_SERVICE_FLAG	cust_stat	Gas Available
SAINT JOHN'S RIVERSIDE HOSPITAL	967 NORTH BROADWAY	Yonkers	NY	10701	NY	Not In Territory		
STEINWAY & SONS - QUEENS FACILITY	1 STEINWAY PL	Long Island City	NY	11105-2601	NY	Not In Territory		
VERNON BLVD PLANT	42-30 VERNON BLVD	Queens	NY	11101	NY	Not In Territory		
YONKERS JOINT WWTP	1 FERNBROOK ST |LUDLOW DOCK SOUTH	Yonkers	NY	10705	NY	Not In Territory		
AMERICAN SUGAR REFINING INC	1 Federal St	Yonkers	NY	10705	NY	Not in Territory		
ASTORIA ENERGY LLC & ASTORIA ENERGY II LLC	17-10 Steinway St	Astoria	NY	11105-1097	NY	Not in Territory		
ASTORIA GAS TURBINE POWER	31-01 20th Ave	Astoria	NY	11105	NY	Not in Territory		
ASTORIA GENERATING STATION	18-01 20th Ave	Astoria	NY	11105	NY	Not in Territory		
BRONX ZOO	2300 Suthern Blvd	Bronx	NY	10460	NY	Not in Territory		
HARLEM RIVER YARDS PLANT	E 132ND ST & ROBERT F KENNEDY BRIDGE APPR|APPR 688 EAST 132ND STREET	Bronx	NY	10458	NY	Not in Territory		
HELL GATE	960 E 134TH ST	Bronx	NY	10458	NY	Not in Territory		
RAVENSWOOD STEAM PLANT	38-54 VERNON BLVD	Long Island City	NY	11101	NY	Not in Territory		
ALLEGANY GENERATING STATION	11537 ST RTE 19A	Hume	NY	14536	UNY	Not in Territory		
AMERICAN PACKAGING CORP	777 DRIVING PARK AVE	Rochester	NY	14613	UNY	Not in Territory		
ANCHOR GLASS CONTAINER CORP	151 E McCanns Blvd	Elmira Heights	NY	14903	UNY	Not in Territory		
AVERY DENNISON RIS, LLC.	524 St RTE 303	Orangeburg	NY	10962	UNY	Not in Territory		
BORGER STATION	219 Ellis Hollow CK Rd	Ithaca	NY	14850	UNY	Not in Territory		
FORT DRUM	85 FIRST ST WEST	Fort Drum	NY		UNY	Not In territory		
GLOBALFOUNDRIES EAST FISHKILL FACILITY	2070 ST RTE 52	Hopewell Junction	NY	12533	UNY	Not In territory		
HIGH ACRES LANDFILL & RECYCLING CENTER	425 PERINTON PKWY	Fairport	NY	14450	UNY	Not In territory		
SENECA ENERGY LFGTE FACILITY	ST RTE 414|RENEWABLE RESOURCES PARK	Seneca Falls	NY	13148	UNY	Not In territory		
ATHENS GENERATING PLANT	9300 US RTE 9W	Athens	NY	12105	UNY	Not inTerritory		
CALPINE JFK ENERGY CENTER								

Janet M. Audunson, P.E., Esq.
Senior Counsel II

April 30, 2018

VIA ELECTRONIC DELIVERY

Honorable Kathleen H. Burgess
Secretary
New York State Public Service Commission
Three Empire State Plaza, 19th Floor
Albany, New York 12223-1350

RE: Case 16-G-0058 – Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of KeySpan Gas East Corporation d/b/a National Grid for Gas Service

Case 16-G-0059 – Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of The Brooklyn Union Gas Company d/b/a National Grid NY for Gas Service

NATIONAL GRID: GAS DEMAND RESPONSE REV DEMONSTRATION PROJECT – Q1 2018 REPORT

Dear Secretary Burgess:

KeySpan Gas East Corporation d/b/a National Grid and The Brooklyn Union Gas Company d/b/a National Grid NY (collectively "National Grid") hereby submit for filing the quarterly report for the Gas Demand Response REV Demonstration Project covering the period of January 1, 2018 to March 31, 2018 ("Q1 2018 Report").

Please direct any questions regarding this filing to:

Arunkumar Vedhathiri
Director, New Energy Solutions
National Grid
1125 Broadway
Albany, New York 12204
Tel.: 518-433-5013
Mobile: 518-423-5738
Email: arunkumar.vedhathiri@nationalgrid.com

National Grid looks forward to continuing to work collaboratively with the New York State Department of Public Service Staff as it proceeds with the implementation of the Gas Demand Response REV Demonstration Project.

Respectfully submitted,

/s/ Janet M. Audunson

Janet M. Audunson, P.E., Esq.
Senior Counsel II

Enc.

cc: Marco Padula, DPS Staff, w/enclosure (via electronic mail)
Robert Cully, DPS Staff, w/enclosure (via electronic mail)
Cynthia McCarran, DPS Staff, w/enclosure (via electronic mail)
John Sano, DPS Staff, w/enclosure (via electronic mail)
Davide Maioriello, DPS Staff, w/enclosure (via electronic mail)
Cathy Hughto-Delzer, w/enclosure (via electronic mail)
Kate Grant, w/enclosure (via electronic mail)
Carlos Nouel, w/enclosure (via electronic mail)
Christopher Cavanagh, w/enclosure (via electronic mail)
Owen Brady, w/enclosure (via electronic mail)

nationalgrid

Gas Demand Response
REV Demonstration Project
in
New York City and Long Island

Q1 2018 Report

April 30, 2018

Table of Contents

1.0 Executive Summary....1
2.0 Highlights since Implementation Plan Filing....1
2.1 Marketing and Customer Engagement....1
2.2 Customer-site Systems....1
2.2.1 Gas Usage Measurement....1
2.2.2 Remote Device Control....2
2.3 Management Software....2
2.3.1 Communication Platform....2
2.3.2 Energy Insights & Actionable Information....2
2.4 Pricing....3
2.5 Program Design....3
2.6 Load Relief Analysis and Operational Impact....3
2.7 Program Benefit-Cost Analysis....3
2.8 Season 1 - Dry Run and Activations....4
2.9 Challenges, Changes, and Lessons Learned....5
3.0 Next Quarter Forecast....6
3.1 Checkpoints/Milestone Progress....6
3.1.1 Summary....6
3.1.2 Checkpoints....7
4.0 Work Plan and Budget Review....8
4.1 Updated Work Plan....8
4.2 Updated Budget....9
Appendix A: Customer-Site Usage Graphs....10
DR Event Reduction....10
Variety of Customer Usage Profiles....11
Appendix B: System Pressure Graphs....12
ND 21 – KEDLI....12
Gov 4 – KEDNY....16
Gov 126 – KEDNY....18

1.0 Executive Summary

The Gas Demand Response REV Demonstration Project (the "Project") being implemented by The Brooklyn Union Gas Company d/b/a National Grid NY and KeySpan Gas East Corporation d/b/a National Grid (collectively, "National Grid" or the "Companies" and individually, "KEDNY" and "KEDLI") will test a customer-centric, voluntary gas demand response ("DR") program targeting large commercial, firm gas customers in locations prioritized by predicted distribution constraints. Current interruptible ("IT") or temperature controlled ("TC") customers may also be included in the Project. The introduction of a gas DR program should allow the Companies to operate their assets in the most efficient way possible.

The current quarter (Q1 2018) marked the first season for activating events in the Project following the hardware installation and commissioning in the last quarter (Q4 2017). The following are key implementation highlights in Q1 2018:

- First event was called on January 17.
- A total of five (5) events were called, with two (2) events being exclusive to KEDLI customers.
- 100% (16 out of 16) of enrolled facilities participated in this first season.
- Initial results indicate the customers were able to curtail the loads as planned.
- The DR payment processing is underway and is expected to be completed by April 30, in line with the dates from the Project's Implementation Plan.
- The analysis completed on the data from this first season appears to show a positive impact on gas system pressures, though more data will be needed to verify this result.

2.0 Highlights since Implementation Plan Filing

The following sections highlight key activities accomplished during Q1 2018, as well as key activities planned for Q2 2018.

2.1 Marketing and Customer Engagement

No additional marketing efforts were undertaken during Q1 2018 and no new customers were added. The season progressed with the same list of customers outlined in the Q4 2017 Report.

2.2 Customer-site Systems

2.2.1 Gas Usage Measurement

No changes were made to the gas measurement approach outlined in the Q4 2017 Report.

2.2.2 Remote Device Control

In addition to the work outlined in the Q4 2017 Report, the Project's DR team worked with National Grid's Meter Operations group to better understand the Zener barrier[1] and available options. Research revealed a different model Zener Barrier, specifically the MTL7710+, which would work with a 10VDC supply voltage being provided to the "safe" side of the barrier. The Project vendor, IPKeys Technologies, LLC ("IPKeys"), confirmed with the manufacturer (Eaton) that the MTL7710+ should be compatible with the EISS® Box controller and National Grid verified that it will provide the same level of protection. This MTL7710+ unit was ordered and will be tested by IPKeys for possible future use.

2.3 Management Software

Q1 2018 changes to the management software aspect of the project are outlined below:

2.3.1 Communication Platform

The AutoGrid Systems, Inc. ("AutoGrid") demand response optimization and management system ("DROMS™") successfully performed all required functions, including aggregating customer data and communicating DR events to participating customers. Given that DR for gas is relatively new, there were several aspects of the program that lay outside the standard functionality of AutoGrid. The AutoGrid team was very responsive in terms of working with the Project team to create a product that met the Project's needs. If gas DR scales up, there are additional modifications to the communication platform that could be made to increase the functionality.

One area that presented a challenge during the first season was the need for coordination between AutoGrid and IPKeys. The need for secure transmission of data means that each vendor has security certificates in place. Twice during the course of this first season, the vendors updated security certificates at different times, which created issues for the EISS® Box controllers. This process has since been updated with checks in place to ensure device stability going forward.

2.3.2 Energy Insights & Actionable Information

As mentioned above, the AutoGrid team continued to work with the Project team to provide a useful platform that provides insight into the data that was acquired from the Project. In addition to being able to view usage for individual customers, a different module of the program allows the Project team to view the whole portfolio of DR customers in one aggregated view. The latter view shows "real-time" usage and usage over a larger time scale to show trends for the whole customer class. During DR events, there is a significant reduction in the total consumption, which would indicate the reduction in demand/usage.

[1] Zener barriers are a type of intrinsically safe, electric barriers that prevent a hazardous level of voltage reaching the gas meter. They are used regularly by National Grid and were used for the Project due to the connection between the KYZ pulse output device on the gas meter and the IPKeys system.



Figure 1: Example Aggregated View of Customer Usage (non-DR day and a subset of total customers)

Looking at customers on an individual basis, it becomes possible to identify trends in their usage. This has significant benefits for system planning beyond the Project itself. Traditional system planning has characterized customers using two factors, temperature dependent (heating) and temperature independent (base). These factors are generally quite accurate if the customer's usage is either strongly influenced by temperature or strongly independent of temperature. For those customers that vary their usage based on other factors (*i.e.*, day of week, season, supply chain changes, market factors, etc.) a modeling challenge can exist. The level of data resolution achieved in the Project shows the potential insight that can be gained through the rollout of advanced metering infrastructure ("AMI") for the gas system. See Appendix A to this report for the pressure graphs.

2.4 Pricing

No changes were made to the pricing model since the Q4 2017 Report.

2.5 Program Design

The design of the Project has not materially changed since the filing of the Project's Implementation Plan.

2.6 Load Relief Analysis and Operational Impact

The Project team has been analyzing the impact of the DR events on the distribution system now that the season has concluded and data is available for all events. Depending on the day and the state of the distribution system, the effect is more or less pronounced. There are a number of days, including during the coldest event day on Jan 17, where the system pressure is impacted positively. This effect is most pronounced during the peak hours, as was expected. More data is needed before the causal relationship can be characterized.

2.7 Program Benefit-Cost Analysis

An analysis of the performance and benefits of the Project will be completed as part of the annual review. This review will be completed by the end of July each year (beginning in 2018) so that any lessons learned can be incorporated into the planning for the upcoming season. At the end of the three-year term of the Project, a final report documenting the findings will be produced.

While the simple benefit-cost analysis ("BCA") (dollars invested vs benefit achieved) has been completed, the true impact of the Project extends quite a bit further. Understanding the total value

afforded by gas DR will require an understanding of how this program could impact other National Grid efforts. The Project team is working closely with National Grid's long-term planning and supply groups to validate the operational benefits of gas DR. The Synergi model (described in the Project's Q4 2017 Report), which operates in a static state, will be updated to reflect the new peaks for participating customers, which will then be used to produce new system maps.

Another benefit that is being explored is the possibility to refine customer modeling, as described above. Currently, new customers submit an application for gas service that includes 100% of their theoretical load based on the nameplate information of their connected equipment. This is then assessed using a Gas Capacity Request ("GCR"), a process by which the engineering group adjusts the 100% load based on the anticipated usage of the equipment. After two (2) years on the system, sufficient usage data for that customer will exist for a linear regression to be performed. This produces the base and heating factors referenced earlier in this report. These factors, in theory, provide better insight into the anticipated usage of the customer. If the predicted peak day usage varies a great deal from what was predicted by the GCR, the system will be updated to reflect this new peak for planning. Having more insight into customer data, either through participation in a program like gas DR or through the rollout of AMI, would make it possible to validate the existing estimates and to inform new models.

Finally, the DR team is participating in the IT/TC Collaborative,[2] which is exploring the non-firm rates for National Grid. It is possible that gas DR could be used as a system planning tool for situations involving non-firm customers. Depending on how it is used, this could improve customers' overall level of service.

2.8 Season 1 - Dry Run and Activations

Based on the budget for the year and the incentive prices agreed to by the participants, the Project team planned to call three (3) events in the KEDNY service territory and six (6) events in the KEDLI territory. The first DR event was on January 17 and it was called for customers in both service territories. Members of the Project team visited customer sites during the event to gather customer perspectives. Customers were engaged and interested in the Project. Some even participated at levels that exceeded their commitments to test how their systems would respond. However, due to connectivity issues, only 11 out of 16 the sites were activated by the direct load control ("DLC") system for this event. Together, they represented 167 DTh/hr. (87% of the Project total). The connectivity issues were subsequently resolved with both vendors.

The next two events, which occurred on February 13 and 23, were only called for KEDLI customers. The remaining two events (KEDNY and KEDLI) of the season were called on February 27 and 28. One hundred percent of customers participated in the events called during February. Despite the cold weather at the beginning of the year, the weather was mild in February, which made it difficult to call events on days

[2] *See* Cases 16-G-0058, *et al*., *Proceeding on Motion of the Commission as to the Rates, Charges, Rules and Regulations of Key Span Gas East Corporation d/b/a National Grid for Gas Service*, Order Adopting Terms of Joint Proposal and Establishing Gas Rate Plans (issued December 16, 2016), pp. 49-50, where National Grid committed, under section VI.10.7 of the Joint Proposal, to commence a collaborative to address such IT/TC issues as the structure of services, rates and alternative fuel documentation requirements, followed by a report to the Commission with recommendations and/or positions of the parties for Commission decision.

that would be deemed representative of peak days. The days selected were cool in the morning but not cold enough that IT/TC customers were activated, which changes the strain placed on the system and increases the total load, thereby requiring a larger demand reduction. The goal for the next season will be to call events on days when non-firm customers are activated to assess system impact under those conditions.

Service Territory	Type	Units*	% of Total Gas Demand	Q1 2018 performance – No. of events participated
Mineola (KEDLI)	Manufacturing	9.4	50%	4**
Brooklyn (KEDNY)	Heating	70.0	50%	3
Brooklyn (KEDNY)	Heating	33.6	79%	3
Queens (KEDNY)	Heating	25.6	50%	3
Brooklyn (KEDNY)	Heating	19.4	32%	3
Brooklyn (KEDNY)	Heating/Laundry	33.4	25%	3
Queens (KEDNY)	Heating	29.2	50%	3
Staten Island (KEDNY)	Heating	6.98	34%	2**
Queens (KEDNY)	Heating	10.4	50%	3
Brooklyn (KEDNY)	Heating/ Dehumidification	25.1	100%	2**
Queens (KEDNY)	Heating	9.8	100%	3
Staten Island (KEDNY)	Heating	6.0	75%	2**
Kings Point (KEDLI)	Cooking	4.0	100%	5
Kings Point (KEDLI)	Heating	74.0	100%	5
Kings Point (KEDLI)	Heating	24.5	100%	5
Port Jefferson (KEDLI)	Heating	4.0	20%	4**
*1 Unit = 500 cubic feet per hour ("cfh") = 500,000 BTU/hr. ** Customers were ready and willing to participate but communication challenges rendered the DLC system nonresponsive at these sites.				

Table 1: List of Participating Customer Sites

The images below represent three (3) pressure governor points on the gas system that are physically close to customers participating in the Project. The graphs correspond to three (3) pressure points on the National Grid system:

1) ND 21 – KEDLI
2) Gov 4 – KEDNY
3) Gov 216 – KEDNY

See Appendix B to this report for these graphs.

2.9 Challenges, Changes, and Lessons Learned

At this juncture in the Project, one additional lesson learned was identified:

Qtr.	Issue or Change	Resulting Change to Project Scope/Timeline?	Strategies to Resolve	Lessons Learned
Q1 2018	Need to establish software update schedule with vendors	None	N/A	When working with multiple vendors, ensure that processes are established to verify that updates by any one vendor will not adversely affect others.

3.0 Next Quarter Forecast

During Q2 2018, the Project team will issue incentive payments, make any necessary upgrades to the DR system (comprised of the AutoGrid "DROMS™ and equipment installed at customer sites), complete the BCA, and determine the plan for year two of the Project.

3.1 Checkpoints/Milestone Progress

3.1.1 Summary

Checkpoint/Milestone	Anticipated Start-End Date	Revised Start-End Date	Status
National Grid's Strategic Sales and Gas Sales Support to sign-up customers for the Project	6/1/17-11/30/17	6/1/17-9/30/17	✓
Set up processes to issue payment within 90 days of the conclusion of the DR season	6/1/17-11/30/17	No Change	✓
Coordinate communication methods with National Grid's Gas Control and the selected software for the Project.	6/1/17-9/30/17	No Change	✓
Enhance and automate interfaces within National Grid's systems to support interactions with selected customer sites	6/1/17-11/30/17	No Change	✓
Install necessary site equipment	9/15/17-11/1/17	9/30/17-1/19/18	✓
Work with National Grid's Advanced Data Analytics group on BCA and end-of-season incentive payment	3/1/17-11/30/17	3/1/17-3/1/18	✓

Key

- On-Track
- Delayed start, at risk of on-time completion, or over-budget
- Terminated/abandoned checkpoint
- ✓ Milestone Completed

Work Stream

Work Stream	Future Milestones	Status
Information Systems ("IS")	**Perform penetration test on EISS® BOX 3.0 controller**	●
	Oversee migration of DR System from 3G to 4G LTE	●
	Develop plan for supporting scaled Project	●
Operations	**Support participating customers during DR season**	✓
	Complete site checks to verify successful operation and participation in the gas DR program by customers	✓
Gas DR Team	**Contact customers following dry run to ensure success and to make any needed modifications before DR season**	✓
	Develop plan for timing of activations	✓
	Summarize findings based on sales experience	✓
	Summarize findings from site enablement process to develop lessons learned for years 2 and 3	●

3.1.2 Checkpoints

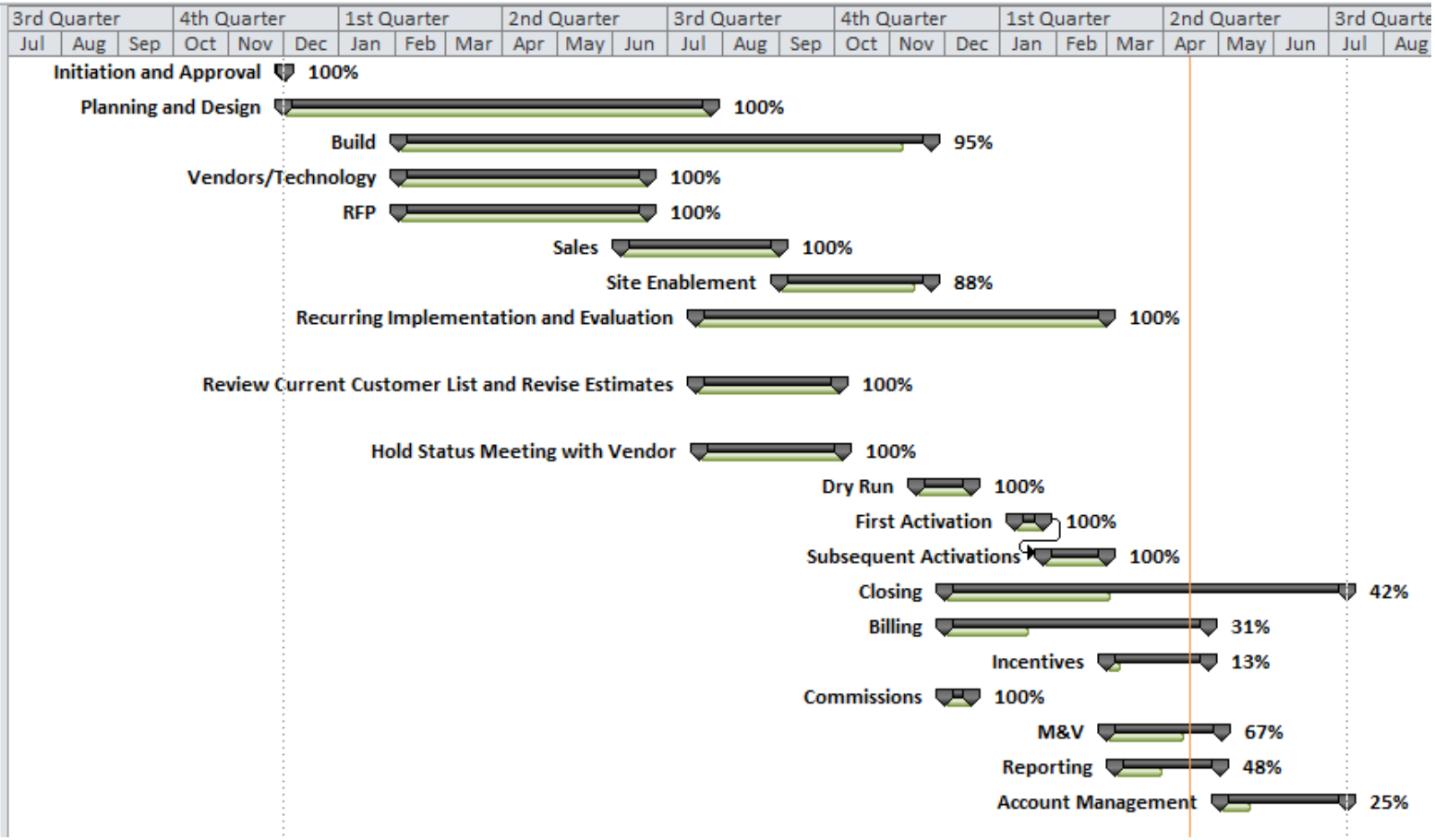
Scenario	Description	Target	Status (if known)
$ expanded per "Unit" of demand reduction	Per unit present value ("PV") of costs and payments ($/DTH/hr.)	Cap the Project below normal winter distribution charge.	46% under budget across all customers
Actual load reduction potential per participating customer for process and heating loads	% reduction of customer's historical winter peak	Average of 25% process and 10% heating loads.	Exceeded the target for reduction per customer.
Predicted market penetration rate	# of participants, expressed as a percentage, per SIC[3] code	15% of customers in each SIC code	Will be determined during annual review.
Significance of demand reduction potential	Gross Potential in DTh/hr.	140 total (80 DTh/hr. in KEDNY and 60 DTh/hr. in KEDLI)	192 DTh/hr. committed
Potential impact on gas system planning	Per unit PV of costs and payments ($/DTH-hr.)	Comparable to per unit growth capital	Will be determined during annual review.
Satisfaction with gas DR compared to existing programs	Percent of participants "satisfied" or better.	Exceed satisfaction level of Interruptible and TC customers - similar to electric.	Will be determined during customer survey

[3] Standard Industrial Classification.

4.0 Work Plan and Budget Review

4.1 Updated Work Plan

Start	Finish
Thu 12/1/16	Thu 12/1/16
Thu 12/1/16	Mon 7/24/17
Fri 2/3/17	Thu 11/23/17
Fri 2/3/17	Mon 6/19/17
Fri 2/3/17	Mon 6/19/17
Mon 6/5/17	Thu 8/31/17
Fri 9/1/17	Thu 11/23/17
Mon 7/17/17	Wed 2/28/18
Mon 7/17/17	Tue 10/3/17
Wed 7/19/17	Thu 10/5/17
Thu 11/16/17	Fri 12/15/17
Tue 1/9/18	Wed 1/24/18
Thu 1/25/18	Wed 2/28/18
Fri 12/1/17	Tue 7/10/18
Fri 12/1/17	Wed 4/25/18
Thu 3/1/18	Wed 4/25/18
Fri 12/1/17	Fri 12/15/17
Thu 3/1/18	Wed 5/2/18
Mon 3/5/18	Tue 5/1/18
Wed 5/2/18	Tue 7/10/18



4.2 Updated Budget

Changes to the structure of the Project have resulted in the need to modify the anticipated allocations of funds for the Project. However, it has been possible to do this while staying within the budget for Year 1, despite high Op Ex costs for establishing the necessary software systems. It is forecasted that the three-year budget for the Project will be able to be maintained.

Year 1 Budget report:

Project Task	4th Quarter Actual Spend	Project Total Spend to Date	Project Budget[4]	Remaining Balance
CapEx				
Total	$122,563.91	$130,255.20	$114,000	($16,255.20)
Grants Credited Against Incremental Capital Costs				
(Only if applicable)	$	$	$	($)
OpEx				
Total	$467,836.92	$467,836.92	$663,102.00	$195,265.08
Project Total	**$590,400.83**	**$598,092.12**	**$777,102.00**	**$179,009.88**

[4] National Grid updated the Project budget to reflect incremental costs and account for costs that may have originally been characterized as capital or operating expenses, but now, because of changed circumstances (*e.g.*, leasing instead of owning software), should be categorized differently.

Appendix A: Customer-Site Usage Graphs

DR Event Reduction

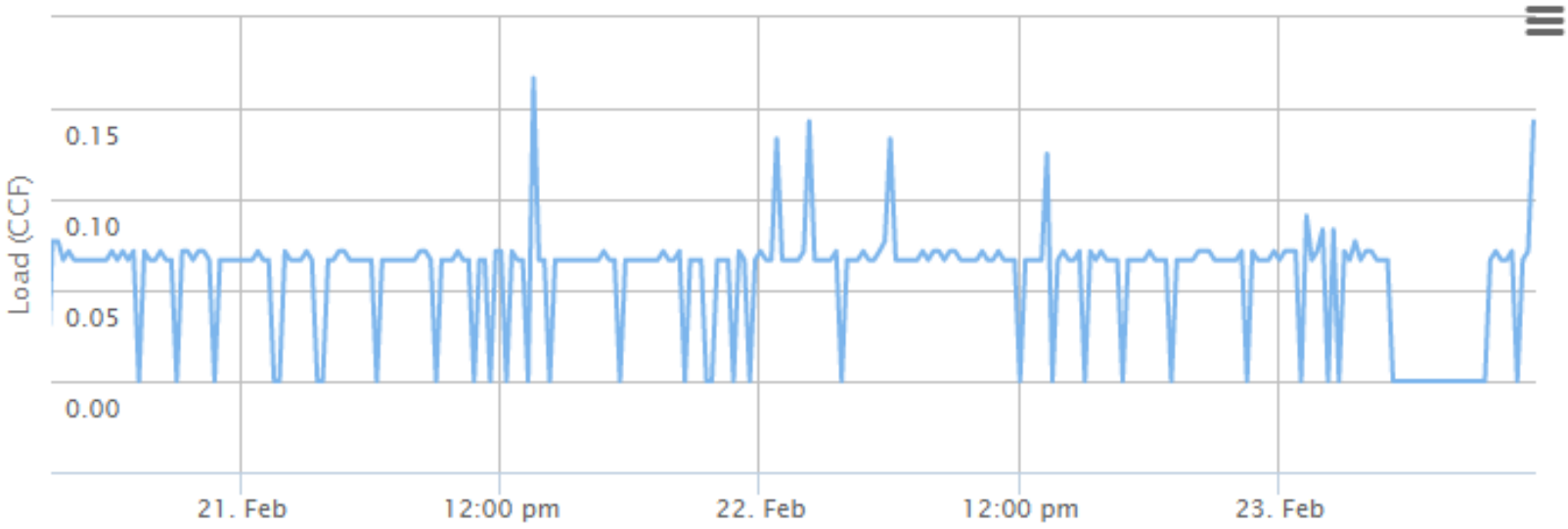


Figure 2: Customer usage profile during gas DR event. Usage reduction due to the DR event can be seen at the right-hand side of the image

Variety of Customer Usage Profiles

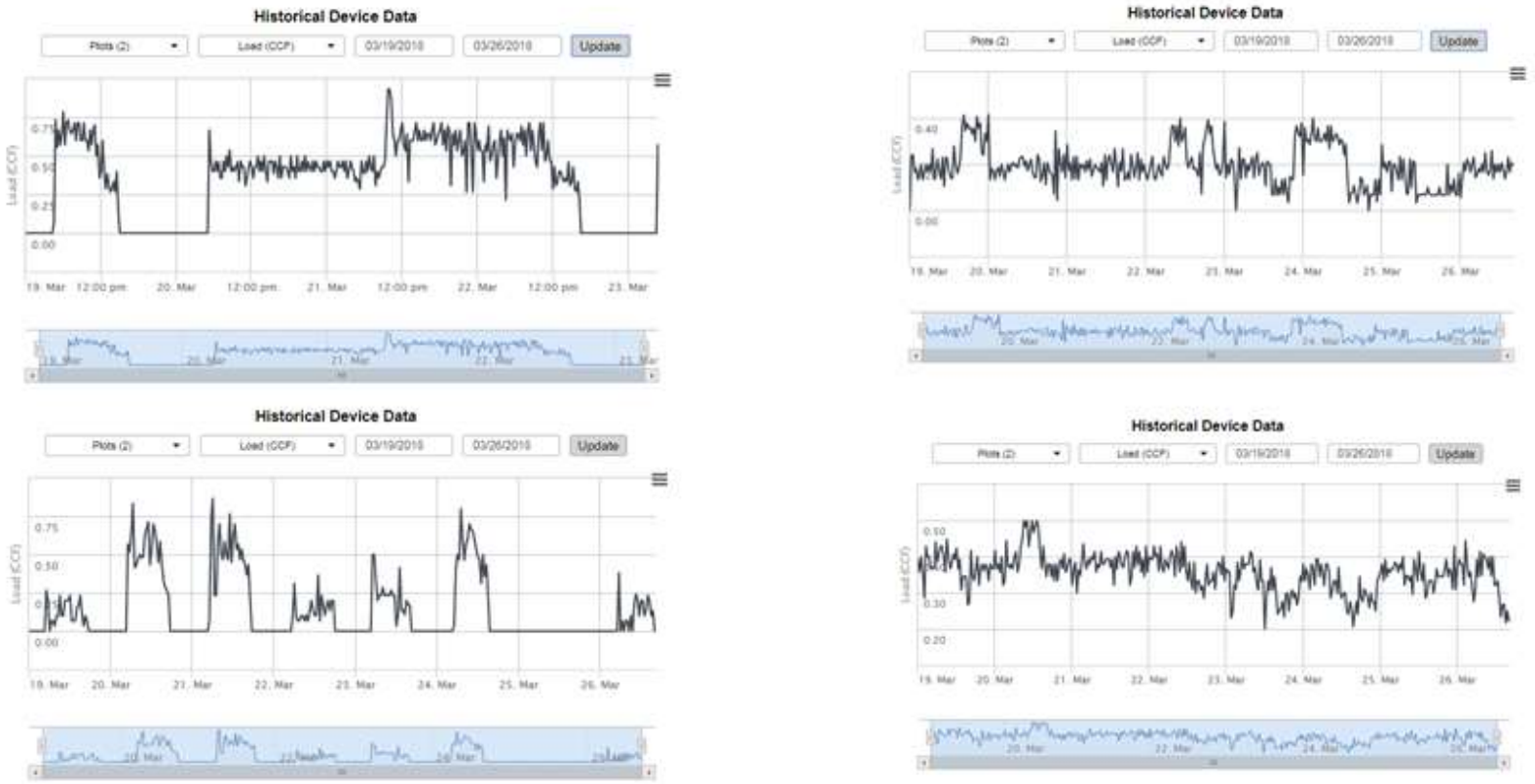
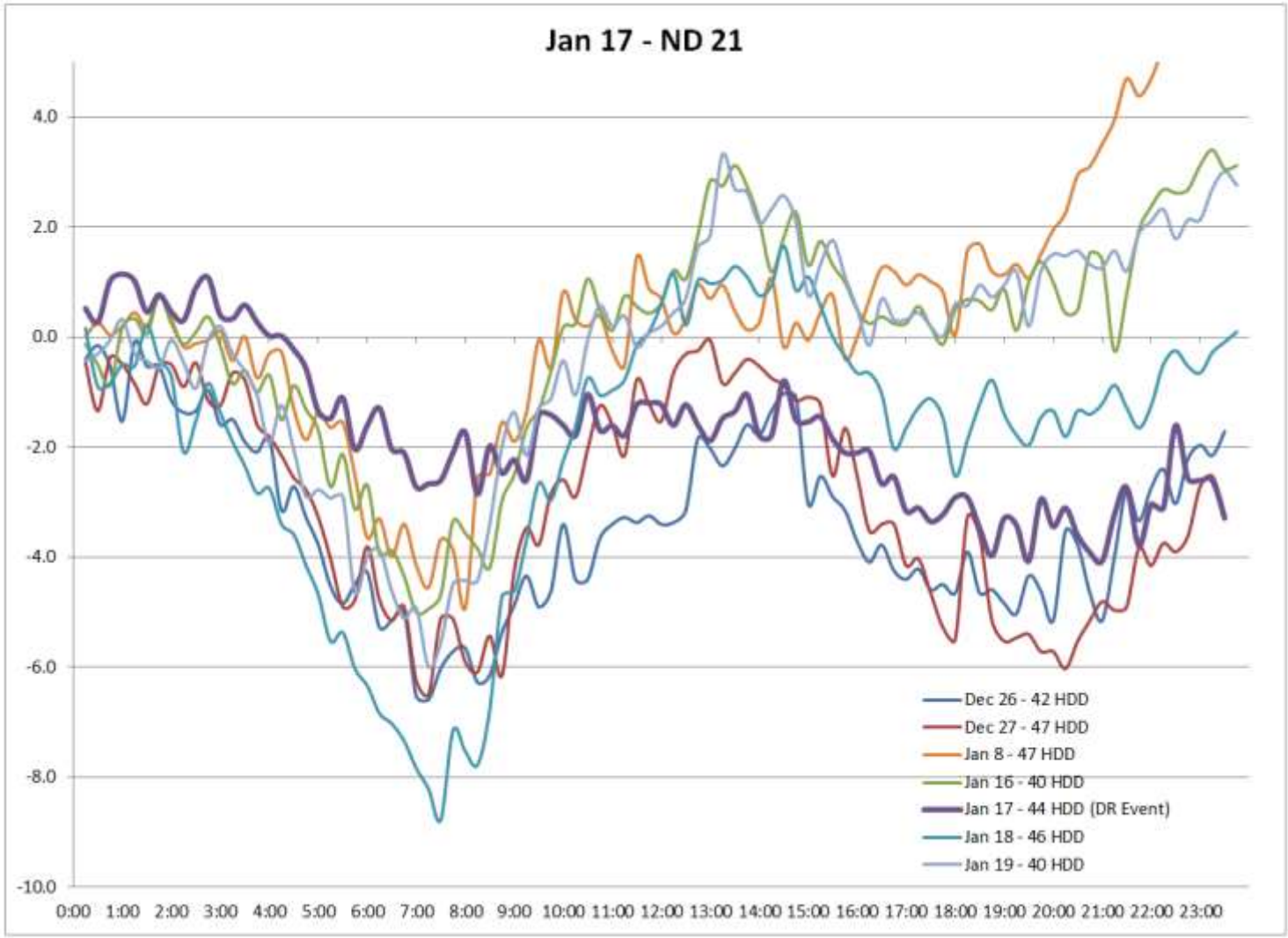
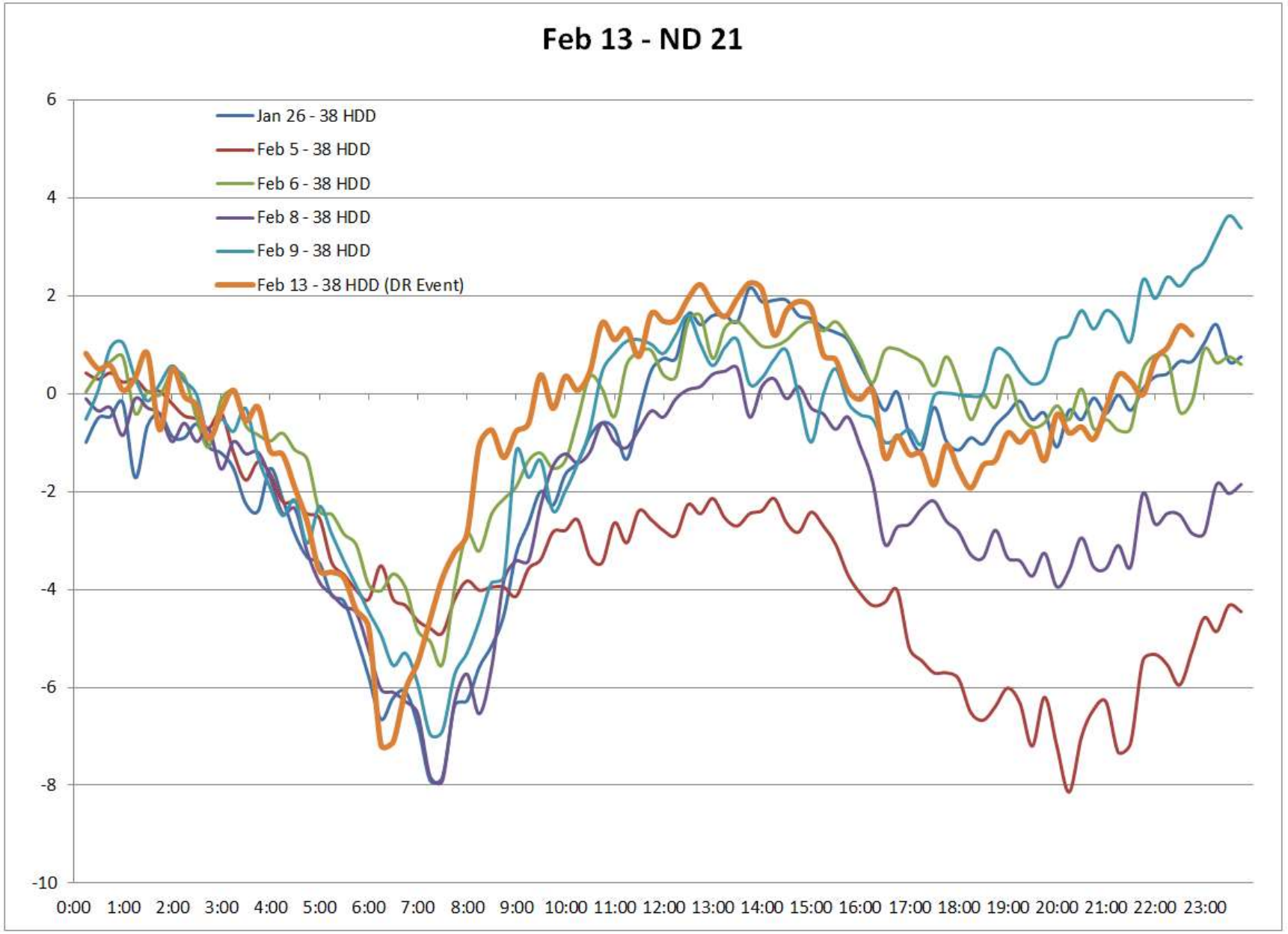


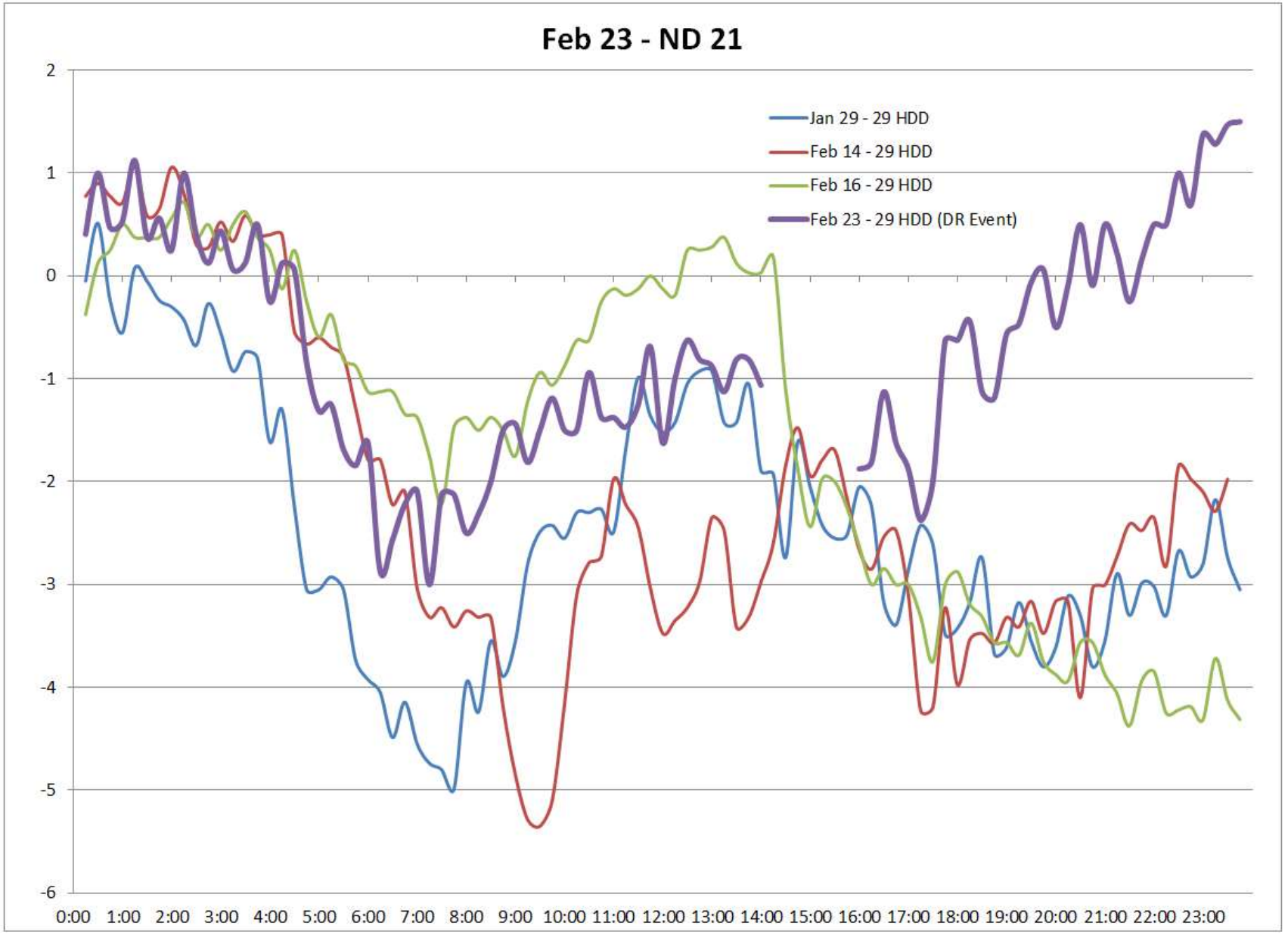
Figure 3: Individual customer usage trends showing the diverse possible usage profiles. These were recorded at the end of March (non-DR event days)

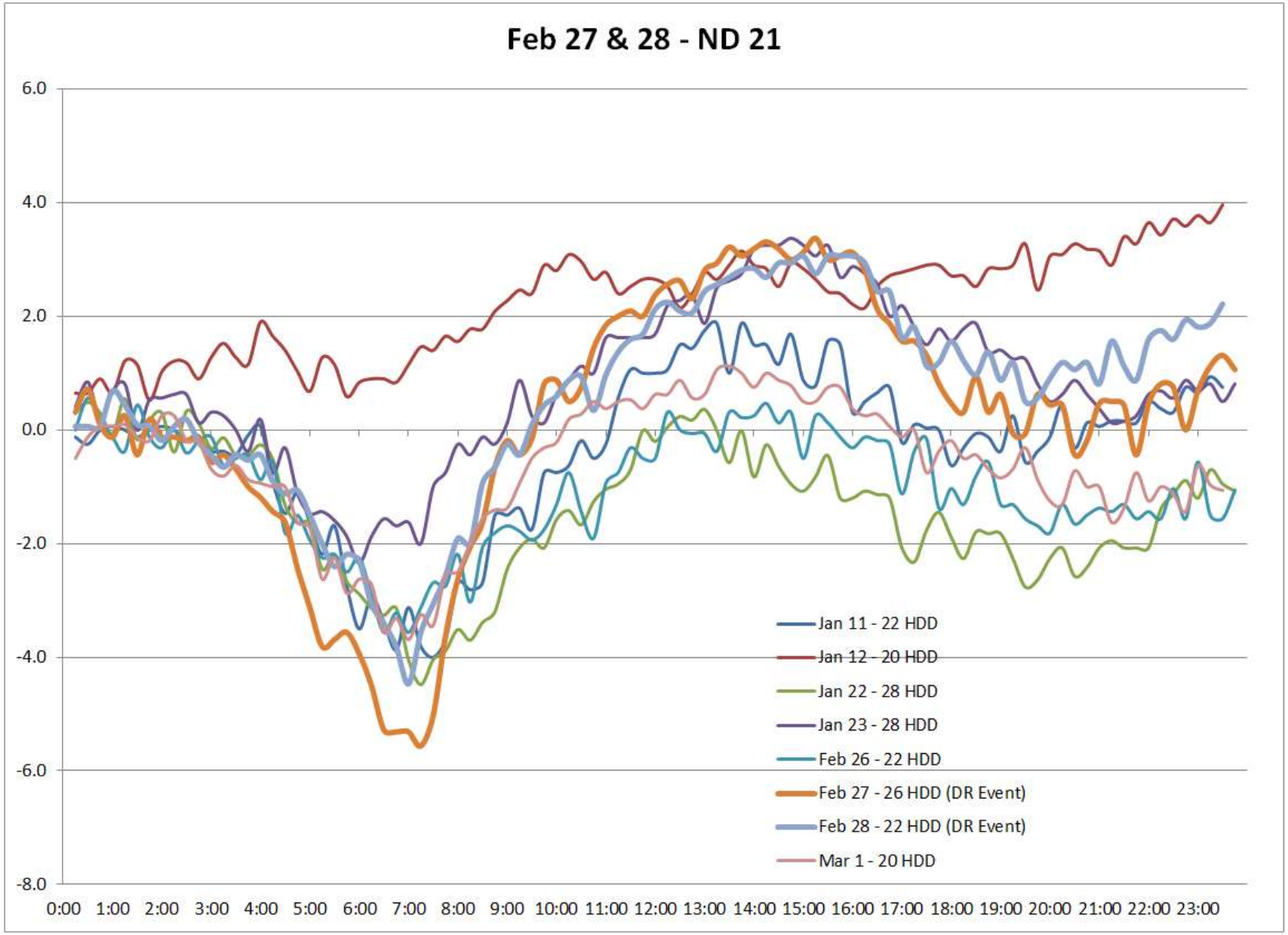
Appendix B: System Pressure Graphs

ND 21 – KEDLI

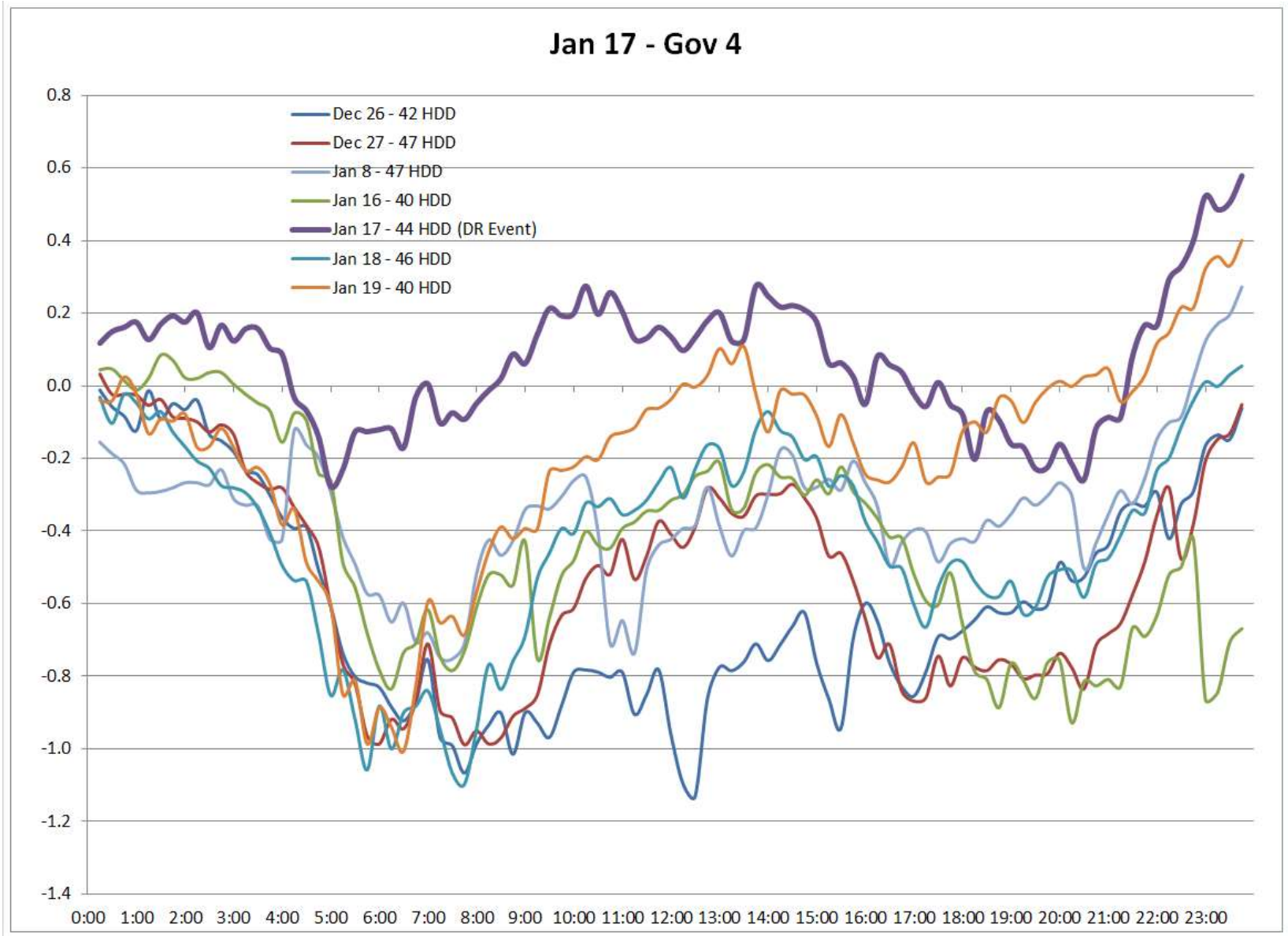


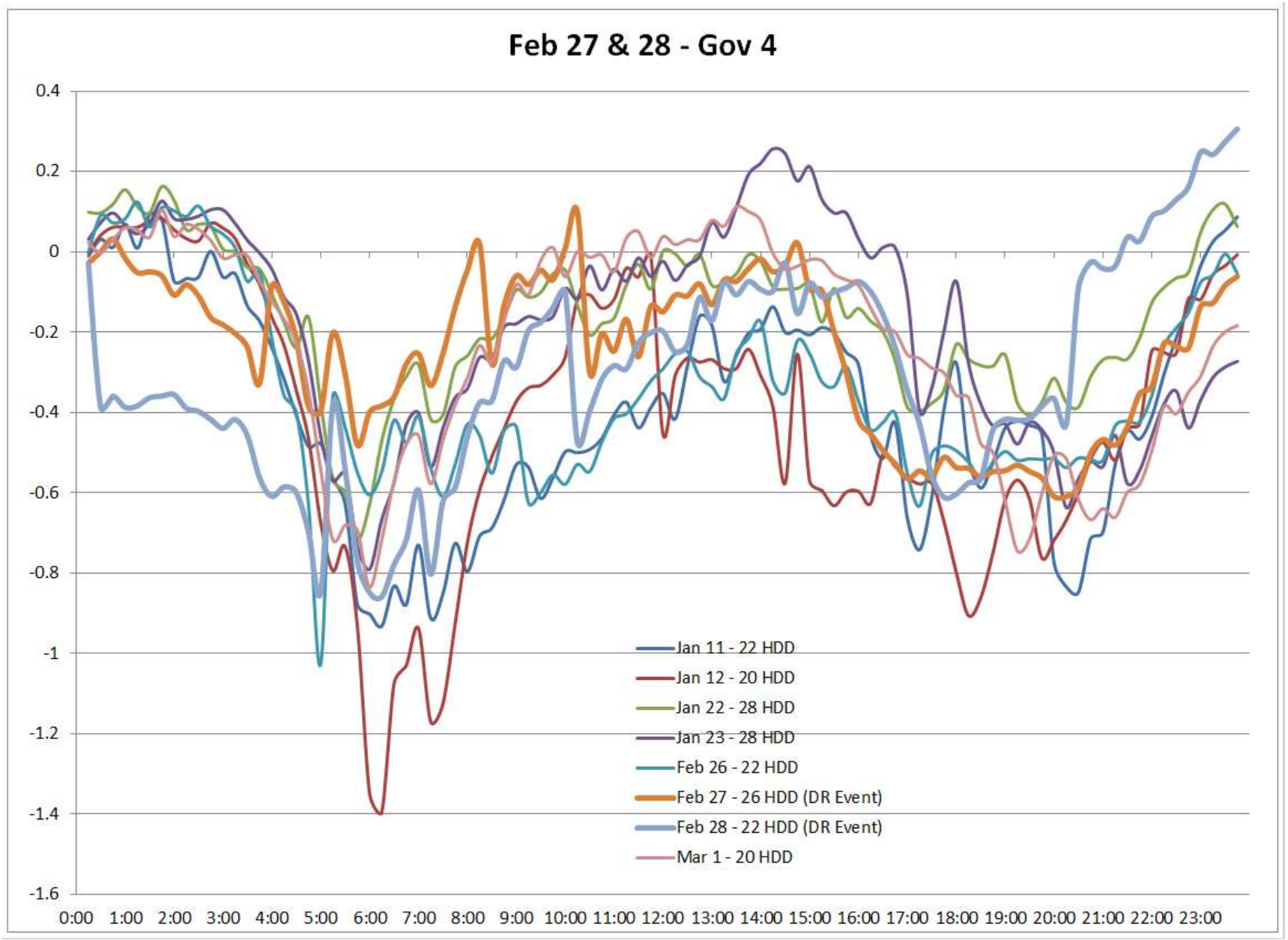

Feb 13 - ND 21
Jan 26 - 38 HDD
Feb 5 - 38 HDD
Feb 6 - 38 HDD
Feb 8 - 38 HDD
Feb 9 - 38 HDD
Feb 13 - 38 HDD (DR Event)
6
4
2
0
-2
-4
-6
-8
-10
0:00 1:00 2:00 3:00 4:00 5:00 6:00 7:00 8:00 9:00 10:00 11:00 12:00 13:00 14:00 15:00 16:00 17:00 18:00 19:00 20:00 21:00 22:00 23:00


Feb 23 - ND 21
Jan 29 - 29 HDD
Feb 14 - 29 HDD
Feb 16 - 29 HDD
Feb 23 - 29 HDD (DR Event)
2
1
0
-1
-2
-3
-4
-5
-6
0:00 1:00 2:00 3:00 4:00 5:00 6:00 7:00 8:00 9:00 10:00 11:00 12:00 13:00 14:00 15:00 16:00 17:00 18:00 19:00 20:00 21:00 22:00 23:00

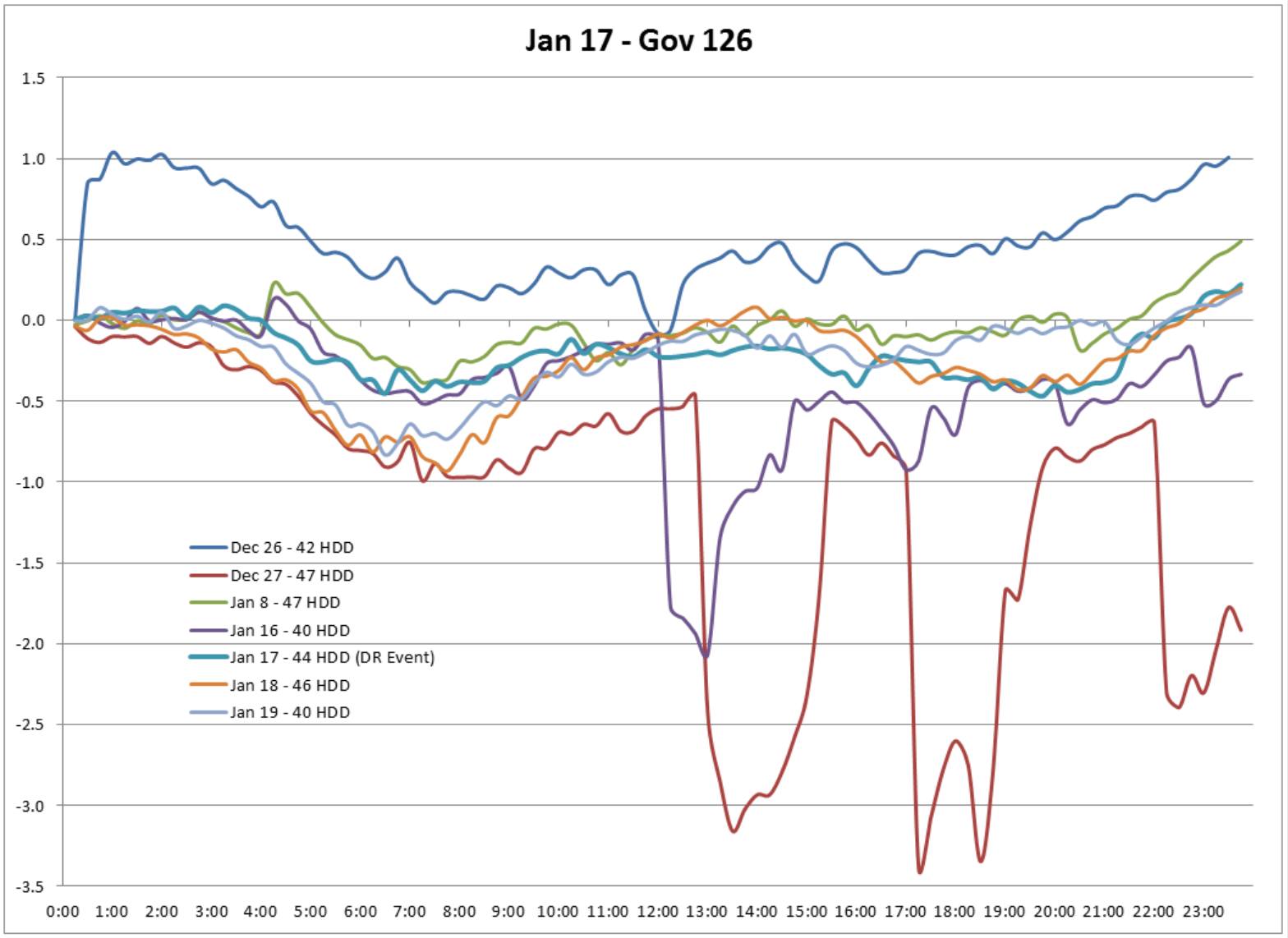

Feb 27 & 28 - ND 21
6.0
4.0
2.0
0.0
-2.0
-4.0
-6.0
-8.0
0:00 1:00 2:00 3:00 4:00 5:00 6:00 7:00 8:00 9:00 10:00 11:00 12:00 13:00 14:00 15:00 16:00 17:00 18:00 19:00 20:00 21:00 22:00 23:00
Jan 11 - 22 HDD
Jan 12 - 20 HDD
Jan 22 - 28 HDD
Jan 23 - 28 HDD
Feb 26 - 22 HDD
Feb 27 - 26 HDD (DR Event)
Feb 28 - 22 HDD (DR Event)
Mar 1 - 20 HDD

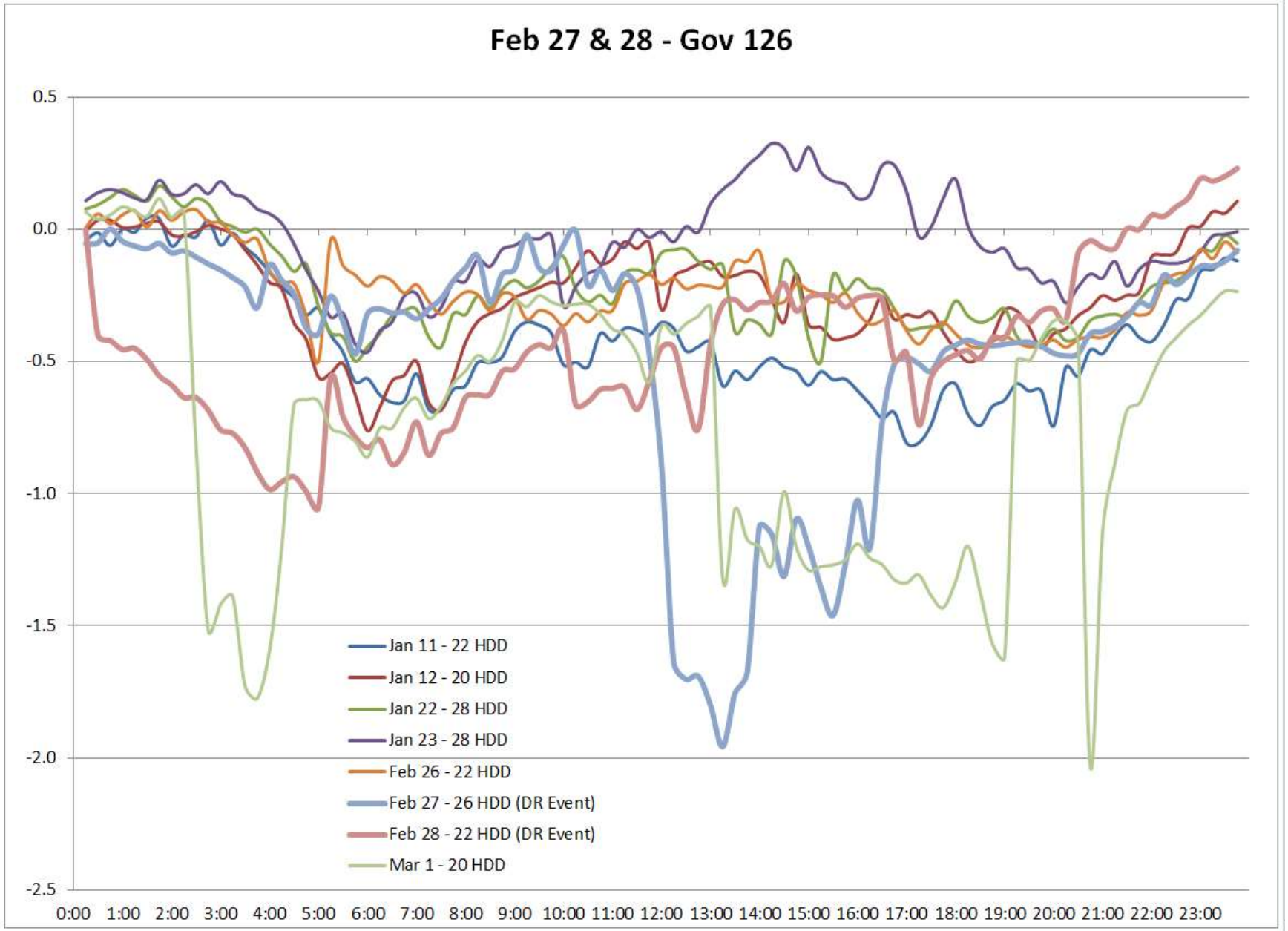
Gov 4 – KEDNY


Jan 17 - Gov 4
Dec 26 - 42 HDD
Dec 27 - 47 HDD
Jan 8 - 47 HDD
Jan 16 - 40 HDD
Jan 17 - 44 HDD (DR Event)
Jan 18 - 46 HDD
Jan 19 - 40 HDD
0.8
0.6
0.4
0.2
0.0
-0.2
-0.4
-0.6
-0.8
-1.0
-1.2
-1.4
0:00 1:00 2:00 3:00 4:00 5:00 6:00 7:00 8:00 9:00 10:00 11:00 12:00 13:00 14:00 15:00 16:00 17:00 18:00 19:00 20:00 21:00 22:00 23:00


Feb 27 & 28 - Gov 4
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
-1.2
-1.4
-1.6
0:00 1:00 2:00 3:00 4:00 5:00 6:00 7:00 8:00 9:00 10:00 11:00 12:00 13:00 14:00 15:00 16:00 17:00 18:00 19:00 20:00 21:00 22:00 23:00
Jan 11 - 22 HDD
Jan 12 - 20 HDD
Jan 22 - 28 HDD
Jan 23 - 28 HDD
Feb 26 - 22 HDD
Feb 27 - 26 HDD (DR Event)
Feb 28 - 22 HDD (DR Event)
Mar 1 - 20 HDD

Gov 126 – KEDNY




Feb 27 & 28 - Gov 126
0.5
0.0
-0.5
-1.0
-1.5
-2.0
-2.5
0:00 1:00 2:00 3:00 4:00 5:00 6:00 7:00 8:00 9:00 10:00 11:00 12:00 13:00 14:00 15:00 16:00 17:00 18:00 19:00 20:00 21:00 22:00 23:00
Jan 11 - 22 HDD
Jan 12 - 20 HDD
Jan 22 - 28 HDD
Jan 23 - 28 HDD
Feb 26 - 22 HDD
Feb 27 - 26 HDD (DR Event)
Feb 28 - 22 HDD (DR Event)
Mar 1 - 20 HDD

ENGINEERING SERVICES REIMBURSEMENT AGREEMENT

This Engineering Services Reimbursement Agreement ("*Agreement*"), effective as of this _____ day of _____ ("*Effective Date"),* is by and between _____ ("*Customer*"), a _____ organized and existing under the laws of _____ and _____ ("*Company*"), a corporation organized and existing under the laws of the State of New York.

WHEREAS, Customer is proposing to build an anaerobic digester within a _____ located in _____, New York that will recover digester methane gas from _____, with excess gas to be sent to Company's natural gas distribution system (the "*Project*"); and

WHEREAS, Customer desires to have Company perform certain engineering services (as specified below) in connection with the Project, and Company has agreed to perform such services upon the terms and conditions set forth below;

NOW, THEREFORE, in consideration of the mutual promises and covenants contained herein, and other good and valuable consideration, the receipt and sufficiency of which are hereby acknowledged, the parties entering into this Agreement (each a "*Party*", and collectively, the "*Parties*"), with the intent to be bound, agree as follows:

ARTICLE I – SERVICES

Section 1 - Scope of Services

Company will perform those services specified in Exhibit A attached hereto and hereby incorporated herein ("*Services*"). No goods, equipment, or materials will be provided under this Agreement.

This Agreement does not provide for generation interconnection service, procurement of equipment, installation or construction, or transmission service.

Section 2 - Customer's Responsibilities

Customer shall provide:

1. Complete and accurate information regarding requirements for Services, including, without limitation, constraints, space requirements and relationships, special equipment, systems, site requirements, underground or hidden facilities and structures, and all applicable drawings and specifications;

2. If and to the extent applicable, Company access to the site where Services will be performed;

3. A project manager who will be given the authority to coordinate all aspects of the Project between Customer and Company;

4. If and to the extent applicable, adequate parking for the vehicles of Company personnel performing the Services; and

5. Other responsibilities and access deemed necessary by, and in the sole discretion of, Company to facilitate performance of the Services.

Customer shall reasonably cooperate with Company as required to facilitate Company's performance of the Services. Other express Customer responsibilities, if any, shall be as specified in Exhibit A attached hereto.

Anything in this Agreement to the contrary notwithstanding, Company shall have no responsibility or liability under this Agreement for any defective performance or nonperformance to the extent such defective performance or nonperformance is caused by the inability or failure of (i) Customer to cooperate or to perform any of the tasks or responsibilities contemplated to be performed or undertaken by Customer in Exhibit A or elsewhere in this Agreement, or (ii) Customer and Company to reach agreement on any matter requiring their mutual agreement as contemplated in Exhibit A or elsewhere in this Agreement.

Section 3 - Unknown Conditions

Customer represents, warrants and covenants that all information provided by Customer is accurate and complete and acknowledges and agrees that Company may and will rely on this representation, warranty and covenant in performing under this Agreement. If, as a result of additional, different, or previously unknown information, any changes in Services are required that will result in an increase or decrease in the cost or time of performance under the Agreement, the Price, schedule and other affected provisions of this Agreement shall be equitably adjusted and this Agreement shall be amended in writing to memorialize such changes.

Section 4 - Changes and Extras

Customer may request changes in Services in writing. If any such changes will result in an increase or decrease in the cost or time of performance under this Agreement, the Price, schedule and other affected provisions of the Agreement shall be equitably adjusted and this Agreement shall be amended in writing to memorialize such changes. Company may make changes in Services with the prior written approval of Customer (which approval shall not be unreasonably withheld, conditioned, or delayed).

Section 5 - Governmental Requirements

Changes in Services may be necessary in order to meet the requirements of governmental authorities, laws, regulations, ordinances, Good Utility Practice (as such term is defined in Article V, Section 1, below) and/or codes. After Customer's approval (which shall not be unreasonably withheld, conditioned, or delayed), Company will make changes in Services as it deems necessary, in its sole discretion, to conform to such requirements. If any such changes will result in an increase or decrease in the cost or time of performance under this Agreement, the

Price, schedule and other affected provisions of this Agreement shall be equitably adjusted and this Agreement shall be amended in writing to memorialize such changes. If Customer withholds its approval, and in Company's sole and exclusive judgment the withholding of approval by Customer is not reasonable, then, at Company's election, this Agreement may be immediately terminated upon written notice to Customer. Nothing in this Agreement shall relieve Customer of the responsibility to comply with requirements of ISO-NE or other utilities with regard to the Project and the Services.

ARTICLE II – PRICE, TAXES, AND PAYMENT

Section 1 - Price

The price for the Services to be paid by Customer shall be the actual costs and expenses incurred by the Company and its affiliates in connection with performance of the Services or otherwise incurred by Company in connection with this Agreement, and shall include, without limitation, any such costs that may have been incurred by Company prior to the Effective Date (the "*Price*").

The Price shall include, without limitation, the actual costs and expenses for the following to the extent incurred in connection with performance of the Services: labor (including, without limitation, internal labor); materials; subcontracts; equipment; travel, lodging, and per diem paid in accordance with Company policy; copying and reproduction of materials, overnight delivery charges, certified mailing charges, first class mailing charges and similar types of incidental charges; transportation; carrying charges and surcharges; all applicable overheads including an Administrative and General (A&G) expense charge at Company's current rate at the time of invoicing; all federal, state and local taxes incurred; all costs and fees of outside experts, consultants, counsel and contractors; all other third-party fees and costs; and all costs of obtaining any required consents, releases, approvals, or authorizations. All invoiced sums will include applicable expenses, surcharges, and federal, state and local taxes.

If Customer claims exemption from sales tax, Customer agrees to provide Company with an appropriate, current and valid tax exemption certificate, in form and substance satisfactory to Company, relieving Company from any obligation to collect sales taxes from Customer ("*Sales Tax Exemption Certificate*"). During the term of this Agreement, Customer shall promptly provide Company with any modifications, revisions or updates to the Sales Tax Exemption Certificate or to Customer's exemption status. If Customer fails to provide an acceptable Sales Tax Exemption Certificate for a particular transaction, Company shall add the sales tax to the applicable invoice to be paid by Customer.

Section 2 – Payment

Customer shall provide Company with an initial prepayment in the amount of _____ US dollars ($_____.00) ("*Initial Prepayment*"). Company shall not be obligated to commence performance of Services until it has received the Initial Prepayment. If, during the performance of the Services, Company determines that one or more additional prepayments are required before completing the Services, Company may, but is not required to, request additional

prepayment from Customer; any such requests will be in writing. If an additional prepayment is requested and is not received from Customer on or before the date specified in each such request, or if no date is specified, within 30 days of receipt of the written request, Company may cease work upon the depletion of the Initial Prepayment and any other prepayments made by Customer to date, as applicable. Upon Company's receipt of the additional requested prepayment from Customer (such prepayment to be additional to the Initial Prepayment and any other prepayments made by Customer to date), Company will continue to perform the Services. The Initial Prepayment and the additional prepayments (if any) represent estimates only.

Company is not required to request additional prepayments from Customer and may elect, in its sole discretion, to continue performing Services hereunder after the depletion of the Initial Prepayment, or any other prepayments made by Customer to date, as applicable, without additional prepayments and invoice Customer for such Services at a later date. Customer shall be responsible to pay Company the total Price for completing the Services actually performed by Company whether or not any additional prepayments were made at Company's request. Any election by Company to seek or defer additional prepayments in one instance shall not obligate the Company to seek or defer additional prepayments in any other instance.

Company will invoice Customer for all sums owed under this Agreement. With the exception of additional prepayments required under the first paragraph of this Section 2 of Article II, in which case the due date provided in such paragraph shall apply, payment shall be due in full within thirty (30) days of Company's submittal of an invoice, without regard to claims or off-sets. Payment shall be made in immediately available funds transmitted by the method specified in the invoice. A continuing late payment charge of 1.5% per month will be applied on any late payments.

If Company's Price for completing the Services is less than the Initial Prepayment plus any such additional prepayments paid by Customer under this Article ("*Total Prepayment*"), Company will refund the remaining unused portion of the Total Prepayment to Customer.

ARTICLE III - SCHEDULE, DELAYS, AND FORCE MAJEURE

Company will use reasonable efforts to commence the Services promptly following its receipt of all of the following: a fully executed Agreement, the Initial Prepayment, and all information required by this Agreement to be supplied by Customer prior to commencement of the Services.

If Company's performance of the Agreement is delayed by Customer, an equitable adjustment shall be made for any increase in the cost and/or time of performance caused by the delay.

Any delays in, or failure of, performance by Customer or Company, other than payment of monies, shall not constitute default and shall be excused hereunder, if and to the extent such delays or failures of performance are caused by occurrences beyond the reasonable control of Customer or Company, as applicable, including, but not limited to, acts of God, Federal and/or state law or regulation, sabotage, explosions, acts of terrorism, unavailability of personnel, equipment, supplies, or other resources for utility-related duties, delays by governmental authorities in granting licenses, permits or other approvals necessary in connection with Services,

compliance with any order or request of any governmental or judicial authority, compliance with Company's public service obligations, storms, fires, inclement or adverse weather, floods, riots or strikes or other concerted acts of workers, and accidents.

ARTICLE IV – INTELLECTUAL PROPERTY

Any drawings, specifications or other documents (i) prepared or used by Company, or (ii) prepared by Customer for Company in connection with this Agreement, shall be the proprietary, confidential information and sole property of Company at no cost to Company (collectively "*Materials*").

Excluding third-party owned documents and software, Customer is granted an irrevocable, nontransferable, and non-assignable license to use such Materials solely in connection with the Project. No commercialization of such Materials by Customer is authorized. Customer shall not disclose any of the Materials to any third party, in whole or in part, without the prior written consent of Company.

The obligations imposed by this Article IV shall survive the completion, cancellation, or termination of this Agreement.

ARTICLE V – PERFORMANCE

Section 1 -- Performance

Company shall perform the Services in a manner consistent with "Good Utility Practice" (as such term is defined below); provided, however, that Company shall have no responsibility or liability in connection with (i) any items or services provided by Customer or its third party contractors or representatives whether or not such items or services are incorporated in the Services, (ii) any items or services provided, manufactured or licensed by third parties whether or not such items or services are incorporated in the Services, or (iii) any defects in Services that result from the acts or omissions of persons other than Company or accidents not caused by Company.

"*Good Utility Practice*" shall mean the practices, methods and acts engaged in or approved by a significant portion of the electric utility industry during the relevant time period, or any practices, methods and acts which, in the exercise of good judgment in light of the facts known at the time the decision was made**,** would have been reasonably expected to accomplish the desired result consistent with good business practices, safety, and law. Good Utility Practice is not intended to require or contemplate the optimum practice, method or act, to the exclusion of all others, but rather to be reasonably acceptable practices, methods, or acts generally accepted in the region in which the Services are to be performed.

Prior to the expiration of one (1) year following the date of completion of a Service, Customer shall have the right to give Company written notice that some or all of such Service was not performed in compliance with the first paragraph of this Section 1, and the Company shall, at the option of Company, either (i) re-perform or repair the defective portion of such Service, or (ii)

refund the amount of money paid by the Customer to Company attributable to the defective portion of such Service. The remedy set forth in this Section 1 of Article V is the sole and exclusive remedy granted to Customer for any failure of Company to meet the performance standards or requirements set forth in this Agreement.

ARTICLE VI – INSURANCE

From the commencement of the Agreement through its expiration, each Party shall provide and maintain, at its own expense, insurance policies issued by reputable insurance companies with an A. M. Best rating of at least B+ (collectively, the "*Required Insurance Policies*"). The Required Insurance Policies shall, at a minimum, include the following coverages and limitations:

Workers' Compensation and Employers Liability Insurance, as required by the State in which the work activities under this Agreement will be performed. If applicable, coverage will include the U.S. Longshoremen's & Harbor Workers' Compensation Act, and the Jones Act. If a Party is a qualified self-insurer by the State, Excess Workers' Compensation coverage shall be maintained in lieu of the Workers' Compensation coverage.

Public Liability, including Contractual Liability and Products/Completed Operations coverage, covering all operations to be performed under this Agreement, with minimum limits of:

Bodily Injury	$1,000,000 per occurrence Property Damage $1,000,000 per occurrence

Automobile Liability, covering all owned, non-owned and hired vehicles used under or in connection with this Agreement, with minimum limits of:

Bodily Injury	$500,000 per occurrence
Property Damage	$500,000 per occurrence
OR	
Combined Single Limit	$1,000,000 per occurrence

If requested, each Party will provide evidence to the other Party that it maintains the Required Insurance Policies required under this Article.

Either Party may elect to self-insure to the extent authorized or licensed to do so under the applicable laws of the State of New York, provided, that, the electing Party provides written notice of any such election to the other Party. Company hereby notifies Customer that it is a qualified self-insurer under the applicable laws of the State of New York and that it elects to self-insure to satisfy its obligations under this Article.

ARTICLE VII – TERM AND TERMINATION

The term of this Agreement shall expire one (1) year from the Effective Date. As of the expiration of this Agreement or, if earlier, its termination, the Parties shall no longer be bound by the terms and provisions hereof, except (a) to the extent necessary to enforce the rights and

obligations of the Parties arising under this Agreement before such expiration or termination (including, without limitation, with respect to payment of all amounts due and payable hereunder), and (b) such terms and provisions that expressly or by their operation survive the termination or expiration of this Agreement.

Either Party may terminate this Agreement for convenience by delivery of written notice to the other Party, such termination to be effective on the tenth (10th) day following delivery of such written notice, or upon payment in full of all amounts due and payable hereunder, whichever is later. On or before the effective termination date of this Agreement, Customer shall pay Company all amounts due and payable as the Price for that portion of the Services performed to the effective date of termination ("*Amount Outstanding*"), including, without limitation, all costs and expenses incurred, less the Total Prepayment. In the event that the Total Prepayment exceeds the Amount Outstanding, Company shall remit the balance to Customer.

ARTICLE VIII – MISCELLANEOUS PROVISIONS

Section 1 - Assignment and Subcontracting

Customer agrees that Company has the right, but not the obligation, to (i) use the services of its affiliated companies in connection with the performance of Services, and (ii) issue contracts to third parties for, or in connection with, the performance of Services hereunder, without the prior consent of Customer, and that the costs and expenses of such affiliated companies or third parties charged or chargeable to Company shall be paid by Customer as part of the Price.

Section 2 – No Third-Party Beneficiary

Nothing in this Agreement is intended to confer on any person, other than the Parties, any rights or remedies under or by reason of this Agreement.

Section 3 – Amendment; Equitable Adjustments

This Agreement shall not be amended, superseded or modified, except in a writing signed by both Parties. In any circumstance in which this Agreement contemplates an equitable adjustment to Price, schedule or any other term of this Agreement, Company shall have no obligation to continue performance hereunder until and unless such equitable adjustment has been mutually agreed to by both Parties in writing.

Section 4 – Notices

Any notice given under this Agreement shall be in writing and shall be hand delivered, sent by registered or certified mail, delivered by a reputable overnight courier, or sent by facsimile (fax) with electronic confirmation of receipt, to the party's representatives as follows:

Customer:

[_____]
ATTN: [_____]
[_____]
[_____]
Phone: [_____]
Fax: [_____]
Email: [_____]

Company:

[_____]
ATTN: [_____]
[_____]
[_____]
Phone: [_____]
Fax: [_____]
Email: [_____]

Section 5 - Waiver

No term of this Agreement may be waived except in a writing signed by an authorized representative of the Party against whom the amendment, modification, or waiver is sought to be enforced. Waiver of any provision herein shall not be deemed a waiver of any other provision herein, nor shall waiver of any breach of this Agreement be construed as a continuing waiver of other breaches of the same or other provisions of this Agreement.

Section 6 - Approvals

It is understood that Company may be required to obtain, regulatory, and other third- party approvals and releases in connection with the provision of the Services. If so, this Agreement shall be effective subject to the receipt of any such approvals and releases, in form and substance satisfactory to Company in its sole discretion, and to the terms thereof.

Section 7 - Laws

This Agreement shall be interpreted and enforced according to the laws of the State of New York and not those laws determined by application of the State of New York's conflicts of law principles. Venue in any action with respect to this Agreement shall be in the State of New York; each Party agrees to submit to the personal jurisdiction of courts in the State of New York with respect to any such actions.

Section 8 - Severability

To the extent that any provision of this Agreement shall be held to be invalid, illegal or unenforceable, it shall be modified so as to give as much effect to the original intent of such provision as is consistent with applicable law and without affecting the validity, legality or enforceability of the remaining provisions of the Agreement.

Section 9 - Integration and Merger; Entire Agreement

Customer and Company each agree that there are no understandings, agreements, or representations, expressed or implied, with respect to the subject matter hereof other than those expressed herein. This Agreement supersedes and merges all prior discussions and understandings with respect to the subject matter hereof, and constitutes the entire agreement between the Parties with respect to such subject matter.

Section 10 – Authority

Each Party represents to the other that the signatory identified beneath its name below has full authority to execute this Agreement on its behalf.

Section 11 – Information and Coordination Contact

[____ Name, contact information _____] or such other representative as Company may designate, will be the point of contact for Customer to submit the information required for Company to perform the Services stated in this Agreement.

[_____Name, contact information_____] or such other representative as Customer may designate, will be the point of contact for Company to request additional information from Customer, if required.

Section 12 – Counterparts

This Agreement may be executed in multiple counterparts, each of which shall be considered an original, and all of which together shall constitute one and the same agreement. The exchange of copies of this Agreement and of signature pages by facsimile or other electronic transmission (including, without limitation, by e-mailed PDF) shall constitute effective execution and delivery of this Agreement as to the Parties and may be used in lieu of the original Agreement for all purposes. Signatures of the Parties transmitted by facsimile or other electronic means (including, without limitation, by e-mailed PDF) shall be deemed to be their original signatures for all purposes.

IN WITNESS WHEREOF, the Parties hereto have caused this Agreement to be executed by their duly authorized representatives as of the Effective Date.

[_____Customer Name_____]
By: [_____]
Name: [_____]
Title: [_____]

[_____Company Name_____]
By: [_____]
Name: [_____]
Title: [_____]

EXHIBIT A - Scope of Services

Company's scope of Services shall be:

- Assign a Project Engineer and Project Manager to provide technical support for the Project;
- Arrange and schedule periodic Project meetings;
- Provide standards for Customer to follow in order to design metering equipment in accordance with Company specifications;
- Provide the specifications for the meters to be installed and determine the size and quantity of meters required;
- Provide technical assistance as needed by Customer in reviewing the design and layout for analytical equipment to be installed by Customer in accordance with manufacturer's recommendations;
- Provide technical assistance as needed by Customer in reviewing the design and layout for odorant equipment to be installed by Customer in accordance with applicable health and safety codes for the storage of odorant, including DEC, DEP, and Suffolk County Department of Health;
- Review drawings and specifications created by Customer for the equipment set forth below. Company reserves the right to make changes to the design in order to meet National Grid standards; and
- Provide engineering services to assist Customer in design and development of specifications for the work to purchase and install the equipment and facilities set forth below; and
- Assign process safety engineers to ensure National Grid safety criteria are met at the interconnect point and to perform any required process safety assessments.

Equipment and Facilities Required for Project (to be provided by Customer):

- Gas service and associated metering equipment for back up supply from Company
- Gas outlet system tie-in and associated metering equipment for gas produced on site
- Remote Terminal Unit (RTU) to transmit gas quality and flow data to Company's Gas Control Room
- Gas Chromatograph (10 component) to measure BTU, inerts (CO2, N2), oxygen of digester gas
- Odorant Chromatograph to measure mercaptans, total sulfur, and H2S in the digester gas
- Moisture Meter to measure amount of H2O in the digester gas
- Remote control valve to enable remote shut-in of Customer's outlet in cases where gas from the plant is out of specification as listed in Table below.
- Odorant injection system with sight glass diffusion probe, storage tank(s) with dike
- Gas filters with differential gages on plant outlet line
- Analyzer Building – prefab concrete building to house RTU and all analytical equipment with electric service and Power Conditioning, and Battery Back Up system, gas detector(s)
- Odorant Building – negative pressure concreted building to house odorant equipment with electric service and gas detector(s), charcoal filter, blower, fire suppression and monitoring equipment (as required by _____ Fire Marshall).

Assumptions and Conditions:

Any dates, schedules or cost estimates resulting from the Services are preliminary projections/estimates only and shall not become or give rise to any binding commitment.

The Services contemplated by this Exhibit and this Agreement do not include any construction, relocations, alterations, modifications, or upgrades with respect to any facilities ("Construction"), nor does Company make any commitment to undertake such Construction. If the Parties elect, in their respective sole discretion, to proceed with any Construction: (i) such Construction would be performed pursuant to a separate, detailed, written, and mutually acceptable Cost Reimbursement Agreement to be entered into by the Parties prior to the commencement of any such Construction, and (ii) payment of all actual costs incurred by Company or its Affiliates in connection with or related to such Construction shall be the responsibility of Customer and Customer shall reimburse Company for all such costs.

For the avoidance of doubt: This Agreement does not provide for generation interconnection service, procurement of equipment, installation or construction. The Company shall not have any responsibility for seeking or acquiring any real property rights in connection with the Services or the Project including, without limitation, licenses, consents, permissions, certificates, approvals, or authorizations, or fee, easement or right of way interests. Neither this Agreement nor the Services include securing or arranging for Customer or any third party to have access rights in, through, over or under any real property owned or controlled by the Company.

EXHIBIT B – Labor Rates

New York	**Rate Per Hr Total Salary**
Sr. Engineer	$ 96.43
Principle Engineer	$ 123.64
Engineer Manager	$ 128.19
Associate Project Manager	$ 64.23
Project Manager	$ 79.45
Sr. Project Manager	$ 96.43
Lead Project Manager	$ 109.90
Principle Project Manager	$ 123.64

Note: These fully-loaded labor rates for the specified titles are provided as examples only. The labor rates actually incurred may vary from the above rates based upon the actual salaries and titles of the individuals providing the services.

GAS SALES (INTERCONNECT) AGREEMENT

This Digester Gas Sales Agreement ("*Agreement"*), effective as of this _____ day of _____ ("*Effective Date*"), is by and between _____ ("Buyer"), a corporation organized and existing under the laws of the State of New York, and _____ ("Seller"), a _____ organized and existing under the laws of _____.

WHEREAS, Seller owns an anaerobic digester within a _____ located in _____, New York that recovers digester methane gas from _____; and

WHEREAS, Buyer is a regulated natural gas distribution company which owns and operates a natural gas distribution system in _____ counties; and

WHEREAS, Seller desires to sell and deliver Renewable Natural Gas ("RNG" or Biomethane) to Buyer, and Buyer desires to purchase and accept such RNG from Seller; and

WHEREAS, Buyer has agreed to operate and maintain certain of the facilities required in connection with the delivery of RNG, and Seller has agreed to reimburse Buyer for performing such operation and maintenance services; and

NOW, THEREFORE, in consideration of the mutual promises and covenants contained herein, and other good and valuable consideration, the receipt and sufficiency of which are hereby acknowledged, the parties entering into this Agreement (each a "*Party*", and collectively, the "*Parties*"), with the intent to be bound, agree as follows:

ARTICLE 1: DEFINITIONS

1.1 The term "Btu" means British Thermal Unit, and shall be the quantity of heat required to raise the temperature of one (1) pound of water one (1) degree Fahrenheit at sixty (60) degrees Fahrenheit at a pressure of 14.73 psia.

1.2 The term "dekatherm" means a unit of heat energy equal to 1,000,000 BTUs.

1.3 The term "Day" means a period of twenty-four (24) consecutive hours beginning and ending at 9:00 AM Central Standard Time.

1.4 The term "Delivery Point" shall mean the point of interconnection between the facilities of Seller and Buyer at or near the facility where RNG will be sold and delivered by Seller to Buyer under this Agreement, as shown on Exhibit "A" hereto.

[Insert Schematic drawing]

1.5 Facilities" means those facilities that will be maintained by the Buyer pursuant to this Agreement and other facilities utilized in connection with the delivery of RNG.

1.6 The term “Maximum Daily Quantity” (or "MDQ') is the maximum amount of RNG that Buyer is obligated to purchase on any Day during the term of this Agreement.

1.7 The term “MMbtu” means one million Btu.

1.8 The term “mdth” means one thousand dekatherms.

1.9 The term “Month” means a period beginning at 9:00 AM Central Standard Time on the first Day of any calendar month and ending at 9:00 AM Central Time on the first Day of the next succeeding calendar month (per NAESB and FERC).

1.10 The term “Plant” means the digester and processing facilities operated by Seller.

1.11 The term “RNG” means the gas produced by Seller at the Plant.

1.12 “Services” has the meaning set forth in Article 8 of this Agreement.

1.13 “Pipeline Quality” has the meaning defined in latest version of AGA Report 4A.

ARTICLE 2: EFFECTIVE DATE AND TERM

2.1 The term of the Agreement shall commence as of the date first written above and shall remain in effect through _____, and from month to month thereafter unless terminated by either Party on no less than thirty (30) days prior written notice to the other.

2.2 Upon the termination of this Agreement for any reason, any monies due and owing Seller or Buyer shall be paid pursuant to the terms hereof, and any corrections or adjustments to payments previously made shall be determined and made at the earliest possible time. The provisions of this Agreement shall remain in effect until the obligations under this paragraph have been fulfilled.

ARTICLE 3: SALE AND PURCHASE OBLIGATIONS

3.1 Subject to the terms and conditions of this Agreement, Seller agrees to sell and deliver, and Buyer agrees to purchase and receive, each Day during the term of this Agreement, at the Delivery Point, a quantity of RNG equal to the lesser of (a) the quantity of RNG produced by the Plant on such Day or (b) the MDQ for such Day.

3.2 As of the effective date of this Agreement, the MDQ shall be___MMBtu.

3.3 Seller shall tender RNG for delivery at a substantially uniform rate of flow throughout each Day, at a minimum of 0 mdth/day and a maximum of _____ mdth/day. If Seller becomes aware that the rate of delivery or the total quantity of RNG that Seller will deliver for any Day will differ by more than twenty-five percent (25%) (positive or negative) from that achieved the previous Day, Seller shall so notify Buyer's Gas Control

Center at the contact set forth in Section 13.10 below. Seller also shall notify Buyer's Gas Control Center at least twenty-four (24) hours in advance of any suspension of RNG deliveries under this Agreement necessitated by Seller's maintenance of its Plant.

ARTICLE 4: PRICE OF GAS

4.1 The price paid for each MMBtu of RNG sold and purchased under this Agreement in any Month shall be equal to the New York Mercantile Exchange (NYMEX) natural gas futures contract last day settle price for such Month.

ARTICLE 5 TITLE TO GAS

5.1 Seller hereby warrants good and merchantable title to all RNG delivered hereunder, free and clear of all liens, encumbrances and claims whatsoever. Seller will indemnify Buyer and hold it harmless from any and all suits, actions, debts, accounts, damages, costs, losses, and expenses arising from or out of adverse title claims of any and all persons to said RNG.

5.2 Title to all RNG received by Buyer shall pass to Buyer at the Delivery Point. As between the Parties hereto, Seller shall be deemed to be in exclusive control and possession of the RNG deliverable hereunder and responsible for any damage or injury caused thereby until the same shall have been delivered to Buyer at the Delivery Point; thereafter Buyer shall be deemed to be in exclusive control and possession of such gas and responsible for any damage or injury caused thereby.

ARTICLE 6: GAS PRESSURE, TEMPERATURE AND QUALITY

6.1 Seller shall tender RNG for delivery to Buyer under this Agreement at the Delivery Point at pressures sufficient for such RNG to enter Buyer's facilities at such point, but in no event in excess of the maximum allowable operating pressure on Buyer’s system which, at the time of execution of this Agreement, is _____ psig. Buyer shall promptly notify Seller of any changes in the maximum operating pressure of the Buyer's system.

6.2 Seller shall tender RNG for delivery to Buyer under this Agreement at the Delivery Point at a temperature no less than _____ degrees Fahrenheit and no greater than _____ degrees Fahrenheit. Should Seller tender RNG to Buyer at the Delivery Point at a temperature colder or warmer than such range and Buyer's meter is damaged as a result, then in addition to and without limitation of any other remedy Buyer may have, Buyer shall be entitled to receive from Seller an amount equal to Buyer's cost to repair or replace such meter and any other related equipment affected.

6.3 Seller agrees that it will exercise reasonable care and diligence in tendering RNG for delivery to Buyer under this Agreement, and warrants that all RNG when tendered for delivery to Buyer hereunder at the Delivery Point shall:

a. be compatible and interchangeable with pipeline gas as defined in 16 NYCRR 229;

b. be within the limits set forth in Table 1
c. be monitored as to conformity with all of the foregoing criteria by manual test or by mutually acceptable continuous monitoring equipment; and Buyer will require quarterly random grab sampling to ensure gas is free of objectionable materials with analytical costs to be reimbursed by the Seller.

Table 1: Gas Quality Specifications

Gas Quality Specification	Low	High
BTU Content (Heat Content) [BTU/scf]		
Wobbe Number		
Relative Density		
Water Vapor Content [lb/MMscf]		
Mercaptans (as Odorant) [lb/MMscf]		
Hydrocarbon Dew Point, [°F] CHDP		
Hydrogen Sulfide (H_2S)		
Total Sulfur		
Total Diluent Gases **Carbon Dioxide (CO_2)** **Nitrogen (N_2)** **Oxygen (O_2)**		
Hydrogen	-	
Total Bacteria	-	Not Detectable
Mercury	-	Not Detectable
Other Volatile Metals (including arsenic)	-	Not Detectable
Siloxanes	-	Not Detectable
Ammonia	-	Not Detectable
Non-Halogenated Semi-Volatile and Volatile Compounds	-	Not Detectable
Halocarbons	-	Not Detectable
Aldehyde/Ketones	-	Not Detectable

NOTES:

1. *Not-detectable for purposes of this specification is defined as a value less than the lowest detectable level for a mutually agreeable standard industry analytical test method*
2. *BTU = commonly referred to as Higher Heating Value (HHV)*
3. *Wobbe = Interchangeability parameter; ratio of BTU content to specific gravity*
4. *In addition to the specified limits above, gas received into Buyer's pipeline system shall be pipeline quality and as such remain commercially free of objectionable materials and merchantable as defined in latest edition of AGA Report 4A "Natural Gas Contract Measurement and Quality Clauses"*

6.4 Seller shall maintain in good working order its facilities at the Plant that enable it to ensure that the pressure, temperature and quality of the RNG it tenders for delivery under this Agreement fully conform with the criteria set forth in this Agreement.

6.5 In addition to any and all other remedies that it may have, Buyer shall have the right to reject as non-conforming any RNG Seller tenders for delivery under this Agreement that fails to comply with the pressure, temperature or quality specifications set forth in this Agreement, and will maintain suitable equipment at Seller's premise in order to remotely monitor and shut off Seller's supply should it not meet such specifications.

6.6 The Parties shall develop a facility start-up gas quality sampling and testing plan (the "Plan") to ensure all equipment is functioning as and intended in order to provide RNG conforming to the quality specifications set forth in Table 1 above. The Plan shall include provisions regarding frequency of initial testing.

ARTICLE 7: GAS MEASUREMENT

7.1 The quantity of RNG delivered hereunder shall be measured according, to Boyle's and Charles' Laws for the measurement of gas under varying temperatures and pressures and shall be determined as follows:

a. the sales unit of the RNG delivered shall be one (1) MMBtu of gas measured as HHV on a real, dry, basis at standard temperature and pressure;
b. the unit of weight for the purpose of measurement shall be one (1) pound mass of gas;
c. the average absolute atmospheric pressure shall be assumed to be 14.73 pounds per square inch; and
d. the temperature of gas passing through the meter shall be determined by the continuous use of a temperature measuring device; the arithmetic averages of the temperature recorded each twenty-four (24) hour Day shall be used in computing gas volumes or continuous instantaneous temperature measurements may be applied to metering instruments to provide the volume computation.

7.2 The metering equipment shall be sealed and the seals shall be broken only upon occasions when the meters are to be inspected, tested or adjusted, and representatives of Seller shall be afforded at least twenty-four (24) hour notice and reasonable opportunity to be present upon such occasions. Buyer shall use reasonable efforts to give Seller more than twenty-four (24) hour notice of such inspections, tests or adjustments.

7.3 Periodic tests of such metering equipment, at intervals not to exceed two times per year, will be made at any reasonable time upon request there for by Seller. If, as a result of any such additional test, the metering equipment is found to be defective or inaccurate, it will be restored to a condition of accuracy or replaced. If an additional test of the metering equipment is made at the request of Seller with the result that said metering equipment is found to be registering correctly or within two percent (2%) plus or minus of one hundred

percent (100%) accuracy, Seller shall bear the expense of such additional test. If such additional test shows an error greater than two percent (2%) plus or minus of one hundred percent (100%) accuracy, then Buyer shall bear the expense of such additional test and any necessary repair or replacement.

7.4 All meters shall be adjusted as close as practical to one hundred percent (100%) accuracy at time of installation and testing. If any of the metering equipment tests provided for herein disclose that the error for such equipment exceeds two percent (2%) plus or minus of one hundred percent (100%) accuracy, and the period of inaccuracy cannot be reasonably ascertained, then the period of inaccuracy will be assumed to have begun at the midpoint in time between the discovery of the inaccuracy and the previous meter test.

7.5 Any correction in billing resulting from such correction in meter records shall be made in the next monthly invoice rendered by Buyer after the inaccuracy is discovered. Should any metering equipment fail to register the gas delivered or received during any period of time, the amount of RNG delivered or received during such period will be estimated by the Parties according to the amounts previously delivered or received during similar periods under substantially similar conditions, and upon mutual agreement of the Parties shall be used as the basis for billing for that period.

ARTICLE 8: OPERATION and MAINTENANCE SERVICES, EQUIPMENT REPLACEMENT COSTS

8.1 **SCOPE** – During the term of this Agreement the Buyer will perform, or cause to be performed, in a prudent and workman like manner the Services set forth in Section 8.2 below. Upon the mutual agreement of the Parties, the Buyer may perform additional Services (the "Unscheduled Services") in connection with the Facilities. In the case of emergencies that render the Facilities unsafe, the Buyer may perform emergency services that it deems necessary to make the Facilities safe (the "Emergency Services"), including shutting off gas supply and the gas delivery. The Buyer shall attempt to notify Seller prior to commencing any such Emergency Services, however if prior notification is impractical, the Buyer shall have the right to commence the Emergency Services immediately and to notify Seller within 24 hours thereafter.

8.2 **SERVICES** – During the term of this Agreement, the Buyer shall provide the labor and materials necessary to operate and maintain the gas meters, gas regulators, odorant system, gas chromatographs, telephone lines and other ancillary equipment required by the Company in connection with the delivery of RNG pursuant to this Agreement (the "Services"). The Services do not include repairs for damages, malfunctions or failures caused by or occurring as the result of: (a) repairs, adjustments or any other actions performed by persons other than the Buyer's authorized representatives; (b) failure of components not serviced by the Buyer's authorized representatives; (c) abuse, misuse or negligent acts of Seller or others; or (d) an event of force majeure as defined in Article 11 hereof. Installation of the equipment described above is the Seller's responsibility.

8.3 **COST OF SERVICES** – Seller shall reimburse the Company for the fully loaded cost incurred by the Company in performing the Services, Unscheduled Services and/or Emergency Services.

8.4 **EQUIPMENT REPLACEMENT AT END OF LIFE** – Seller shall reimburse the Company for the fully loaded cost to replace gas meters, gas regulators, odorant system, gas chromatographs, telephone lines and other ancillary equipment when such equipment reaches the end of its service life.

ARTICLE 9: BILLING AND PAYMENT

9.1 On or before the fifth (5th) day of each Month, Buyer shall notify Seller of the quantity of RNG delivered by Seller to Buyer during the preceding Month. Seller shall render a written statement to Buyer on or before the fifteenth (15th) day of such succeeding Month which, upon verification by Buyer, shall be paid by Buyer by the twenty-fifth (25th) day of such Month. If the twenty-fifth (25th) day of any Month falls on a weekend or bank holiday, payment by Buyer shall be due on the next succeeding business day.

9.2 The fully loaded costs incurred by the Buyer in performing any Services, Unscheduled Services, and/or Emergency Services will be applied as an offset to the amount invoiced by Seller pursuant to Section 9.1 above.

9.3 **AUDITS** – Each Party shall have the right at its own expense to examine and audit at a reasonable time and upon reasonable prior notice the books, records and charts of the other Party relevant to this Agreement. Each Party shall use reasonable efforts to make available such records as may be necessary to verify the accuracy of any statements or charges made under or pursuant to any of the provisions of this Agreement. A formal audit of accounts shall not be made more than once each calendar year.

ARTICLE 10: ACCESS TO PREMISES

10.1 Seller agrees during the term of this Agreement that it will provide access as may be required by the Buyer's authorized representatives for the performance of its obligations hereunder. Upon 24 hours' notice, Seller shall grant access to, or obtain access for, the Buyer's authorized representatives for performance of the Services and the Unscheduled Services. Furthermore, Seller shall grant or obtain immediate access for the Buyer's authorized representatives for the performance of Emergency Services.

ARTICLE 11: FORCE MAJEURE

11.1 The term force majeure as employed herein shall mean acts of God, strikes, lockouts or other industrial disturbances, acts of the public enemy, wars, blockades, insurrections, riots, epidemics, landslides, lightning earthquakes, fires, storms, floods, washouts, arrests, the order of any court of governmental authority having jurisdiction while the same is in force and effect, civil disturbances, explosions, breakage, accidents to machinery or lines or pipe, freezing of or damage to facilities, inability to obtain or unavoidable delay in

obtaining material, equipment, and any other cause whether of the kind herein enumerated or otherwise, not reasonably within the control of the Party claiming suspension and which by the exercise of due diligence such Party is unable to prevent or overcome.

11.2 In the event of either Party being rendered unable, wholly or in part, by force majeure to carry out its obligations (other than the continuing obligation set forth herein below), it is agreed that on such Party's giving notice and full particulars of such force majeure in writing or by telegraph or telecopy to the other Party within a reasonable time (not to exceed five (5) days) after occurrence of the cause relied on, the obligations of both Parties, so far as they are affected by such force majeure, shall be suspended during such period of force majeure, but for no longer period, and such cause shall so far as possible be remedied with all reasonable dispatch.

11.3 Neither Party shall be liable in damages to the other for any act, omission or circumstance occasioned by, or in consequence of, force majeure, as herein defined. Such causes or contingencies affecting the performance by either Party, however, shall not relieve it of liability unless such Party shall give notice and full particulars of such cause or contingency in writing, to the other Party at the address set forth in Section 13.10 within a reasonable time after the occurrence relied upon, nor shall such causes or contingencies affecting the performance by either Party relieve it of liability in the event of its failure to use due diligence to remedy the situation and remove the cause with all reasonable dispatch, nor shall such causes or contingencies affecting the performance relieve Buyer from its obligation to make payments of amounts in respect of RNG delivered.

11.4 To the extent that, in Buyer's sole judgment. Buyer's ability to receive, measure monitor and/or odorize RNG is impaired by conditions on its system including, but not limited to, the performance of routine maintenance or repairs, then Buyer's obligation to purchase and receive such RNG shall be suspended for the duration of such condition.

ARTICLE 12: EVENTS OF DEFAULT

12.1 **EVENTS OF DEFAULT** – The occurrence of anyone or more of the following shall be an "Event of Default" under this Agreement:

a. Failure by a party to pay/reimburse any amount when due and payable that is required to be paid by the terms of this Agreement.

b. Failure by a party to perform any covenant, condition or agreement required to be performed by it by the terms of this Agreement that continues for a period of ten (10) days after the required date of performance.

12.2 **REMEDIES ON DEFAULT**

a. The non-defaulting party shall have the right, upon written notice to the defaulting party, to terminate this Agreement upon any Event of Default.

b. Upon any Event of Default by the Buyer, Seller, or a designee of Seller, may cure any breach or default of the Company under this Agreement that resulted in an Event of Default (including the failure to perform Services), in which case the full cost thereof shall be reimbursed to Seller by the Buyer.

ARTICLE 13: MISCELLANEOUS

13.1 Except as provided hereinafter, neither this Agreement nor any rights or obligations hereunder may be assigned or transferred, by operation of law or otherwise by either Party without the prior written consent of the other Party, which consent shall not be unreasonably withheld. Notwithstanding the foregoing, Buyer may assign this Agreement and all of its rights and obligations to an affiliate of Buyer at any time upon 30 days prior written notice to Seller.

13.2 Seller shall provide, at no cost to Buyer, all of the electricity and compressed air required for Buyer to operate the facilities that will measure, regulate and odorize the RNG gas delivered by Buyer to Seller under this Agreement at Buyer's facilities for such purposes located at or near the Delivery Point.

13.3 The sale and delivery of RNG by Seller and the purchase and receipt thereof by Buyer are subject to all valid legislation with respect to the subject matter hereof and to all valid present and future orders, rules and regulations of duly constituted authorities having jurisdiction. Neither Buyer nor Seller shall be liable to the other for failure to perform any obligation hereunder where such failure is due to compliance with such valid laws, orders, rules or regulations. If any statute, order, rule, or regulation of a duly constituted authority having jurisdiction over a Party or the performance of this Agreement prevents Seller from charging or collecting the price or prices payable hereunder or prevents Buyer from recovering costs representing the price or prices payable hereunder, the following shall apply notwithstanding any other provision of this Agreement:

a. If Buyer is prevented from recovering any costs representing all or a portion of the price or prices payable hereunder, or Buyer’s recovery of such costs is made subject to refund, Buyer may, at its option, terminate this Agreement by written notice to Seller, effective not less than sixty (60) days after delivery thereof;
b. If Seller is prevented from charging or collecting all or any part of the price or prices payable hereunder, or Seller's collection of such prices is made subject to refund, Seller may, at its option, terminate this Agreement by written notice to Buyer, effective not less than sixty (60) days after delivery thereof.

13.4 This Agreement sets forth all understandings between the Parties respecting the terms and conditions of this transaction. All other agreements, understandings and representations by and between the Parties hereto prior to this Agreement, whether consistent or inconsistent, oral or written, concerning this transaction are merged into and superseded by this written Agreement.

13.5 All headings appearing herein are for convenience only and shall not be considered a part

of this Agreement for any purpose.

13.6 The Parties may, by mutual agreement, waive any provision herein; however, a waiver shall not be construed to constitute a continuing waiver hereunder and furthermore, a waiver by either Party of any one or more defaults by the other Party in performance of any provision of this Agreement shall not operate or be construed as a waiver of future default or defaults, whether of a like or different character.

13.7 Seller hereby agrees to indemnify and hold harmless Buyer from damage to Buyer's or third parties' property or injury to persons (including death) to the extent resulting from the negligence of Seller, its servants, agents or employees, while engaged in activities under this Agreement. Buyer shall indemnify and hold harmless Seller from damage to Seller's or third parties' property or injury to persons (including death) to the extent resulting from the negligence of Buyer, its servants, agents or employees while engaged in activities under this Agreement except to the extent Buyer's Schedule for Gas Service (as filed with and approved by the Public Service Commission of the State of New York), limits Buyer's liability. The obligations under this Section shall survive termination of this Agreement.

13.8 This agreement shall be governed by and construed in accordance with the laws of the state of New York, without regard to any rules governing conflicts of laws that would require application of the laws of another jurisdiction.

13.9 This Agreement may be executed in several counterparts, each of which is an original and all of which constitute one and the same instrument.

13.10 Unless otherwise specified, any notice, request, demand, statement, bill or other payment provided for in this Agreement, or any notice which a Party may desire to give to the other, shall be considered duly delivered as of the earlier of the date of the receipt by the addressee or three (3) business days after the postmark date when mailed by ordinary mail or given to the addressee at the addresses listed below:

CONTACT INFORMATION

Buyer Notices:

[_____]
ATTN: [_____]
[_____]
[_____]
Phone: [_____]
Fax: [_____]
Email: [_____]

Buyer Billings:

[_____]
ATTN: [_____]
[_____]
[_____]
Phone: [_____]
Fax: [_____]
Email: [_____]

Buyer Gas Control Center:

[_____]
ATTN: [_____]
[_____]
[_____]
Phone: [_____]
Fax: [_____]
Email: [_____]

Seller: Notices and Billings

[_____]
ATTN: [_____]
[_____]
[_____]
Phone: [_____]
Fax: [_____]
Email: [_____]

IN WITNESS WHEREOF, the Parties hereto have caused this Agreement to be executed by their duly authorized representatives as of the Effective Date.

[_____Buyer_____]
By: [_____]
Name: [_____]
Title: [_____]

[_____Seller_____]

By: [_____]
Name: [_____]
Title: [_____]

Exhibit 32
List of Electric Generators by Territory

Generator	Fed From
KEDLI	
Bayswater Far Rockaway	Distribution System
Bethpage (Calpine)	Transmission System
Bethpage Energy Center	Transmission System
Bethpage Peaker	Transmission System
Caithness	Transmission System
Caithness Merchant	Transmission System
Edgewood (PPL)	Transmission System
EF Barrett Power Station (Steam)	Transmission System
EF Barrett Gas Turbines	Transmission System
Freeport Eqqus	Transmission System
Village of Freeport	Transmission System
Glenwood Energy Center	Transmission System
Holtsville (NYPA)	Transmission System
Nissequogue NCP	Transmission System
Northport Power Station (Steam)	Transmission System
Pilgrim (NYPA)	Transmission System
Pinelawn	Transmission System
Port Jefferson Power Station (Steam)	Transmission System
Port Jefferson Energy Center	Transmission System
TBG	Transmission System
Trigen NDEC	Transmission System
KEDNY	
Narrows Facility (US Power)	Transmission System
Gowanus Facility	Transmission System
Arthur Kill Facility (NRG)	Transmission System
23rd & 3rd Avenue Facility (NYPA)	Transmission System
Grand Avenue (Kent) Facility (NYPA)	Transmission System
Pouch Terminal Facility (NYPA)	Transmission System
KIAC Facility (Calpine)	Transmission System
Brooklyn Navy Yard	Transmission System
NIMO	
Burrstone Energy LLC	Distribution System
Carr St Generating	Transmission System
Carthage Energy	Transmission System
Castleton Power LLC	Transmission System
Indeck Corinth	Transmission System
Indeck Energy Oswego	Transmission System
NRG Oswego Harbor Power	Transmission System
Oneida Indian Nation	Distribution System
Rensselaer Cogeneration, LLC	Transmission System
Rensselaer Polytechnic Institute	Distribution System
Sithe Independence	Transmission System
Sterling Power	Transmission System
WPS Syracuse Generating	Transmission System

Case 18-M-0272

Exhibit 33
The Brooklyn Union Gas Company
Table 8a Backup

Units ($ per dt)	Nov-2016	Dec-2016	Jan-2017	Feb-2017	Mar-2017	Total
Normalized Use per Customer - dt per month	10	16	18	15	13	**72**
Cost of Gas (after losses)						
Average Demand Cost of Gas	$ 1.72093	$ 1.72093	$ 1.78409	$ 1.77286	$ 1.77286	
Average Variable Cost of Gas	$ 1.58377	$ 2.45184	$ 3.04514	$ 2.80784	$ 2.06114	
Gross Monthly Cost of Gas	**$ 3.30470**	**$ 4.17277**	**$ 4.82923**	**$ 4.58070**	**$ 3.83400**	
Annual Reconciliation	$ 0.08922	$ 0.08922	$ 0.25000	$ 0.25000	$ 0.25000	
Monthly Cost of Gas Adjustment	$ -	$ 0.30000	$ -	$ 0.50000	$ 1.10000	
Pipeline Refund Provision	$ -	$ -	$ -	$ -	$ -	
Research & Development Charge	$ -	$ -	$ -	$ -	$ -	
Merchant Function Charge (MFC)	$ 0.25067	$ 0.25067	$ 0.16670	$ 0.17137	$ 0.16865	
Annual MFC Reconciliation	$ (0.03938)	$ (0.03938)	$ 0.01021	$ 0.01021	$ 0.01021	
System Performance Adjustment	$ 0.01632	$ 0.01632	$ 0.00120	$ 0.00120	$ 0.00120	
Net Monthly Cost of Gas	**$ 3.62153**	**$ 4.78960**	**$ 5.25735**	**$ 5.51348**	**$ 5.36406**	
Check	**$ -**	**$ -**	**$ -**	**$ -**	**$ -**	
Normalized Monthly Gas Costs	$ 36.88	$ 75.06	$ 93.26	$ 84.78	$ 69.90	**$ 359.87**
w/GRT and MTA	$ 37.83	$ 77.00	$ 95.66	$ 86.97	$ 71.70	**$ 369.17**

Units ($ per dt)	Nov-2017	Dec-2017	Jan-2018	Feb-2018	Mar-2018	Total
Normalized Use per Customer - dt per month	10	15	18	15	13	**72**
Cost of Gas (after losses)						
Average Demand Cost of Gas	$ 2.00929	$ 2.00929	$ 2.00200	$ 2.00200	$ 2.00200	
Average Variable Cost of Gas	$ 1.91302	$ 2.57610	$ 2.36804	$ 2.69601	$ 2.23833	
Gross Monthly Cost of Gas	**$ 3.92231**	**$ 4.58539**	**$ 4.37004**	**$ 4.69801**	**$ 4.24033**	
Annual Reconciliation	$ 0.73841	$ 0.73841	$ 0.03973	$ 0.03973	$ 0.03973	
Monthly Cost of Gas Adjustment	$ -	$ 0.20000	$ 0.20000	$ 1.00000	$ 1.37000	
Pipeline Refund Provision	$ -	$ -	$ -	$ -	$ -	
Research & Development Charge	$ -	$ -	$ 0.01317	$ 0.01317	$ 0.01317	
Merchant Function Charge (MFC)	$ 0.16303	$ 0.17904	$ 0.16209	$ 0.18301	$ 0.18138	
Annual MFC Reconciliation	$ 0.01021	$ 0.01021	$ 0.00820	$ 0.00820	$ 0.00820	
System Performance Adjustment	$ 0.00120	$ 0.00120	$ 0.01889	$ 0.01889	$ 0.01889	
Net Monthly Cost of Gas	**$ 4.83516**	**$ 5.71425**	**$ 4.81212**	**$ 5.96101**	**$ 5.87170**	
Check	**$ -**	**$ -**	**$ -**	**$ -**	**$ -**	
Normalized Monthly Gas Costs	$ 49.28	$ 88.41	$ 85.71	$ 92.04	$ 76.83	**$ 392.27**
w/GRT and MTA	$ 50.59	$ 90.76	$ 87.98	$ 94.48	$ 78.87	**$ 402.67**

Case 18-M-0272

Exhibit 34
KeySpan Gas East Corporation
Table 8b Backup

Units ($ per dt)	Nov-2016	Dec-2016	Jan-2017	Feb-2017	Mar-2017	Total
Normalized Use per Customer - dt per month	10.2	15.7	17.7	15.3	13.1	**72**
Cost of Gas (after losses)						
Average Demand Cost of Gas	$ 1.53198	$ 1.53198	$ 1.66722	$ 1.65608	$ 1.65613	
Average Variable Cost of Gas	$ 1.60374	$ 2.48276	$ 3.02141	$ 2.78596	$ 2.04507	
Gross Monthly Cost of Gas	**$ 3.1357**	**$ 4.0147**	**$ 4.6886**	**$ 4.4420**	**$ 3.7012**	
Annual Reconciliation	$ (0.33724)	$ (0.33724)	$ 0.00616	$ 0.00616	$ 0.00616	
Monthly Cost of Gas Adjustment	$ -	$ 0.30000	$ -	$ 0.26000	$ 0.30000	
Pipeline Refund Provision	$ -	$ -	$ -	$ -	$ -	
Research & Development Charge	$ -	$ -	$ -	$ -	$ -	
Merchant Function Charge (MFC)	$ 0.15345	$ 0.15345	$ 0.14363	$ 0.14387	$ 0.13136	
Annual MFC Reconciliation	$ (0.03240)	$ (0.03240)	$ (0.00506)	$ (0.00506)	$ (0.00506)	
Net Monthly Cost of Gas	**$ 2.9195**	**$ 4.0985**	**$ 4.8334**	**$ 4.8470**	**$ 4.1337**	
Check	**$ -**	**$ -**	**$ -**	**$ -**	**$ -**	
Normalized Monthly Gas Costs	$ 29.90	$ 64.18	$ 85.53	$ 74.39	$ 53.97	**$ 307.97**
w/GRT and MTA	$ 30.25	$ 64.93	$ 86.53	$ 75.26	$ 54.60	**$ 311.56**

Units ($ per dt)	Nov-2017	Dec-2017	Jan-2018	Feb-2018	Mar-2018	Total
Normalized Use per Customer - dt per month	10.2	15.5	17.8	15.4	13.1	**72**
Cost of Gas (after losses)						
Average Demand Cost of Gas	$ 1.88147	$ 1.88147	$ 1.87791	$ 1.87791	$ 1.87791	
Average Variable Cost of Gas	$ 1.89812	$ 2.55602	$ 2.35368	$ 2.67966	$ 2.22475	
Gross Monthly Cost of Gas	**$ 3.7796**	**$ 4.4375**	**$ 4.2316**	**$ 4.5576**	**$ 4.1027**	
Annual Reconciliation	$ 0.00616	$ 0.00616	$ 0.08043	$ 0.08043	$ 0.08043	
Monthly Cost of Gas Adjustment	$ -	$ 0.20000	$ 0.20000	$ 0.90000	$ 1.33000	
Pipeline Refund Provision	$ -	$ -	$ -	$ -	$ -	
Research & Development Charge	$ -	$ -	$ -	$ -	$ -	
Merchant Function Charge (MFC)	$ 0.13182	$ 0.14714	$ 0.14479	$ 0.16310	$ 0.16265	
Annual MFC Reconciliation	$ (0.00506)	$ (0.00506)	$ 0.00631	$ 0.00631	$ 0.00631	
Net Monthly Cost of Gas	**$ 3.9125**	**$ 4.7857**	**$ 4.6631**	**$ 5.7074**	**$ 5.6821**	
Check	**$ -**	**$ -**	**$ (0)**	**$ (0)**	**$ (0)**	
Normalized Monthly Gas Costs	$ 40.10	$ 74.00	$ 82.86	$ 87.95	$ 74.48	**$ 359.40**
w/GRT and MTA	$ 40.58	$ 74.89	$ 84.01	$ 89.15	$ 75.49	**$ 364.13**

Exhibit 35
Niagara Mohawk Power Coperation
Table 8c Backup

In Therms	Nov-2016	Dec-2016	Jan-2017	Feb-2017	Mar-2017	Total
Forecast Therms per Customer	106.1	151.9	174.1	149.3	132.0	**713**
Actual Therms per Customer	102.7	149.8	149.7	124.1	146.5	**673**
Average Demand Cost of Gas	$ 0.07792	$ 0.07572	$ 0.07591	$ 0.07358	$ 0.07391	
Average Commodity Cost of Gas	$ 0.14776	$ 0.23494	$ 0.28515	$ 0.26929	$ 0.21184	
Total Cost of Gas	$ 0.22567	$ 0.31065	$ 0.36107	$ 0.34287	$ 0.28575	
Factor of Adjustment	1.0184	1.0184	1.0184	1.0184	1.0184	
Gross Monthly Cost of Gas Per Therm	**$ 0.22983**	**$ 0.31637**	**$ 0.36771**	**$ 0.34918**	**$ 0.29101**	
Annual Reconciliation Per Therm	$ (0.03007)	$ (0.03007)	$ 0.00109	$ 0.00109	$ 0.00109	
Monthly Imbalance Surcharge or Refund (after LAUF):	$ -	$ -	$ 0.00509	$ 0.02546	$ 0.06620	
System Performance Adjustment (SPA)	$ 0.00129	$ 0.00129	$ (0.00078)	$ (0.00078)	$ (0.00078)	
Net Monthly Cost of Gas Per Therm	**$ 0.20105**	**$ 0.28759**	**$ 0.37311**	**$ 0.37495**	**$ 0.35752**	
Check	**$ -**	**$ -**	**$ -**	**$ -**	**$ -**	
Reconciliation Dollars- Normalized	$ (3.19)	$ (4.57)	$ 0.19	$ 0.16	$ 0.14	**($7.26)**
Reconciliation Dollars- Actual	$ (3.09)	$ (4.50)	$ 0.16	$ 0.14	$ 0.16	**($7.13)**
Actual - Total Winter Bill (w GRT)	$67.34	$95.04	$108.75	$97.58	$105.50	**$474.19**
Normalized - Total Winter Bill (w GRT)	$68.14	$95.72	$119.98	$108.32	$97.53	**$489.69**

In Therms	Nov-2017	Dec-2017	Jan-2018	Feb-2018	Mar-2018	Total	Percentage
Budgeted Therms per Customer	106.1	151.9	174.1	149.3	132.0	**713**	
Average Demand Cost of Gas	$ 0.09076	$ 0.09092	$ 0.08932	$ 0.08841	$ 0.08875		
Average Variable Cost of Gas	$ 0.19265	$ 0.24715	$ 0.24474	$ 0.28257	$ 0.22197		
Total Cost of Gas	$ 0.28341	$ 0.33807	$ 0.33406	$ 0.37098	$ 0.31072		
Factor of Adjustment	1.0184	1.0184	1.0184	1.0184	1.0184		
Gross Monthly Cost of Gas Per Therm	**$ 0.28863**	**$ 0.34429**	**$ 0.34021**	**$ 0.37781**	**$ 0.31644**		
Annual Reconciliation Per Therm	$ 0.00109	$ 0.00109	$ (0.00607)	$ (0.00607)	$ (0.00607)		
Monthly Imbalance Surcharge or Refund (after LAUF):	$ -	$ -	$ 0.03055	$ 0.07129	$ 0.14258		
System Performance Adjustment (SPA)	$ (0.00078)	$ (0.00078)	$ 0.00080	$ 0.00080	$ 0.00080		
Net Monthly Cost of Gas Per Therm	**$ 0.28894**	**$ 0.34460**	**$ 0.36549**	**$ 0.44383**	**$ 0.45375**		
	-	-	-	-	-		
Reconciliation Dollars- Normalized	$ 0.12	$ 0.17	$ (1.06)	$ (0.91)	$ (0.80)	**($2.48)**	
Normalized - Total Winter Bill (w GRT)	$78.89	$106.23	$120.34	$120.54	$112.08	**$538.08**	
Normalized - Total Winter Bill (w GRT)	$78.89	$106.23	$120.34	$120.54	$112.08	**$538.08**	**100%**
Commodity :							
Gas Cost Factor	$30.65	$52.34	$63.63	$66.29	$59.90	**$272.80**	**51%**
Annual Reco (GAC)	$0.00109	$0.00109	($0.00607)	($0.00607)	($0.00607)	**($0.02)**	**0.00%**
MFC	$2.35	$3.72	$4.40	$4.26	$3.82	**$18.56**	**3%**
Total Commodity	**$33.00**	**$56.06**	**$68.03**	**$70.55**	**$63.72**	**$291.35**	
Check	$0.00	$0.00	$0.00	$0.00	$0.00		
Delivery :							
Delivery w GRT	$42.03	$44.96	$46.60	$44.98	$43.86	**$222.43**	**41%**
GRT	$0.92	$1.00	$1.05	$1.00	$0.97	**$4.93**	**1%**
SBC	$1.62	$2.33	$2.52	$2.16	$1.91	**$10.53**	**2%**
RDM	$1.36	$1.95	$2.24	$1.92	$1.70	**$9.17**	**1.7%**
18A	$0.00	$0.00	$0.00	$0.00	$0.00	**$0.00**	**0%**
NRS	($0.05)	($0.07)	($0.08)	($0.07)	($0.06)	**($0.34)**	**-0.1%**
Total Delivery	**$45.89**	**$50.17**	**$52.31**	**$49.99**	**$48.37**	**$246.73**	
Check	$0.00	$0.00	$0.00	$0.00	$0.00		

nationalgrid

Celia B. O'Brien
Assistant General Counsel and Director

February 20, 2019

VIA HAND DELIVERY AND ELECTRONIC MAIL

Luly E. Massaro, Commission Clerk
Rhode Island Public Utilities Commission
89 Jefferson Boulevard
Warwick, RI 02888

RE: Docket 4816 - Gas Long-Range Resource and Requirements Plan for the Forecast Period 2017/18 to 2026/27
Joint Memorandum

Dear Ms. Massaro:

On behalf of National Grid[1] and the Division of Public Utilities and Carriers (Division), I enclose for filing with the Public Utilities Commission (PUC) in the above-referenced docket ten (10) copies of the Company's and the Division's Joint Memorandum outlining the parties' joint recommendations for improving the Gas Long-Range Resource and Requirements Plan as it relates to the annual Gas Cost Recovery proceeding. This filing is made in compliance with the PUC's October 30, 2018 Open Meeting decision in Docket No. 4872.

Thank you for your attention to this filing. If you have any questions concerning this transmittal, please contact me at 781-907-2153 or Rob Humm at 401-784-7415.

Very truly yours,

Celia B. O'Brien

Celia B. O'Brien

Enclosures

cc: Docket 4816 Service List
Kevin Lynch, Division
Jonathan Schrag, Division
Tom Kogut, Division
John Bell, Division
Leo Wold, Esq.

[1] The Narragansett Electric Company d/b/a National Grid (National Grid or the Company).

Certificate of Service

I hereby certify that a copy of the cover letter and any materials accompanying this certificate was electronically transmitted to the individuals listed below.

The paper copies of this filing are being hand delivered to the Rhode Island Public Utilities Commission and to the Rhode Island Division of Public Utilities and Carriers.



Joanne M. Scanlon

February 20, 2019
Date

Docket No. 4816 – National Grid's Gas Long-Range Resource Plan
Service List as of 2/5/2019

Name/Address	E-mail	Phone
Robert Humm, Esq. National Grid 280 Melrose St. Providence, RI 02907	Robert.humm@nationalgrid.com; Celia.obrien@nationalgrid.com; Joanne.scanlon@nationalgrid.com;	401-784-7415
Elizabeth D. Arangio Theodore Poe Nancy Culliford National Grid 40 Sylvan Road Waltham, MA 02541	Theodore.poe@nationalgrid.com; Elizabeth.Arangio@nationalgrid.com; Nancy.culliford@nationalgrid.com;	
Leo Wold, Esq. Division of Public Utilities & Carriers	Leo.Wold@dpuc.ri.gov; John.bell@dpuc.ri.gov; Jonathan.schrag@dpuc.ri.gov; Ronald.Gerwatowski@dpuc.ri.gov; dmacrae@riag.ri.gov; Al.mancini@dpuc.ri.gov; MFolcarelli@riag.ri.gov ;	401-780-2177
Greg Lander, President Skipping Stone, LLC 83 Pine St., Suite 101 West Peabody, MA 01960	GLander@skippingstone.com;	978-717-6140
File an original & nine (9) copies w/: Luly E. Massaro, Commission Clerk Margaret Hogan, Commission Counsel	Luly.massaro@puc.ri.gov; Alan.nault@puc.ri.gov; Patricia.lucarelli@puc.ri.gov;	401-780-2107

Public Utilities Commission 89 Jefferson Blvd. Warwick RI 02888	Margaret.hogan@puc.ri.gov; Sharon.ColbyCamara@puc.ri.gov; Todd.bianco@puc.ri.gov;	
Office of Energy Resources Christopher Kearns Nicholas Ucci	Christopher.Kearns@energy.ri.gov; Nicholas.ucci@energy.ri.gov;	

JOINT MEMORANDUM OF
THE NARRAGANSETT ELECTRIC COMPANY d/b/a NATIONAL GRID
AND THE DIVISION OF PUBLIC UTILITIES AND CARRIERS

Introduction

On October 30, 2018 in The Narragansett Electric Company d/b/a National Grid's (National Grid or the Company) 2018 Gas Cost Recovery (GCR) proceeding, Docket No. 4872, the Public Utilities Commission (PUC) ordered that National Grid and the Division of Public Utilities and Carriers (Division) (collectively, the Parties and, individually, a Party) submit a joint memorandum in Docket No. 4816 outlining each of their recommendations for improving the Long Range Gas Supply Plan (LRP) as it relates to the annual GCR filing. The Parties submit this Joint Memorandum in compliance with the PUC's October 30, 2018 ruling in Docket No. 4872.

In addition to the outline of joint recommendations below, the Parties also believe it is helpful to provide a "problem statement" and summarize the underlying causes of the problem.

Problem Statement

How can the current regulatory review processes be revised to:

(i) provide the Company assurance that it has the support of its regulators before it makes substantial financial commitments that place the Company at prudency risk from after-the-fact regulatory challenges; and

(ii) provide regulators assurance that an unreasonable financial risk is not being placed on customers to bear the financial responsibility for long-term commitments that may turn out to be too conservative or unnecessary, when other reasonable alternatives at lower cost may have been available?

Two competing interests drive the problem statement. On the one hand, the Company seeks to obtain the Division's support for commitment decisions in advance of the commitments being made. It is completely understandable to the Division why regulatory support is important, when the net present value of the commitments involve tens of millions of dollars and could put the Company at prudency risk if the Division later disagrees with the commitment decision after it is too late for the Company to shift course. On the other hand, the Division desires to have insight into the rationale and justification for these commitments, to assure that customers are not being over-committed, and stranded contract costs are not being created for the future. But the current processes do not provide enough time for the Division to adequately review the decisions in advance and the Division has not been comfortable with the level of detail provided by the Company to support the decisions in advance.

Causes of the Problem

The markets in New England have changed significantly because of capacity issues that impact the reliability and cost of gas supply. This is a very different market than what was in existence over the past two decades. In the past, the key procurement decisions have revolved around gas supply purchases that rarely gave rise to pipeline constraint issues. As a result, the types of decisions that needed to be made by the Company annually tended to relate to supply contracts and the question of whether the Company should be making short- or medium-term supply commitments, given the prevailing market conditions at the time.

However, the rising demand for natural gas has resulted in winter capacity constraints that have changed the Company's procurement decisions. As can be seen from the recent GCR dockets, as well as the Company's efforts to find a longer-term solution to substitute for the loss of the liquefied natural gas (LNG) tank in Cumberland, the Company has been faced with procurement decisions that contemplate long-term contractual commitments that will result in significant costs to customers.

In the past, the LRP filings were not controversial and tended to raise few complicated issues. But now, the Company needs to plan in a way that assures adequate capacity and delivery security under supply contracts, the magnitude and implications of which have grown substantially. As a result, the current framework and template for the Company's long-range planning is no longer sufficient for an appropriate regulatory review. Further, the short time allowed for review in the GCR docket is not conducive to a complete and reasonable review. Left in its current state of regulatory processes, the GCR could become an annual contentious process with the Division compelled to oppose Company cost-recovery on the grounds that the Company has not adequately supported its decisions. Such an annual contentious process is not in the best interest of customers or the Company.

Additionally, there are two more specific items directly affecting the Company that stem from the regional capacity constraints. First, the Company's Capacity Exempt Transportation-only customers who rely on gas suppliers to deliver firm gas supply on interstate pipeline capacity that is held by third parties now assume more risk because their gas suppliers do not have access to interstate pipeline capacity due to regional constraints. Third-party suppliers are typically unwilling to make long-term (20-year) commitments to interstate pipeline companies that are necessary to build new pipeline projects. Second, due to on-system limitations, gas growth on the Company's system has resulted in the need to deliver gas to specific take stations fed by either Tennessee Pipeline Company or Algonquin Gas Transmission. This poses challenges that need to be addressed, such as limitations for gas supply options to meet gas growth.

Outline of Potential Solution

The Parties provide the following outline to address the LRP requirements and review and the annual GCR docket:

(1) The LRP filing should take place after the winter period, using the same forecasts that will be used for the GCR docket in that year;

(2) The LRP should no longer be limited to a foundation for planning that shows how the Company plans, but should also include concrete information about how the Company is planning to address supply and capacity needs over the five-year period;

(3) The LRP should be subject to approval by the PUC;

(4) If there is a material change to the LRP after approval, the Company should be required to make a supplemental filing with the PUC with notice to the Division;

(5) A new requirement should be established through which the Company is required to seek advance approval through a filing and proceeding at the PUC for long-term commitments that meet certain triggering criteria relating to the net present value of the cost and term of commitment; and

(6) To the extent that the larger-impact commitments are addressed through the new pre-approval process and the official approval of the LRP, this should reduce the number of litigated issues that occur in the GCR. In other words, the GCR should become a proceeding that effectively reconciles costs from known and supported commitments, rather than first-impression review of decisions that create an "all-or-nothing" financial risk for either the Company or customers.

Specific elements of this proposed solution are described in more detail below.

Long-Term Commitments:

The Parties strongly believe that item No. 5 above is a critical step in providing resolution to the long-term planning issue. The Parties believe that adoption of a "Review and Approval" mechanism in connection with long-term gas supply and/or gas transportation commitments would be beneficial. Such a mechanism would:

- Allow the Company to provide the PUC and Division with a detailed description of the proposed transaction, what gives rise to the propose transaction, what alternatives have or have not been studied and why, prior to commitment;

- Provide for formal discovery so that the PUC and Division have an opportunity to fully understand the proposed transaction;

- Provide for approval by the PUC and consent from the Division (to the extent deemed to be prudent and in the best interests of the Company's customers); and

- Upon receipt of such approval, provide the Company with assurance of recovery of the proposed costs and price structure associated with such transaction.

In furtherance of the development of such a mechanism, the Parties propose the following process:

Criteria Applicable to Commitments Greater than One Year in Duration – The Parties propose that the Review and Approval mechanism would be applicable to any gas supply and/or gas transportation commitment that will have a duration in excess of one year.

Filing – Prior to committing to any such transaction, the Company shall submit a filing to the PUC, seeking approval, and to the Division, seeking consent, to any such transaction. Such filing shall include (1) a detailed description of such transaction (including term and estimated cost); (2) a description of the customer need that such transaction is intended to address; (3) a description of the range of viable alternatives that could address the customer need; (4) a description of the alternative transactions that the Company evaluated with the results of the evaluations; and (5) such other information as may be useful to the PUC and Division in connection with their evaluation of the proposed transaction.

Discovery – Following submission of the filing, the Company shall respond to discovery requests from the PUC and Division.

Timing – The Company shall make its filing at least six months prior to the date by which it seeks approval of any transaction. Discovery shall occur, at the discretion of the PUC and Division, any time following the date of such filing until the date that is one month prior to the requested approval date. The Company shall provide written responses to the discovery requests on a rolling basis as soon as possible, and no later than 14 business days of receipt of any such request. The Division shall decide on the Company's request for consent by the requested date. The PUC shall rule on the Company's request for approval by the requested date.

Short-Term Commitments:

Notification of Short Term Commitments – Any gas supply or gas transportation commitments that have a duration of one year or less and have a reservation charge or demand charge that is $1 million or greater will be submitted to the Division, accompanied by a brief description of the context for the commitment, within 15 business days of the commitment being made, to give the Division time to commence its review prior to the annual GCR filing.

Comprehensive LRP:

The Company's bi-annual LRP filing submitted in March 2018 (Docket No. 4816) needs to be fully reviewed and approved to be able to move forward with certainty on long-term planning. For example, design planning standards and related forecasting are fundamental drivers to long-term planning, and they can potentially lead to significant cost decisions. The Company and the Division will meet to review its design planning and related forecasting methodologies. If these can be approved in a timely manner, such standards and related forecasts will be used in the Company's 2019 LRP and GCR submissions. Otherwise, the Company will use its current standards and related forecasting methods in its 2019 LRP and GCR submissions while the Company and Division continue their discussions, and any modifications or updates will appear in the Company's LRP and GCR submissions in 2020 and beyond, subject to each Party's right to take whatever position it deems appropriate in any related PUC proceedings if agreement cannot be reached.

Another item that needs to be reviewed is the impact to the portfolio from the Company's largest customers (FT-1), including those that the Company partially plans for as well those that the Company does not plan for (capacity exempt customers). These important items need to be fully vetted so that both the Company and the Division are comfortable using them in the forecasting and planning process going forward.

Once the forecasting and planning process is fully reviewed and vetted, the Company will be able to incorporate the agreed-upon results into the future annual process described below, resulting in a comprehensive LRP (Comprehensive LRP).

Subsequent Annual LRP Filings:

Once the Comprehensive LRP is fully vetted and approved, the Company will incorporate all elements of the Comprehensive LRP into subsequent annual LRP filings, which will be scaled-down versions of the Comprehensive LRP, but will include concrete information about how the Company is planning to address supply and capacity needs for the upcoming winter season as well as what the Company is pursuing for the remaining four-year period.

The annual LRP filings would be submitted in June, as soon as practical, following the release of the Company's annual forecast, permitting the Company to base its annual forecast on the most recent customer usage data and prior to the Company's annual GCR filing. These annual LRP filings will include such information as:

- Retail volume forecast by rate group for normal weather;
- Retail meter count forecast by rate group for normal weather;
- Rhode Island Economic Forecast variables for normal weather;
- Wholesale volume forecast by rate group for normal and design weather;
- SENDOUT® forecasts (normal and design weather) for capacity planning purposes for volumes and costs;
- Updated portfolio information showing all changes to the portfolio (capacity/supply/LNG), including:
    - Updated Chart IV-C-2 (schematic) if any changes have occurred;
    - Updated Chart IV-C-3 (a description of the contracts within the portfolio, including expiration date and evergreen provisions);
    - A consolidated version of Sections IV.C. (Available Resources), IV.C.2. (Underground Storage Services), and IV.C.3. (Peaking Resources); and
    - A consolidated version of Section IV.C.3.b. (e.g., LNG and/or CNG Contracts);
- Detailed information on needs for upcoming winter season, including SENDOUT® analysis showing derivation of need;
- Discussion of subsequent four-years and associated need and what the Company is pursuing with potential suppliers and pipelines to meet customer requirements, as well as expected costs of options;
- Provide historic (5-10 years) and projected (out 5 years) annual wholesale load duration curves showing the following:
- Stack existing supply resources (by path) against the daily wholesale load duration curve for historic period;
    - Stack proposed supply resources (by path) against the daily wholesale load duration curves for the projected periods;
    - Stack existing supply resources (by path) against the daily wholesale load duration curves for the historic November-March period;
    - Stack proposed supply resources (by path) against the wholesale load duration curves for the projected November-March periods; and
    - The Company will endeavor to develop equivalent hourly wholesale load duration curves;

- For individually metered high load factor Transportation customers, the Company will develop aggregated annual historic (5-10 years) and projected (out 5 years) load duration curves. For those customers with hourly metering, the Company will endeavor to provide the historic (5 years) aggregated hourly load duration curve;

- The Company will provide fixed cost of existing supply resources on a dollar per dekatherm (Dth) per day basis (annualized). Once individualized, then the Company will provide the same annualized information by path;

- For each existing supply resource (by path), the Company will provide an estimated effective Fixed Cost (on a Dth per day basis) (i.e., taking into account load factor utilization) for the current period and forecasted time periods for both its normal and design weather scenario, which is the basis of the Company's decision-making;

- For each proposed supply resource (by path), the Company will provide an estimated effective Fixed Cost (on a Dth per day basis) (i.e., taking into account load factor utilization) for the current period and forecasted time periods both for its normal and design weather scenario, which is the basis of the Company's decision-making; and

- For the gas commodity for each of the next five years of projected periods (annual and November through March), the Company will provide, by month for each projected period, the dollar per Dth for the gas estimated to be used on each path under both normal and design weather. The Company will also present the effective citygate gas (variable) cost by month of each path accounting for usage rates and fuel under both normal and design weather.

Subsequent to the annual LRP submission, the Company and the Division will jointly review the LRP submission, and the Company will keep the Division abreast of its plans for the portfolio for the upcoming GCR year.

With a firm basis founded in the review of the Comprehensive LRP filing, the Parties believe that these annual LRP filings would satisfy the statutory requirement of biannual submissions and will provide the Division with sufficient time to review the GCR filing without the need to seek additional time past the GCR hearing to investigate gas costs.

GCR Filing:

The annual GCR filing will reflect the final costs and volumes that are derived from the annual LRP filings. The Company will prepare a comparison of volumes and costs presented in its GCR filing in the same form (i.e., presentation format) as its annual LRP filing from June of the same year and identify any differences. By the time the GCR is filed, these items found in the Company's LRP submission will have already been fully vetted, and the Division will only

need to review any changes that have occurred in the interim or are projected by Company to occur during the upcoming GCR period, subject to the Division's right to review and dispute any costs in the GCR that were not approved in accordance with the process identified in this Joint Memorandum or otherwise.

Company Name: Con Edison
Case Description: 2019 Con Ed Electric & Gas Rate Filings
Case: 19-E-0065, 19-G-0066

Response to EDF Interrogatories – Set EDF-6
Date of Response: 8/2/2019
Responding Witness: Gas Infrastructure Operations & Supply Panel

Question No. : 48

Please refer to the Queens Transmission Upgrade White Paper (NESE Denied), which asserts that, if the Williams Northeast Supply Enhancement ("NESE") project is not built, "with the forecasted increase in customer demand, there would not be enough pipeline supplies to meet the combined New York Facilities system demand except from the existing supply-line in lower Manhattan by the winter of 2020/21…."

a) When did Con Edison first identify that the ability to serve load growth in New York City depended on approval of the NESE project?
b) When did Con Edson first notify the Commission and/or Staff that the ability to serve load growth in New York City depended on approval of the NESE project?
c) Has Con Edison conducted an analysis of the connection between the Company's ability to serve load growth in New York City and the NESE project? If yes, please provide that analysis.
d) Please provide all documents and/or analyses addressing the connection between the Company's ability to serve load growth in New York City and the NESE project.
e) Has Con Edison conducted an analysis to assess whether contracting for NESE capacity would or would not be lower cost than the Queens Transmission Upgrade project? If yes, please provide that analysis.

Response

a) First, the Company notes that its decision to propose additional projects (*i.e.*, both the Queens and Manhattan projects) did not solely depend upon whether Williams Northeast Supply Enhancement ("NESE") project was approved. The Company has also made clear that the timing and nature of these projects would change depending upon whether NESE was approved or denied.

Each year, the Company and NGRID (together "Companies") prepare joint hydraulic models of the NY Facilities System. Starting with the models prepared in the Spring/Summer of 2018, the Companies prepared models for the forward five winter periods, with and without the NESE project entering service for the winter 2020/2021, in accordance with Williams' schedule. The

Spring/Summer of 2018 modelling exercise showed that without NESE it would be difficult to meet firm peak gas demand in NYC, but the planning assumption was that NESE would be in service in accordance with Williams' schedule.

The assumption was based on the expectation that NESE would receive the required permits on a timely basis. Con Edison is not a shipper on this project, and based its expectations on its discussions with NGRID and Williams.

In early 2019, however, this outlook began to change. NGRID communicated to the Company that it had begun taking steps to limit firm customer demand growth should the NESE project not be approved, because of growing uncertainty around the approval of state water permits.

In addition, when FERC issued its final Environmental Impact Statement for NESE at the end of January, parties opposed to NESE increased their focus on the pending state approvals in both NJ and NY. At the same time, New York State in particular announced its intention to pursue more aggressive policies geared toward reducing greenhouse gas emissions.

As a result, the Company began an intensive engineering review of how it could postpone the need for a moratorium in the NYC area under a scenario that assumed NESE did not enter service but NGRID's future peak demand growth ceased. The projects mentioned by the Company in the April Preliminary Update were the result of that review.

The ability of these projects to enable the Company to meet the most recent forecast for firm peak demand by procuring more delivered services on existing pipelines until the announced Iroquois compressor project enters service for the winter 2023-2024 was confirmed by the preliminary joint hydraulic models of the NY Facilities system prepared by CECONY and NGRID in the past month.

b) The Company has continually indicated in publicly filed documents since September 2017 that moratoriums may be necessary to continue meeting firm peak demand if a particular region was projected to exceed available supply. The Company has been keeping DPS Staff informed with respect to the potential for moratoriums contemporaneous with its public filings.

c) Please see response to part a).

d) The joint hydraulic models referenced in the response to subpart b) are confidential critical infrastructure information. Please see the Company's rebuttal testimony and associated white papers discussing the Queens and Manhattan Transmission Projects.

e) Given the Company and NGRID jointly own the NY Facilities system, the impact of NESE on the downstate Gas Supply versus firm peak demand imbalance is unchanged by which Company contracts for the capacity. Accordingly, even if the project volume had been split between Con Edison and NGRID, the Queens project would still be required.

Company Name: Con Edison
Case Description: 2019 Con Ed Electric & Gas Rate Filings
Case: 19-E-0065, 19-G-0066

Response to EDF Interrogatories – Set EDF-6
Date of Response: 8/2/2019
Responding Witness: Gas Infrastructure Operations & Supply Panel

Question No. : 50

Please refer to the Gas Infrastructure, Operations and Supply Panel Update/Rebuttal Testimony at page 136, which states that each year "Con Edison and the two downstate National Grid companies create a joint hydraulic flow model to determine the citygate supply deliveries required to meet peak firm demand under design conditions."

a) Does the Company submit the results of this modeling exercise to the Commission and/or Staff for review and/or approval?
b) Has this joint modeling exercise of the New York Facilities identified that there may not be enough pipeline supplies to meet the combined New York Facilities system demand? If yes, please provide such analysis and the date of such analysis.
c) Is the planning model referenced in this testimony the same model referred to in Con Edison's White Paper on the Queens Transmission Upgrade (NESE denied) White Paper at page 1, which notes that the modeling and forecasting of the combined New York Facilities demand will be completed this summer? If no, please explain the differences between these two modeling processes.

Response

a) Under the revised NY Facilities agreement recently entered into between Con Edison and NGRID ("Companies"), the Companies will file an updated appendix annually with the PSC. In addition, the Company reviews the results of hydraulic modeling of the NY Facilities system as well as its high, medium and low pressure systems with DPS Staff from time to time at their request.
b) Please see responses to EDF-6-48(a) & (d).
c) Yes.